# The Continuum Limit of the Correlation Functions of Wilson Loops in Four-Dimensional Lattice Yang–Mills Theory


### Abstract

We consider pure Yang–Mills theory with a compact connected simple gauge group on a four-dimensional periodic lattice of spacing $\varepsilon$, with the bare coupling fixed by Bałaban's renormalisation group so that the running coupling at a fixed unit scale has a prescribed small value. For every finite family of pairwise disjoint simple closed lattice loops, in any planes, each with the character of an arbitrary irreducible representation, we divide the expectation of the product of their Wilson loops by an explicit number for each loop, the exponential of a perimeter term and of a term for each corner, whose coefficients do not depend on the loop. The coupling is fixed independently of the representations, which enter only through the scale at which the number is evaluated. We prove that the resulting renormalised expectation is bounded uniformly in $\varepsilon$ and converges as $\varepsilon \to 0$ along the sequence $\varepsilon = uL^{-K}$. The limits are reflection positive, and they do not vanish for loops whose lengths lie below a bound which depends on their representations and tends to infinity as the coupling tends to zero.


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# 1 Introduction

The expectation of a Wilson loop in lattice Yang–Mills theory is not expected to have a limit as the lattice spacing $\varepsilon$ tends to zero. Already at the level of perturbation theory the loop acquires a self-energy proportional to its length in lattice units, so that one expects a factor of the form $\exp(-m(\varepsilon)P/\varepsilon)$ for a loop of physical length $P$, together with one further factor for each of its corners. The question that we address here is whether these are the only divergences: if one divides the expectation by a number of this form, with coefficients that do not depend on the loop, does the quotient converge?

We answer this question for the pure gauge theory on a four-dimensional torus, at small values of the renormalised coupling. The construction on which everything rests is Bałaban's renormalisation group for lattice gauge theories [8, 10, 9, 11, 12], which controls the effective densities obtained from the lattice measure by successive block averaging, uniformly in the number of steps. Bałaban's results concern the partition function and the effective actions, however, and not the expectations of observables, so that the Wilson loops have to be carried through the construction. This is what we do: we insert the loops into the bare density and follow them from step to step, as a line defect whose effect on the density, after each step, is the product of a number and of a function of the block field that is close to the Wilson loop of the coarser lattice. The numbers, collected over the steps, are the normalisation by which we divide.

## 1.1 The Lattice Theory

Let $G$ be a compact connected simple Lie group in a fixed unitary realisation $G \subset U(N)$, and write $\mathrm{tr} \coloneqq N^{-1}\,\mathrm{Tr}$ for the normalised trace. We fix an odd integer $L$, a length $u > 0$, and an integer $m_T \geq 0$. The torus is $\mathbb{T} \coloneqq (\mathbb{R}/T\mathbb{Z})^4$ with the period $T \coloneqq 2L^{m_T}u$, and for an integer $K \geq 0$ the lattice of spacing

$$\varepsilon_K \coloneqq uL^{-K} \tag{1.1}$$

is $\mathbb{T}_K \coloneqq (\varepsilon_K\mathbb{Z}/T\mathbb{Z})^4 \subset \mathbb{T}$. A bond of $\mathbb{T}_K$ is an ordered pair $b = (x, x \pm \varepsilon_K e_\mu)$ of neighbouring sites, with $e_0, \dots, e_3$ the coordinate unit vectors, and $b^{-1}$ denotes the reversed bond. A gauge field is a map $U$ from the bonds to $G$ with $U(b^{-1}) = U(b)^{-1}$. For a path $\gamma = (b_1, \dots, b_n)$ of consecutive bonds we write $U(\gamma) \coloneqq U(b_1)\cdots U(b_n)$, and $\partial p$ is the boundary of a plaquette $p$. The lattice theory at the bare coupling $g_0 > 0$ is the probability measure

$$\mathrm{d}\mu_{K,g_0}(U) \coloneqq \frac{1}{Z_{K,g_0}} \exp\Bigl(-\frac{1}{g_0^2}\sum_p (1 - \mathrm{Re}\,\mathrm{tr}\,U(\partial p))\Bigr)\,\mathrm{d}U\ , \tag{1.2}$$

where the sum runs over the unoriented plaquettes of $\mathbb{T}_K$ and $\mathrm{d}U$ is the product of the normalised Haar measures over the positive bonds.

The bare coupling is not a free parameter in the limit $K \to \infty$. Bałaban's renormalisation group assigns to $g_0$ a sequence of running couplings $g_1, \dots, g_K$, the coupling $g_k$ belonging to the lattice of spacing $L^k\varepsilon_K$, and we fix $g_0$ by the requirement that the running coupling at the unit scale $u$ has a prescribed value $\bar{g}$. More precisely, Theorem 4.1 provides, for every sufficiently small $\bar{g} > 0$ and every $K$, a bare coupling $g_0(K, \bar{g})$ for which $g_K = \bar{g}$, and constants $0 < b \leq b'$ and $C_W$ such that

$$\bar{g}^{-2} + bK - C_W \leq g_0(K,\bar{g})^{-2} \leq \bar{g}^{-2} + b'K + C_W\ . \tag{1.3}$$

The bare coupling thus tends to zero like $(\log(1/\varepsilon_K))^{-1/2}$, which is asymptotic freedom in the form in which it enters here. We write $\mathbb{E}_K$ for the expectation with respect to $\mu_{K,g_0(K,\bar{g})}$.

### 1.2 Loop Ensembles

The loops that we consider are closed lattice paths without self-intersections. Since the lattices $\mathbb{T}_K$ are nested, a path of bonds of $\mathbb{T}_q$ is also a path of bonds of $\mathbb{T}_K$ for every $K \geq q$, and it is in this sense that a loop is fixed while the cutoff is removed.

**Definition 1.1** (Loops and Ensembles)**.** Let $q \geq 0$ be an integer.

(i) A *loop at level* $q$ is a closed path $C$ of bonds of $\mathbb{T}_q$ which visits no site twice and whose $\ell^\infty$-diameter $\operatorname{diam}(C)$, measured along a lift of $C$ to $\mathbb{R}^4$, satisfies $3\operatorname{diam}(C) < T$; its *corners* are the sites at which it changes direction. A loop carries its orientation.

(ii) A *loop ensemble at level* $q$ is a finite family $\mathcal{C} = ((C_1, \chi_1), \dots, (C_n, \chi_n))$, $n \geq 1$, of pairwise vertex-disjoint loops at level $q$, each with a character $\chi_i = (\dim \pi_i)^{-1} \operatorname{Tr} \pi_i$ of an irreducible unitary representation $\pi_i$ of $G$.

The loops need not be planar, and no assumption is made on their relative position beyond disjointness. A loop at level $q$ is also a loop at every level $q' \geq q$. For a gauge field $U$ on $\mathbb{T}_K$, $K \geq q$, the Wilson loop of $(C, \chi)$ is $W_\chi(C; U) \coloneqq \chi(U(C))$, which does not depend on the site at which the closed path $C$ is started, and we write

$$W_{\mathcal{C}}(U) \coloneqq \prod_{i=1}^{n} W_{\chi_i}(C_i; U) \tag{1.4}$$

for the observable of an ensemble.

Two quantities measure the shape of a loop, and all our constants depend on the loops only through them and through the perimeters. The first is the chord–arc constant $\Lambda(C) \geq 2$ of Definition 2.4, the largest ratio between the distance of two points along the loop and their $\ell^\infty$-distance. The second is the ratio $P(C)/r(C)$ of the perimeter to the corner isolation $r(C)$ of Definition 2.2, which is the smallest of half the shortest straight segment, the least distance between two non-adjacent segments, and an eighth of the perimeter. Both are invariant under dilations, and for a square $\Lambda(C) = 2$ and $P(C)/r(C) = 8$. For an ensemble we put $\Lambda(\mathcal{C}) \coloneqq \max_i \Lambda(C_i)$.

*Remark* 1.2 (The Chord–Arc Constant and Self-Contact). The chord–arc constant measures how close a loop comes to touching itself, relative to the length of the loop between the two places which come close. Near a corner two points are close both in space and along the loop, which is why $\Lambda(C) \geq 2$ for every loop, and the constant becomes large only when two stretches of $C$ which are far apart along the loop come close to each other in space, as the two long sides of a thin rectangle do: for a rectangle with sides $a \geq b$ one finds $\Lambda(C) = \max(2, 1 + a/b)$. Since $\Lambda(C)$ does not change under dilations, the bound $\Lambda(C) \leq \Lambda_*$ keeps a loop away from contact with itself uniformly on all scales, whatever its size. This is the property which our estimates use: any two points of $C$ in a cube of side $s$ are joined by an arc of $C$ of length at most $\Lambda(C)s$, so that the loop cannot leave a cube and return to it except along a short arc, and the number of its bonds which meet the cube is at most $2\Lambda(C)(s + 1)$ (Lemma 5.3).

The restriction is also natural from the point of view of renormalisation, although nothing in the proofs depends on what follows. The perimeter factor and one factor for each corner are what perturbation theory predicts for a loop which does not meet itself [17, 13]. A loop which passes twice through the same point is not renormalised by factors of this kind: in perturbation theory its expectation mixes with those of the loops obtained by rerouting the path at the point of self-intersection [13], so that no number attached to the loop alone, of the

kind by which we divide, removes its divergences. A lattice loop in the sense of Definition 1.1 never meets itself, but as $\Lambda(C)$ grows it comes closer, relative to its own length, to a loop which does. The constants of our estimates grow with $\Lambda(C)$, and so does the lower bound $L_0(\Lambda_*)$ on the blocking factor below; we do not show that this growth is necessary.

### 1.3 The Normalisation and the Results

The renormalisation group is run from the lattice $\mathbb{T}_K$ down to a lattice whose spacing does not depend on $K$, on which the loops of an ensemble at level $q$ are paths of a fixed number of bonds. This lattice has the spacing $uL^{-j_q}$, where the *stopping index* $j_q \geq q$ of Definition 3.13 depends on $q$, on $\bar{g}$, and on the constants of the construction, and not on $K$. The run thus consists of $K' \coloneqq K - j_q$ steps. Each step $k \leq K'$ contributes to a loop $C$ a factor $\kappa_k(C,\chi) \neq 0$, and the *normalisation* of $(C,\chi)$ at the cutoff $K$ is the product

$$\kappa_K^{(q)}(C,\chi) \coloneqq \prod_{k=1}^{K-j_q} \kappa_k(C,\chi) \,, \tag{1.5}$$

defined in Definition 7.12. Its structure is the one described at the beginning: $\log \kappa_k(C,\chi)$ is the sum of a term proportional to the total length of the segments of $C$ of each orientation and of a term for each corner of $C$, with coefficients that are numbers of the step $k$ and of the character and depend neither on the loop nor on the torus.

**Definition 1.3** (Renormalised Expectation)**.** For a loop ensemble $\mathcal{C}$ at level $q$ and $K > j_q$,

$$W_K(\mathcal{C}) \coloneqq \frac{\mathbb{E}_K[W_{\mathcal{C}}]}{\prod_{i=1}^n \kappa_K^{(q)}(C_i,\chi_i)} \,. \tag{1.6}$$

Since a loop at level $q$ is also a loop at every level $q' \geq q$, the level is part of the data of an ensemble: it fixes the stopping index $j_q$ and with it the number of step factors in (1.5), while the step factors themselves do not depend on it.

The blocking factor $L$ and the other constants of the construction are fixed first, in the order of Appendix A, and the coupling $\bar{g}$ after them. These choices decide which loops the results cover, and they decide it loop by loop and through the shape alone: a loop is covered if it is not too irregular for the chosen $L$, whatever its size and its representation, and then any number of such loops may be combined. The condition on the shape is a bound on the chord–arc constant: there is an increasing function $L_0$ on $[2,\infty)$, defined in (C.1), and we fix $\Lambda_* \geq 2$ with $L \geq L_0(\Lambda_*)$.

The representations enter in another way. A character is normalised, $|\chi| \leq 1$, but the insertions which a loop produces are built from the derivatives of $\pi(U(C))$, and their size is measured by

$$c_\pi \coloneqq \sup\{\|\mathrm{d}\pi(X)\| \,|\, X \in \operatorname{Lie} G,\ \|X\| \leq 1\} \,, \tag{1.7}$$

where $\mathrm{d}\pi$ is the differential of $\pi$ on the Lie algebra $\operatorname{Lie} G \subset \mathfrak{u}(N)$ and $\|\cdot\|$ is the operator norm. This number grows with the highest weight of $\pi$: for $G = SU(2)$ and the representation of spin $j$, $c_\pi = 2j$, and $\chi(\operatorname{diag}(e^{i\phi}, e^{-i\phi})) = \sin((2j+1)\phi)/((2j+1)\sin\phi)$ is close to 1 on fluctuations $\phi$ of the order of a coupling $g$ only if $(2j+1)g$ is small. Every expansion of the construction in which a loop takes part therefore requires $c_\pi$ times the running coupling of its step to be small, and at a fixed coupling $\bar{g}$ no single bound on the running couplings achieves this for every $\pi$. The loops, however, take part only in the steps $k \leq K'$, which integrate the scales finer than the stopping scale $uL^{-j_q}$, and at these scales the running coupling is smaller than

$\bar g$ by asymptotic freedom: Section 3.6 gives $g_k \le \gamma_q$ for $k \le K'$, with a number $\gamma_q$ which does not depend on $K$ and tends to zero as $q \to \infty$, at every $\bar g$. A large representation is therefore accommodated by reading its loop at a fine enough level, that is, by stopping the run and the product of the step factors at a fine enough scale, while the coupling $\bar g$, the construction, and the step factors stay the same. The estimates see a representation only through $c_\pi$, by the bound $\|\mathrm{d}\pi(X)\| \le c_\pi \|X\|$ and its consequences for the transports $\pi(U(J))$ at complex fields, while the normalised trace of a product is bounded by the product of the operator norms; every constant of the paper which depends on the representations of an ensemble is therefore a nondecreasing function of $c_{\mathcal C} \coloneqq \max_i c_{\pi_i}$. The size of a loop at level $q$ is measured by the number $P_q(C)$ of its bonds in $\mathbb{T}_q$, so that its physical length is $P_q(C) u L^{-q}$; it enters the constants of our estimates and none of their conditions.

**Definition 1.4** (Admissible Loops)**.** Let $L$ and $\Lambda_*$ be fixed. A loop $C$ at level $q$ is *admissible* if

$$\Lambda(C) \le \Lambda_* \,. \tag{1.8}$$

Admissibility is thus a condition on the shape alone. The size of a loop enters no condition because every estimate of the paper which involves a loop is a sum, over its traversals at some scale, of quantities which are small uniformly in the loop, and only these quantities are ever required to be small: a long loop makes the sums, and with them the constants, large, but it makes none of the summands larger. Our main result is the following.

**Theorem 1.5** (Uniform Bound and Limit)**.** *Let $L$ and $\Lambda_*$ be as above. There are $\gamma_0 > 0$ and a function $q_\sharp \colon [0,\infty) \times (0,\gamma_0] \to \{0,1,2,\dots\}$, nondecreasing in each argument and, for every $c$, equal to 0 for all sufficiently small $\bar g$, which depend only on the constants of the construction and on $\Lambda_*$, such that the following holds for every $\bar g \in (0,\gamma_0]$. For every loop ensemble $\mathcal C$ at a level*

$$q \ge q_\sharp(c_{\mathcal C}, \bar g) \tag{1.9}$$

*whose loops are admissible, whatever its number $n$ of loops and the irreducible representations $\pi_i$ of its characters, there are an integer $K_0$ and $B < \infty$ such that*

$$\sup_{K \ge K_0} |W_K(\mathcal C)| \le B \qquad \text{and} \qquad \sum_{K \ge K_0} |W_{K+1}(\mathcal C) - W_K(\mathcal C)| < \infty \,. \tag{1.10}$$

*In particular the limit*

$$W(\mathcal C) \coloneqq \lim_{K \to \infty} W_K(\mathcal C) \tag{1.11}$$

*exists. The numbers $K_0$ and $B$ depend on the constants of the construction, on $\bar g$, $q$, and $T/u$, and on $\mathcal C$ through $n$, the characters, the ratios $P(C_i)/r(C_i)$, and the numbers $P_q(C_i)$.*

The level enters the theorem only through the normalisation, and for the continuum limit it is a choice without physical consequence. The loops are fixed paths in the torus, and the level of an ensemble decides only the scale $uL^{-j_q}$ at which the product (1.5) of the step factors is stopped, that is, the scale at which the normalisation is fixed. Writing $W_K^{(q)}$ for the renormalised expectation with the normalisation of the level $q$, we have, for levels $q' \ge q$ and every $K > j_{q'}$,

$$W_K^{(q')}(\mathcal C) = W_K^{(q)}(\mathcal C) \prod_{i=1}^{n} \prod_{k=K-j_{q'}+1}^{K-j_q} \kappa_k(C_i, \chi_i) \,, \tag{1.12}$$

by (1.5) and because the step factors do not depend on the level. The product runs over the $j_{q'} - j_q$ steps between the two stopping scales, a number which does not depend on $K$, and

each of its factors is the exponential of a perimeter term and of a term for each corner, with coefficients which do not depend on the loop. A change of the level is therefore a finite renormalisation of the perimeter and corner terms, whose finite parts are a matter of convention in any renormalisation of Wilson loops, and a quantity in which the normalisation cancels does not depend on the level at all: for instance $W_K(\mathcal{C})/\prod_i W_K((C_i,\chi_i))$ is the quotient $\mathbb{E}_K[W_{\mathcal{C}}]/\prod_i \mathbb{E}_K[W_{\chi_i}(C_i;\cdot)]$ whenever the denominators do not vanish. Since an ensemble at a level $q_0$ is an ensemble at every level $q \geq q_0$, the condition (1.9) restricts only this choice and not the ensembles, and the theorem covers, at every coupling $\bar{g} \leq \gamma_0$, every ensemble of admissible loops with arbitrary irreducible representations. The level required is bounded in the coupling, $q_\sharp(c,\bar{g}) \leq q_\sharp(c,\gamma_0)$, and it is given by (11.28). The restriction to irreducible representations concerns the normalisation only: the character of a reducible representation is the convex combination of the characters of its irreducible constituents, weighted by their dimensions, and the theorem applies to each term with its own normalisation.

The requirements on the coupling in Sections 3–12 are of two kinds. Those which involve no representation are upper bounds on $\bar{g}$ in terms of the constants of the construction and of $\Lambda_*$, and $\gamma_0$ is the largest number below which all of them hold. Those which involve the representations of an ensemble $\mathcal{C}$ concern only the steps $k \leq K'$ at which its loops are carried, and they are imposed as upper bounds on the number $\gamma_q$ of (3.46) in terms of $c_{\mathcal{C}}$, of $\Lambda_*$, and of the constants of the construction; we then say that $\gamma_q$ is small, depending on $c_{\mathcal{C}}$. A coupling and a level "as in Theorem 1.5" for $\mathcal{C}$ are a coupling $\bar{g} \in (0,\gamma_0]$ and a level $q$ at which $\gamma_q$ satisfies all the requirements of the second kind. The function $q_\sharp$ is computed from these requirements in Section 11.7.

The bound and the limit are proved by different means. The bound follows from an exact identity, Theorem 8.1, which expresses $W_K(\mathcal{C})$ as a convex combination, over the terms of Bałaban's expansion of the density at the stopping scale, of expectations of functions which are bounded uniformly in $K$. The limit requires a comparison of the two runs with $K$ and $K+1$ steps, which share all their lattices but the finest one, and this comparison, Theorem 11.1, is the longest part of the paper.

The limits inherit the symmetries of the lattice measures, as far as the normalisations share them. The torus has the same period in the four directions, and for $K \geq q$ the lattice $\mathbb{T}_K$ is mapped to itself by every map $h(x) = Ox + t$ of $\mathbb{T}$ in which $O$ is a signed permutation matrix and $t \in uL^{-q}\mathbb{Z}^4$. We call these maps the *symmetries of level* $q$, and we write $\mathfrak{G}_q$ for the group of those whose matrix $O$ is diagonal. Thus $\mathfrak{G}_q$ is generated by the translations by vectors of $uL^{-q}\mathbb{Z}^4$ and by the reflections $\theta_\mu$, $\mu \in \{0,\dots,3\}$, which replace $x_\mu$ by $-x_\mu$, and it contains, for every $\mu$ and every $a$ with $2a \in uL^{-q}\mathbb{Z}$, the reflection $\theta_{\mu,a}$ which replaces $x_\mu$ by $2a - x_\mu$ modulo $T$; the symmetries of level $q$ which do not belong to $\mathfrak{G}_q$ are those which exchange two axes. A symmetry $h$ of level $q$ is an isometry of $\mathbb{T}_q$ for the distance $|\cdot|_\infty$ which maps segments to segments and corners to corners, and it therefore maps a loop $C$ at level $q$ to a loop $hC$ at level $q$, with the image orientation, $\Lambda(hC) = \Lambda(C)$, $P(hC)/r(hC) = P(C)/r(C)$, and $P_q(hC) = P_q(C)$. For an ensemble $\mathcal{C}$ we write $h\mathcal{C} \coloneqq ((hC_1,\chi_1),\dots,(hC_n,\chi_n))$, with the same characters, and $\bar{\mathcal{C}}$ for the ensemble of the loops of $\mathcal{C}$ with the conjugate characters $\bar{\chi}_i$. The character $\bar{\chi}$ is that of the conjugate representation $\bar{\pi}$, whose differential $\mathrm{d}\bar{\pi}(X)$ is the complex conjugate of the matrix $\mathrm{d}\pi(X)$ and has the same operator norm, so that $c_{\bar{\pi}} = c_\pi$ and $c_{\bar{\mathcal{C}}} = c_{\mathcal{C}}$. Finally $C^{-1}$ denotes the loop $C$ with the opposite orientation.

**Corollary 1.6** (Symmetries of the Limits)**.** *Let $L$, $\Lambda_*$, $\gamma_0$, and $q_\sharp$ be as in Theorem 1.5, let $\bar{g} \in (0,\gamma_0]$, and let $\mathcal{C}$ be a loop ensemble at a level $q \geq q_\sharp(c_{\mathcal{C}},\bar{g})$ whose loops are admissible. Then the limit $W(\mathcal{C}')$ exists for each of the ensembles $\mathcal{C}'$ below, and:*

*(i)* $W(\mathcal{C})$ *does not depend on the order of the loops of* $\mathcal{C}$*;*

*(ii)* $W(g\mathcal{C}) = W(\mathcal{C})$ *for every* $g \in \mathfrak{G}_q$*;*

*(iii)* $W(\bar{\mathcal{C}}) = \overline{W(\mathcal{C})}$*, and* $W(\mathcal{C})$ *does not change if a loop* $C_i$ *of* $\mathcal{C}$ *is replaced by* $C_i^{-1}$ *and its character by* $\bar{\chi}_i$*;*

*(iv)* $W(h\mathcal{C}) = W(\mathcal{C})$ *for every symmetry* $h$ *of level* $q$ *for which there are* $g_1, \dots, g_n \in \mathfrak{G}_q$ *with* $hC_i = g_i C_i$ *for every* $i$*.*

*The same identities hold for* $W_K$ *at every* $K > j_q$*.*

Statement (ii) is the covariance under translations and under the reflections in coordinate hyperplanes, with unchanged characters, and statement (iii) is the invariance under charge conjugation. Since an ensemble at level $q$ is an ensemble at every level $q' \geq q$, statement (ii) gives the covariance under every translation of the dense group $\bigcup_q uL^{-q}\mathbb{Z}^4$, at a level at which it is a symmetry; the normalisations of two levels differ by the factors (1.12), which are the same for an ensemble and its translate. Statement (iv) applies, for instance, to the reflection which exchanges $x_\mu$ and $x_\nu$ and to an ensemble of squares in planes spanned by $e_\mu$ and $e_\nu$, since the image of such a square is also its image under $\theta_\mu$ followed by a translation. For a general ensemble we do not know whether the limits are covariant under the exchange of two axes: the lattice measures are invariant under it, but the normalisations are not shown to be, for the reason explained in Section 12.3.

The limits also inherit the reflection positivity of the lattice measures, for the reflections $\theta_{\mu,a}$ with $a \in uL^{-q}\mathbb{Z}$, which we call the *reflections of level* $q$. The points fixed by such a reflection form the two hyperplanes $\{x_\mu \equiv a\}$ and $\{x_\mu \equiv a + T/2\}$, coordinates being taken modulo $T$, and their complement consists of two open half-tori, $\{a < x_\mu < a + T/2\}$ and $\{a + T/2 < x_\mu < a + T\}$, which the reflection exchanges. The pairing of reflection positivity combines the reflection of a loop with the conjugation of its character, and so involves the ensembles $\theta\bar{\mathcal{D}}$ of the reflected loops $\theta C$, with the image orientations and the conjugate characters; we have $c_{\theta\bar{\mathcal{D}}} = c_{\mathcal{D}}$. We admit the empty ensemble, with $W(\emptyset) \coloneqq 1$.

**Corollary 1.7** (Reflection Positivity)**.** *Let* $L$*,* $\Lambda_*$*,* $\gamma_0$*, and* $q_\sharp$ *be as in Theorem 1.5, let* $\bar{g} \in (0, \gamma_0]$ *and* $q \geq 0$*, let* $\theta$ *be a reflection of level* $q$*, and let* $H_+$ *be one of the two open half-tori bounded by the points fixed by* $\theta$*. Let* $\mathcal{D}_1, \dots, \mathcal{D}_m$ *be loop ensembles at level* $q$*, possibly empty, whose loops are admissible and lie in* $H_+$*, and assume* $q \geq q_\sharp(c, \bar{g})$*, where* $c$ *is the largest of the numbers* $c_\pi$ *of their representations. Then the limits* $W(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b)$ *exist for* $0 \leq a, b \leq m$*, with* $\mathcal{D}_0 \coloneqq \emptyset$*, and:*

*(i) the matrix* $(W(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b))_{1 \leq a,b \leq m}$ *is Hermitian and positive semidefinite;*

*(ii)* $W(\theta\bar{\mathcal{D}}_a) = \overline{W(\mathcal{D}_a)}$ *for every* $a$*, and the matrix* $(W(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) - W(\mathcal{D}_a)\overline{W(\mathcal{D}_b)})_{1 \leq a,b \leq m}$ *is Hermitian and positive semidefinite.*

The statement (ii) is positivity for the connected functions, and it is not empty: at small coupling the limits do not vanish on the ensembles whose loops are short compared with a length which grows without limit as $\bar{g} \to 0$ (Proposition 12.5), and for a loop $C$ in $H_+$ to which this applies the ensemble $\mathcal{D} \coloneqq ((C, \chi))$ has $W(\mathcal{D} \cup \theta\bar{\mathcal{D}}) \geq |W(\mathcal{D})|^2 \geq 1/4$.

Finally we ask whether the limits are new, that is, whether they could be the Wilson-loop functions of a Gaussian field. The question has to be posed with some care, because the normalisation is a convention: a function which differs from $W$ by a factor $\exp(\sum_v \lambda_v N_v(C) +$

$\sum_c \zeta_{v_c,w_c}$) for each loop, where $N_v(C)$ is the total length of the segments of $C$ of orientation $v$, the second sum runs over the corners, and the coefficients do not depend on the loop, is the same object with another normalisation. For an abelian Gaussian field the answer is then contained in a single number. If $\mathsf{B}$ is a real Gaussian 1-form and $\mathsf{g}$ a charge, then $\log|\mathbb{E}[e^{i\mathsf{g}\oint_C \mathsf{B}}]| = -\frac{1}{2}\mathsf{g}^2 \operatorname{Var} \oint_C \mathsf{B}$ is a quadratic function of the covariance of $\mathsf{B}$, a change of normalisation adds to it a function which is linear in the lengths and in the numbers of corners of each type, and a suitable signed combination of these logarithms over a few rectangles and pairs of rectangles annihilates both, whatever the covariance and whatever the normalisation.

To state this precisely, we fix a level $q$, a coordinate plane spanned by $e_\mu$ and $e_\nu$, lengths $t, s \in uL^{-q}\mathbb{N}$ with $3\max(4t, s) < T$, a point $o \in uL^{-q}\mathbb{Z}^4$, and a nontrivial irreducible unitary representation $\pi$ of $G$ with the character $\chi$. For $a \in \{0, \dots, 3\}$ and $n \in \{1, \dots, 4\}$ let $I_a$ and $R_n$ be the boundaries of the rectangles $o + [at, (a+1)t]e_\mu + [0, s]e_\nu$ and $o + [0, nt]e_\mu + [0, s]e_\nu$, traversed in the sense which runs from $e_\mu$ to $e_\nu$, so that $R_1 = I_0$, and let $R_n^{(1)} \coloneqq R_n + te_\mu$ for $n \in \{2, 3\}$. The *eleven diagrams* are the ensembles $((C, \chi))$ of one loop, for $C$ one of the nine rectangles $I_0, \dots, I_3$, $R_2$, $R_3$, $R_4$, $R_2^{(1)}$, $R_3^{(1)}$, and the pairs $P_k \coloneqq ((I_0, \chi), (I_k, \chi))$ for $k \in \{2, 3\}$, whose loops are at distance $(k-1)t \geq t$; we write $C$ also for the ensemble $((C, \chi))$. They are ensembles at level $q$. For a function $W$ on the eleven diagrams with nonzero values let

$$\mathfrak{Y}(W) \coloneqq \log\Big|\frac{W(R_1)\,W(R_3)^3}{W(R_2)^3\,W(R_4)}\Big| - \log\Big|\frac{W(P_2)}{W(R_1)^2}\Big| + \log\Big|\frac{W(P_3)}{W(R_1)^2}\Big| \,. \tag{1.13}$$

A *Gaussian charge model* is a triple $(\mathsf{g}, \mathsf{X}, \kappa)$ of a number $\mathsf{g} \in \mathbb{R}$, a real Gaussian vector $\mathsf{X} = (\mathsf{X}_0, \dots, \mathsf{X}_3)$ of any mean and covariance, and a function $\kappa$ on loops with values in $\mathbb{C} \setminus \{0\}$ such that $\log|\kappa(C)| = \sum_v \lambda_v N_v(C) + \sum_c \zeta_{v_c,w_c}$ with real numbers $\lambda_v$ and $\zeta_{v,w}$ which do not depend on $C$. Its function on the eleven diagrams is $W(\mathcal{D}) \coloneqq \mathbb{E}[e^{i\mathsf{g}\mathsf{X}(\mathcal{D})}]/\prod_{C\in\mathcal{D}} \kappa(C)$, where $\mathsf{X}(\mathcal{D})$ is the sum of the $\mathsf{X}_a$ over the indices $a$ for which $I_a$ is one of the rectangles that tile the rectangle $\mathcal{D}$, or one of the two loops of the pair $\mathcal{D}$. The models with $\mathsf{X}_a = \oint_{I_a} \mathsf{B}$ for a real Gaussian 1-form $\mathsf{B}$, on a lattice which contains the rectangles or on the continuum torus, are of this kind, with $\mathsf{X}(C) = \oint_C \mathsf{B}$ for each of the nine rectangles, since the interior sides of a tiling are traversed once in each direction.

**Theorem 1.8** (The Limits Are Not Gaussian)**.** *Let $q$, the plane, $t$, $s$, $o$, and $\pi$ be as above, let $\Lambda_* \geq 1 + \max(s/t, 4t/s)$, and let $L$ be as in Theorem 1.5. There is $\bar{g}_0 > 0$, depending on $q$, $t/u$, $s/u$, $T/u$, $\pi$, and the constants of the construction, such that for every $\bar{g} \in (0, \bar{g}_0]$ the following hold.*

*(i) The limits $W(\mathcal{D})$ exist on the eleven diagrams, $|W(\mathcal{D})| \geq 1/2$, $W(I_a) = W(I_0)$, and $W(R_n^{(1)}) = W(R_n)$.*

*(ii) $\mathfrak{Y}(W) \geq \frac{1}{2} c_O \bar{g}^2$, with $c_O \coloneqq C_2(\pi) s^2 t^3/(672 e^2 T^5)$ and the Casimir number $C_2(\pi) > 0$ of (??), and $\mathfrak{Y}(W)$ is the same for the normalisations of every level $q' \geq q$.*

*(iii) No Gaussian charge model has the function $W$ on the eleven diagrams, and $W$ is not the limit on the eleven diagrams of the functions of a sequence of Gaussian charge models.*

The number $\mathfrak{Y}$ is a third difference in the width of the rectangles, corrected by the two *pair terms* $\log|W(P_k)/W(R_1)^2|$. In an abelian Gaussian model all three are of the order $\mathsf{g}^2$, and they cancel: the covariances of separated rectangles which make up the third difference also make up the pair terms. In the limits of Yang–Mills theory the pair terms are of the order $\bar{g}^4$, up to logarithms, because the term of first order of a non-abelian Wilson loop has trace zero, while the third difference is, at the order $\bar{g}^2$, the number $C_2(\pi)$ times that of the

free field, which is positive. The same mechanism makes the theorem silent on Gaussian fields with values in the Lie algebra and path-ordered holonomies, which agree with Yang–Mills theory at the second order (Section 13.2).

### 1.4 The Argument

Let us describe how the proof is organised and where its difficulties lie. Bałaban's construction is a sequence of steps, each of which integrates the fluctuations of the gauge field at one scale. The integrand is first decomposed by test functions, which decide in every region of the lattice whether the field is small there, and the outcome is a sum of terms, labelled by the regions in which some test function has failed. On the regions where all of them pass, the fluctuation integral is performed by a cluster expansion around a background field, the minimiser of the action with given block averages. On the other regions no expansion is available, and the term carries instead a small factor from the action, which a separate operation, Bałaban's **R**-operation [11, 12], later turns into terms of the effective action.

The first difficulty is one of consistency and is independent of the loops. The thresholds of the test functions are functions of the running couplings, while the running couplings are defined by the construction, and it is not known whether the coupling of Bałaban's construction as printed is a continuous function of the bare coupling, which is what one needs in order to solve $g_K = \bar{g}$. We therefore run a variant of the construction, in which every threshold, every cube size, and the depth of the **R**-operation are functions of the target $(K, \bar{g})$ alone, and in which the plaquette test functions are continuous rather than characteristic functions. Section 3 defines this variant and shows that Bałaban's estimates hold for it, and Section 4 proves the continuity of the couplings and solves for the bare coupling.

The loops enter in Section 5, which treats the first step, and in Section 6, which treats the general one. After $k$ steps the contribution of an ensemble to a term of the density is a *dressed loop*: the product of the normalisations $\kappa_1 \cdots \kappa_k$ of its loops, of the exponential of a small local functional, and of the sum of the Wilson loops of the background field and of a series of insertions along the loops, with activities that decay in the size of their supports. The heart of the paper is the statement that this form is reproduced by a step, with constants that do not depend on $k$, including on the terms with large fields and through the **R**-operation. The perimeter and corner coefficients are extracted in Section 7 from the part of the loop's contribution that survives at the trivial background, and it is here that the geometry of a general loop is used: away from its corners a loop cannot be told from a straight line, and near a corner it cannot be told from two perpendicular rays, up to errors that decay exponentially in the corner isolation.

Stopping the induction at the index $K'$ gives the identity and the bound of Section 8. For the limit we have to compare the dressed loops, and the densities themselves, of the runs with $K$ and $K + 1$ steps on their common lattices. In the regions of small fields this is a comparison of two expansions whose data differ by the one additional step, and the difference decays geometrically in the number of steps that separate a scale from the finest one. The terms with large fields require a different argument, since their test functions are evaluated on different fields in the two runs, and it is for this comparison that the test functions were made continuous. Sections 10 and 9 prove the two inputs of the comparison, and Section 11 assembles them. Section 12 derives the two corollaries and the lower bound on the limits, Section 13 proves Theorem 1.8, and Section 14 records where our reading of Bałaban's papers departs from the printed text.

The results of Bałaban's which we use in the setting in which he proved them are cited

by page and equation. Those which we use outside his hypotheses, or of which he printed no proof, are proved again in a supplement to this paper, cited as "Supplement, Proposition S.$n$".

## 2 Loops and Their Geometry

All statements of this section are about closed paths on the lattice $\mathbb{Z}^4$ with the distance $|x-y|_\infty := \max_\mu |x_\mu - y_\mu|$. They will be applied at every scale of the construction, with the lattice unit equal to the spacing of that scale, and their purpose is to replace the elementary facts about a square that such an argument uses without comment: that a short piece of it is straight or contains one corner, that few of its bonds lie in a small box, and that a piece containing half of it is large.

### 2.1 Segments and Corners

Let $\mathcal{V} := \{\pm e_0, \pm e_1, \pm e_2, \pm e_3\}$ be the set of the eight orientations. A *closed path* of length $n$ is a sequence of sites $x_0, x_1, \dots, x_n = x_0$ of $\mathbb{Z}^4$ with $v_i := x_{i+1} - x_i \in \mathcal{V}$ for $0 \le i < n$, and it is *simple* if $x_0, \dots, x_{n-1}$ are distinct. We read the indices modulo $n$. A *loop* is a simple closed path, and two loops which differ by a cyclic shift of the indices are the same loop. The bonds of the loop $C$ are the pairs $b_i = (x_i, x_{i+1})$, the *position* of $b_i$ is its initial site $x_i$, and $|C| \subset \mathbb{R}^4$ is the union of the closed unit intervals $[x_i, x_{i+1}]$. The number $P(C) := n$ of bonds is the *perimeter*.

A site $x_i$ of $C$ is a *corner* if $v_i \neq v_{i-1}$, and $\mathcal{K}(C)$ is the set of the corners. A *segment* of $C$ is a maximal run of consecutive bonds of one orientation, together with its two end sites, so that the segments are the pieces of $C$ between consecutive corners. At a corner $c = x_i$ we write $v_c := v_{i-1}$ and $w_c := v_i$ for the incoming and the outgoing orientation.

**Lemma 2.1** (Corners)**.** *Let $C$ be a loop.*

*(i) For every $v \in \mathcal{V}$, $C$ has as many bonds of orientation $v$ as of orientation $-v$. In particular $P(C)$ is even.*

*(ii) At every corner $c$ the orientations $v_c$ and $w_c$ are perpendicular.*

*(iii) $C$ has at least four segments, and as many corners as segments.*

*(iv)* $\operatorname{diam}(C) \le P(C)/2$*, where* $\operatorname{diam}(C)$ *is the $\ell^\infty$-diameter of $|C|$.*

*Proof.* For (i) we sum the orientations around the loop. Since $\sum_{i<n} v_i = x_n - x_0 = 0$, the coefficient of each $e_\mu$ in this sum vanishes, and this coefficient is the number of bonds of orientation $e_\mu$ minus the number of bonds of orientation $-e_\mu$.

For (ii), $w_c \neq v_c$ by the definition of a corner. If $w_c = -v_c$ at $c = x_i$, then $x_{i+1} = x_i - v_{i-1} = x_{i-1}$, so that a site is visited twice, which is excluded because $n \ge 3$ for a simple closed path with $v_i \in \mathcal{V}$. Two elements of $\mathcal{V}$ which are neither equal nor opposite are perpendicular.

For (iii), the orientations $v_i$ are not all equal, since otherwise $x_n \neq x_0$. So $C$ has a corner, and by (ii) at least two coordinate axes carry bonds of $C$. By (i) each of these axes carries bonds of both of its orientations, so that at least four orientations occur, and each of them is the orientation of at least one segment. The segments and the corners alternate around the loop, which gives the last assertion.

For (iv), let $R_\mu$ be the difference between the largest and the smallest value of the coordinate $x_\mu$ on $C$. The path passes from a site with the smallest value to a site with the largest value and back, and each bond changes $x_\mu$ by at most one, so that $P(C) \geq 2R_\mu$. Since $\operatorname{diam}(C) = \max_\mu R_\mu$, this is (iv). *q.e.d.*

Two segments are *adjacent* if they are consecutive around the loop, that is, if they share a corner. By Lemma 2.1(iii) two distinct segments share at most one corner, and every segment has a segment which is not adjacent to it.

**Definition 2.2** (Lengths)**.** For a loop $C$, the *shortest segment* $s(C)$ is the least number of bonds of a segment of $C$, the *separation* $\delta(C)$ is the least $\ell^\infty$-distance between two distinct non-adjacent segments of $C$, as closed subsets of $\mathbb{R}^4$, and the *corner isolation* is

$$r(C) := \min\Big(\frac{s(C)}{2},\, \delta(C),\, \frac{P(C)}{8}\Big)\,. \tag{2.1}$$

The separation is positive, and it is at least as large as the mesh of any lattice which contains the corners. This is the content of the next lemma, which also separates the loops of an ensemble from each other.

**Lemma 2.3** (Separation)**.** *Let $a \geq 1$ be an integer and $x_* \in \mathbb{Z}^4$.*

*(i) If all corners of a loop $C$ lie in $x_* + a\mathbb{Z}^4$, then every segment of $C$ has a number of bonds which is a positive multiple of $a$, and $s(C) \geq a$ and $\delta(C) \geq a$.*

*(ii) If $C$ and $C'$ are loops without a common site, all of whose corners lie in $x_* + a\mathbb{Z}^4$, then the $\ell^\infty$-distance between $|C|$ and $|C'|$ is at least $a$.*

*Proof.* We may take $x_* = 0$. A segment joins two corners which differ in one coordinate only, by a nonzero multiple of $a$, which gives the first assertion of (i) and $s(C) \geq a$.

Let now $A$ and $B$ be two closed axis-parallel intervals of $\mathbb{R}^4$ with end points in $a\mathbb{Z}^4$, $A$ parallel to $e_\mu$ and $B$ parallel to $e_\nu$. We claim that, if $A \cap B = \emptyset$, then their $\ell^\infty$-distance is at least $a$. On each of the two intervals all coordinates but one are constant and lie in $a\mathbb{Z}$. If a coordinate $x_\lambda$ with $\lambda \notin \{\mu, \nu\}$ takes different values on $A$ and on $B$, these values differ by at least $a$, and the claim holds. Assume that this is not the case. If $\mu = \nu$, then $A$ and $B$ lie on one line, where they are disjoint intervals with end points in $a\mathbb{Z}$, and so they are at least $a$ apart. If $\mu \neq \nu$, let $b_\mu$ be the value of $x_\mu$ on $B$ and $a_\nu$ the value of $x_\nu$ on $A$. The intervals meet if and only if $b_\mu$ lies in the range of $x_\mu$ on $A$ and $a_\nu$ lies in the range of $x_\nu$ on $B$. Since they do not meet, one of these fails, say the first, and then $b_\mu \in a\mathbb{Z}$ lies at distance at least $a$ from that range, whose end points are in $a\mathbb{Z}$. So $|x_\mu - y_\mu| \geq a$ for all $x \in A$ and $y \in B$, which proves the claim.

It remains to see that the claim applies. If two axis-parallel intervals with end points in $\mathbb{Z}^4$ meet, then they have a site of $\mathbb{Z}^4$ in common: on a common line two closed intervals with integer end points which meet share an integer point, and two perpendicular ones meet in the point with the coordinates $b_\mu$ and $a_\nu$ above, which is a site. A site which lies on two segments of the loop $C$ is a site $x_i$ of $C$, and, since $C$ is simple, it lies on two segments only if it is the corner between them. Two non-adjacent segments of $C$ are therefore disjoint, and the claim gives $\delta(C) \geq a$. In (ii) a segment of $C$ and a segment of $C'$ have no site in common by hypothesis, so they are disjoint, and the claim gives (ii). *q.e.d.*

### 2.2 The Chord–Arc Constant

For two points $x, y \in |C|$ let $d_C(x, y)$ be the length of the shorter of the two arcs of $|C|$ which join them. Along an arc every coordinate changes by at most the length of the arc, so that

$$|x - y|_\infty \le d_C(x, y) \le \frac{P(C)}{2} \,. \tag{2.2}$$

The constant which bounds $d_C$ from above by the $\ell^\infty$-distance is the one geometric quantity through which the shape of a loop enters most of our counts.

**Definition 2.4** (Chord–Arc Constant)**.** For a loop $C$,

$$\Lambda(C) := \sup\Big\{ \frac{d_C(x, y)}{|x - y|_\infty} \,\Big|\, x, y \in |C|,\ x \ne y \Big\} \,. \tag{2.3}$$

**Lemma 2.5** (Bounds on the Chord–Arc Constant)**.** *For every loop $C$,*

$$2 \le \Lambda(C) \le \max\Big(2,\, \frac{P(C)}{2\delta(C)}\Big) \le \max\Big(2,\, \frac{P(C)}{2r(C)}\Big) \,, \qquad |\mathcal{K}(C)| \le \frac{P(C)}{s(C)} \le \frac{P(C)}{2r(C)} \,. \tag{2.4}$$

*For a square, $\Lambda(C) = 2$ and $P(C)/r(C) = 8$.*

*Proof.* For the lower bound let $c$ be a corner, and let $x$ and $y$ be the sites of $C$ before and after $c$. They differ by one in each of two coordinates, by Lemma 2.1(ii), so that $|x-y|_\infty = 1$, while the arc through $c$ has length 2 and the other arc has length $P(C) - 2 \ge 2$, because $P(C) \ge 4$ by Lemma 2.1(iii). Hence $d_C(x, y) = 2$.

For the upper bound let $x \ne y$ be points of $|C|$. If they lie on one segment, the part of the segment between them is an arc of length $|x - y|_\infty$. If they lie on two adjacent segments, which meet at the corner $c$, let $\alpha$ and $\beta$ be their distances from $c$. The two segments are perpendicular, so that $|x - y|_\infty = \max(\alpha, \beta)$, and the arc through $c$ has length $\alpha + \beta \le 2\max(\alpha, \beta)$. In the remaining case they lie on two non-adjacent segments, so that $|x - y|_\infty \ge \delta(C)$, while $d_C(x, y) \le P(C)/2$ by (2.2). The third inequality holds because $\delta(C) \ge r(C)$. Every segment has at least $s(C) \ge 2r(C)$ bonds, and there are as many segments as corners, which gives the bound on $|\mathcal{K}(C)|$.

For a square of side $s$ we have $\delta(C) = s$ and $P(C) = 4s$, so that $\Lambda(C) = 2$ by the two bounds, and $r(C) = \min(s/2, s, s/2) = s/2$. *q.e.d.*

All the quantities of Definition 2.2 and the perimeter are multiplied by $m$ when $C$ is replaced by its image $mC$ under the dilation $x \mapsto mx$, $m \ge 1$ an integer, and so are $d_C$ and the $\ell^\infty$-distance. Hence $\Lambda(mC) = \Lambda(C)$ and $P(mC)/r(mC) = P(C)/r(C)$: the two shape constants of a loop do not depend on the scale at which it is read.

### 2.3 Arcs around a Set

The dressed loops of the later sections are sums over finite sets of positions on a loop, and what one needs to know about such a set is how much of the loop it takes to connect it. On a square, a set of positions of diameter $D < s/2$ lies in an arc of length at most $4D$. The next lemma is the form of this statement for a general loop. By an *arc* we mean a connected closed subset of $|C|$, and a *gap* of a set $S \subset |C|$ is a connected component of $|C| \setminus S$.

**Lemma 2.6** (Arcs around a Set)**.** *Let $C$ be a loop, $S$ a nonempty set of sites of $C$ of $\ell^\infty$-diameter $D$, and $x_0 \in S$. Put $\Lambda := \Lambda(C)$ and $A := \{x \in |C| \,|\, d_C(x_0, x) \le \Lambda D\}$.*

*(i)* $S \subset A$, *and* $S$ *has at most* $2\Lambda D + 1$ *elements.*

*(ii) If* $4\Lambda D < P(C)$, *then* $S$ *has exactly one gap of length greater than* $P(C)/2$. *Its complement* $I(S)$ *is the unique shortest arc which contains* $S$. *It lies in* $A$, *its length is at most* $2\Lambda D$, *and each of its points is within* $\ell^\infty$*-distance* $\Lambda D$ *of* $x_0$.

*(iii) If* $4\Lambda D \geq P(C)$, *then every point of* $|C|$ *is within* $\ell^\infty$*-distance* $2\Lambda D$ *of every point of* $S$.

*Proof.* For $x \in S$ the definition of $\Lambda$ gives $d_C(x_0, x) \leq \Lambda |x - x_0|_\infty \leq \Lambda D$, so that $S \subset A$. The sites of $C$ in $A$ are those at arc distance at most $\Lambda D$ from $x_0$ on either side, at most $2\lfloor \Lambda D \rfloor + 1$ in number if $2\Lambda D < P(C)$, and otherwise all the $P(C) \leq 2\Lambda D$ sites of $C$. This is (i).

For (ii), $A$ is the arc of length $2\Lambda D < P(C)/2$ centred at $x_0$, and its complement is an open arc $G_0$ of length greater than $P(C)/2$ which contains no point of $S$ by (i). Let $G$ be the gap of $S$ which contains $G_0$. The gaps of $S$ are pairwise disjoint, so that every other gap has length less than $P(C)/2$, and $G$ is the only gap longer than $P(C)/2$. An arc which contains $S$ and is not all of $|C|$ is the complement of an open arc inside a gap, and it is shortest when that open arc is the longest gap. So $I(S) \coloneqq |C| \setminus G$ is the unique shortest arc containing $S$, and $I(S) \subset |C| \setminus G_0 = A$. The last assertion follows from (2.2).

For (iii), two points of $|C|$ are at $\ell^\infty$-distance at most $\operatorname{diam}(C) \leq P(C)/2$ by Lemma 2.1(iv), and $P(C)/2 \leq 2\Lambda D$ by hypothesis. *q.e.d.*

In both cases of the lemma the shortest arc containing $S$, taken to be all of $|C|$ in the case (iii), has length at most $4\Lambda D$ and lies within $\ell^\infty$-distance $2\Lambda D$ of $S$. For a square these are the constants $8D$ and $4D$.

A cube which is smaller than the separation sees at most one corner of the loop, and the counts become those of a bent line.

**Lemma 2.7** (Small Cubes)**.** *Let* $C$ *be a loop and* $Q$ *a closed cube of side* $t < \delta(C)$ *with corners in* $\mathbb{Z}^4$. *Then* $Q \cap |C|$ *is empty or an arc contained in one segment or in two adjacent segments. It contains at most* $2t + 1$ *sites of* $C$, *and at most* $2t + 2$ *bonds of* $C$ *meet* $Q$.

*Proof.* Two segments which meet $Q$ are within $\ell^\infty$-distance $t < \delta(C)$ of each other, so they are equal or adjacent. Three distinct segments cannot be pairwise adjacent, because $C$ has at least four segments and adjacency is the cyclic order, so that at most two segments meet $Q$. A segment meets the convex set $Q$ in an interval, which has length at most $t$ and hence at most $t+1$ sites. If two segments meet $Q$, in points $x$ and $y$, let $c$ be their common corner, and let the segments be parallel to $e_\mu$ and $e_\nu$. Then $c$ has the coordinate $x_\nu$ of $x$, the coordinate $y_\mu$ of $y$, and the other coordinates in common with both, so that $c$ lies in the coordinate box spanned by $x$ and $y$, which is contained in the cube $Q$. Both intervals thus contain $c$, their union is an arc, and it has at most $2(t + 1) - 1$ sites. Finally, a bond which meets $Q$ has an end site in $Q$, since $Q$ has integer corners, and an arc with $m$ sites of $C$ is met by at most $m + 1$ bonds of $C$. *q.e.d.*

The next two statements are used for the pieces of a loop which are long. The first bounds the number of cubes of a partition which a piece meets, and the second shows that a piece which contains half of the loop cannot be small.

**Lemma 2.8** (Cubes along an Arc)**.** *Let $C$ be a loop, $M' \geq 1$ an integer, and consider the closed cubes $M'z + [0, M']^4$, $z \in \mathbb{Z}^4$. An arc of $n$ bonds of $C$ consists of at most $n/s(C) + 2$ straight pieces, and it meets at most*

$$\frac{8n}{M'} + \frac{16n}{s(C)} + 32 \tag{2.5}$$

*of these cubes.*

*Proof.* The straight pieces of an arc are, in the order of the arc, a part of a segment, a number of whole segments, and a part of a segment. The whole segments have at least $s(C)$ bonds each, so there are at most $n/s(C)$ of them. A straight piece of $n_i$ bonds parallel to $e_\mu$ occupies an interval $[\tau, \tau + n_i]$ of the coordinate $x_\mu$, which meets the interval $[M'j, M'(j+1)]$ only for the integers $j$ with $\tau/M' - 1 \leq j \leq (\tau + n_i)/M'$, of which there are at most $n_i/M' + 2$. Each of its three other coordinates is constant and lies in at most two of the intervals $[M'j, M'(j+1)]$. The piece therefore meets at most $8(n_i/M' + 2)$ cubes, and the sum over the at most $n/s(C) + 2$ pieces, with $\sum_i n_i = n$, is at most (2.5). *q.e.d.*

**Lemma 2.9** (Long Arcs)**.** *Let $C$ be a loop. The set of the positions of at least $P(C)/2$ consecutive bonds of $C$, together with the final site of the last of these bonds, has $\ell^\infty$-diameter at least $P(C)/(2\Lambda(C))$.*

*Proof.* By Lemma 2.1(i) the number $P(C)/2$ is an integer, so that the set contains two sites $x$ and $y$ which are $P(C)/2$ bonds apart along the loop. Both arcs between them have length $P(C)/2$, hence $d_C(x, y) = P(C)/2$, and $|x - y|_\infty \geq d_C(x, y)/\Lambda(C)$ by the definition of $\Lambda(C)$. *q.e.d.*

### 2.4 The Neighbourhood of a Corner

The coefficients of the normalisation are defined on two model paths: the bi-infinite straight line, and the bent path which consists of two perpendicular rays. We now make precise the sense in which a loop coincides with one of them near each of its bonds. For a corner $c$ of $C$ let $S_{\text{in}}(c)$ and $S_{\text{out}}(c)$ be the segments which end and begin at $c$, and let the *bent path* at $c$ be

$$B_c \coloneqq \{c - tv_c \,|\, t \in \mathbb{Z},\ t \geq 0\} \cup \{c + tw_c \,|\, t \in \mathbb{Z},\ t \geq 0\}\ , \tag{2.6}$$

traversed from the first ray to the second. It contains the sites of $S_{\text{in}}(c)$ and of $S_{\text{out}}(c)$. We distribute the bonds of $C$ among its corners: of a segment of $\ell$ bonds, the first $\lceil \ell/2 \rceil$ bonds are *assigned* to its initial corner and the last $\lfloor \ell/2 \rfloor$ bonds to its final corner. Every bond of $C$ is assigned to exactly one corner.

**Lemma 2.10** (Corner Isolation)**.** *Let $C$ be a loop, $c$ a corner of $C$, and $b$ a bond of $C$ assigned to $c$, with position $x$. Every site of $C$ or of $B_c$ which lies neither on $S_{\text{in}}(c)$ nor on $S_{\text{out}}(c)$ is at $\ell^\infty$-distance at least $\min(s(C)/2, \delta(C)) \geq r(C)$ from $x$.*

*Proof.* Let $\ell$ and $\ell'$ be the numbers of bonds of $S_{\text{in}}(c)$ and $S_{\text{out}}(c)$, let $c'$ be the initial corner of $S_{\text{in}}(c)$ and $c''$ the final corner of $S_{\text{out}}(c)$, and write $v \coloneqq v_c$ and $w \coloneqq w_c$. Since $C$ has at least four segments, the segment before $S_{\text{in}}(c)$ is not adjacent to $S_{\text{out}}(c)$, and the segment after $S_{\text{out}}(c)$ is not adjacent to $S_{\text{in}}(c)$.

Assume first that $b$ is one of the last $\lfloor \ell/2 \rfloor$ bonds of $S_{\text{in}}(c)$, so that $x = c' + \alpha v$ with $\alpha \geq \ell - \lfloor \ell/2 \rfloor = \lceil \ell/2 \rceil$. The segment before $S_{\text{in}}(c)$ is perpendicular to $v$ and contains $c'$,

so that all its sites have the same coordinate along $v$ as $c'$, which differs from that of $x$ by $\alpha \geq s(C)/2$. All other segments of $C$ apart from $S_{\text{out}}(c)$ are not adjacent to $S_{\text{in}}(c)$, and their sites are at distance at least $\delta(C)$ from $x$. The sites $c' - tv$, $t > 0$, of the first ray of $B_c$ beyond $c'$ differ from $x$ by $\alpha + t$ in the coordinate along $v$. The sites $c + tw$, $t > \ell'$, of the second ray beyond $c''$ differ from $x$ by $t > \ell' \geq s(C)$ in the coordinate along $w$, which is constant on $S_{\text{in}}(c)$.

Assume now that $b$ is one of the first $\lceil \ell'/2 \rceil$ bonds of $S_{\text{out}}(c)$, so that $x = c + iw$ with $0 \leq i \leq \lceil \ell'/2 \rceil - 1$. The segment after $S_{\text{out}}(c)$ is perpendicular to $w$ and contains $c'' = c + \ell' w$, so that its sites differ from $x$ by $\ell' - i \geq \lfloor \ell'/2 \rfloor + 1 > s(C)/2$ in the coordinate along $w$, and the sites $c + tw$, $t > \ell'$, of $B_c$ differ from it by more. All other segments of $C$ apart from $S_{\text{in}}(c)$ are not adjacent to $S_{\text{out}}(c)$, which gives the distance $\delta(C)$. The sites $c - tv$, $t > \ell$, of the first ray beyond $c'$ differ from $x$ by $t > \ell \geq s(C)$ in the coordinate along $v$, which is constant on $S_{\text{out}}(c)$. *q.e.d.*

### 2.5 A Summation over the Scales

The lengths of a loop grow by the factor $L$ from each scale to the next finer one, and the errors of the corner analysis decay exponentially in these lengths. Their sum over the scales is controlled by the following elementary bound, which we record here because it is used for every such sum.

**Lemma 2.11** (Sums over Scales)**.** *For $x > 0$, $c > 0$, and a real number $L \geq 2$,*

$$\sum_{m \geq 0} x L^m e^{-cxL^m} \leq x e^{-cx} \Big(1 + \frac{2}{cx}\Big) \qquad \text{and} \qquad \sum_{m \geq 0} e^{-cxL^m} \leq e^{-cx} \Big(1 + \frac{2}{cx}\Big) \,. \tag{2.7}$$

*Proof.* Put $y_m \coloneqq xL^m$. For $m \geq 1$ we have $y_m - y_{m-1} = y_m(1 - L^{-1}) \geq y_m/2$, and $e^{-ct} \geq e^{-cy_m}$ on the interval $[y_{m-1}, y_m]$, so that

$$y_m e^{-cy_m} \leq 2(y_m - y_{m-1}) e^{-cy_m} \leq 2 \int_{y_{m-1}}^{y_m} e^{-ct} \, \mathrm{d}t \,. \tag{2.8}$$

The intervals $[y_{m-1}, y_m]$, $m \geq 1$, are disjoint up to their end points and lie in $[x, \infty)$, so that the terms with $m \geq 1$ sum to at most $2 \int_x^\infty e^{-ct} \, \mathrm{d}t = (2/c) e^{-cx}$. Adding the term $x e^{-cx}$ of $m = 0$ gives the first bound. The second follows from the first, because $e^{-cy_m} \leq (y_m/x) e^{-cy_m}$ for $y_m \geq x$. *q.e.d.*

## 3 The Construction

This section defines the renormalisation group that the paper runs: Bałaban's steps, with every threshold, cube size, and depth fixed by the target $(K, \bar{g})$, and with continuous plaquette test functions. It shows that Bałaban's estimates hold for this construction, and it fixes the stopping index of an ensemble.

### 3.1 The Tube of a Target

Bałaban's construction is governed by a small number of parameters, each of which he defines as a function of the running coupling of its scale. The plaquette threshold of the scale $j$ is

$$\varepsilon_j = g_j p_0(g_j) \,, \qquad p_0(g) \coloneqq A_0 (\log g^{-2})^{p_0} \,, \tag{3.1}$$

and the size of the cubes of that scale is $MR_j$, where $R_j$ is the smallest power of $L$ with $R_j \geq (\log g_j^{-2})^r$ [9, (2.4), (2.5), p. 255]. Here $A_0$ is a large constant, $r \geq 2$ and $p_0 \geq 5r$ are exponents [9, pp. 245–246], and we keep his notation, in which $p_0$ denotes both the function and its exponent. The running couplings, in turn, are produced by the construction: $g_{j+1}$ is computed from the effective action which the first $j$ steps have generated. As long as one studies a given bare coupling this causes no difficulty. We, however, have to find the bare coupling for which $g_K = \bar{g}$, and for this we need the map $g_0 \mapsto g_K$ to be continuous, which is not clear when a cube size, an integer, is a function of $g_j$.

Bałaban mentions a way out. In the first step he notes that there are "two convenient choices" of the threshold, the second of which replaces the coupling inside the logarithm by an upper bound which is known in advance, and that this choice is "more convenient in some nonperturbative aspects of the procedure, but it is technically more difficult", and he assumes the first [9, p. 246]. We take this second route, in a stronger form: every parameter of the scale $j$ is evaluated at a number $\hat{g}_j$ which is fixed by the target and which the running coupling $g_j$ is shown, afterwards, to follow within a constant factor.

Throughout, $\beta_0 \in (0, \sqrt{2}-1]$ is a fixed number, the constant of Bałaban's inequalities (2.6), and $0 < b_* \leq b^*$ are fixed numbers, which Section 4 takes to be multiples of the first coefficient of the $\beta$-function.

**Definition 3.1** (Tubes)**.** Let $\gamma > 0$. A *tube* consists of an integer $K \geq 1$ and real numbers $b$, $b'$, $C_W$, and $\bar{g} > 0$ such that, with $t \coloneqq \bar{g}^{-2}$,

$$b_* \leq b \leq b' \leq \min(\tfrac{3}{2}b,\, b^*)\,, \qquad C_W \geq 0\,, \qquad t \geq \max(\gamma^{-2} + C_W,\, 4C_W/\beta_0)\,. \tag{3.2}$$

For every integer $k \leq K$ we put

$$\ell_k \coloneqq t + b(K-k) - C_W\,, \qquad u_k \coloneqq t + b'(K-k) + C_W\,, \qquad \hat{g}_k \coloneqq u_k^{-1/2}\,, \tag{3.3}$$

and for $0 \leq k \leq K$ we let $H_k$ be the set of the $(x_0, \dots, x_k) \in \mathbb{R}^{k+1}$ with $\ell_j \leq x_j \leq u_j$ for $0 \leq j \leq k$ and

$$b(n-m) - C_W \leq x_m - x_n \leq b'(n-m) + C_W \tag{3.4}$$

for $0 \leq m < n \leq k$. A point of $H_k$ is a *history* of the tube, with the couplings $g_j \coloneqq x_j^{-1/2}$. The *ratio* of the tube is

$$\rho \coloneqq \max\Big(\frac{b'}{b},\, \frac{t + C_W}{t - 2C_W}\Big)\,. \tag{3.5}$$

The set $H_k$ is closed and convex, being defined by finitely many weak linear inequalities. Since $\beta_0 < 1/2$, the last condition of (3.2) gives $C_W \leq \beta_0 t/4 < t/8$, so that $(t + C_W)/(t - 2C_W) < (9/8)/(6/8)$ and therefore

$$1 \leq \rho \leq \tfrac{3}{2}\,. \tag{3.6}$$

The numbers $\hat{g}_k$ increase with $k$, and $\hat{g}_K \leq \bar{g}$. The role of $x_j$ is that of the inverse square of the running coupling of the scale $j$, and the tube is the set in which we shall confine it: (3.4) says that the inverse square of the coupling decreases, from each scale to a coarser one, by an amount between $b$ and $b'$ per step, up to the additive constant $C_W$.

The lemmata of this subsection require $\gamma$ to be small. With $p^* \geq 1$ an upper bound for the exponents $r$, $p_0$, $p_1$, $q_0$, $q_1$ of the construction, the conditions are

$$\begin{gathered} b^*\gamma^2 \leq \frac{\beta_0}{2}\,, \qquad \log\gamma^{-2} \geq \frac{p^*}{\beta_0}\,, \qquad \Big(1 + \frac{\log(3/2)}{\log\gamma^{-2}}\Big)^{p^*} \leq 2\,, \\ \Big(1 + \frac{b^*\gamma^2}{\log\gamma^{-2}}\Big)^{r} < L\,, \qquad \Big(1 + \frac{\log(1+b^*)}{\log\gamma^{-2}}\Big)^{r} \leq 1 + \frac{1}{L}\,. \end{gathered} \tag{3.7}$$

Each of them holds for all sufficiently small $\gamma$, once $\beta_0$, $b^*$, $p^*$, $r$, and $L$ are fixed.

**Lemma 3.2** (Couplings of a History)**.** *Let $\gamma$ satisfy* (3.7)*, and let $(x_0, \dots, x_k)$ be a history of a tube, with the couplings $g_j = x_j^{-1/2}$. Put $\beta' \coloneqq b' + C_W$.*

*(i) For $0 \le j \le k$,*

$$\hat g_j \le g_j \le \rho^{1/2} \hat g_j \,, \qquad g_j \le \gamma \,, \qquad \log g_j^{-2} \le \log \hat g_j^{-2} \le \log g_j^{-2} + \log \rho \,. \tag{3.8}$$

*(ii) For $0 \le m < n \le k$,*

$$g_n \le (1+g_n^2\beta'(n-m))^{1/2} g_m \le (1+\beta_0)(n-m)^{1/2} g_m \qquad \textit{and} \qquad g_m \le (1+\beta_0) g_n \,, \tag{3.9}$$

*and, for every real $p \in (0, p^*]$,*

$$(\log g_n^{-2})^p \le (1+\beta_0)(\log g_m^{-2})^p \qquad \textit{and} \qquad (\log g_m^{-2})^p \le (1+g_n^2\beta'(n-m))^{\beta_0} (\log g_n^{-2})^p \,. \tag{3.10}$$

*(iii) The inequalities* (3.9) *and* (3.10) *hold for the numbers $\hat g_j$ in place of the $g_j$, with $b'$ in place of $\beta'$ and for all integers $m < n \le K$, and moreover $\hat g_m \le \hat g_n$.*

The inequalities (3.9) and (3.10) are Bałaban's (2.6) and (2.7) [9, p. 255], which he states for the couplings of his construction and for positive integers $p$. The lemma thus says that a history of a tube, and also the sequence $(\hat g_j)$ itself, satisfy the hypotheses which he places on a sequence of couplings.

*Proof.* (i) Since $x_j \le u_j$ we have $\hat g_j \le g_j$. With $s \coloneqq K - j \ge 0$,

$$\frac{u_j}{x_j} \le \frac{u_j}{\ell_j} = \frac{(t + C_W) + b's}{(t - C_W) + bs} \le \max\Big(\frac{t + C_W}{t - C_W},\, \frac{b'}{b}\Big) \le \rho \,, \tag{3.11}$$

where the first step uses $x_j \ge \ell_j > 0$, the second that a quotient $(A + A')/(B + B')$ of sums of positive numbers lies between $A/B$ and $A'/B'$, and the third the definition (3.5). Hence $g_j^2 = x_j^{-1} \le \rho u_j^{-1} = \rho \hat g_j^2$. Moreover $x_j \ge \ell_j \ge t - C_W \ge \gamma^{-2}$ by (3.2), so that $g_j \le \gamma$, and the last chain of (3.8) is the logarithm of the first.

(ii) The upper bound of (3.4) and $n - m \ge 1$ give $x_m \le x_n + \beta'(n - m)$, that is, $g_m^{-2} \le g_n^{-2}(1 + g_n^2\beta'(n - m))$, which is the first inequality of (3.9). For the second we use $1+y(n-m) \le (1+y)(n-m)$ for $y \ge 0$, and bound $y = g_n^2\beta'$: by (i) and (3.2), $g_n^2 \le (t-C_W)^{-1}$ with $C_W \le \beta_0 t/4$ and $t \ge \gamma^{-2}$, so that

$$g_n^2 \beta' \le \frac{b' + C_W}{t - C_W} \le \frac{b^*\gamma^2 + \beta_0/4}{1 - \beta_0/4} \le \frac{3\beta_0/4}{1 - \beta_0/4} \le \beta_0 \,, \tag{3.12}$$

by the first condition of (3.7) and $\beta_0 \le 1$. Thus $(1 + g_n^2\beta')^{1/2} \le 1 + \beta_0$. For the third inequality, the lower bound of (3.4) gives $x_n \le x_m + C_W$, and $C_W \le \beta_0 t/4 \le (\beta_0/3) x_m$, because $x_m \ge t - C_W \ge (1 - \beta_0/4)t \ge 3t/4$. Hence $g_m^2 \le (1 + \beta_0/3) g_n^2$.

We turn to (3.10). Put $\lambda_j \coloneqq \log g_j^{-2}$, which is at least $\log \gamma^{-2} \ge p/\beta_0 \ge 2p$ by (i), the second condition of (3.7), and $\beta_0 \le 1/2$. From $g_m \le (1+\beta_0) g_n$ we get $\lambda_n \le \lambda_m + 2\log(1+\beta_0)$, and therefore

$$\Big(\frac{\lambda_n}{\lambda_m}\Big)^p \le \Big(1 + \frac{2\log(1+\beta_0)}{\lambda_m}\Big)^p \le (1+\beta_0)^{2p/\lambda_m} \le 1 + \beta_0 \,, \tag{3.13}$$

using $1 + x \le e^x$ and $2p \le \lambda_m$. In the same way the first inequality of (3.9) gives $\lambda_m \le \lambda_n + \log(1 + y)$ with $y := g_n^2 \beta'(n - m)$, so that $(\lambda_m/\lambda_n)^p \le (1 + y)^{p/\lambda_n} \le (1 + y)^{\beta_0}$, because $p/\lambda_n \le \beta_0$.

(iii) Since $u_m = u_n + b'(n-m)$, the numbers $\hat{g}_j$ are increasing and $\hat{g}_m^{-2} = \hat{g}_n^{-2}(1 + \hat{g}_n^2 b'(n - m))$, which is the first inequality of (3.9) with equality. The remaining inequalities follow as in (ii), with $\hat{g}_n^2 b' \le b^* \gamma^2 \le \beta_0$ in place of (3.12) and $\log \hat{g}_j^{-2} \ge \log \gamma^{-2}$, which holds because $u_j \ge t \ge \gamma^{-2}$. *q.e.d.*

We can now define the parameters of the construction for a tube. They are Bałaban's functions of the coupling, evaluated at $\hat{g}_j$.

**Definition 3.3** (Parameters of a Tube)**.** For a tube and every integer $j \le K$:

(i) $R_j$ is the smallest power of $L$ with $R_j \ge (\log \hat{g}_j^{-2})^r$;

(ii) $\varepsilon_j := \hat{g}_j p_0(\hat{g}_j)$, with $p_0(g) = A_0(\log g^{-2})^{p_0}$, and $\vartheta_j := (A_1/A_0)\varepsilon_j$;

(iii) $\delta'_j := \hat{g}_j A_1 (\log \hat{g}_j^{-2})^{p_1}$;

(iv) $\alpha_{0,j} := \hat{g}_j C_0^\alpha (\log \hat{g}_j^{-2})^{q_0}$ and $\alpha_{1,j} := \hat{g}_j C_1^\alpha (\log \hat{g}_j^{-2})^{q_1}$.

The printed definitions are those of (3.1) for $\varepsilon_j$ and $R_j$; $\delta_k = g_k(A_1/A_0)p_0(g_k)$ for the threshold of the fluctuation field, which we write $\vartheta_k$ [9, p. 265, after (3.4)]; $\delta'_j = g_j A_1 p_1(g_j)$ with $p_1(g) = (\log g^{-2})^{p_1}$ and $p_1 < p_0$ for the threshold of the **R**-operation [11, p. 183]; and (2.28) for the radii $\alpha_{0,j}$, $\alpha_{1,j}$ of the domains of analyticity, with integers $q_0, q_1 > 1$ and large constants which he writes $C_0$, $C_1$ [9, p. 259]. A parameter of the scale $j$ of a tube depends on $K$ and $j$ only through the difference $K - j$, since $u_j$ does. This will be essential in Section 11: the runs for the targets $(K, \bar{g})$ and $(K + 1, \bar{g})$ have the same parameters on each of the lattices which they share.

**Lemma 3.4** (Cube Sizes of a Tube)**.** *Let $\gamma$ satisfy* (3.7)*. For every tube and every history $(x_0, \dots, x_k)$ of it:*

*(i)* $(\log g_j^{-2})^r \le R_j < L(\log \hat{g}_j^{-2})^r \le 2L(\log g_j^{-2})^r$ *for* $0 \le j \le k$*;*

*(ii)* $R_{j+1} \in \{R_j, L^{-1}R_j\}$ *for every integer* $j < K$*, so that* $R_n \le R_m$ *for* $m < n$*;*

*(iii) for all integers* $m < n \le K$*,*

$$R_m \le L(1 + \hat{g}_n^2 b'(n - m))^{\beta_0} R_n \qquad \text{and} \qquad R_m \le (L + 1)(n - m)^{\beta_0} R_n \,. \tag{3.14}$$

Part (i) is Bałaban's (2.5), with a second inequality in which the factor $2L$ stands where his definition would give $L$, and (ii) and (iii) contain his (2.9) [9, pp. 255–256], since $\hat{g}_n^2 b' \le g_n^2 \beta'$.

*Proof.* (i) The lower bound is the definition of $R_j$ and $\hat{g}_j \le g_j$. Since $\log \hat{g}_j^{-2} \ge \log \gamma^{-2} > 1$, the smallest power of $L$ which is at least $(\log \hat{g}_j^{-2})^r$ is less than $L$ times this number. Finally $\log \hat{g}_j^{-2} \le \log g_j^{-2} + \log \rho$ by Lemma 3.2(i), with $\rho \le 3/2$ and $\log g_j^{-2} \ge \log \gamma^{-2}$, so that the third condition of (3.7) gives $(\log \hat{g}_j^{-2})^r \le 2(\log g_j^{-2})^r$.

(ii) The numbers $u_j$ decrease, hence $R_{j+1} \le R_j$. Moreover $u_j = u_{j+1} + b' \le (1 + b^*\gamma^2)u_{j+1}$, because $u_{j+1} \ge \gamma^{-2}$, so that

$$R_j < L(\log u_j)^r \le L\Big(1 + \frac{\log(1 + b^*\gamma^2)}{\log u_{j+1}}\Big)^r (\log u_{j+1})^r < L^2 R_{j+1} \,, \tag{3.15}$$

by the fourth condition of (3.7) and $(\log u_{j+1})^r \le R_{j+1}$. The quotient $R_j/R_{j+1}$ is a power of $L$ in $[1, L^2)$, so it is 1 or $L$.

(iii) Put $s := n - m$ and $\lambda := \log \gamma^{-2}$. As in (i), $R_m < L(\log u_m)^r$ and $(\log u_n)^r \le R_n$, and $u_m = u_n(1 + \hat g_n^2 b' s)$, so that

$$\frac{R_m}{R_n} < L\Big(1 + \frac{\log(1 + \hat g_n^2 b' s)}{\log u_n}\Big)^r \,. \tag{3.16}$$

For the first inequality we bound the bracket by $(1 + \hat g_n^2 b' s)^{r/\log u_n}$ and use $r/\log u_n \le r/\lambda \le \beta_0$. For the second we use $\hat g_n^2 \le \gamma^2$, $b' \le b^*$, and $\log u_n \ge \lambda$, which bound the bracket by $(1 + \log(1 + b^* s\gamma^2)/\lambda)^r$, and we distinguish two cases. If $s \le \gamma^{-2}$, this is at most $(1 + \log(1 + b^*)/\lambda)^r \le 1 + 1/L$, by the last condition of (3.7), and $1 \le s^{\beta_0}$. If $s > \gamma^{-2}$, then $1 + b^* s\gamma^2 \le (1 + b^*)s\gamma^2$, and with $y := \log s > \lambda$ the bracket is at most $\varphi(y) := ((y + \log(1 + b^*))/\lambda)^r$. We have $\varphi(\lambda) \le 1 + 1/L$, and the logarithmic derivative of $\varphi$ is $r/(y + \log(1 + b^*)) \le r/\lambda \le \beta_0$, so that $\varphi(y) \le (1 + 1/L)e^{\beta_0(y-\lambda)} \le (1 + 1/L)s^{\beta_0}$. In both cases $R_m/R_n < L(1 + 1/L)s^{\beta_0}$. *q.e.d.*

**Lemma 3.5** (Thresholds of a Tube)**.** *Let $\gamma$ satisfy* (3.7)*, and let $\tau_j$ be one of the parameters $\varepsilon_j$, $\vartheta_j$, $\delta'_j$, $\alpha_{0,j}$, $\alpha_{1,j}$ of a tube, so that $\tau_j = \tau(\hat g_j)$ with $\tau(g) = cg(\log g^{-2})^p$ for a constant $c > 0$ and an exponent $p \in (0, p^*]$.*

*(i) For every history of the tube and $0 \le j \le k$,*

$$\rho^{-1/2}\tau(g_j) \le \tau_j \le \tau(g_j) \,. \tag{3.17}$$

*(ii) For all integers $m < n \le K$,*

$$\tau_n \le (1+\beta_0)(n-m)^{1/2}\tau_m \qquad \text{and} \qquad \tau_m \le (1+\hat g_n^2 b'(n-m))^{\beta_0}\tau_n \le (1+\beta_0)(n-m)^{\beta_0}\tau_n \,. \tag{3.18}$$

For $\tau = \varepsilon$ the inequalities (3.18) are Bałaban's (2.8) [9, p. 256]. The upper bound of (3.17) says that the thresholds of a tube are the smaller ones: a statement which uses a threshold only as an upper bound on a field which has passed a test function holds for the tube with the printed value. The lower bound controls the loss in the statements which use a threshold from below.

*Proof.* (i) The function $\tau$ is increasing on $(0, \gamma]$, since its logarithmic derivative is $g^{-1}(1 - 2p/\log g^{-2})$ and $\log g^{-2} \ge \log \gamma^{-2} \ge p^*/\beta_0 > 2p$. As $\hat g_j \le g_j \le \gamma$ by Lemma 3.2(i), this gives the upper bound. For the lower bound we write $\tau_j = (\hat g_j/g_j)\, cg_j(\log \hat g_j^{-2})^p$ and use $\hat g_j \ge \rho^{-1/2} g_j$ and $\log \hat g_j^{-2} \ge \log g_j^{-2}$ from the same lemma.

(ii) We have $\tau_n/\tau_m = (\hat g_n/\hat g_m)(\log u_n/\log u_m)^p \le \hat g_n/\hat g_m$, because $u_n \le u_m$, and $\hat g_n/\hat g_m \le (1 + \beta_0)(n - m)^{1/2}$ by Lemma 3.2(iii). In the other direction $\tau_m/\tau_n = (\hat g_m/\hat g_n)(\log u_m/\log u_n)^p \le (\log u_m/\log u_n)^p$, which is at most $(1 + \hat g_n^2 b'(n - m))^{\beta_0}$ by the second inequality of (3.10) for the sequence $(\hat g_j)$. The last step is $1 + y(n-m) \le (1+y)(n-m)$ with $y = \hat g_n^2 b' \le \beta_0$ and $(1 + \beta_0)^{\beta_0} \le 1 + \beta_0$. *q.e.d.*

### 3.2 The Density after $k$ Steps

We recall the form which Bałaban gives to the density of the block field after $k$ steps, since every later section works inside it. The description is his inductive assumption [9, Sect. 2], and we cite each of its parts by page and equation. Nothing in this subsection is new.

At the step $k$ we use his units, in which the lattice $T^{(k)}$ of the block field $V_k$ has the spacing one and the fine lattice $T_\eta$, which is our $\mathbb{T}_K$, has the spacing $\eta = L^{-k}$. A *scale* $j \le k$ refers to the lattice of spacing $L^{-(k-j)}$ in these units, the lattice on which the block field of the step $j$ lives. For the localisation one fixes $M = L^m$ and, at each scale $j$, the family $\pi_j$ of the closed cubes of side $M$ of that scale. A *localisation domain* of the scale $j$ is the union of a finite family of cubes of $\pi_j$ which is connected, in the sense that any two of its cubes are joined by a chain of cubes of the family in which consecutive cubes have a common wall. The class of these domains is $\mathbf{D}_j$, and the *linear size* $d_j(X)$ of $X \in \mathbf{D}_j$ is the length of a shortest tree graph in $X$ which meets all the cubes of $X$, divided by $M$ [8, p. 257].

*The terms of the expansion.* A term of the density is labelled by three sequences of sets. The first two are sequences of localisation domains $\Omega_j, \Lambda_j \in \mathbf{D}_j$ with

$$\Omega_1 \supset \Lambda_1 \supset \Omega_2 \supset \Lambda_2 \supset \cdots \supset \Omega_k \supset \Lambda_k \,, \tag{3.19}$$

and the third consists of sets $S_j \subset \Omega_j \cap \Lambda_j^c$, each of which is empty or a union of cubes of side $LM_2R_j$ of the scale $j$ [9, (2.1), p. 254]. Roughly speaking, $\Omega_j$ is the region in which the field of the scale $j$ has passed the test functions of the step $j$, so that the step could be performed there, and $\Lambda_j$ is the smaller region in which the fluctuation field of that step was small as well. The complements

$$Z_j \coloneqq \Lambda_j^c \tag{3.20}$$

are the *large-field regions* [9, (2.3), p. 255]. The sequence $\{\Omega_j\}$ determines on which lattice the field is still an integration variable at each point: with

$$\Gamma_0 \coloneqq \Omega_1^c \,, \qquad \Gamma_j \coloneqq \Omega_j^{(j)} \setminus \Omega_{j+1}^{(j)} \quad (1 \le j < k) \,, \qquad \Gamma_k \coloneqq \Omega_k^{(k)} \,, \qquad \mathbf{B} \coloneqq \bigcup_{j=0}^{k} \Gamma_j \,, \tag{3.21}$$

where the superscript $(j)$ denotes the points, bonds, and plaquettes of the scale $j$ in a set, the *determining set* $\mathbf{B}$ carries the field $V$ which equals the block field $V_j$ of the step $j$ on $\Gamma_j$ [9, (2.2), (2.10), pp. 255–256]. The domains $\Omega_j$ and $\Lambda_j$ which are determined by the step $j$, and not by an $\mathbf{R}$-operation, are unions of cubes of side $MR_j$ of the scale $j$, and the distance between their boundaries is at least $2MR_j$ [9, p. 256].

*The background field.* Let $M_{\mathbf{B}}$ be the averaging operation which maps a gauge field $U$ on $T_\eta$ to the field on $\mathbf{B}$ equal to the $j$-fold block average $M^j(U)$ on $\Gamma_j$. A regular field $V$ on $\mathbf{B}$ determines the set of the minima of the action $U \mapsto A^\eta(U)$ on the regular fields $U$ with $M_{\mathbf{B}}(U) = V$, which is a single gauge orbit, and a minimal configuration is denoted by $U_k(V)$, or by $U_k$ [9, (2.11), (2.12), p. 256]. The existence, the uniqueness up to gauge transformations, and the regularity of $U_k(V)$ are the subject of [7]. Every term of the effective action depends on $V$ through this *background field*.

*The expansion.* For a cube $\square$ of the partition of $T_\eta$ into cubes of side $LM_2R_k$ let $U_{k,\square}(V_k)$ be the minimal configuration of [9, (2.16), p. 257], which is determined by the block field in a neighbourhood of $\square$. The test function of the step $k$ on $\Omega_k$ is the product

$$\chi_k(\Omega_k) \coloneqq \prod_{\square \subset \Omega_k} \chi\Big( \Big\{ \sup_{p \subset \square^\sim} |U_{k,\square}(V_k, \partial p) - 1| < \varepsilon_k \eta^2 \Big\} \Big) \tag{3.22}$$

of the characteristic functions of the sets on which the plaquettes of these local minimisers are below the threshold $\varepsilon_k$, in the units of the fine lattice, and the density is

$$\varrho_k(V_k) = \sum_{\{\Omega_j\},\{\Lambda_j\}} \chi_k(\Omega_k)\, \mathbf{T}_k(\{\Omega_j\},\{\Lambda_j\}) \exp A_k\Big(\frac{1}{g_k^2}, U_k\Big) \tag{3.23}$$

[9, (2.17), (2.18), p. 257]. The sum runs over the admissible sequences of domains, the sum over the sequences $\{S_j\}$ is included in the operation $\mathbf{T}_k$, and the effective action $A_k$ depends on all three sequences.

*The operations on the large-field regions.* The operation $\mathbf{T}_k$ consists of the integrations which the earlier steps have left: over the block fields $V_j$ on $\Omega_{j+1}^c$, and over the fluctuation fields $A_j$ on $Z_{j+1} \cap \Omega_{j+1}$. It factorises over the connected components $X_1, \dots, X_n$ of $Z_k$, and each factor is an ordered product over the scales,

$$\mathbf{T}_k(Z_k) = \prod_{i=1}^{n} \mathbf{T}_k(X_i)\,, \qquad \mathbf{T}_k(X) = \prod_{j=k-1}^{0} \mathbf{T}^{(j)}(Z_{j+1} \cap X)\,, \tag{3.24}$$

where $\mathbf{T}^{(j)}$ comes from the step $j+1$ and, if no **R**-operation has changed it, has the form of [9, (2.21), p. 258]: the integral over $V_j$ against the $\delta$-function of the block averaging and the function $\zeta(\Omega_{j+1}^c)$, which contains the characteristic functions of the large fields, and the integral over $A_j$ against a characteristic function and a Gaussian weight [9, (2.19), (2.20), p. 258].

*The effective action.* The effective action has the form

$$A_k\Big(\frac{1}{g_k^2}, U_k\Big) = -A\Big(\frac{1}{g_k^2}, U_k\Big) + \mathbf{E}_k(U_k) + \mathbf{R}_k(U_k) + \mathbf{B}_k(U_k, A) - E_k \tag{3.25}$$

[9, (2.23), p. 258], in which $A = \{A_{j-1}\}$ stands for the fluctuation fields in the last but one term. Its five terms are the following.

(a) The first is the Wilson action of the background field with a coupling which depends on the position, defined by

$$\frac{1}{g_{j-1}^2(x)} = \frac{1}{g_j^2(x)} + \beta_j(g_{j-1})\phi_j(x)\,, \tag{3.26}$$

where $\phi_j \in C_0^\infty(\Lambda_j)$ equals one on $\Lambda_j^{\sim -1}$, the domain obtained from $\Lambda_j$ by removing a layer of cubes, so that $g_j(x) = g_j$ there [9, (2.24), p. 259]. The numbers $\beta_j(g_{j-1})$ are the coefficients of the coupling flow, the subject of Section 4.

(b) The *regular part* is

$$\mathbf{E}_k(U_k) = \sum_{j=1}^{k} [\mathbf{E}^{(j)}(\Lambda_j, g_{j-1}, U_k) - \mathbf{E}^{(j)}(\Lambda_j, g_{j-1}, 1) - \beta_j(g_{j-1}) A(\phi_j, U_k)]\,, \tag{3.27}$$

with $\mathbf{E}^{(j)}(\Lambda_j, U_k) = \sum_{z\in\Lambda_j^0} \sum_{X\ni z} \mathbf{E}^{(j)}(X, U_k, z)$, the inner sum over the $X \in \mathbf{D}_j$ contained in $\Lambda_j$ and $\Lambda_j^0$ obtained from $\Lambda_j^{(j)}$ by removing one layer of the cubes of side $MR_j$ [9, (2.25)–(2.27), p. 259]. Each term $\mathbf{E}^{(j)}(X, U_k, z)$ depends on $U_k$ restricted to $X$; it extends to an analytic function of a pair $(\mathbf{U}, \mathbf{J})$ in the space $U_j^c(X, \alpha_{0,j}, \alpha_{1,j})$; the extension is invariant under gauge transformations; and it satisfies the bound

$$|\mathbf{E}^{(j)}(X, (\mathbf{U}, \mathbf{J}), z)| \le E_0 \exp(-\kappa d_j(X)) \tag{3.28}$$

of [8, (1.18), p. 263]. The space $U_j^c(X, \alpha_0, \alpha_1)$ consists of the gauge orbits of the pairs of a $G^c$-valued field $\mathbf{U} = U'U$ and a $\mathfrak{g}^c$-valued field $\mathbf{J}$ on $X$ in which $U$ is a real field with plaquettes below $\alpha_0$, $U' = \exp i\xi A'$ with $A'$ and its covariant derivative below $\alpha_1$, and $\mathbf{J}$ is below $\alpha_0$, together with the corresponding bounds for the minimisers built from the block averages of $\mathbf{U}$ [8, Definition, (1.10)–(1.16), p. 262]. It is a complex neighbourhood of the real regular fields on $X$, of a width set by $\alpha_1$.

(c) The **R**-*part* is $\mathbf{R}_k(U_k) = \sum_{j=1}^k [\mathbf{R}^{(j)}(\Lambda_j, U_k) - \mathbf{R}^{(j)}(\Lambda_j, 1)]$, where $\mathbf{R}^{(j)}(\Lambda_j, U_k)$ is a sum of terms $\mathbf{R}^{(j)}(X, U_k)$ over the domains $X \in \mathbf{D}_j$ contained in $\Lambda_j^{\sim -1}$ which are unions of cubes of side $MR_j$. These terms have the first three properties of the terms of the regular part and satisfy

$$|\mathbf{R}^{(j)}(X, (\mathbf{U}, \mathbf{J}))| \le g_j^{\kappa_0} \exp(-\kappa d_j(X)) \tag{3.29}$$

[9, (2.30), (2.31), p. 260]. They are produced by the **R**-operation from terms with large fields, and they are so small that they need no counterterm.

(d) The *boundary part* is $\mathbf{B}_k = \sum_{j=1}^k \mathbf{B}^{(j)}(U_k, A, \{S_i\})$, where $\mathbf{B}^{(j)}$ is a sum of terms $\mathbf{B}^{(j)}(X, U_k, A, \{S_i \cap X\})$ over the $X \in \mathbf{D}_j$ with $X \cap \Omega_j \neq \emptyset$ and $X \cap Z_j^{\sim} \neq \emptyset$. Each of them depends on $U_k$ and $A$ restricted to $X$, extends analytically to the space $\tilde{U}_j^c(X, \tilde{\alpha}_0, \tilde{\alpha}_1)$, is invariant under $G$-valued gauge transformations, and satisfies

$$|\mathbf{B}^{(j)}(X, (\mathbf{U}, \mathbf{J}), A, \{S_i \cap X\})| < B_0 \exp(-\kappa d_j(X)) \tag{3.30}$$

[9, (2.40)–(2.42), p. 261]. The space $\tilde{U}_j^c(X, \tilde{\alpha}_0, \tilde{\alpha}_1)$ is the version of $U_j^c$ for a domain which meets several of the layers $\Omega_n \setminus \Omega_{n+1}$: its conditions (2.34)–(2.39) are those of $U_j^c$ with the radii $\alpha_{0,n}$, $\alpha_{1,n}$ of the scale $n$ on the layer $n$, each multiplied by $1-\beta(1-2^{-(j-n)})$ with a constant $\beta$, for which Bałaban suggests $\beta = 1/4$ [9, p. 261].

(e) $E_k$ is a constant, which depends on $\{\Omega_j\}$ and $\{\Lambda_j\}$ [9, p. 262].

The terms which a step has just created are defined on slightly larger spaces and decay faster, with $(1 + 4\beta)\kappa$ in place of $\kappa$, and this margin is consumed by the **R**-operation [9, p. 262].

*Bałaban's theorems.* The main result of [9] is that the form just described is reproduced by a step.

**Theorem 3.6** (Bałaban [9, Theorem 1, p. 262])**.** *There exist constants, introduced in the above description, such that if the sequence of coupling constants $\{g_k\}$, determined by the recursive renormalisation group equations (0.18), (0.20) of [8], satisfies the inequalities (0.31) of [8], then the sequence of densities $\{\varrho_k\}$, generated by successive applications of the operations $\mathbf{R}T$ to the density $\varrho_0 = \exp[-(1/g_0^2)A - E]$, satisfies all the inductive assumptions.*

The proof of this theorem occupies [9], for the step $T$, and [11, 12], for the **R**-operation. Two consequences are used throughout. The first is a bound on the regular and the **R**-parts in a region $\Omega$, [9, Theorem 2, (2.43), (2.44), p. 263]: with a constant $E_1$ independent of $j$, $k$, $\Omega$, the sequences of domains, and the torus,

$$\Big| \sum_{z \in \Lambda_j^0 \cap \Omega} [\mathbf{E}^{(j)}(\Lambda_j, U_k, z) - \mathbf{E}^{(j)}(\Lambda_j, 1, z)] - \beta_j(g_{j-1}) A(\phi, U_k) \Big| \le E_1 \sum_{n=j}^k (L^{j-n})^\beta |\Gamma_n \cap \Omega| \tag{3.31}$$

for $\beta < 1$, and the corresponding sum of the **R**-terms is at most $R_1 g_j^{\kappa_0} \sum_{n=j}^k |\Gamma_n \cap \Omega|$. Here $\Omega \in \mathbf{D}_k$ is a domain contained in the union of the blocks of $\Lambda_j^0$, $\phi \in C_0^\infty(\Omega^{\sim})$ equals one on

$\Omega$, the field is restricted by the test functions of (3.23), and $|\Gamma_n \cap \Omega|$ is the number of points of the scale $n$. The second is ultraviolet stability, [9, Corollary 3, (2.50), p. 264]: there are constants $E_\pm$, independent of $\eta$ and of the torus but dependent on $g_k$, with

$$\chi_k(T_\eta) \exp\Big[-\frac{1}{g_k^2} A(U_k(V_k)) - E_- |T_\eta|\Big] \le \varrho_k(V_k) \le e^{E_+ |T_\eta|} \,. \tag{3.32}$$

These statements are for Bałaban's parameters, which are functions of the running couplings, and for characteristic functions as test functions. The construction of this paper differs from his in both respects, and the remainder of this section shows that the description and the bounds hold for it.

### 3.3 Soft Test Functions

The domains of a term are produced by decompositions of unity. In each step the lattice is divided into cubes $\square$, a nonnegative function $f_\square$ of the fields near $\square$ is compared with a threshold $t$, and the integrand is multiplied by

$$1 = \sum_P \prod_{\square \not\subset P} \chi(\{f_\square < t\}) \prod_{\square \subset P} \chi(\{f_\square \ge t\}) \,, \tag{3.33}$$

the sum running over the unions $P$ of cubes. In the term $P$ we say that the test function of a cube has *passed* if the cube lies outside $P$, and that it has *failed* if the cube lies in $P$. Bałaban's steps contain two kinds of such decompositions. In the first, $f_\square$ is the supremum of $|U(\partial p) - 1|$ over the plaquettes $p$ near $\square$, for the fine field in the first step and for a local minimiser $U_{k+1,\square}$ of the action in the later ones, and $t$ is a plaquette threshold $\varepsilon$ in the units of the plaquette [9, (1.1), (1.4), p. 246; (2.17), p. 257; (3.2), p. 265]. In the second, $f_\square$ is the supremum over the bonds $b$ near $\square$ of the deviation $|V_k(b)(V_\square^{(k)}(b))^{-1} - 1|$ of the field from a reference configuration, the block average of a local minimiser, and $t = 2\vartheta_k$ [9, (1.8), p. 247; (3.3), p. 265]. We call the factors of (3.33) *test functions*, of the plaquettes and of the reference configuration respectively.

A characteristic function is the natural choice as long as one run of the construction is studied. For the limit $K \to \infty$ we shall compare two runs, with $K$ and $K+1$ steps, on the lattices which they share, and there the functions $f_\square$ of the two runs differ by a small amount, since they are computed from minimisers on two different fine lattices. A characteristic function can change by one under an arbitrarily small change of its argument, and the set on which this happens, a thin shell around the level set $\{f_\square = t\}$, has no reason to be of small measure for the densities at hand. We therefore replace the characteristic functions by continuous functions, whose change is proportional to that of the argument.

**Definition 3.7** (Soft Test Functions)**.** Let $\kappa_s > 0$. For $x \in [0, \infty]$ put

$$\bar\chi(x) \coloneqq \min\Big(1, \max\Big(0, \frac{1 + \kappa_s - x}{\kappa_s}\Big)\Big) \,, \tag{3.34}$$

so that $\bar\chi(x) = 1$ for $x \le 1$, $\bar\chi(x) = 0$ for $x \ge 1 + \kappa_s$, and $\bar\chi$ is affine in between. The *soft* form of the decomposition (3.33) is

$$1 = \sum_P \prod_{\square \not\subset P} \bar\chi\Big(\frac{f_\square}{t}\Big) \prod_{\square \subset P} \Big(1 - \bar\chi\Big(\frac{f_\square}{t}\Big)\Big) \,. \tag{3.35}$$

**Lemma 3.8** (Properties of the Soft Decomposition)**.** *Let $(f_\square)$ be a finite family of functions with values in $[0, \infty]$ and $t > 0$.*

*(i) The identity* (3.35) *holds, and each of its terms has values in* $[0,1]$.

*(ii) The term* $P$ *of* (3.35) *vanishes unless* $f_\square < (1+\kappa_s)t$ *for every* $\square \not\subset P$ *and* $f_\square > t$ *for every* $\square \subset P$. *In particular it is bounded by the term* $P$ *of the decomposition* (3.33) *in which the passed test functions are taken at the threshold* $(1+\kappa_s)t$ *and the failed ones at the threshold* $t$, *with strict inequality.*

*(iii) For* $x, y \in [0,\infty)$, $|\bar\chi(x) - \bar\chi(y)| \le |x-y|/\kappa_s$. *Consequently, for two families* $(f_\square)$, $(f'_\square)$ *and with* $\Pi_P$, $\Pi'_P$ *their terms* $P$ *of* (3.35),

$$|\Pi_P - \Pi'_P| \le \frac{1}{\kappa_s t} \sum_\square |f_\square - f'_\square| \,. \tag{3.36}$$

*Proof.* (i) Put $a_\square \coloneqq \bar\chi(f_\square/t) \in [0,1]$. Expanding the product $\prod_\square (a_\square + (1-a_\square))$, which equals one, gives the sum over the subsets of the family of the products $\prod_{\square\notin P} a_\square \prod_{\square\in P}(1-a_\square)$, and each of these is a product of numbers in $[0,1]$.

(ii) The factor $a_\square$ vanishes for $f_\square \ge (1+\kappa_s)t$, and the factor $1-a_\square$ vanishes for $f_\square \le t$. Since all factors are at most one, the term is bounded by the product of the characteristic functions of the sets $\{f_\square < (1+\kappa_s)t\}$, $\square \not\subset P$, and $\{f_\square > t\}$, $\square \subset P$.

(iii) The function $\bar\chi$ is piecewise affine with slopes 0 and $-1/\kappa_s$, which gives the first assertion. For the second we replace the factors of $\Pi_P$ by those of $\Pi'_P$ one at a time. Each replacement changes one factor by at most $|f_\square - f'_\square|/(\kappa_s t)$, both for a factor $a_\square$ and for a factor $1-a_\square$, while the other factors lie in $[0,1]$. *q.e.d.*

The first property says that nothing is lost: the lattice integrals are the same whichever decomposition is used to expand them, and only the terms of the expansion change. The second is the form in which the soft test functions enter every estimate. A bound which Bałaban derives from a passed test function is a bound on a field below the threshold, and it holds for the soft term with the threshold enlarged by the factor $1+\kappa_s$; a bound which he derives from a failed test function, the small factor which the action gives on a region of large fields, holds unchanged. The third property is used only in Section 10.

In the construction of this paper all the test functions of the plaquettes and of the reference configuration are soft, with

$$\kappa_s \coloneqq \frac{\beta}{30} \,, \tag{3.37}$$

$\beta$ being the constant of the spaces of the boundary terms in Section 3.2. This value is dictated by the margins which Bałaban's inequalities leave between a threshold and the bound that is drawn from it, and which the factor $1+\kappa_s$ must not exhaust. The characteristic functions which restrict the fluctuation fields are not test functions in the above sense, and they are left as they are.

### 3.4 The Depth of the R-Operation

The **R**-operation of the step $k$ acts on those connected components of the large-field region $Z_k$ which are small and old: a component must be contained in a cube of side $100MR_k$, and in the preceding $N$ steps no new large-field region may have been created inside it, the earlier regions inside it being small in the same sense on their scales [11, p. 177, (i), (ii)]. The number $N$ is the *depth* of the operation. Bałaban does not fix it. He writes that "conditions on $N$ will be formulated in constructions of this section" [11, p. 177], and the conditions which he then states are the following four.

(N1) $N > N_0$, where $N_0$ is the smallest positive integer with $L^{-N_0+1} M R_{k-N_0+1} = M$ [11, p. 179].

(N2) $N \le O(1)(\log g_k^{-2})^\nu$ for a positive integer $\nu$ with $\nu/2 \le p_0 - p_1 - 1$, and $N$ is so large that $L^{-N} N^{1/2} O(1) B_3^3 B_5 M^6 \le \alpha/2$ [11, (1.94), p. 198].

(N3) $N \le R_k$ [12, p. 361].

(N4) $L^{-N} \le (\log g_k^{-2})^{-\nu}$ for a sufficiently large $\nu$, with $\nu \ge 2p_0 + dr$ [12, p. 363].

The last condition recurs as "the usual conditions on $N$", which make $p_0(g_k) q_1(g_k) R_k^7 L^{-N}$ small [12, (1.47), p. 368], and as "our assumptions on $N$", which make $R_j^{1+\beta_0}(p_0(g_j)/p_1(g_j)) L^{-N}$ small [12, p. 382]. In the large-field induction of [12, p. 384] the conditions (i), (ii) are taken "with $N = R_j$". We make a different choice, which satisfies (N1)–(N4) with a depth of the order of $\log\log g_k^{-2}$ only. The reason is that in Section 6 the factor $L^N$ occurs, and has to be bounded by a power of $\log g_k^{-2}$.

**Definition 3.9** (Depth)**.** Put $\nu_L := p_0 + q_1 + 7r + 1$. For a tube and an integer $k \le K$:

(i) $N_0(k)$ is the smallest positive integer $N_0$ with $L^{N_0-1} = R_{k-N_0+1}$;

(ii) $N_k'$ is the least integer with $L^{-N_k'} \le (\log \hat g_k^{-2})^{-\nu_L}$;

(iii) the *depth* is $N_k := \max(N_0(k) + 1,\, N_k')$.

For the next lemma we assume, in addition to (3.7), that $p^* \ge \nu_L$ and that

$$\log_L(16L^4(L+1)^2) + \nu_L \log_L \lambda \le \lambda^r \qquad \text{for every } \lambda \ge \log \gamma^{-2}\,, \tag{3.38}$$

which holds for $\gamma$ small because $r \ge 2$.

**Lemma 3.10** (Bounds on the Depth)**.** *Let $\gamma$ satisfy* (3.7) *with $p^* \ge \nu_L$ and* (3.38)*, and let $L \ge 3$. For every tube and every integer $k \le K$:*

*(i) $N_0(k)$ exists, $N_0(k) \ge 2$, and $N_0(k)$, $N_k'$, and $N_k$ are nonincreasing in $k$;*

*(ii) $L^{-N_k} \le (\log \hat g_k^{-2})^{-\nu_L}$ and $L^{N_k} \le 4L^4(L+1)^2(\log \hat g_k^{-2})^{\nu_L}$;*

*(iii) for every history $(x_0, \dots, x_{k'})$ of the tube and $0 \le k \le k'$, with $C_{\text{dep}} := 16L^4(L+1)^2$,*

$$L^{-N_k} \le (\log g_k^{-2})^{-\nu_L}\,, \qquad L^{N_k} \le C_{\text{dep}} (\log g_k^{-2})^{\nu_L}\,, \qquad N_k \le R_k\,. \tag{3.39}$$

*Proof.* (i) Put $n := K - k \ge 0$, and for an integer $t \ge 0$ let $\varrho(t) := \log_L R_{K-t}$ and $h(t) := t - \varrho(t)$. By Lemma 3.4(i) and (ii), $\varrho$ takes integer values at least 1 and $\varrho(t+1) - \varrho(t) \in \{0, 1\}$, so that $h(t+1) - h(t) \in \{0, 1\}$ and $h(0) \le -1$. Moreover $h(t) \to \infty$, because $\varrho(t) \le 1 + r \log_L \log u_{K-t}$ grows like $\log\log t$. A nondecreasing integer-valued function with steps 0 or 1 which starts below $n$ and is unbounded takes the value $n$, and we let $T(n)$ be the least $t$ with $h(t) = n$. With $t = n + N_0 - 1$, so that $K - t = k - N_0 + 1$, the equation $L^{N_0-1} = R_{k-N_0+1}$ reads $\varrho(t) = t - n$, that is, $h(t) = n$. Its solutions have $N_0 = \varrho(t) + 1 \ge 2$, and the smallest is $N_0(k) = T(n) - n + 1$. Since $h$ is nondecreasing and $h(T(n)) = n$, we have $h(t) \le n$ for $t \le T(n)$, hence $T(n+1) \ge T(n) + 1$, and $T(n) - n$ is nondecreasing in $n$. So $N_0(k)$ is nonincreasing in $k$. The same holds for $N_k'$, because $\hat g_k$ increases with $k$, and therefore for $N_k$.

(ii) The first inequality is the definition of $N'_k \le N_k$. By the minimality of $N'_k$ we have $L^{N'_k} < L(\log \hat g_k^{-2})^{\nu_L}$. For $N_0 = N_0(k)$ we use the second inequality of (3.14) with $m = k - N_0 + 1$ and $n = k$: with $x := N_0 - 1 \ge 1$,

$$L^x = R_{k-x} \le (L+1)x^{\beta_0} R_k \le (L+1)(x+2)R_k \le 2(L+1)L^{x/2}R_k \,, \tag{3.40}$$

where the last step is the inequality $x+2 \le 2L^{x/2}$, which holds for $x \ge 0$ and $L \ge 3$, since the two sides agree at $x = 0$ and the derivative $L^{x/2}\log L$ of the right side is at least $\log 3 > 1$. Hence $L^{x/2} \le 2(L+1)R_k$, and with $R_k < L(\log \hat g_k^{-2})^r$,

$$L^{N_0+1} = L^2 L^x \le 4L^2(L+1)^2 R_k^2 < 4L^4(L+1)^2(\log \hat g_k^{-2})^{2r} \,. \tag{3.41}$$

Since $\nu_L \ge 2r$ and $\log \hat g_k^{-2} \ge 1$, both $L^{N_0+1}$ and $L^{N'_k}$ are at most $4L^4(L+1)^2(\log \hat g_k^{-2})^{\nu_L}$.

(iii) By Lemma 3.2(i), $\log g_k^{-2} \le \log \hat g_k^{-2} \le \log g_k^{-2} + \log \rho$, so that the first inequality follows from (ii), and the second from (ii) and $(\log \hat g_k^{-2})^{\nu_L} \le 2(\log g_k^{-2})^{\nu_L}$, which is the third condition of (3.7) with $\nu_L \le p^*$. Taking logarithms, $N_k \le \log_L C_{\mathrm{dep}} + \nu_L \log_L \log g_k^{-2}$, which is at most $(\log g_k^{-2})^r \le R_k$ by (3.38) and Lemma 3.4(i). *q.e.d.*

The depth $N_k$ satisfies Bałaban's four conditions. (N1) holds by definition and (N3) by (3.39). For (N4), (3.39) gives the exponent $\nu_L$, and $\nu_L - (2p_0 + dr) = q_1 - p_0 + 3r + 1 > 0$ at $d = 4$, because $q_1 \ge p_0$ (Appendix A). The two further uses of (N4) hold as well: with $R_k \le 2L(\log g_k^{-2})^r$, the product $p_0(g_k)q_1(g_k)R_k^7 L^{-N_k}$ is at most a constant times $(\log g_k^{-2})^{p_0+q_1+7r-\nu_L} = (\log g_k^{-2})^{-1}$, and $R_k^{1+\beta_0}(p_0(g_k)/p_1(g_k))L^{-N_k}$ is at most a constant times $(\log g_k^{-2})^{r(1+\beta_0)+p_0-p_1-\nu_L}$, whose exponent is negative. For (N2), the first restriction holds with $\nu = \lceil r \rceil$, because $N_k \le R_k \le 2L(\log g_k^{-2})^r$, under the condition $\lceil r \rceil/2 \le p_0 - p_1 - 1$ on the exponents, and the second holds for $\gamma$ small, because $L^{-N_k} N_k^{1/2} \le (2L)^{1/2}(\log g_k^{-2})^{r/2-\nu_L}$ tends to zero. That these are all the places at which the depth enters the two papers, and that each of them holds for $N_k$, is Supplement, Proposition S.3.

### 3.5 The Construction of This Paper

We can now say which construction we run. It is Bałaban's, with the parameters of a tube and with soft test functions.

**Definition 3.11** (The Construction)**.** Let a tube be given, with $\gamma$ as in Lemma 3.10, and let $g_0 > 0$. The *construction for the tube started at* $g_0$ is the sequence of densities $\varrho_0, \varrho_1, \dots$ obtained from $\varrho_0 = \exp[-(1/g_0^2)A - E]$ by Bałaban's operations $\mathbf{R}T$ of [9, 11, 12], with the following changes.

(i) Every parameter of the scale $j$ which is defined there as a function of the coupling $g_j$ is that function at $\hat g_j$: the cube size $R_j$, the thresholds $\varepsilon_j$, $\vartheta_j$, $\delta'_j$ and the radii $\alpha_{0,j}$, $\alpha_{1,j}$ of Definition 3.3, and every other threshold, radius, or support of a cutoff function of that scale. The depth of the $\mathbf{R}$-operation of the step $k$ is $N_k$.

(ii) The characteristic functions which restrict a fluctuation field are functions of the unscaled field: the fluctuation field $A_j$ of the step $j+1$ is restricted by $|g_j A_j(b)| < \vartheta_j$, which is the form of [9, (1.18), p. 250] and [11, (1.9), p. 178], and the field $B_j$ of [11, (1.55), p. 188] by $|g_j B_j(b)| < \delta'_j$; the bump functions of the changes of variables [9, (1.22), p. 251] are functions of the unscaled field as well, with supports fixed by the thresholds (Supplement, Definition 3.5).

(iii) The test functions of the plaquettes and of the reference configuration are the soft test functions of Definition 3.7, with the width (3.37); the cascade of test functions of the **R**-operation [11, (1.22)–(1.24), p. 181] is soft with the nested widths of Supplement, Definition 2.2, and so is the cutoff $\chi(\Lambda_i)$ of [11, (1.101), p. 201].

The *running couplings* are $g_k = x_k^{-1/2}$ with $x_{k+1} \coloneqq x_k - \beta_{k+1}$, where $\beta_{k+1} = \beta_{k+1}(g_0, \dots, g_k)$ is the coefficient which Bałaban's formula [9, (3.61), p. 282] assigns to the regular part of the effective action of the step $k+1$, the coefficient of (3.26). The construction is defined up to the step $k \le K$ as long as $(x_0, \dots, x_k)$ is a history of the tube.

In (ii) the factor $g_j$ which converts the field of the Gaussian integral into the unscaled field is the running coupling, and only the threshold is fixed by the target. This is what makes the construction continuous in the couplings: no cube size, no depth, and no set on which a characteristic or test function is supported depends on $g_0, \dots, g_k$, which enter only through the action, the Gaussian weights, and the terms of the effective action generated before.

**Proposition 3.12** (Bałaban's Description for the Construction)**.** *Let $\gamma$ be sufficiently small. For every tube, every $g_0$, and every $k \le K$ up to which the construction for the tube started at $g_0$ is defined, the density $\varrho_k$ has the expansion* (3.23)*, with soft test functions in* (3.22)*, and its terms have the properties (a)–(e) of Section 3.2, with constants $E_0$, $B_0$, $\kappa$, $\kappa_0$ which do not depend on the tube, on $g_0$, on $k$ and $K$, or on the torus. The bounds* (3.31) *and* (3.32) *hold, with constants which are independent of the same quantities, and so do the results of [11, 12] on the* **R***-operation.*

*Proof.* Supplement, Propositions S.1, S.2, and S.3. Proposition S.1 goes through the places at which a parameter enters [9, 11, 12] and the papers on which they rest, and shows that each of them is of one of four kinds: a relation among the parameters alone, which holds for the sequence $(\hat g_j)$ by Lemmata 3.2–3.5; a lower bound drawn from the action, which holds with the constant in the exponent divided by $\rho \le 3/2$, because the coefficient of the action is at least $\hat g_m^{-2}/\rho$ at the points which carry a field of the scale $m$ (Supplement, Lemma 2.1); an upper bound relative to the action, which holds because the thresholds of a tube are the smaller ones; or a conversion between a scaled and an unscaled field, which is the change (ii). Proposition S.2 goes through the places at which a test function enters and shows that every bound drawn from a passed test function holds at the threshold $(1+\kappa_s)t$ of Lemma 3.8(ii). Proposition S.3 is the corresponding statement for the depth. *q.e.d.*

### 3.6 The Stopping Index

The loops are drawn on a lattice $\mathbb{T}_q$ whose spacing $uL^{-q}$ does not depend on $K$, and the construction is stopped on a lattice which is finer than $\mathbb{T}_q$ by a factor that does not depend on $K$ either. This factor has to be so large that, at every step, the straight segments of a loop are long compared with the cubes of the construction. Since the parameters of a tube at the scale $K-n$ depend on $n$ and not on $K$, we may write

$$R^{[n]} \coloneqq R_{K-n} \,, \tag{3.42}$$

the smallest power of $L$ which is at least $(\log(\bar g^{-2} + b'n + C_W))^r$.

**Definition 3.13** (Stopping Index)**.** Let $M = L^m$. For an integer $j \ge 0$ put

$$\bar q(j) \coloneqq 3 + m + \log_L R^{[j+1]} \,, \qquad \text{so that} \qquad L^{\bar q(j)} = L^3 M R^{[j+1]} \,. \tag{3.43}$$

The *stopping index* $j_q$ of the level $q \ge 0$ is the least integer $j$ with $j \ge q + \bar q(j)$. For a target $(K, \bar g)$ with $K > j_q$, the *stopping step* is $K' \coloneqq K - j_q$.

**Lemma 3.14** (The Stopping Index)**.** *Let $\gamma$ be as in Lemma 3.4. For every $q \geq 0$ the stopping index $j_q$ exists. It depends on $q$, $L$, $M$, $r$, $\bar{g}$, $b'$, and $C_W$ only, and*

$$j_q = q + \bar{q}(j_q) \,, \qquad L^{j_q - q} = L^3 M R_{K'-1} \geq L^3 M R_{K'} \,. \tag{3.44}$$

*Consequently, for a loop $C$ at level $q$, read on the lattice of the step $k \leq K'$, whose spacing is $uL^{k-K}$: the corners of $C$ lie in a coset of $a_k \mathbb{Z}^4$ with $a_k := L^{j_q - q} L^{K'-k}$, and, in the notation of Section 2 for this lattice,*

$$P_k(C) = P_q(C)\, a_k \,, \qquad s_k(C) \geq a_k \,, \qquad \delta_k(C) \geq a_k \,, \qquad a_k \geq L^3 M R_{K'} L^{K'-k} \geq L^2 M R_k \,. \tag{3.45}$$

*Two loops of an ensemble at level $q$ are at $\ell^\infty$-distance at least $a_k$ on that lattice.*

*Proof.* The function $\bar{q}$ is nondecreasing, because $R^{[n]}$ is nondecreasing in $n$ by Lemma 3.4(ii), and $\bar{q}(j) \leq 4 + m + r \log_L \log(\bar{g}^{-2} + b'(j+1) + C_W)$ by Lemma 3.4(i). Hence $j - q - \bar{q}(j) \to \infty$ as $j \to \infty$, and the least $j$ with $j \geq q + \bar{q}(j)$ exists. It is determined by $q$ and the function $\bar{q}$, which gives the dependence. Since $\bar{q} \geq 3$, we have $j_q > q$, so that $j_q - 1 \geq q$ and, by minimality, $j_q - 1 < q + \bar{q}(j_q - 1) \leq q + \bar{q}(j_q)$. Together with $j_q \geq q + \bar{q}(j_q)$ this gives the first identity of (3.44). The second is (3.43) with $R^{[j_q+1]} = R_{K - j_q - 1} = R_{K'-1}$, and $R_{K'-1} \geq R_{K'}$ by Lemma 3.4(ii).

The lattice of the step $k$ has the spacing $uL^{k-K}$, and $uL^{-q} = L^{K-k-q} uL^{k-K}$ with $K - k - q = (j_q - q) + (K' - k)$. So one bond of $\mathbb{T}_q$ is $a_k$ bonds of that lattice, which gives the statement on the corners and the first identity of (3.45). The bounds on $s_k(C)$ and $\delta_k(C)$ are Lemma 2.3(i) with $a = a_k$. For two loops of an ensemble we apply Lemma 2.3(ii) to a lift of the first and to all the translates of a lift of the second by the periods of the torus, which are multiples of $a_k$ because $T/(uL^{-q}) = 2L^{m_T + q}$ is an integer: these lifts have no common site, since the loops are vertex-disjoint on the torus, and their corners lie in one coset of $a_k \mathbb{Z}^4$. For the last chain we use $L^{j_q - q} \geq L^3 M R_{K'}$ and, by Lemma 3.4(ii) applied $K' - k$ times, $R_k \leq L^{K'-k} R_{K'}$. *q.e.d.*

The inequality $a_k \geq L^2 M R_k$ says that every segment of a loop, and every gap between two of its segments or between two loops, is at least $L^2$ cubes of the step $k$ long, at every step up to the stopping step. On the stopping lattice itself a loop at level $q$ has $P_{K'}(C) = P_q(C) L^3 M R_{K'-1}$ bonds, a number which does not depend on $K$.

The steps $k \leq K'$ are those at which the loops will be carried, and their couplings are bounded by a number which does not depend on $K$. Indeed, for a history $(x_0, \dots, x_k)$ of the tube with $k \leq K'$ we have $x_k \geq \ell_k = t + b(K - k) - C_W \geq t + b j_q - C_W$ by (3.3), because $K - k \geq K - K' = j_q$, and $\hat{g}_k \leq g_k$ by Lemma 3.2(i). With

$$\gamma_q := (\bar{g}^{-2} + b j_q - C_W)^{-1/2} \tag{3.46}$$

we therefore have $\hat{g}_k \leq g_k \leq \gamma_q$ for $0 \leq k \leq K'$. The number $\gamma_q$ is small if $\bar{g}$ is small and also if $q$ is large: since $\bar{g}^{-2} - C_W \geq \gamma^{-2}$ by (3.2) and $j_q \geq q$ by (3.44),

$$\gamma_q \leq (\bar{g}^{-2} + bq - C_W)^{-1/2} \leq \min\Big( (\bar{g}^{-2} - C_W)^{-1/2},\ (\gamma^{-2} + bq)^{-1/2} \Big) \,, \tag{3.47}$$

and $(\bar{g}^{-2} - C_W)^{-1/2} \leq (1 - \beta_0/4)^{-1/2} \bar{g} \leq (1 + \beta_0) \bar{g}$, because $C_W \leq \beta_0 \bar{g}^{-2}/4$ by (3.2) and $(1 + \beta_0)^2 (1 - \beta_0/4) \geq 1$ for $0 < \beta_0 \leq 1$. Every requirement of the paper which involves the representations of an ensemble concerns the steps $k \leq K'$ only, and it is imposed as an upper bound on $\gamma_q$; by (3.47) such a bound holds at every coupling $\bar{g} \leq \gamma$ once the level $q$ is large enough.

## 4 The Coupling Flow

The construction of Section 3 takes a bare coupling and returns the running couplings. What the lattice theory of Section 1.1 needs is the converse: for a prescribed value $\bar{g}$ of the coupling at the unit scale and for every number $K$ of steps, a bare coupling $g_0$ with $g_K = \bar{g}$, together with bounds on the couplings in between which do not depend on $K$. Bałaban states this as a theorem for $G = SU(2)$ and writes that "a proof of this theorem, based on perturbative calculations, will be given in a separate paper" [8, Theorem 2, p. 259]; in [9] the bounds on the couplings are a hypothesis, as Theorem 3.6 shows. In this section we prove the statement, for a simple group and for the construction of this paper.

**Theorem 4.1** (Coupling Flow)**.** *There are $\gamma' > 0$ and constants $0 < b \le b'$ and $C_W \ge 0$ such that, for every integer $K \ge 1$ and every $\bar{g} \in (0, \gamma']$, there is a bare coupling $g_0 = g_0(K, \bar{g})$ for which the running couplings $g_0, g_1, \dots, g_K$ of the construction for the target $(K, \bar{g})$ satisfy $g_K = \bar{g}$ and, for $0 \le k \le l \le K$,*

$$\bar{g}^{-2} + b(K-k) - C_W \le g_k^{-2} \le \bar{g}^{-2} + b'(K-k) + C_W \qquad \text{and} \qquad g_k^{-2} \ge g_l^{-2} - C_W \,. \tag{4.1}$$

*The constants are $b = 2c \ln L - \delta'$ and $b' = 2c \ln L + \delta'$, where $c$ is the number (4.3), the first coefficient of the $\beta$-function, and $0 \le \delta' \le (2/5)c \ln L$.*

The *target* $(K, \bar{g})$ stands for the tube of Definition 3.1 with these $b$, $b'$, and $C_W$. The proof has three inputs of different kinds. The first is a computation at one loop: at zero coupling the inverse square of the coupling decreases by $2c \ln L$ per step, up to an error which stays bounded when it is summed over any number of steps (Theorem 4.4). The second is perturbative as well, but concerns Bałaban's expansions: the true decrease differs from the one at zero coupling by a multiple of $g_k^2$, uniformly in the step and in the earlier couplings (Proposition 4.3(i), (ii)). With these two facts every bare coupling would produce a sequence that satisfies (4.1) up to its last step, but the last coupling would be whatever it turns out to be. To prescribe it we need the third input, the continuity of the couplings as functions of the bare coupling (Proposition 4.3(iii)), and it is for this that the parameters of the construction were detached from the running couplings in Section 3.1. Given the three inputs, the bare coupling is found by the intermediate value theorem.

### 4.1 The Coefficients of the Flow

We first say what the coefficient $\beta_{k+1}$ of Definition 3.11 is. Let $\mathfrak{g} \subset \mathfrak{u}(N)$ be the Lie algebra of $G$, a space of anti-Hermitian matrices, with the inner product $\langle X, Y \rangle \coloneqq -\operatorname{tr}(XY)$ and the Killing form $\mathrm{K}(X, Y) \coloneqq \operatorname{Tr}(\operatorname{ad} X \operatorname{ad} Y)$.

**Lemma 4.2** (Invariant Forms)**.** *Every bilinear form on $\mathfrak{g}$ which is invariant under $\mathrm{Ad}(G)$ is a real multiple of the inner product. In particular there is a number $\kappa > 0$, depending on $G$ and on its realisation, with*

$$\mathrm{K}(X, Y) = \kappa \operatorname{tr}(XY) \,, \tag{4.2}$$

*and $\kappa = 2N^2$ for $G = SU(N)$ in its defining representation.*

*Proof.* An invariant bilinear form can be written as $(X, Y) \mapsto \langle SX, Y \rangle$ with a linear map $S$ of $\mathfrak{g}$ which commutes with $\mathrm{Ad}(G)$, hence with every $\operatorname{ad} X$, and we treat its symmetric and antisymmetric parts separately, each of which commutes with ad as well, because the maps $\operatorname{ad} X$ are antisymmetric for the invariant inner product. A symmetric $S$ has a real eigenvalue $\lambda$, and $\ker(S - \lambda)$ is a nonzero ideal of $\mathfrak{g}$, since $S[X, Y] = [X, SY]$. As $\mathfrak{g}$ is simple, $S = \lambda$. For

an antisymmetric $S$ the square $S^2$ is symmetric and commutes with ad, so that $S^2 = -\mu^2$ with $\mu \geq 0$ by what has been shown and $\langle S^2X, X\rangle = -|SX|^2$. If $\mu$ were positive, $J := S/\mu$ would be a complex structure on $\mathfrak{g}$ with $J[X,Y] = [X, JY]$, so that $\mathrm{ad}(JX) = J\,\mathrm{ad}\,X$ and $\mathbf{K}(JX, JX) = \mathrm{Tr}(J\,\mathrm{ad}\,X\,J\,\mathrm{ad}\,X) = \mathrm{Tr}(J^2(\mathrm{ad}\,X)^2) = -\mathbf{K}(X,X)$. This is impossible, because $\mathbf{K}(X,X) = -\,\mathrm{Tr}((\mathrm{ad}\,X)^T\,\mathrm{ad}\,X)$ is negative for every $X \neq 0$, the centre of a simple Lie algebra being trivial. Hence $S = 0$. The Killing form is invariant and negative definite, which gives (4.2) with $\kappa > 0$, and for $\mathfrak{su}(N)$ one has $\mathbf{K}(X,Y) = 2N\,\mathrm{Tr}(XY) = 2N^2\,\mathrm{tr}(XY)$. *q.e.d.*

The number

$$c := \frac{11}{48\pi^2}\,\kappa \tag{4.3}$$

is the first coefficient of the $\beta$-function in the normalisation of the action (1.2): for $SU(N)$ and the usual coupling $g_{\mathrm{s}}$, with $g^{-2} = 2Ng_{\mathrm{s}}^{-2}$, the statement that $g^{-2}$ decreases by $2c\ln L$ when the lattice spacing is multiplied by $L$ is $\mathrm{d}g_{\mathrm{s}}^{-2}/\mathrm{d}\ln\varepsilon = -11N/(24\pi^2)$. From now on the numbers $b_*$ and $b^*$ of Section 3.1 are

$$b_* := c\ln L\,, \qquad b^* := 3c\ln L\,, \tag{4.4}$$

and a tube is always one that satisfies (3.2) with them.

*The polarisation coefficient.* We use the units of the step $k+1$, in which the block field of that step lives on the unit lattice. Let $F$ be a function of the background field of the step $k+1$ on the lattice $\mathbb{Z}^4$, which is invariant under gauge transformations and under the symmetries of the lattice and is a sum of local terms of the kind (3.27), so that its derivatives at the trivial field are summable. For a block field $V = \exp iB$, with $B$ a field on the bonds with values in $i\mathfrak{g}$ and components $B^a_\mu(x)$ in an orthonormal basis, the second derivative of $B \mapsto F(U_{k+1}(\exp iB))$ at $B = 0$ with respect to $B^a_\mu(x)$ and $B^{a'}_\nu(x')$ is, for fixed $\mu$, $x$, $\nu$, $x'$, an invariant bilinear form on $\mathfrak{g}$, because $F$ is invariant under constant gauge transformations. By Lemma 4.2 and translation invariance it equals $\delta^{aa'}\Pi_{\mu\nu}(F; x - x')$, and we put

$$\Lambda(F) := \sum_{x\in\mathbb{Z}^4} \Pi_{\mu\nu}(F;x)\,x_\mu x_\nu \tag{4.5}$$

for a pair $\mu \neq \nu$, on which the value does not depend. This is Bałaban's definition of the $\beta$-function from the vacuum polarisation [8, (1.20)–(1.22), p. 264], where the independence of the pair is part of the statement and the limit of the infinite lattice is taken with the help of the local representation. The coefficient of the step $k+1$ is $\Lambda$ of the regular part which that step generates on the whole lattice,

$$\beta_{k+1} := \Lambda(\mathbf{E}^{(k+1)})\,, \tag{4.6}$$

and [9, (3.57)–(3.64), pp. 281–283] shows that this is the number for which the counterterm $-\beta_{k+1}A(\phi_{k+1}, U)$ of (3.27) removes the marginal part of $\mathbf{E}^{(k+1)}$. The regular part contains the logarithm of the normalisation factor of the Gaussian measure of the fluctuation field of the step,

$$Z^{(k)}(U) := \int \delta(\tilde{Q}B)\exp[-\tfrac{1}{2}\langle B, \Delta^{(k)}(U)B\rangle]\,\mathrm{d}B \tag{4.7}$$

[8, (1.3), (1.4), p. 260], which is its part of order zero in the couplings, and the rest is the fluctuation integral proper, $\mathbf{E}^{(k+1)} = \log Z^{(k)} + \mathbf{E}^{(k+1)}_{\mathrm{fl}}$ up to a constant. Since $\Lambda$ is linear,

$$\beta_{k+1} = b_{k+1} + \Lambda(\mathbf{E}^{(k+1)}_{\mathrm{fl}})\,, \qquad b_{k+1} := \Lambda(\log Z^{(k)})\,. \tag{4.8}$$

The numbers $b_1, b_2, \dots$ are the *coefficients at zero coupling*. They are determined by the quadratic forms $\Delta^{(k)}(U)$ of [8, (1.5), p. 261], that is, by $L$, $G$, the averaging operations, and $k$, and they involve no coupling, threshold, or tube.

*Histories which are not flows.* For the argument that will produce the bare coupling we need $\beta_{k+1}$ as a function on the whole set $H_k$ of Definition 3.1, whereas the construction started at $g_0$ only visits the points of $H_k$ with $x_{j+1} = x_j - \beta_{j+1}$ for all $j < k$. We therefore extend the construction. For $h = (x_0, \dots, x_k) \in H_k$ the densities $\varrho_0^h, \dots, \varrho_k^h$ are defined inductively: $\varrho_0^h := \exp[-x_0 A - E]$, and $\varrho_{j+1}^h$ is obtained by applying the operations $\mathbf{R}T$ of Definition 3.11 to $\varrho_j^h$, with $g_j = x_j^{-1/2}$ as the coupling of the scale $j$, and by multiplying each term of the resulting expansion (3.23) by

$$\exp[-(x_{j+1} - x_j + \beta_{j+1}(x_0, \dots, x_j))A(U_{j+1})] \,, \tag{4.9}$$

where $A(U_{j+1})$ is the Wilson action of the background field of the term and $\beta_{j+1}(x_0, \dots, x_j)$ is the coefficient (4.6) of the step just performed. The effect is that the Wilson term of every term of $\varrho_{j+1}^h$ has the coefficient function

$$\frac{1}{g_{j+1}^2(x)} = x_{j+1} + \sum_{i=1}^{j+1} \beta_i(x_0, \dots, x_{i-1})(1 - \phi_i(x)) \,, \tag{4.10}$$

which is $x_{j+1}$ on the domain $\Lambda_{j+1}^{\sim -1}$ where all the $\phi_i$ equal one. Indeed, by (3.26) the step turns the coefficient function $y$ of $\varrho_j^h$ into $y - \beta_{j+1}\phi_{j+1}$, and (4.9) adds the constant $x_{j+1} - x_j + \beta_{j+1}$. If $h$ is a flow, the factor (4.9) is one at every step and $\varrho_j^h$ is the density of the construction started at $g_0 = x_0^{-1/2}$. In all cases $\beta_{k+1}(h)$ denotes the coefficient (4.6) of the step $k+1$ applied to $\varrho_k^h$.

**Proposition 4.3** (The Coefficients on a Tube)**.** *There are constants $C_\Lambda$, $C_3$, $C_{\text{rem}}$, $C_R$, and $\gamma_1 > 0$ such that, for every $\gamma \in (0, \gamma_1]$, every tube, every $k < K$, and every history $h = (x_0, \dots, x_k) \in H_k$, with $g_k = x_k^{-1/2}$, the following holds. The densities $\varrho_0^h, \dots, \varrho_k^h$ are defined and have the properties stated in Proposition 3.12, and:*

*(i)* $|\beta_{k+1}(h) - b_{k+1}| \le \delta'$*, where* $\delta' := C_\Lambda C_3 \gamma p_0(\gamma) + C_R \gamma^{\kappa_0 - 4}$*;*

*(ii)* $|\beta_{k+1}(h) - b_{k+1}| \le (C_{\text{rem}} + C_R \gamma^{\kappa_0 - 6})\, g_k^2$*;*

*(iii) the function $\beta_{k+1}$ is continuous on $H_k$.*

*The constants depend on those of the construction and not on $\gamma$, the tube, $k$, or $h$.*

*Proof.* Supplement, Propositions S.1, S.4, and S.5. Let us say where the difficulty lies in each part. That the densities of a history which is not a flow have Bałaban's properties is part of Proposition S.1, since the coefficient function (4.10) differs from that of a flow only through coefficients which obey the same bounds. By (4.8) the difference in (i) and (ii) is $\Lambda$ of the fluctuation integral, whose local terms Bałaban bounds by a multiple of the threshold $\varepsilon_1$ of the fluctuation field [10, (2.41), p. 21]. In the construction of this paper that threshold is $\vartheta_k$, which in the units of the step is at most $\gamma p_0(\gamma)$, and since $\Lambda$ is bounded on local terms with exponential decay this gives the first term of $\delta'$. The second term accounts for the **R**-terms of the earlier steps, which enter the regular part through the fluctuation integral and which the bound (3.29) makes small. The improvement to $g_k^2$ in (ii) rests on a symmetry: when $g_k$ is replaced by a complex number $z$ in the exponent of the fluctuation integral and the

threshold is held fixed, the integral is an even analytic function of $z$ which vanishes at $z = 0$ up to a Gaussian tail. For (iii) one has to follow every term of the density through the step and to check that it is continuous in the history at each configuration, which is where the construction of Section 3 is used: no set on which a characteristic function or a test function is supported moves with the history. *q.e.d.*

The exponent $\kappa_0$ is that of (3.29), and Bałaban takes $\kappa_0 \geq 7$ [9, (2.46), p. 263], so that $\delta' \to 0$ as $\gamma \to 0$. The first input of the proof, on the other hand, concerns the numbers $b_k$ alone.

**Theorem 4.4** (The Coefficients at Zero Coupling)**.** *There is a constant $C_W$, depending on $L$, $G$, and the averaging operations only, such that for all integers $0 \leq m < n$*

$$\Big| \sum_{j=m+1}^{n} b_j - 2c\,(n-m) \ln L \Big| \leq C_W \ . \tag{4.11}$$

*Proof.* Supplement, Proposition S.6. The statement is the content of Bałaban's Theorem 2 of [8] at zero coupling, and we describe the structure of its proof, since the number $c$ is computed there. It suffices to show that $\Phi_k := \sum_{j=1}^{k} b_j$ satisfies $|\Phi_k - 2ck \ln L| \leq C_W/2$ for every $k$. By (4.8) and the local representation of the terms of order zero [8, (0.28)–(0.30), p. 258], $\Phi_k$ is, up to a bounded error, the coefficient of the polarisation of the Gaussian factor $\sum_{j<k} \log Z^{(j)}(U_k)$ of $k$ steps, measured against the polarisation of the action of the background field, whose symbol is bounded above and below at a fixed nonzero momentum [2, (1.67), p. 29]. The Gaussian factor of $k$ steps is then written exactly as a difference $F_k - F_0$ of two functionals, each of which is one half of the logarithm of the determinant of a gauge-invariant quadratic form, of the action at the level $k$ for $F_k$ and of the action at the level 0 for $F_0$, divided by the determinant of the corresponding gauge operator, plus a sum of terms which come from the current $J$ of [8, (1.5), (1.8), p. 261]. The identity is linear algebra (a Schur complement and the matrix determinant lemma), and it telescopes over the steps. Three estimates of symbols then give the theorem: the polarisation of the terms of the current is bounded uniformly in $k$, because gauge covariance makes it vanish to second order at zero momentum; the polarisation of $F_k$ is bounded uniformly in $k$ at a fixed momentum, because it is the one-loop polarisation of a theory on a lattice of unit spacing with a cutoff at unit scale; and the polarisation of $-F_0$, the one-loop polarisation of the bare lattice theory seen from the coarse lattice, is $c \ln(1/\eta_k^2)$ times the action of the background up to a bounded error, with $\eta_k = L^{-k}$. The last estimate is the computation of the one-loop coefficient of the lattice theory in a background field, in which the logarithm comes from the region of momenta between the two cutoffs, and the coefficient $11/48\pi^2$ is the one of the continuum computation, $\kappa$ entering through Lemma 4.2. *q.e.d.*

We now show how the coupling flow follows.

## 4.2 Proof of the Coupling Flow

*Proof of Theorem 4.1. The constants.* Let $C_W$ be the constant of Theorem 4.4. We fix $\gamma \in (0, \gamma_1]$ so small that the conditions of Section 3 hold with (4.4) and that

$$\delta' \leq \tfrac{2}{5} c \ln L \qquad \text{and} \qquad \gamma^{-2} \geq \Big( \frac{4}{\beta_0} - 1 \Big) C_W \ , \tag{4.12}$$

and we put $b := 2c \ln L - \delta'$, $b' := 2c \ln L + \delta'$, and $\gamma' := (\gamma^{-2} + C_W)^{-1/2}$. Then $b \geq (8/5) c \ln L \geq b_*$, $b' \leq (12/5) c \ln L \leq b^*$, and $b'/b \leq 3/2$, and for $\bar{g} \leq \gamma'$ the number $t = \bar{g}^{-2}$ satisfies

$t \geq \gamma^{-2} + C_W \geq 4C_W/\beta_0$. So $(K, b, b', C_W, \bar{g})$ is a tube in the sense of Definition 3.1, for every $K \geq 1$ and $\bar{g} \in (0, \gamma']$. The choice is not circular, since the constants of Proposition 4.3, and hence $\delta'$, are the same for all tubes.

*The window of the coefficients.* Let $0 \leq m < n \leq K$ and let $h_j \in H_j$ be arbitrary histories, $m \leq j < n$. By Proposition 4.3(i) each $\beta_{j+1}(h_j)$ differs from $b_{j+1}$ by at most $\delta'$, and by (4.11) the sum of the $b_{j+1}$ differs from $2c(n-m)\ln L$ by at most $C_W$. Hence

$$b(n-m) - C_W \leq \sum_{j=m}^{n-1} \beta_{j+1}(h_j) \leq b'(n-m) + C_W \; . \tag{4.13}$$

*The projected flow.* The set $H_k$ is closed, convex, and nonempty, since it contains $(u_0, \dots, u_k)$. Let $P_k$ be the Euclidean projection of $\mathbb{R}^{k+1}$ onto it, which does not increase distances and is therefore continuous. For $x_0 \in \mathbb{R}$ we define $y_0 \coloneqq x_0$ and

$$y_{k+1} \coloneqq y_k - \beta_{k+1}(P_k(y_0, \dots, y_k)) \tag{4.14}$$

for $0 \leq k < K$. This sequence is defined for every $x_0$, whether or not it stays in the tube, and each $y_k$ is a continuous function of $x_0$, by induction on $k$ and Proposition 4.3(iii). Moreover the arguments of the coefficients in (4.14) are histories, so that (4.13) applies and gives, for $0 \leq m < n \leq K$,

$$b(n-m) - C_W \leq y_m - y_n \leq b'(n-m) + C_W \; . \tag{4.15}$$

*The bare coupling.* With $m = 0$ and $n = K$, (4.15) says that $y_K$ lies between $x_0 - b'K - C_W$ and $x_0 - bK + C_W$. At $x_0 = \ell_0 = t + bK - C_W$ this gives $y_K \leq t$, and at $x_0 = u_0 = t + b'K + C_W$ it gives $y_K \geq t$. By the intermediate value theorem the continuous function $x_0 \mapsto y_K$ takes the value $t$ on $[\ell_0, u_0]$; its set of solutions there is closed, and we let $g_0(K, \bar{g})^{-2}$ be the smallest of them.

*The projection is inactive.* For this $x_0$, (4.15) with $n = K$ and $y_K = t$ gives $\ell_m \leq y_m \leq u_m$ for every $m < K$, by the definition (3.3) of $\ell_m$ and $u_m$, and (4.15) itself is the condition (3.4). So $(y_0, \dots, y_k) \in H_k$ for every $k \leq K$, the projection $P_k$ leaves it fixed, and (4.14) reads $y_{k+1} = y_k - \beta_{k+1}(y_0, \dots, y_k)$: the sequence is a flow, the densities $\varrho_k^h$ of its histories are those of the construction for the tube started at $g_0$, and its running couplings are $g_k = y_k^{-1/2}$, with $g_K = t^{-1/2} = \bar{g}$. The first part of (4.1) is $\ell_k \leq y_k \leq u_k$, and the second follows from the lower bound of (4.15), since $y_k - y_l \geq b(l-k) - C_W \geq -C_W$ for $k < l$. *q.e.d.*

Three remarks are in order. First, the construction depends on the target: the parameters, the densities, the coefficients, and the bare coupling are those of the tube of $(K, \bar{g})$, and objects of different targets are not identified. What is uniform are the constants. Secondly, by Lemma 3.2 the running couplings of the theorem satisfy Bałaban's inequalities (2.6), which he derives from the renormalisation group equations and uses in their place [9, p. 255]; the window (4.1) is the inequality (0.31) of [8] up to the additive constant $C_W$, with $\beta = b/\ln L$ and $\beta' = b'/\ln L$. Thirdly, the bare coupling window (1.3) of the introduction is (4.1) at $k = 0$.

## 5 Dressed Loops and the First Step

Bałaban's construction acts on densities, and an observable has to be carried along with them. This section says in what form the loops of an ensemble are carried, and proves that the first step puts them into that form.

### 5.1 Two Integrals

Let $\mathcal{C} = ((C_1, \chi_1), \dots, (C_n, \chi_n))$ be a loop ensemble at level $q$, let $K > j_q$, and let $g_0 = g_0(K, \bar{g})$ be the bare coupling of Theorem 4.1. With $\varrho_0 \coloneqq \exp[-g_0^{-2} A - E]$ the bare density of Section 3.2, the expectation of the observable (1.4) is the quotient

$$\mathbb{E}_K[W_{\mathcal{C}}] = \frac{Z_K(\mathcal{C})}{Z_K} , \qquad Z_K(\mathcal{C}) \coloneqq \int W_{\mathcal{C}}(U) \varrho_0(U) \, \mathrm{d}U , \qquad Z_K \coloneqq \int \varrho_0(U) \, \mathrm{d}U , \tag{5.1}$$

in which the constant $E$ cancels. We apply the construction to the numerator and to the denominator separately. Each operation of the construction preserves the integral of a density: the block averaging $T$ does, because $\int \delta(\bar{U}(c) V(c)^{-1}) \, \mathrm{d}V(c) = 1$ for the Haar measure, and the **R**-operation is defined so that "integrals of the densities are unchanged" [11, p. 176]; the density it produces "may not be equal to $\varrho_k$, but is equivalent to it, in the sense that they have equal integrals" [11, p. 200]. Hence $Z_K = \int \varrho_k(V_k) \, \mathrm{d}V_k$ for every $k \le K$, and for the numerator we shall construct densities $N_k$ with

$$Z_K(\mathcal{C}) = \int N_k(V_k) \, \mathrm{d}V_k , \qquad N_0 \coloneqq W_{\mathcal{C}} \, \varrho_0 , \tag{5.2}$$

for $0 \le k \le K' = K - j_q$. No law of the block variables enters: the quotient of the two integrals at the step $K'$ is the expectation at the cutoff $K$. The point of the construction is the form of $N_k$, which is that of $\varrho_k$ with one more factor in each term.

### 5.2 The First Step of the Construction

The first step integrates the fine field with its block averages $V$ on the lattice of the step 1 held fixed. Bałaban decomposes the integrand by test functions into terms labelled by two domains, the domain $\Omega_1$ on which the test functions of the plaquettes have passed and the domain $\Lambda_1 \subset \Omega_1$ on which the fluctuation field is small as well, and after the fluctuation integral on $\Lambda_1$ and the renormalisation of the coupling he obtains

$$\varrho_1(V) = \sum_{\{\Omega_1, \Lambda_1\}} \chi_1(\Omega_1) \, \mathbf{T}_1(\{\Omega_1, \Lambda_1\}) \exp A_1\Big(\frac{1}{g_1^2}, U_1\Big) \tag{5.3}$$

[9, (1.28), p. 253]. Here $\mathbf{T}_1$ is the operation which holds the integrals over the bare field $V_0$ on $\Omega_1^c$ and over the fluctuation field $A$ on $\Omega_1 \cap \Lambda_1^c$, with the Gaussian weight of the step and the characteristic functions $\chi(\Omega_1 \cap \Lambda_1^c, S_1)$ of the large fluctuations [9, (1.29), p. 254], and the effective action is

$$\begin{aligned} A_1\Big(\frac{1}{g_1^2}, U_1\Big) = -A\Big(\frac{1}{g_1^2}, U_1\Big) + \{\mathbf{E}^{(1)}(\Lambda_1, g_0, U_1) - \mathbf{E}^{(1)}(\Lambda_1, g_0, 1) - \beta_1(g_0) A(\phi_1, U_1)\} \\ + \mathbf{B}^{(1)}(U_1, A, S_1) - E_1 \end{aligned} \tag{5.4}$$

[9, (1.30), p. 254], with the coupling of [9, (1.27), p. 253]. These integrals stay pending inside $\varrho_1$: Bałaban leaves the operations $\mathbf{T}_k$ of the later steps "undefined, only their basic general properties are formulated" [9, p. 254]. A *labelled term* of $\varrho_1$ is a term of (5.3) with given labels. For the numerator we use the refinement of the labels of [1, Section 9], in which the domain $\Lambda_1$ is further restricted near the loop by a retreat of the carrier of the loop's integral; the refined labels are again labels of Bałaban's form, and on each of them the operation $\mathbf{T}_1$ and the action $A_1$ are those of (5.3) (Supplement, Proposition S.7).

In the construction of this paper the thresholds of the first step are those of the tube at $\hat{g}_0$, and the test functions of the plaquettes are soft (Definition 3.11). The first step with the

loop inserted is the subject of the companion manuscript [1], for one square loop, Bałaban's thresholds at $g_0$, and characteristic functions as test functions, and for two disjoint squares in [1, Section 10]. We use its results for a general ensemble and for the construction of this paper, and the changes this requires are proved in Section 5.4.

### 5.3 Dressed Loops

Let us describe the form in which the ensemble is carried, beginning with a heuristic. After $k$ steps the fine field has been integrated out down to the scale $k$, and a loop $C_i$ is a path of $P_k(C_i)$ bonds of that scale. If the block field were exactly flat, the loop would contribute to each term of the density a number, the product $\kappa^{(k)}(C_i, \chi_i)$ of the numbers which the steps $1, \dots, k$ extract from it; since it is not, the loop also contributes the Wilson loop of the background field, corrected by insertions along the loop which are small where the background is close to flat and decay away from the regions of large fields. The insertions interact with each other through the excluded volume of the polymer expansions which produced them, and that interaction is resummed into a factor attached to each family of compatible insertions. The precise form is as follows.

Fix $k \geq 1$ and a labelled term of $\varrho_k$, with the labels of the construction at every scale. A *traversal* at the step $k$ is an oriented bond of the lattice of the scale $k$ which is contained in one of the loops $C_i$, traversed in the orientation of $C_i$; by Lemma 3.14 each loop is a union of $P_k(C_i) = P_q(C_i)a_k$ traversals. For a field $U$ on the fine lattice and an arc $J$ of $C_i$, that is, a set of consecutive traversals, $U(J)$ denotes the product of the transports along the fine bonds of $J$. The objects which make up a dressed loop are the following.

(a) *Groups.* A group $W$ is a finite set of localisation domains of the scales $\leq k$, together with a set $\mathcal{I}(W)$ of pairwise disjoint arcs of the loops, its *intervals*; the set $\mathbb{L}(W)$ of the loops which carry an interval of $W$ is nonempty. Its *size* is $\sigma_k(W) \coloneqq \#_k(W) + n_k(W)/M$, where $\#_k(W)$ is the number of cubes of $\pi_k$ in its domains and $n_k(W)$ the number of traversals in its intervals. A group is a *cross group* if $|\mathbb{L}(W)| \geq 2$. Two groups are *compatible* if their domains are disjoint and so are their intervals, and a *family* is a finite set $\mathfrak{B}$ of pairwise compatible groups.

(b) *Insertions.* Each group $W$ carries an *insertion* $Y_W$, a linear map of $\bigotimes_{i \in \mathbb{L}(W)} V_{\pi_i}$, where $V_{\pi_i}$ is the space of the representation $\pi_i$. It is a function of the block field $V_k$ and of the outer variables of the term, the variables which $\mathbf{T}_k$ holds pending.

(c) *Words.* For a family $\mathfrak{B}$, the word $w_k(\mathfrak{B})$ is the normalised trace, on $\bigotimes_i V_{\pi_i}$, of the product over the loops of the ordered products along each $C_i$, in which the transport $\pi_i(U_k(J))$ along every arc $J$ outside the intervals of $\mathfrak{B}$ alternates with the insertions $Y_W$ at their intervals. Here $U_k$ is the background field of the term, and on a large-field region crossed by a loop the transports also contain outer variables. For the empty family $w_k(\emptyset) = \prod_i f_k^{(i)}$ with the *leading words* $f_k^{(i)} \coloneqq \chi_i(U_k(C_i))$.

(d) *Interactions.* $Q_k(\mathfrak{B}) = \exp(-\sum_\Gamma \varphi_k(\Gamma))$, the sum over the clusters $\Gamma$ of a hard-core gas of groups, the *reference gas* of the step, which are incompatible with a member of $\mathfrak{B}$, with numbers $\varphi_k(\Gamma)$.

(e) *The scalar exponent.* $\mathcal{E}_k = \sum_{j \leq k} \sum_X e_X^{(j)} + E_{\text{const},k}$, a sum of local terms $e_X^{(j)}$ indexed by connected unions $X$ of boxes of the lattice integrated at the step $j$, and a number $E_{\text{const},k}$.

The bounds of the class are stated with the following quantities. The small parameters of the step $k$ are, with the radius $r_k \coloneqq A_1(\log g_k^{-2})^{p_0}$ of the fluctuation cutoff in units of the coupling,

$$\tau_k \coloneqq C_\tau[g_k(1+r_k)^3 + e^{-c_\tau r_k^2}] \,, \qquad \rho_k \coloneqq C_\rho g_k^2(1+r_k)^4 \,, \qquad \rho_{t,k} \coloneqq C_t g_k(1+2r_k) \,, \quad (5.5)$$

with constants fixed in Appendix A. For a traversal $\mathfrak{n}$ and $t \geq 1$ let $\mathcal{P}_t(\mathfrak{n})$ be the set of the fine plaquettes of $\Omega_k$ within distance $C_{\mathrm{box}}M(t+1)$ of $\mathfrak{n}$, the distance measured in units of the scale $k$. The *deviation* of the background near $\mathfrak{n}$ is

$$\delta_{\mathfrak{n}}^*(U) \coloneqq L^{2k}\sum_{t\geq 1}(t+1)^3 e^{-\eta_w t}\max\{|U(\partial p)-1| \,|\, p \in \mathcal{P}_t(\mathfrak{n})\} \,, \quad (5.6)$$

the maximum over the empty set being 0; the factor $L^{2k}$ converts the deviation of a fine plaquette into that of a plaquette of the scale $k$. Finally $d_k(\mathfrak{n})$ is the distance, in units of the scale $k$, from $\mathfrak{n}$ to the large-field data of the scale $k$ (the large-field region $Z_k$ of the term), and $N_{\min}^{(k)}$ is the smallest period of the torus in units of the scale $k$.

**Definition 5.1** (Dressed Loops)**.** Let $k \geq 1$. A *dressed loop of the step $k$* for the ensemble $\mathcal{C}$ on a labelled term of $\varrho_k$ is a function of $V_k$ and of the outer variables of the term of the form

$$\mathcal{W}_k = \prod_{i=1}^{n}\kappa^{(k)}(C_i,\chi_i)\cdot e^{\mathcal{E}_k}\Big[\prod_{i=1}^{n} f_k^{(i)} + \sum_{\mathfrak{B}\neq\emptyset} Q_k(\mathfrak{B})\, w_k(\mathfrak{B})\Big] \,, \qquad \kappa^{(k)}(C,\chi) \coloneqq \prod_{j=1}^{k}\kappa_j(C,\chi) \,, \quad (5.7)$$

the sum over the nonempty families, absolutely convergent, with the objects (a)–(e) and the step factors $\kappa_j$ of Definition 7.8, such that the following hold.

(i) *Interactions.* $|Q_k(\mathfrak{B})| \leq \prod_{W\in\mathfrak{B}} e^{\theta(W)}$ with $0 \leq \theta(W) \leq \nu\sigma(W)/16$, where $\sigma(W)$ is the size of $W$ at the step at which it was made.

(ii) *Insertions.* For every traversal $\mathfrak{n}$, the sum $\nu_{\mathfrak{n}}^{(k)} \coloneqq \sum_{W\ni\mathfrak{n}} e^{\theta(W)}\|Y_W\|$ over the groups of the term made at the step $k$, and over the groups of earlier steps which are not attached to a large-field region of their step, satisfies

$$\nu_{\mathfrak{n}}^{(k)} \leq C_\chi\tilde{\rho}_k\delta_{\mathfrak{n}}^*(U_k) + C\hat{\rho}_k e^{-c_{\mathrm{att}}d_k(\mathfrak{n})/M} + C\rho_{t,k}\,\mathfrak{m}_k(\mathfrak{n}) + \sum_{j=1}^{k} CL^{k-j}\hat{\rho}_j e^{-c_{\mathrm{att}}N_{\min}^{(j)}/M} \,, \quad (5.8)$$

where $\tilde{\rho}_k \coloneqq \rho_k/\min(\alpha_{0,k},\alpha_{1,k})$ for $k \geq 2$ and $\tilde{\rho}_1 \coloneqq \rho_1$, $\hat{\rho}_k \coloneqq C\rho_k$, and $\mathfrak{m}_k(\mathfrak{n})$ is the sum of the marks of the large-field data of the step $k$ near the boundary of $Z_k$, weighted by $e^{-c'_*\,\mathrm{dist}(\cdot,\mathfrak{n})/M}$. The groups attached at an earlier step $j<k$ to a component $Z$ of the large-field region of the step $j$ satisfy, summed over all traversals, the bound $C\hat{\rho}_j\sum_{\mathfrak{n}_j} e^{-c_{\mathrm{att}}d_j^Z(\mathfrak{n}_j)/M}$, the sum over the traversals of the step $j$, and the corresponding bound with $\rho_{t,j}$ for those attached to the boundary layer of $Z$. The groups of the *deviation species*, whose insertions vanish at pure-gauge data on their arcs, satisfy the first term of (5.8) with weights: $\sum_{W\ni\mathfrak{n}} e^{\theta(W)+\nu\sigma(W)}\|Y_W\| \leq C_\chi\tilde{\rho}_k\delta_{\mathfrak{n}}^*(U_k)$, the sum over these groups, with $\sigma(W)$ the size of $W$ at the step at which it was made.

(iii) *Analyticity.* For real $U_k$ in the support of the indicators of the term, the factors which depend on the background extend analytically to $\mathbf{U} = e^{iL^{-k}A'}U_k$, for complex $A'$ in the polydisc of the radii $(1-\beta(1-2^{-(k-m)}))\alpha_{1,m}L^{-m}$ per fine bond on $\Omega_m\setminus\Omega_{m+1}$ (and on $\Omega_k$ for $m=k$), and for $k\geq 2$ with the gradient condition of [9, (2.39), p. 261] and in the variable $\mathbf{J}$ of [9, (2.36)]; they are covariant under $G$-valued gauge transformations,

and (5.8) holds there with the deviation of $\mathbf{U}$ from $U_k$ added to $\delta^*_{\mathfrak{n}}$. The same factors are defined, and analytic in the same polydiscs, at the *reference data* of the term, the trivial field 1 in place of $U_k$, with $\mathbf{J} = 0$ and the outer variables trivial, whether or not the reference data lie in the support of the indicators.

(iv) *Scalar parts.* $Q_k(\mathfrak{B})$ and $\mathcal{E}_k$ are numbers on each labelled term, independent of $V_k$ and of the outer variables, and

$$\sum_{X \ni x} |e^{(j)}_X| e^{\kappa_e |X|} \le C \tau_j^2 e^{-c_{\text{th}} d^{(j)}(x)} , \qquad |E_{\text{const},k}| \le \sum_{j \le k} \epsilon_j(\mathcal{C}, T) , \tag{5.9}$$

where $d^{(j)}(x)$ is the distance of the box $x$ of the scale $j-1$ from the complement of the carrier of the loop's integral at the step $j$, and $\epsilon_j(\mathcal{C}, T)$ is the remainder bound of the step $j$ defined in (7.31).

(v) *Uniformity.* The constants depend on the constants of the construction, the characters, and the shape constants $\Lambda(C_i)$, and not on $k$, $K$, the torus, the labelled term, or the number of loops; for an ensemble with admissible loops they may be taken to be nondecreasing functions of $c_{\mathcal{C}}$ which depend on $\Lambda_*$ in place of the $\Lambda(C_i)$, because $\Lambda(C_i)$ enters only through the counts of Lemma 5.3, which increase with it, and a character only through the number $c_{\pi_i}$ of (1.7) (Section 1.3).

(vi) *Torus consistency.* On the trivial term and at its reference data, let $T \subset T'$ be two tori of the family $\mathfrak{T}$ of Section 7.1. For every insertion and every cluster weight $\varphi_k(\Gamma)$ of an object whose dependence set has diameter less than $\iota_{T,k}/4$, the values on $T$ and on $T'$, at fields which coincide under the covering on the $\iota_{T,k}/4$-neighbourhood of a lift of that set, differ by amounts whose sums rooted at a traversal are at most $C_{\text{rel}} e^{-\iota_{T,k}/(32M)}$ times the right side of (5.8) for the insertions, and at most $C_{\text{rel}} e^{-\iota_{T,k}/(32M)} \mathfrak{a}_k$ for the cluster weights.

The ensemble is *dressed at the step $k$* if $N_k = \sum \chi_k(\Omega_k) \mathbf{T}_k[\mathcal{W}_k \exp A_k]$, the sum over the labelled terms of $\varrho_k$, with a dressed loop $\mathcal{W}_k$ of the step $k$ on each of them.

Three comments explain the definition. First, the bound (5.8) separates two kinds of insertions. The first term, the *deviation part*, is small because the background is close to flat; it vanishes when $U_k$ is a pure gauge, by Schur's lemma, which is why it is proportional to $\delta^*$. The other terms are *attached* parts, which carry no deviation and are instead small through the decay away from the large-field regions, the boundary layers, and the wrapping of the torus. Secondly, no smallness of $\sum_W \|Y_W\|$ over a whole loop is asserted: the sum over families is controlled by the product $\prod_W (1 + e^{\theta(W)} \|Y_W\|)$, whose logarithm is a sum over the traversals of (5.8), so that it costs a factor which is exponential in the number of traversals with a small rate. Thirdly, the step factors $\kappa_j$ are defined in Section 7 by an explicit formula, and the requirement that the normalisation in (5.7) be exactly $\prod_j \kappa_j$ is what makes the class an induction for the normalisation of Definition 1.3.

### 5.4 The First Step with a Loop Ensemble

**Theorem 5.2** (The First Step)**.** *Let $L$ and $\Lambda_*$ be as in Theorem 1.5, let $\mathcal{C}$ be a loop ensemble at level $q$ whose loops are admissible, and let $\bar{g}$ and $q$ be as in Theorem 1.5 for $\mathcal{C}$. For every $K$ larger than an index $K_{\text{tor}}(\mathcal{C})$, which depends on $\mathcal{C}$ only through $\max_i P_q(C_i)$ and on $q$ and $T/u$, the density $N_1$ obtained from $N_0 = W_{\mathcal{C}} \varrho_0$ by the first step of the construction for the target $(K, \bar{g})$ satisfies* (5.2)*, and the ensemble is dressed at the step* 1*.*

The proof uses the first-step manuscript [1] for the analysis of one square loop and of two squares, and consists in checking that each of its statements holds for a general ensemble and for the construction of this paper. We first collect the properties of a general loop that replace those of a square in that analysis; they are consequences of Section 2. Throughout, $C$ is a loop of $\mathcal{C}$, read in fine units on the lattice $\mathbb{T}_K$, with $P_0(C)$ bonds, and $\ell$ is a length which is at most a quarter of the separation $a_0 \geq L^3 M R_{K'} L^{K'}$ of Lemma 3.14.

**Lemma 5.3** (Square Properties of a General Loop)**.** *Let $C$ be a loop with chord–arc constant $\Lambda(C)$.*

*(i) The bonds of $C$ which meet a closed $\ell^\infty$-cube of side $s$ are at most $\Lambda(C) \cdot 2(s+1)$, and the sites of $C$ in a set of $\ell^\infty$-diameter $D$ are at most $2\Lambda(C)(D+1)+1$.*

*(ii) For a set $S$ of $\ell^\infty$-diameter $D$ which meets $C$, the smallest arc of $C$ which contains $S \cap C$ has at most $4\Lambda(C)D + 2$ bonds, or else contains more than half of $C$.*

*(iii) A set of $\ell^\infty$-diameter less than $r_0(C)$ which meets $C$ meets at most two consecutive segments of $C$, and on two consecutive segments the chord–arc constant is 2.*

*Proof.* (i) Two sites $x, y$ of $C$ in the cube have $|x-y|_\infty \leq s$, hence, by Definition 2.4, arc distance at most $\Lambda(C)s$. The sites of $C$ in the cube therefore lie on an arc of at most $\Lambda(C)s$ bonds measured from any one of them in either direction, that is, on at most $2\Lambda(C)s+1$ sites, and a bond which meets the cube has an end among them or is one of the two bonds at each end; this gives at most $2\Lambda(C)(s+1)$ bonds for $s \geq 1$, and the count of sites in a set of diameter $D$ is the same argument. (ii) is Lemma 2.6, and (iii) is Lemma 2.10 together with Lemma 2.5 applied to the path formed by two consecutive segments, which is a bent path with two perpendicular arms. *q.e.d.*

*Proof of Theorem 5.2.* The identity (5.2) at $k = 1$ holds because the first step preserves integrals (Section 5.1). The rest of the proof goes through the analysis of [1] in the order of that paper and states, for each part, the change which a general ensemble or the construction of this paper requires.

*The insertion at a fixed cutoff and the extraction.* The sourced effective interaction of [1, Sections 2–3] is built from the fine bonds of the loop and the local operators of the step, whose ranges are fixed and do not depend on the loop. The loop enters through three counts: the hull of a cluster on the loop, the number of labels which meet a bond, and the number of traversals within a given distance of a point. For a square these are bounded by the elementary geometry of the square; for $C$ they are bounded by Lemma 5.3: the hull constant $32(1+R_*)$ of [1, (5.24)] becomes $16\Lambda(C)(1+R_*)$ by (ii), and the number of source labels of one traversal which meet a bond, which is at most a constant $q^{\mathrm{src}}$ for each of the traversals whose initial sites lie in a set of diameter $2R_*+2$, is at most $q^{\mathrm{src}}(4\Lambda(C)(R_*+1)+1)$ by (i). The constants which depend on these counts are fixed before the coupling, and the conditions on the coupling which contain them depend on $\Lambda(C) \leq \Lambda_*$. Every condition on the coupling in this proof which involves the loops is a condition on $g_0$ or on $g_1$, both of which are at most $\gamma_q$ by (3.46), so that it is a requirement on $\gamma_q$, depending on $c_{\mathcal{C}}$.

*The coefficients.* The perimeter and corner coefficients of [1, Sections 5–6] are defined on the straight line and on the bent path of two perpendicular rays through the coarse sites, in the orientations of the square. We define them for every orientation $v \in \mathcal{V} := \{\pm e_0, \dots, \pm e_3\}$ on the line, and for every ordered pair $(v, w)$ of perpendicular orientations on the bent path (2.6); the proofs of their convergence and of their tail bounds use only that the path is a

line, or two perpendicular rays, through coarse sites, and not which coordinate directions it uses, so they hold with the same constants. The extraction of the regional scalar of the loop is then the following: allocate each cluster of the reference gas to the coarse segment of its source, give each corner $c$ the clusters allocated to the second half of its incoming segment and to the first half of its outgoing segment, and compare the clusters rooted there with those of the bent path $B_c$. By Lemma 2.10 a cluster which distinguishes $C$ from $B_c$ near $c$ reaches a distance at least $r_0(C) - L/2 - 2R_*$ from its root, and by the tail bounds of the cluster expansion such clusters cost at most $C\tau_0^2 e^{-cr_0(C)/L}$ per root. With $\kappa_C$ the exponential of the sum of the line coefficients over the segments and of the bent-path coefficients over the corners, this gives

$$\Big|F_{C,\infty} - \log \kappa_1(C,\chi)\Big| \le C\tau_0^2 \Big(\frac{P_0(C)}{L} + |\mathcal{K}(C)|\Big) e^{-cr_0(C)/L} \,, \tag{5.10}$$

in place of the square's bound, where $F_{C,\infty}$ is the bulk limit of the logarithm of the reference partition function of the loop and $\kappa_1$ is the step factor of Definition 7.8 at $j = 1$. The boundary and periodic comparisons of [1, Section 7 and Appendix F] are made per coarse segment and use the torus only through its smallest period; they hold for $C$ with the sums over the $4n$ coarse segments of a square replaced by the sums over the $P_0(C)/L$ coarse segments of $C$. The bounds of the leading term then hold with $e^{4C\tau_0^2 P_0(C)/L}$ in place of $e^{C\tau^2(9n+1)}$, and the constant $1 + 1/c$, which bounds $(n+1)e^{-cn}$ for a square, is replaced by $1 + 2P(C)/(e\,c\,r(C))$, since $(P_0(C)/L + |\mathcal{K}(C)|)e^{-cr_0(C)/L} \le 2(P(C)/r(C))\,(r_0(C)/L)e^{-cr_0(C)/L}$ by Lemma 2.5.

*The retreated carrier and the labelled terms.* The analysis of [1, Section 9] on the labelled terms uses the loop through the envelopes of the groups, which are arcs of fewer than half of the traversals or the whole loop read from a corner, through the pins of the groups, and through the lifting of the groups to the universal cover. By Lemma 5.3(ii) the number of cubes of side $LM$ which meet an envelope $\mathcal{H}(W)$ is at most $4 + (1 + 2/L)|\mathcal{H}(W)|/(LM)$, in place of $8 + |\mathcal{H}(W)|/(LM)$, so that the constants of the envelopes are unchanged. In the pins, the hull of a component $\gamma$ of a group has at most $(8\Lambda(C)+1)LMn_\gamma$ traversals, in place of $17LMn_\gamma$, and the condition on the contraction rate which V4 imposes becomes $1000\,\eta\,(8\Lambda(C)+2) < 1/128$, which holds for $\eta \le \eta(\mathcal{C}) \coloneqq 1/(1\,200\,000\,\Lambda(\mathcal{C}))$. A small interior group has its contacts within a set of diameter less than $r_0(C)$, so that by Lemma 5.3(iii) they lie on two consecutive segments, and its lifting is the one of the square. The torus conditions of [1] involve the loop through its diameter and through $4ne^{-c\iota_T} \le 1$, which reads $(P_0(C)/L)e^{-c\iota_T} \le 1$; both hold for $K \ge K_{\mathrm{tor}}(\mathcal{C})$, because $3\,\mathrm{diam}(C) < T$ and $\log(P_0(C)/L) \le \log P_q(C) + K \log L$ while $\iota_T$ grows like $L^K$.

*Several loops.* For two squares, [1, Section 10] groups the clusters of the joint expansion into groups of one loop and cross groups, proves Schur's lemma for each loop separately, and bounds the cross terms by the distance between the squares. Its proofs use the geometry of the pair only through the distance between the two loops and through the property that a set of diameter less than that distance meets one loop. For an ensemble, by Lemma 3.14 the loops are at distance at least $a_0$ on the fine lattice, so that a set of diameter less than $a_0$ meets one loop, and the arguments of [1, Section 10] apply to every pair of loops, with the cross exponent the sum over the pairs. The normalised trace on $\bigotimes_i V_{\pi_i}$ of a product of operators is bounded by the product of their operator norms, as for two factors.

*The construction of this paper.* Two features of the construction differ from those assumed in [1]. The thresholds are those of the tube at $\hat{g}_0$: by Lemma 3.5 the radius of the fluctuation cutoff is $t_0 \coloneqq \vartheta_0/g_0 \in [\rho^{-1/2} r_0, r_0]$, and every statement of [1] which uses the radius as an upper bound on a passed field holds a fortiori, while each one which uses it as a lower bound holds with the constant in its exponent divided by $\rho \le 3/2$; the only condition which changes

is the one behind $e^{-c_{\mathrm{ph}} r^2} \le g_0$ in [1, (9.29)(iii)], with $c_{\mathrm{ph}} = 9c_n\alpha_G/128$, which at the radius $t_0$ reads $(9c_n\alpha_G/(64\rho))A_1^2 \log g_0^{-2} \ge 1$ and is a smallness condition on the coupling (Supplement, Proposition S.7). The test functions of the plaquettes are soft: by Lemma 3.8 a term of the decomposition is supported where the passed plaquettes are below $(1+\kappa_s)$ times the threshold, and every bound of [1] drawn from a passed test function holds at the threshold multiplied by $1+\kappa_s$ (Supplement, Proposition S.2).

*Assembly.* With these changes, the main theorem of [1] and its form on the labelled terms give, on every labelled term, the contribution of the ensemble as $\prod_i \kappa_1(C_i,\chi_i) e^{\mathcal{E}_1} [\prod_i f_1^{(i)} + \sum_{\mathfrak{B}\neq\emptyset} Q_1(\mathfrak{B}) w_1(\mathfrak{B})]$, with the insertions bounded per fine traversal by the deviation of the background and the attached parts, the interactions resummed by the cluster expansion against the reference gas, and the scalar exponent the sum of the carrier part, which is a sum of local terms, and of the torus and corner part, which is a constant. The bounds (i)–(iv) of Definition 5.1 at $k=1$ follow by summing the bounds of [1] over the $L$ fine traversals of each traversal of the scale 1, which multiplies the constants by $L$ and converts the decay lengths $LM$ in fine units into $M$ in units of the scale 1, and by the inequality $g_0 \le (1+\beta_0) g_1$ of Lemma 3.2, which converts the parameters at $g_0$ into those at $g_1$. Clause (vi) holds at $k=1$ with vanishing differences: on the trivial term every insertion and every cluster weight of the first step is a finite sum of absolutely convergent walk series and parameter integrals in local data, each term of which is the same expression on a lift [1, Sections E.9.2 and E.10.4]. The constants depend on the loops only through $\Lambda(C_i)$, $P(C_i)/r(C_i)$, and the characters, and on the number of loops not at all, because every count above is a count per traversal or per cube, and a cube of side less than $a_0$ meets one loop. *q.e.d.*

## 6 The Inductive Step

We now show that a renormalisation step maps a dressed loop of the step $k$ to one of the step $k+1$, with the same constants. The difficulty is one of counting. A loop has $P_k(C)$ traversals at the step $k$, and $P_k(C) = P_q(C) a_k$ grows like $L^{K-k}$, so that every quantity which a step contributes in proportion to the perimeter is, summed over the loop, unbounded as $K \to \infty$, even when it is of order $g_k^2$ per traversal. Such quantities can therefore be neither estimated nor neglected; each of them has to be either absorbed exactly into the step factor $\kappa_{k+1}$, or carried along without ever being summed in absolute value over the loop. The deviation parts of the insertions are of the second kind, and what keeps them under control is a contraction: a traversal of the step $k+1$ consists of $L$ traversals of the step $k$, while the deviation of the background, measured in units of the scale $k+1$, is $L^2$ times the same deviation measured in units of the scale $k$, so that with the normalisation $\rho_k \delta^*$ of (5.8) an old insertion contributes to the new bound with a factor of order $L \cdot L^{-2} = L^{-1}$.

**Theorem 6.1** (The Inductive Step)**.** *Let $L$, $\Lambda_*$, $\bar g$, $\mathcal{C}$, and $K \ge K_{\mathrm{tor}}(\mathcal{C})$ be as in Theorem 5.2, and let $1 \le k < K'$. If the ensemble is dressed at the step $k$, then it is dressed at the step $k+1$, with the same constants in Definition 5.1, and* (5.2) *holds at $k+1$.*

**Corollary 6.2** (Dressed Loops at Every Step)**.** *Under the hypotheses of Theorem 5.2, the ensemble is dressed at every step $1 \le k \le K'$, with constants independent of $k$, $K$, the torus, and the number of loops.*

*Proof.* Induction on $k$, with Theorem 5.2 as its base and Theorem 6.1 as its step. *q.e.d.*

The proof of Theorem 6.1 occupies the rest of this section. It follows Bałaban's step [9, Sect. 3] on the terms with small fields, with the dressed loop carried through each of its

operations, his treatment of large fields [12, Sect. 1] for the terms whose large-field regions grow, and the **R**-operation of [11, 12]. We fix a labelled term of $\varrho_k$ and write $U_k$ for its background field, $U_{k+1}$ for the background field of the step $k+1$, and $V^{(k)} \coloneqq M^k(U_{k+1})$ for the block field of the step $k$ determined by $U_{k+1}$.

### 6.1 The Evaluation Point

The fluctuation field of the step $k+1$ is the deviation of $V_k$ from $V^{(k)}$, and every factor of the dressed loop has to be expanded around the configuration at which this deviation vanishes. Two facts make this possible. First, $V^{(k)} = M^k(U_{k+1})$ [8, (2.3), p. 265], and since the minimal orbit with given averages is unique [9, p. 256], $U_k(V^{(k)}) = U_{k+1}$ in the gauge of the step. Secondly, $V^{(k)}$ lies in the support of the test functions of the step $k$ on $\Omega^{\sim}_{k+1}$: by [9, (3.19), p. 268] the block field $V^{(k)}$ is within $\vartheta_k$ of the local reference fields on the cubes of [9, (3.16)], so that it passes the test [9, (3.3), p. 265] with threshold $2\vartheta_k$, and Bałaban concludes that "$\chi_k(\square) = 1$ for $\square \subset \Omega^{\sim}_{k+1}$" [9, p. 266]. For the soft test functions the same holds with the margin of Lemma 3.8. On the cubes of the band $\Omega^{\sim}_{k+1} \setminus \Omega^{\sim -1}_{k+1}$, which lie in no cube of [9, (3.16)], the localisation of the minimiser with constraints of two scales is needed; it is not printed, and it is proved in Supplement, Proposition S.8, from the uniqueness of the minimal orbit [7, Theorem 1].

The regularity of the new background is [9, p. 268]: $|\partial U_{k+1} - 1| < 2B_3\varepsilon_{k+1}(L^{-1}\eta)^2$ on $\Omega_{k+1}$. By the lasso identity, which writes the holonomy around a plaquette of the scale $j$ as an ordered product of conjugates of the holonomies of the $L^{2j}$ fine plaquettes inside it, and by $|\prod_i A_i - 1| \le \sum_i |A_i - 1|$ for unitary $A_i$, a plaquette of the scale $k$ in $\Omega_{k+1}$ has deviation at most $2B_3\varepsilon_{k+1}L^{-2}$ under $U_{k+1}$. Consequently, for the deviation (5.6),

$$\delta^{*,(k+1)}_{\mathfrak{n}'}(U) = L^2\,\delta^{*,(k)}_{\mathfrak{n}'}(U) \qquad \text{and} \qquad \delta^{*,(k+1)}_{\mathfrak{n}'}(U_{k+1}) \le 2c_\sigma B_3\varepsilon_{k+1}\sum_{t\ge 1}(t+1)^3 e^{-\eta_w t}\,, \tag{6.1}$$

where the superscript indicates the scale whose units are used, the first identity holds for the same plaquettes and radii, and $c_\sigma$ converts Bałaban's matrix norm into the norm of the insertions.

### 6.2 The Leading Words and the Small-Field Map

*The leading words.* At the evaluation point the transport of $U_k$ along a traversal is the transport of $U_{k+1}$ multiplied by a function of the fluctuation, which is analytic and equal to one at zero fluctuation [8, p. 265]. Writing $\pi_i(\hat{T}_m) = (1 + Y_m)\pi_i(T_m)$ for each traversal $m$ of $C_i$, with $T_m$ the transport of $U_{k+1}$, and expanding the ordered product, we obtain

$$f^{(i)}_k = \sum_S \dim(\pi_i)^{-1}\,\mathrm{Tr}\Big(\prod_m (Y_m^{[m\in S]})\pi_i(T_m)\Big)\,, \tag{6.2}$$

the sum over the subsets $S$ of the traversals, in which the term $S = \emptyset$ is the leading word $f^{(i)}_{k+1}$ of the step $k+1$ and every other term is a word with *slots* $Y_m$, $m \in S$. The identity is exact. Each slot is a function of the fluctuation which vanishes with it, so that in the fluctuation integral it is either integrated to a scalar (by Schur's lemma at flat background, since the integral commutes with $\pi_i(G)$ by gauge covariance) or produces a fresh insertion of deviation size.

*The small-field map.* On $\Lambda_{k+1}$ the step is Bałaban's: the change of variables of [9, (3.22), p. 269] to the fluctuation field, the Gaussian measure with the covariance $C^{(k)}$ of the

step, and the cluster expansion of the fluctuation integral [10, Sect. 2], whose polymers are localisation domains of the scale $k+1$. The dressed loop enters this integral as a further factor. Its insertions and slots are composed with the change of variables; being analytic in the background on the domains of Definition 5.1(iii), they are analytic in the fluctuation field on the domain of the cluster expansion, and they are covariant under gauge transformations. We expand the product of the dressed loop and of Bałaban's integrand jointly: the polymers of the expansion which meet the support of an insertion or the arc of a slot are merged with it into a *group* of the step $k+1$, whose domains are the merged polymers and whose intervals are the union of the intervals of its members, joined into the smallest arcs which contain them; the remaining polymers form Bałaban's expansion of the effective action. Each group is integrated against the Gaussian measure, which gives the new insertion $Y_W$, and the families of compatible groups are resummed against the gas of the polymers which they exclude, which gives the interactions $Q_{k+1}(\mathfrak{B})$. The identity $N_{k+1} = \sum \chi_{k+1}(\Omega_{k+1}) \mathbf{T}_{k+1}[\mathcal{W}_{k+1} \exp A_{k+1}]$ is the identity of the step for $\varrho_{k+1}$ with this additional factor in every term, and (5.2) holds at $k+1$ because the step preserves integrals.

*The flat parts.* Every group of the step has a *flat value*, its insertion at the reference data of the term, the trivial background with the labels of the term. By gauge covariance under constant gauge transformations, which fix the trivial background, the flat value of a group on one loop commutes with $\pi_i(G)$, and by Schur's lemma it is a multiple $c_W\,\mathrm{id}$ of the identity. The insertion is split as $Y_W = c_W\,\mathrm{id} + (Y_W - c_W\,\mathrm{id})$; the scalars $c_W$ form the reference gas of the step, and the second part vanishes at flat data and is a deviation. Section 7 resums the reference gas along each loop and extracts from it, exactly, the step factor $\kappa_{k+1}$ and the remainder $E^{\circ}_{k+1}$, which joins $\mathcal{E}_{k+1}$. A cross group carries no flat value: its insertion acts on the tensor product of the spaces of several loops, and it is kept as an insertion.

### 6.3 The Contraction of the Deviation Parts

We first carry the insertions which the term inherits from the step $k$. Each traversal $\mathfrak{n}'$ of the step $k+1$ contains $L$ traversals of the step $k$, and an old insertion is split as $Y(U_k) = Y(U_{k+1}) + [Y(U_k) - Y(U_{k+1})]$, into its value at the evaluation point and its fluctuation part.

*The values at the evaluation point.* By (5.8) at the step $k$, the deviation part of the old insertions through a traversal $\mathfrak{n} \subset \mathfrak{n}'$ is at most $C_\chi^{(k)} \tilde\rho_k \delta_{\mathfrak{n}}^{*,(k)}(U_{k+1})$. The neighbourhoods of $\mathfrak{n}$ in (5.6) lie in those of $\mathfrak{n}'$ with the radii of the step $k+1$, because the radii $C_{\mathrm{box}} M(t+1)$ are the same numbers in the units of each step and the units of the step $k+1$ are $L$ times larger. The plaquettes of $\Omega_{k+1}$ give, by (6.1), at most $L^{-2}\delta_{\mathfrak{n}'}^{*,(k+1)}(U_{k+1})$; those of $\Omega_k \setminus \Omega_{k+1}$ give a term which decays with the distance to $\Omega_{k+1}^c$ and is attached to the large-field region of the step $k+1$. Summing over the $L$ traversals in $\mathfrak{n}'$,

$$\sum_{\mathfrak{n}\subset\mathfrak{n}'} C_\chi^{(k)} \tilde\rho_k \delta_{\mathfrak{n}}^{*,(k)}(U_{k+1}) \le L^{-1} \frac{\tilde\rho_k}{\tilde\rho_{k+1}} C_\chi^{(k)} \tilde\rho_{k+1} \delta_{\mathfrak{n}'}^{*,(k+1)}(U_{k+1}) + (\text{attached})\,. \tag{6.3}$$

The ratio is controlled by Lemma 3.2: $g_k \le (1+\beta_0) g_{k+1}$ by (3.9), $(\log g_k^{-2})^p \le (1+\beta_0)(\log g_{k+1}^{-2})^p$ by (3.10), and $\alpha_{i,k+1} \le (1+\beta_0)^2 \alpha_{i,k}$ by (3.18), so that

$$\frac{\rho_k}{\rho_{k+1}} \le (1+\beta_0)^6\,, \qquad \frac{\tilde\rho_k}{\tilde\rho_{k+1}} \le (1+\beta_0)^8\,. \tag{6.4}$$

*The fluctuation parts.* The difference $Y(U_k) - Y(U_{k+1})$ is of order $\tilde\rho_k g_k$ and has mean zero at first order in the fluctuation. Its mean at second order, and its contractions with the other old insertions and with the slots of the leading words, are of order $\tilde\rho_k g_k^2$ per pair and decay

with the distance between the two factors at the rate of the covariance of the step, so that each traversal of the step $k+1$ meets $O(L^2)$ pairs. Gauge covariance makes the leading part of each such contraction scalar. Indeed, if the covariance of the fluctuations at two points is $c\,\mathrm{Ad}(X)$, with $X$ the transport between them, and $T^a = \pi_*(t_a)$ for an orthonormal basis $(t_a)$ of $\mathfrak{g}$, then $XT^bX^{-1} = \sum_c \mathrm{Ad}(X)^{cb}T^c$ and the orthogonality of $\mathrm{Ad}(X)$ give

$$\sum_{a,b} T^a X T^b\,\mathrm{Ad}(X)^{ab} = \sum_{a,b,c} T^a T^c\,\mathrm{Ad}(X)^{cb}\mathrm{Ad}(X)^{ab}\,X = \Big(\sum_a T^a T^a\Big)X = C_2(\pi)\,X\ , \qquad (6.5)$$

where the Casimir operator $C_2(\pi)$ is a scalar by Schur's lemma; the same computation applies to any two families of insertions which transform under $\pi(G)$ like the $T^a$. The scalar part joins the flat parts. The rest is the deviation of the covariance from its transported form, which is of the order of the curvature of the background between the two points, summed over the $O(L^2)$ plaquettes of the scale $k$ in a block of the scale $k+1$, hence of order $L^2\delta^{*,(k)} = \delta^{*,(k+1)}$. Per traversal of the step $k+1$ the non-scalar remainder is therefore at most $C_d L^2 g_k^2 \tilde\rho_k \delta^{*,(k+1)}$, and, against the bound $\tilde\rho_{k+1}\delta^{*,(k+1)}$ of the class, the factor is $C_d L^2 g_k^2 \tilde\rho_k/\tilde\rho_{k+1} \le C_d(1+\beta_0)^8 L^2\gamma_q^2$, using (6.4) and $g_k \le \gamma_q$, which holds by (3.46) because $k < K'$.

*Fresh insertions.* The slots of the leading words (6.2), integrated against the fluctuations of the step, give insertions of size at most $C_{\mathrm{new}}\tilde\rho_{k+1}\delta^{*,(k+1)}_{\mathfrak{n}'}(U_{k+1})$ per traversal: this is the deviation bound of the first step [1, Section 9], which uses only the analyticity and covariance of the slots and the Gaussian bounds of the step, at the scale $k+1$.

*The recursion.* Adding the three contributions, the constant of the deviation part satisfies

$$C_\chi^{(k+1)} \le C_{\mathrm{new}} + aC_\chi^{(k)} + b\,\rho_k (C_\chi^{(k)})^2\ , \qquad a \coloneqq (L^{-1} + C_d L^2\gamma_q^2)(1+\beta_0)^8\ , \qquad (6.6)$$

the quadratic term coming from the contractions of two old insertions. At the steps at which the **R**-operation acts, $C_{\mathrm{new}}$ also contains the source $C^{\mathbf{R}}(g_k) \le C^{\mathbf{R}}(\gamma_q)$ of the deviation insertions which that operation makes, and $a$ contains the further term $C_a M R_k e^{-p_0(g_k)/4}$ (Section 6.4). We take $\beta_0$ so small that $(1+\beta_0)^8 \le L/4$ and $\gamma_q$ so small, depending on $c_{\mathcal{C}}$, that $C_d L^2\gamma_q^2 + C_a M R_k e^{-p_0(g_k)/4} \le 1/L$ for every $k \le K'$; then $a \le 1/2$ at every step. Put $C^* \coloneqq 2C_{\mathrm{new}}/(1-a) \le 4C_{\mathrm{new}}$. If $C_\chi^{(k)} \le C^*$ and $b\rho_k(C^*)^2 \le C_{\mathrm{new}}$, which holds for $\gamma_q$ small, depending on $c_{\mathcal{C}}$, since $\rho_k \le C_\rho\gamma_q^2(1 + A_1(\log\gamma_q^{-2})^{p_0})^4$ for $k \le K'$ by (3.46) and because $x \mapsto x(1 + A_1(\log x^{-1})^{p_0})^4$ is increasing for small $x$, then

$$C_\chi^{(k+1)} \le C_{\mathrm{new}} + aC^* + C_{\mathrm{new}} = (1-a)C^* + aC^* = C^*\ . \qquad (6.7)$$

So the constant of the deviation part stays below $C^*$ at every step, which is the first term of (5.8) at $k+1$ with constants independent of $k$. The same recursion holds for the sums with the weights $e^{\nu\sigma(W)}$ of Definition 5.1(ii): a carried insertion keeps the size and the weight of the step at which it was made, and the fresh insertions and the deviation parts of the new groups, among them the contractions of old insertions with each other and with the slots, are bounded with these weights by the Cauchy estimate below applied to (6.8), which carries them. The analyticity clause is carried in the same way, by Cauchy estimates on the polydiscs of Definition 5.1(iii), whose radii are those of Bałaban's spaces [9, (2.34)–(2.39), p. 261]; the factor $\tilde\rho_k = \rho_k/\min(\alpha_{0,k}, \alpha_{1,k})$ is the price of the Cauchy estimate which converts the bound on a polydisc into a bound on the deviation. The construction of the groups, the counting of the families against the reference gas by the condition of Kotecký and Preiss [15], and the bound $\theta(W) \le \nu\sigma_{k+1}(W)/16$ of the interactions, are the counts of the step $k+1$ per traversal and per cube of $\pi_{k+1}$; by Lemma 5.3(i), applied at the scale $k+1$, a cube of side $s$ meets at most $2\Lambda(C)(s+1)$ traversals, and by Lemma 3.14 it meets one loop when $s < a_{k+1}$, so these

counts are those of a square up to the factor $\Lambda(C)/2$. Their outcome, in the form in which Section 7 uses it, is the size bound

$$\sum_{W \text{ at } p} e^{\theta(W)+\nu\sigma_{k+1}(W)} \sup\|Y_W\| \le 2C_{S'}\rho_{k+1} \tag{6.8}$$

for every traversal $p$ of the step $k+1$ and every cube $p$ of $\pi_{k+1}$, where the sum runs over the groups of the step $k+1$ whose arcs contain $p$, or one of whose domains meets a cube of $\pi_{k+1}$ adjacent to $p$, and the supremum over the polydiscs of Definition 5.1(iii) around the background field of the term and around its reference data.

### 6.4 Large Fields and the R-Operation

The insertions attached to a large-field region carry no deviation, and they cannot be contracted: a segment of the step $n$ which runs parallel to the large-field data of the step $j < n$ contains $L^{n-j}$ traversals of the step $j$, each carrying an attached part of size $C\hat\rho_j$. They are therefore not kept in the per-traversal bound beyond the step after the one at which they were made. They are carried, with the species of their production, as functions which are never summed over the loop, and their total over each component $Z$ of the large-field region of the step $j$ is paid once, by the small factor which Bałaban's large-field estimates attach to $Z$.

*The payment.* Bałaban attaches to every large-field region $Z$ created at the step $j$ the factor $\exp(-\frac12\gamma_0 A_1^2 p_0^2(g_j)(d'_j(Z)+1)-2p_0(g_j))$, with $d'_j(Z)$ the linear size of $Z$, "the fundamental large field factors, which control convergence of the expansion of the effective densities" [12, p. 381], and he shows that its part $\exp(-\kappa_j(Z))$ "controls $K$ renormalization steps, under the assumption that no large fields are created in these steps", in the sense of the inequality [12, (1.80), p. 384]. Two costs have to be paid from it. The first is the total of the attached insertions,

$$A_j(Z) \le C\hat\rho_j N_j(Z) + C\rho_{t,j}\mathfrak{m}_j(Z) \le C_A(d'_j(Z)+1)\,, \tag{6.9}$$

where $N_j(Z) = \sum_{\mathfrak{n}_j} e^{-c_{\mathrm{att}} d_j^Z(\mathfrak{n}_j)/M}$ counts the traversals of the step $j$ near $Z$, which by Lemma 5.3(i) are at most $C\Lambda(C)$ per cube of $Z$, and $\mathfrak{m}_j(Z)$ is the sum of the marks of the data assigned to $Z$, at most a multiple of the number of cubes of $Z$. The second cost recurs at every step $n$ of the lifetime of $Z$: the traversals near $Z$ at the step $n$ are exposed, the expansion of the step is not available on them, and each costs $C_e\hat\tau_n^2$, $\hat\tau_n^2 := \max(\tau_n^2, C\hat\rho_n)$, with a constant $C_e$ of the construction. By Lemma 5.3(i) a cube of side $MR_n$ meets at most $4\Lambda(C)MR_n$ traversals, and along Bałaban's chain for the regions $S^{n-j}(Z)$ which descend from $Z$ [12, pp. 384–385], whose numbers of cubes are at most $C2^{-(n-j)}d'_j(Z)$ until the region fits into a cube of size $64MR_n$, after which at most $R_j$ steps remain, each with a region of at most $64^d$ cubes,

$$D_j(Z) := \sum_n 4C_e\Lambda(C)MR_n\hat\tau_n^2\,\#\mathsf{cubes}_n(S^{n-j}(Z)) \le C_d\Lambda(C)MLR_j^2\,\bar\tau^2\,(d'_j(Z)+1)\,, \tag{6.10}$$

with $\bar\tau^2 := \sup_{n\le K'}\hat\tau_n^2 \le C\gamma_q^2(\log\gamma_q^{-2})^{6p_0}$, by $g_n \le \gamma_q$ for $n \le K'$ from (3.46), the sum running over the steps $n \le K'$ at which the loops are carried, and $R_n \le LR_j$ for $n \ge j$ by Lemma 3.4. Both costs are at most a polylogarithm of $g_j^{-2}$ of fixed degree times $d'_j(Z)+1$, while $\kappa_j(Z)$ is of order $(\log g_j^{-2})^{2p_0}(d'_j(Z)+1)$, so that for $p_0$ large and $\gamma_q$ small, depending on $c_{\mathcal C}$, the factor of $Z$ pays both and leaves a reserve, since $j \le K'$ and $\log g_j^{-2} \ge \log\gamma_q^{-2}$; Bałaban's induction over the merging of regions [12, pp. 384–389] runs with the summand of [12, (1.80)] enlarged by these costs, since both are subadditive under the union of regions up to the slack $2d$ of

[12, p. 386]. That the attached parts are not increased by the later evaluations at $U_{n+1}$, by the fluctuation parts, and by the merging of several parents into one insertion, and that the parts of an attached insertion which reach the region of small fields are paid by the reserve of their component, are proved in Supplement, Proposition S.9.

*The* **R***-operation.* From the step at which the depth condition (N1) first admits a component, the **R**-operation acts on the numerator. It integrates a term over the block field on a domain inside a small and old large-field component and replaces it by the normalised fraction of [11, (1.100), p. 201]; since the fraction integrates to one over the variables which it replaces, the operation has the normalisation property [11, (1.102), p. 201], $\int (\mathbf{R}' f)\, \mathrm{d}V_k = \int f\, \mathrm{d}V_k$, for every integrable $f$ and not only for the density; the removal of the $\delta$-functions on the components of the first class [12, p. 378] is an equivalence which needs the invariance of the integrand under gauge transformations, and the dressed loop is gauge invariant. On the components of the first class the term is expanded by Bałaban's localisation expansion [12, pp. 372–376, (1.66)–(1.67)], which he makes for "any term in the sum" and which applies to the words of the loop, entire functions of their arguments. The part of the resulting polymer gas which is linked to the loop is written exactly as a ratio of two polymer gases, and at flat reference data each group of polymers which interleaves with the loop contracts to a scalar by Schur's lemma, as in Section 6.2. The flat part of the ratio is therefore the cluster series of a one-dimensional gas along each loop; its perimeter and corner parts are added to the step factor $\kappa_k$ (Definition 7.8), and its remainder $E_k^{\circ,R}$, bounded in Proposition 7.10, stays in $\mathcal{E}_k$. The rest is a sum of deviation insertions of the class, each vanishing at pure-gauge data on its hull, with a source of size at most $C^{\mathbf{R}}(g_k) \coloneqq \exp(-A_0(\log g_k^{-2})^{p_0}/4 + \log g_k^{-2} + C \log\log g_k^{-2})$, which is added to $C_{\mathrm{new}}$ in (6.6), and the groups of polymers which contain insertions of the term add $C_a M R_k e^{-p_0(g_k)/4}$ to the coefficient $a$ there. The details, including the depth $N_k$ of Definition 3.9 and the commensurability of the loops with the period $MR_k$ of the gas, are Supplement, Proposition S.10.

### 6.5 Several Loops, the Torus, and the Assembly

*Cross groups.* A group whose intervals lie on two or more loops is a cross group. By Lemma 3.14 the loops are at distance at least $a_{k+1} \geq L^2 M R_{k+1}$ in units of the scale $k+1$, so that a set of diameter less than $a_{k+1}$ meets at most one loop, and a cross group contains a connected chain of domains joining two loops, of total size at least $a_{k+1}/M$. Its insertion is bounded as the others, with the additional factor $e^{-\nu_\times \delta_{\mathfrak{n}}/M}$, where $\delta_{\mathfrak{n}}$ is the distance from the traversal $\mathfrak{n}$ to the other loops; the cross exponent $E_{k+1}^{\times}$, the difference between the logarithm of the reference gas of the ensemble and the sum of those of its loops, is a sum over the traversals of terms which vanish unless a cluster reaches from one loop to another, and

$$|E_{k+1}^{\times}| \leq C_{\mathrm{cr}} \mathfrak{a}_{k+1} \sum_{\mathfrak{n}} e^{-\nu_\times \delta_{\mathfrak{n}}/M} \leq C_{\mathrm{cr}} \mathfrak{a}_{k+1} P_k(\mathcal{C}) e^{-\nu_\times a_{k+1}/M}\,, \tag{6.11}$$

the sum over the traversals of the step $k$, with the amplitude $\mathfrak{a}_{k+1}$ of (7.3) and the constants $C_{\mathrm{cr}}$, $\nu_\times$ of Appendix A. The right side is summable over $k$, because $a_{k+1} = L^{j_q - q} L^{K'-k-1}$ grows geometrically as $k$ decreases. The flat values are taken loop by loop, and a cross group keeps its insertion; no count depends on the number of loops, since every count is per traversal or per cube, and a cube meets one loop.

*The torus and the wrapping.* The remainder of the flat part contains a term from the periodicity of the torus, of size $C \mathfrak{a}_{k+1} P_k(\mathcal{C}) e^{-c\iota_{T,k+1}}$ with $\iota_{T,k} = L\iota_{T,k+1}$, and the insertions whose domains do not lift to the universal cover give the last term of (5.8), which grows by at most a factor $L$ per step. Both are summable over the steps: with $x \coloneqq L^{K-j} \geq 1$ and

$a \coloneqq c'T/(uM) \geq 1$, which is a lower bound on the period, each is at most $Cx\, e^{-ax}$ and

$$\sum_{i\geq 0} L^i e^{-aL^i} \leq e^{-a} \sum_{i\geq 0} L^i e^{-(L^i-1)} \leq 2e^{-a}\,, \tag{6.12}$$

because $L^i e^{-aL^i} = e^{-a} L^i e^{-a(L^i-1)} \leq e^{-a} L^i e^{-(L^i-1)}$ for $a \geq 1$, and the terms of the last series, which decrease in $L$, are at most 1, 0.74, 0.20, 0.01, and so on at $L = 2$.

*Assembly.* The output of the step on every labelled term of $\varrho_{k+1}$ is a dressed loop of the step $k+1$: the normalisation is $\prod_i \kappa^{(k)}(C_i, \chi_i)\kappa_{k+1}(C_i, \chi_i)$ by Section 7; the leading words are the terms $S = \emptyset$ of (6.2); the insertions are the old ones carried as above, the fresh ones, the deviation parts of the new groups, and the attached ones; the interactions are the resummation of the families against the reference gas of the step $k+1$; and the scalar exponent is $\mathcal{E}_k$ plus the remainder $E^{\circ}_{k+1}$ of the flat part, the cross exponent, and the local terms of the step. The clauses (i)–(vi) of Definition 5.1 hold with the same constants by (6.7), by the payment of the attached parts, and by Section 7, which bounds the remainder and carries the torus consistency (vi) from the step $k$ to the step $k+1$. This proves Theorem 6.1, given Supplement, Propositions S.8–S.10 and Section 7.

## 7 The Flat Part and the Normalisation

We now determine the number which a step contributes to each loop, and this is the place at which the length of the loops has to be faced directly. At the trivial background the insertions of the step $k+1$ on one loop are scalars, by Schur's lemma, and the contribution of a loop $C$ becomes the partition function of a hard-core gas of groups on $C$, whose logarithm is a sum over clusters. Every traversal carries clusters of total weight of order $\mathfrak{a}_{k+1}$, so that the logarithm is of order $P_k(C)\mathfrak{a}_{k+1}$, while $P_k(C)$ grows like $L^{K-k}$. This number can be neither bounded nor neglected, and it has to be extracted exactly, by a formula whose coefficients do not depend on the loop. Two properties of the gas make this possible: a cluster sees the loop only through the arcs of its groups, and the gas is invariant under the translations which preserve the partitions of the construction. Consequently a cluster which is small compared with the corner isolation cannot tell a loop from an infinite straight line away from the corners, nor from two perpendicular rays near a corner, and the density of the logarithm along the loop is the periodic density of the line or of the bent path, up to errors which decay exponentially in the corner isolation. The coefficients of the normalisation are the sums of these densities over one period, and the excess of the bent path over its two rays. Since the line and the bent path are infinite, these sums are defined on $\mathbb{Z}^4$, as limits over larger and larger tori, and the third property which we need is that the weights on a torus approach their limits exponentially fast in the period.

### 7.1 The Reference Gas

We fix a step $k+1 \leq K'$ with $k \geq 1$, the step 1 being that of Section 5, and a labelled term $\ell$ of $\varrho_{k+1}$. The *trivial term* $\ell_0$ is the term with $\Omega_j = \Lambda_j = \mathbb{T}$ for every $j \leq k+1$. It has no large-field regions and no outer variables, and its reference data, the trivial field, lie in the support of all its test functions, so that the identities of Section 6 hold there. The traversals in this section are those of the step $k$, and lengths and distances are measured in units of the scale $k$ unless the contrary is said.

*The partitions and the period.* The cubes of the partitions used by the construction have sides which are powers of $L$, and these partitions are compatible with each other [9, p. 245].

That $R_j$, $M$, and $M_2$ are powers of $L$ is Bałaban's choice [9, p. 245], and that $M_1$ is one is a standing condition of [1, (3.1)]. Since every side $s$ is odd, the cubes of a partition can be centred at sites, and we take every partition to consist of the cubes $sz + \{-\frac{s-1}{2}, \dots, \frac{s-1}{2}\}^4$, $z \in \mathbb{Z}^4$, of sites of the scale at which it is used, so that the origin is a site of every lattice and the centre of a cube of every partition; this is compatible with the blocks of the averages, which are centred at the sites of the coarser lattice [8, (0.3), p. 252]. For the families $\pi_j$ it is Bałaban's own convention, whose cubes have their centres at the points of a lattice of spacing $M$ [8, p. 257], and for the other partitions, which [9, p. 245] requires only to be compatible, it is our choice. Let $\Pi_{k+1}$ be the largest side, in units of the scale $k$, of the cubes used at the steps $1, \dots, k+1$: the cubes of the families $\pi_j$, the cubes of side $MR_j$ of which the domains of the step $j$ are unions, the cubes of side $L^2 M_2 R_j$ of the tests [9, (3.2), (3.3), (3.16)], and the cubes of side $R_1 M_1$ of the decoupling [10, p. 3]. Every translation by a vector of $\Pi_{k+1}\mathbb{Z}^4$ maps each of these partitions onto itself.

**Lemma 7.1** (Period Points)**.** *Let $1 \le k < K'$. Then $\Pi_{k+1}$ is a power of $L$ with $\Pi_{k+1} \le L^2 M R_k \le a_k$, so that $\Pi_{k+1}$ divides $a_k$. The corners of the loops of an ensemble at level $q$ lie in $\Pi_{k+1}\mathbb{Z}^4$, the length of every segment is a multiple of $\Pi_{k+1}$, and so are the periods of the torus.*

*Proof.* At the step $j+1$ the cubes listed above have, in units of the scale $j$, the sides $M$ for $\pi_j$, $LM \le MR_j$ for $\pi_{j+1}$, $MR_j$ and $LMR_{j+1} \le LMR_j$ for the domains of the steps $j$ and $j+1$, $L^2 M_2 R_{j+1} \le LMR_j$ for the tests, and $R_1 M_1 \le M$ [10, p. 3]; here we used $R_{j+1} \le R_j$ from Lemma 3.4(ii), $LM_2 < M$ from [1, (3.1)], and $R_j \ge L$, which holds for $\gamma$ small by Lemma 3.4(i). All of them are therefore at most $L^2 MR_j$ in units of the scale $j$, that is, at most $L^{j-k} L^2 MR_j$ in units of the scale $k$. For $j < k$ the second bound of (3.14) gives

$$L^{j-k} R_j \le (L+1)(k-j)^{\beta_0} L^{j-k} R_k \le (1 + L^{-1}) R_k \,, \tag{7.1}$$

because $m^{\beta_0} \le m \le L^{m-1}$ for every integer $m \ge 1$. A side which is a power of $L$ and at most $(1+L^{-1})L^2 MR_k < L^3 MR_k$ is at most $L^2 MR_k$, since $L^2 MR_k$ is a power of $L$ as well. Hence $\Pi_{k+1} \le L^2 MR_k$, and $L^2 MR_k \le a_k$ by (3.45). Both $\Pi_{k+1}$ and $a_k = L^{j_q - q} L^{K'-k}$ are powers of $L$, so the first divides the second.

A loop at level $q$ is a path of bonds of $\mathbb{T}_q$, so that its corners lie in $uL^{-q}\mathbb{Z}^4$, which is $a_k \mathbb{Z}^4$ in units of the scale $k$ by the proof of Lemma 3.14, and its segments, each of which joins two corners along one axis, have lengths which are multiples of $a_k$. The period $T = 2L^{m_T} u$ is $2L^{m_T + q} a_k$ in units of the scale $k$. *q.e.d.*

*The reference gas.* By Section 6.2, the insertion $Y_{\mathcal{G}}$ of a group $\mathcal{G}$ of the step $k+1$ whose intervals lie on one loop takes at the reference data of $\ell$ the value $c_{\mathcal{G}}\,\mathrm{id}$, with a number $c_{\mathcal{G}} = c_{\mathcal{G}}(\ell)$ which is defined by the reference clause of Definition 5.1(iii); for a cross group we put $c_{\mathcal{G}} := 0$. When the groups are formed, the interactions $Q_k(\mathfrak{B})$ of the step $k$ are expanded into the exponential series of their clusters, and the groups which one cluster of the step $k$ meets are merged, together with that cluster, into one group, whose arc is the smallest arc containing their arcs and the traversals of the cluster; the factor of the cluster becomes a factor of the insertion of that group. After this merging two groups interact only through a hard core: they are *compatible* if their arcs are disjoint, their domains are disjoint, and no cluster merged into one of them meets a member of the other. The *reference gas* of $\ell$ is the hard-core gas of these groups with the activities $c_{\mathcal{G}}$, and its partition functions are

$$S^{\circ}(\ell) := \sum_{\mathfrak{E}} \prod_{\mathcal{G} \in \mathfrak{E}} c_{\mathcal{G}} \,, \qquad S^{\circ,[D]}(\ell) := \sum_{\mathfrak{E} \perp D} \prod_{\mathcal{G} \in \mathfrak{E}} c_{\mathcal{G}} \,, \tag{7.2}$$

the first sum over all families of pairwise compatible groups, the second over the families $\mathfrak{E}$ such that $D \cup \mathfrak{E}$ is a family and $D \cap \mathfrak{E} = \emptyset$, for a given family $D$. A *cluster* $\Gamma$ is a finite multiset of groups whose incompatibility graph is connected, and its weight is $\Phi^\circ(\Gamma;\ell) \coloneqq \phi^{\mathrm{U}}(\Gamma)\prod_{\mathcal{G}\in\Gamma} c_{\mathcal{G}}$, with the Ursell function $\phi^{\mathrm{U}}$ of that graph [14, Chapter 5]. The set $J_\Gamma$ of the traversals in the arcs of the groups of $\Gamma$ is nonempty, since every group has an arc, and the *network* $N(\Gamma)$ is the union of the domains, the arcs, and the merged clusters of its groups. On the trivial term we write $\Phi^\circ(\Gamma;T)$ for the weights on the torus $T$. The tori of the family $\mathfrak{T}$ are those whose periods are multiples of the period $T$ of $\mathbb{T}$; on each of them the construction is defined by the same rules, with the partitions of $\mathbb{T}$ lifted, and every bound of the preceding sections holds on them with the same constants, since none of these constants depends on the torus.

**Definition 7.2** (Amplitude)**.** For $j \geq 1$ the *amplitude* of the step $j$ is

$$\mathfrak{a}_j \coloneqq \max(\tau_{j-1}^2,\, 2\rho_j)\;, \tag{7.3}$$

and the *weight* of a group of the step $k+1$ is $a^\circ(\mathcal{G}) \coloneqq (\nu/32)\sigma_{k+1}(\mathcal{G})$, extended additively to clusters.

The factor 2 is the value at $q_1 = q_0$ of the factor $1 + (\log g_k^{-2})^{(q_1-q_0)_+}$ which the exponents of the radii $\alpha_{0,k}$, $\alpha_{1,k}$ contribute to the sizes of the groups, and the construction takes $q_1 = q_0$ (Appendix A). Since $\tau(g)$ and $\rho(g)$ increase for small $g$ and $g_j \leq (1+\beta_0)\bar{g}$ for $j \leq K$ by (3.9) and the window (4.1), the amplitudes are bounded by

$$\bar{\mathfrak{a}} \coloneqq \sup_{j\leq K} \mathfrak{a}_j \leq C\bar{g}^2(\log \bar{g}^{-2})^{6p_0}\;, \qquad \sup_{j\leq K'} \mathfrak{a}_j \leq C\gamma_q^2(\log \gamma_q^{-2})^{6p_0}\;, \tag{7.4}$$

the second because $g_j \leq \gamma_q$ for $j \leq K'$ by (3.46).

**Lemma 7.3** (Root Bound and Tail)**.** *Let $\gamma_q$ be sufficiently small, depending on $c_{\mathcal{C}}$. On every labelled term and every torus of $\mathfrak{T}$, for every traversal $b$ and every $R \geq 0$,*

$$\sum_{\mathcal{G}\colon b\in J_{\mathcal{G}}} |c_{\mathcal{G}}| e^{2a^\circ(\mathcal{G})} \leq C_{S'}\mathfrak{a}_{k+1}\;, \qquad \sum_{\Gamma\colon b\in J_\Gamma} |\Phi^\circ(\Gamma)| e^{a^\circ(\Gamma)} \leq C_{S'}\mathfrak{a}_{k+1}\;, \tag{7.5}$$

*where $J_{\mathcal{G}}$ is the set of the traversals in the arc of $\mathcal{G}$, and*

$$\sum_{\Gamma\colon b\in J_\Gamma,\ \operatorname{diam} N(\Gamma)\geq R} |\Phi^\circ(\Gamma)| \leq C_{S'}\mathfrak{a}_{k+1} e^{-2c_E^T R}\;, \qquad c_E^T \coloneqq \frac{\nu}{64 A_{\mathrm{env}}^* ML}\;, \tag{7.6}$$

*with $A_{\mathrm{env}}^* \coloneqq \max(2, A_{\mathrm{env}}+1)$ and $A_{\mathrm{env}}$ the constant of the envelopes of [1, Lemma E.8].*

*Proof.* For the first bound we note that $c_{\mathcal{G}}$ is the value of $Y_{\mathcal{G}}$ at the reference data, so that $|c_{\mathcal{G}}|$ is at most the supremum in (6.8), and that $e^{2a^\circ} = e^{(\nu/16)\sigma_{k+1}} \leq e^{\theta+\nu\sigma_{k+1}}$ since $\theta \geq 0$. Hence the first sum is at most $2C_{S'}\rho_{k+1} \leq C_{S'}\mathfrak{a}_{k+1}$ by (6.8), applied at the traversal of the step $k+1$ which contains $b$, and by (7.3); the same holds for the sums rooted at a traversal of the step $k+1$ or at a cube of $\pi_{k+1}$.

For the second bound we verify the condition of Kotecký and Preiss with the weight $a^\circ$. A group $\mathcal{G}'$ which is incompatible with $\mathcal{G}$ shares a traversal of the step $k+1$ with the arc of $\mathcal{G}$, or has a domain or a merged cluster which meets a cube of $\pi_{k+1}$ adjacent to a cube of a domain or a merged cluster of $\mathcal{G}$. There are at most $81M\sigma_{k+1}(\mathcal{G})$ such traversals and cubes, since $\sigma_{k+1}$ counts the cubes and, with the weight $1/M$, the traversals of the step $k+1$ of $\mathcal{G}$,

and a cube has 81 cubes adjacent to it or equal to it. By the first bound, rooted at these traversals and cubes,

$$\sum_{\mathcal{G}' \not\sim \mathcal{G}} |c_{\mathcal{G}'}| e^{2a^\circ(\mathcal{G}')} \le 81 M \sigma_{k+1}(\mathcal{G}) C_{S'} \mathfrak{a}_{k+1} \le \frac{\nu}{32} \sigma_{k+1}(\mathcal{G}) = a^\circ(\mathcal{G}) \,, \tag{7.7}$$

where the second inequality is $2592\, M C_{S'} \mathfrak{a}_{k+1} \le \nu$, which holds for $\gamma_q$ small, depending on $c_{\mathcal{C}}$, by the second bound of (7.4), because $k+1 \le K'$. This is the condition of [14, Theorem 5.4] with the functions $a = a^\circ$ and $d = a^\circ$. That theorem gives the convergence of the cluster expansion of the reference gas and of the gases with the activities $c_{\mathcal{G}} \mathbf{1}[\mathcal{G} \perp D]$, and it bounds the clusters which are incompatible with an external polymer by the sum of $|c_{\mathcal{G}}| e^{a^\circ(\mathcal{G}) + d(\mathcal{G})}$ over the groups incompatible with it. With the traversal $b$ as the external polymer, incompatible with the groups whose arcs contain it, this is the second bound of (7.5).

For the tail we bound the diameter of the network by the weight. The groups of a cluster are connected under incompatibility, and two incompatible groups have networks at distance at most one cube side $M$, in units of the scale $k+1$. The network of one group is connected and has diameter at most $M \#_{k+1}(\mathcal{G}) + n_{k+1}(\mathcal{G}) = M \sigma_{k+1}(\mathcal{G})$ in units of the scale $k+1$. Hence, in units of the scale $k$, and since $\sigma_{k+1} \ge 1$ for a group with a domain,

$$\operatorname{diam} N(\Gamma) \le L \sum_{\mathcal{G} \in \Gamma} (M \sigma_{k+1}(\mathcal{G}) + M) \le 2ML \sum_{\mathcal{G} \in \Gamma} \sigma_{k+1}(\mathcal{G}) = \frac{64ML}{\nu}\, a^\circ(\Gamma) \,, \tag{7.8}$$

so that $a^\circ(\Gamma) \ge (\nu/(64ML)) \operatorname{diam} N(\Gamma) \ge 2c_E^T \operatorname{diam} N(\Gamma)$, because $A^*_{\text{env}} \ge 2$. The clusters in (7.6) therefore have $e^{a^\circ(\Gamma)} \ge e^{2c_E^T R}$, and the second bound of (7.5) gives (7.6). *q.e.d.*

The rate $2c_E^T$ is the one which the first step allows: there the network of a cluster has diameter at most $A^*_{\text{env}} LM \sum \sigma$ fine bonds, by the envelopes of [1, Lemma E.8], and the same argument gives (7.6) at the step 1 with the same constant.

**Lemma 7.4** (Locality and Periodicity)**.** *On the trivial term the following hold.*

*(i) Let $\Gamma$ be a cluster of the gas of a loop $C$ on a torus $T$ all of whose groups have arcs of fewer than $P_k(C)/2$ traversals. If $C'$ is a loop on a torus $T' \in \mathfrak{T}$ which contains the arcs of the groups of $\Gamma$, and the dependence set of $\Gamma$ embeds isometrically in $T'$, then $\Gamma$ is a cluster of the gas of $C'$ on $T'$ with the same incompatibility graph, and its weight on $T'$ is the weight on $T$ evaluated on the configuration of $T'$ on the image of the dependence set.*

*(ii) A translation by a vector of $\Pi_{k+1}\mathbb{Z}^4$ maps the reference gas of a loop $C$ onto the reference gas of the translated loop, with the same activities and the same weights.*

*Proof.* (i) The insertion of a group of the step $k+1$ is built from the activities of the polymers of its domains, the slots and the old insertions in its arc, the transports along its arc, and the clusters of the step $k$ merged into it (Section 6.2 and the merging above). None of these depends on the loop outside the arc: the activities are functions of the fields on their polymers, the slots and transports are attached to the traversals of the arc, and the old insertions and old cluster weights have the same property by induction, starting from the first step, where the groups are products along their intervals [1, Definition 3.24]. The length of the loop enters only through the rule which takes the smallest arc containing a set, and for arcs shorter than half the loop this rule gives the same arc on $C$ and on $C'$, by Lemma 2.6(ii).

The hard core depends only on the arcs, the domains, and the merged clusters, so that the incompatibility graph and the Ursell function are the same.

(ii) Every operation of Bałaban's step, the averages, the minimisers, the propagators and their random walk expansions, the test functions, the localisation, and the activities, is defined from the lattice, the partitions, and the fields alone. A translation by a vector of $\Pi_{k+1}\mathbb{Z}^4$ preserves the lattices and the partitions, and it maps the trivial field to itself, so it maps each object to the object of the translated data. For the sums of the regular part and for the **R**-part this covariance is printed [9, (2.29), (2.32), p. 260]; for the localised activities it follows from their definitions, provided the choices which the papers leave free are made by translation covariant rules, which we do (Section 14). *q.e.d.*

The comparison of two tori needs a condition which excludes that a small cluster wraps around the torus. Let $\rho_{\mathrm{dep}} < LMR_1/4$ be the dependence range of the objects of the first step in fine bonds [1, Sections E.9.2 and E.10.4], and put, for $1 \le j \le K'$,

$$\iota_{T,j} := \max\Big(0,\ \frac{N_{\min}^{(j)}}{4} - \rho_{\mathrm{dep}} L^{-j}\Big)\ , \tag{7.9}$$

in units of the scale $j$, so that $\iota_{T,j} = L\iota_{T,j+1}$ whenever both are positive. The dependence set of an object of the step $k+1$ lies within the distance

$$R_D^{(k)} \le 5M + LM + L^2 MR_k + \max_{j<k} L^{j-k} R_D^{(j)} \le 6L^2 MR_k \tag{7.10}$$

of its network: the walk pieces of the propagators depend on the field in the fifth enlargement of their domains by cubes of side $M_1$ [5, Theorem 3.10, p. 416], at most $5M$; the localisation of the step uses the cubes of $\pi_{k+1}$ [10, p. 13], of side $LM$; the tests use cubes of side $L^2 M_2 R_{k+1} \le L^2 MR_k$; and the old insertions have their own ranges. The last inequality follows by iteration from $5M + LM + L^2 MR_k \le 2L^2 MR_k$, from $R_{k-m} \le (L+1)m^{\beta_0} R_k$ by (3.14), and from $(L+1)\sum_{m\ge 1} mL^{-m} = (L+1)L/(L-1)^2 \le 2$ for $L \ge 5$, the range $\rho_{\mathrm{dep}} L^{-k}$ of the first step being smaller than the term $m = k-1$.

**Lemma 7.5** (The Torus Condition)**.** *Let $L \ge 195$. For every loop ensemble at level $q$ and every $1 \le k \le K'$,*

$$\iota_{T,k} \ge 16L \cdot 6L^2 MR_k \ge 16L\, R_D^{(k)}\ . \tag{7.11}$$

*Proof.* A loop at level $q$ has a bond of $\mathbb{T}_q$, so that $uL^{-q} \le \mathrm{diam}(C)$, and $3\,\mathrm{diam}(C) < T = 2L^{m_T} u$ by Definition 1.1; hence $L^{m_T+q} > 3/2$ and $m_T + q \ge 1$. By Lemma 7.1 the smallest period in units of the scale $K'$ is $N_{\min}^{(K')} = 2L^{m_T+q} a_{K'} \ge 2L \cdot L^3 MR_{K'-1} \ge 2L^4 MR_{K'}$, by (3.44) and Lemma 3.4(ii). The first-step range is $\rho_{\mathrm{dep}} L^{-K'} \le L^{1-K'} MR_1/4 \le MR_{K'}/2$, by (3.14) with $m^{\beta_0} L^{-m} \le L^{-1}$ for $m = K' - 1 \ge 1$, and trivially for $K' = 1$. Hence $\iota_{T,K'} \ge (L^4 - 1)MR_{K'}/2$. For $k < K'$ we have $\iota_{T,k} = L^{K'-k}\iota_{T,K'}$ and $R_k \le (L+1)(K'-k)^{\beta_0} R_{K'} \le (1+L^{-1})L^{K'-k} R_{K'}$, so that in all cases

$$\iota_{T,k} \ge \frac{L^4 - 1}{2(1+L^{-1})}\, MR_k \ge 96L^3 MR_k\ , \tag{7.12}$$

where the last inequality is $L^4 - 1 \ge 192L^3 + 192L^2$, which holds for $L \ge 195$. The second inequality of (7.11) is (7.10). *q.e.d.*

A cluster $\Gamma$ *embeds* in $T$ if its dependence set, the $R_D^{(k)}$-neighbourhood of its network, has diameter less than $\iota_{T,k}/4$; the covering map of a torus $T' \in \mathfrak{T}$ over $T$ is then injective on its lifts, and $\tilde{\Gamma}$ denotes a lift.

**Proposition 7.6** (Torus Consistency)**.** *Let $L$ satisfy* (C.1)*, let $\gamma_q$ be sufficiently small, depending on $c_{\mathcal{C}}$, and assume that the ensemble is dressed at the step $k$. On the trivial term, for a loop $C$ of the ensemble, at the reference data:*

*(i) for tori $T \subset T'$ of $\mathfrak{T}$ and every traversal $b$,*

$$\sum_{\Gamma\colon b\in J_\Gamma} |\Phi^\circ(\Gamma;T) - \Phi^\circ(\tilde\Gamma;T')| \le \bar C_Z \mathfrak{a}_{k+1} e^{-c^Z \iota_{T,k+1}} , \qquad c^Z := \frac{\nu}{256 A^*_{\text{env}} M} , \tag{7.13}$$

*where $\Phi^\circ(\Gamma;T) := 0$ for a cluster of $T'$ which is not the lift of a cluster embedding in $T$;*

*(ii) along the tori $T_m \in \mathfrak{T}$ of the periods $L^m T$, the weight of every cluster which embeds in some $T_m$ converges to a limit $\Phi^{\circ,\infty}(\Gamma)$, which does not depend on the sequence of tori, and*

$$\sum_{\Gamma\colon b\in J_\Gamma} |\Phi^\circ(\Gamma;T) - \Phi^{\circ,\infty}(\Gamma)| \le 3\bar C_Z \mathfrak{a}_{k+1} e^{-c^Z \iota_{T,k+1}} ; \tag{7.14}$$

*(iii) the clause (vi) of Definition 5.1 holds at the step $k+1$.*

*Proof.* We split the clusters at $b$ into those which embed in $T$ and those which do not.

*The embedding clusters.* Let $\Gamma$ embed in $T$. The activities of its groups are composed of two kinds of factors. The factors made at the step $k+1$ are the same functions on $T$ and on $T'$ of the configuration on the dependence set: the walk pieces of the propagators with boundary conditions depend on the field in their domains only [5, p. 394; Theorem 3.10, p. 416]; a term of the localised expansion "depends on $\mathbf{U}$, $\mathbf{J}$ restricted to $Y_0 \cup \square_0$" [10, p. 5], and the activity $H(Z)$ of the fluctuation integral "is localized in the interior of $Z$ with respect to the external gauge fields" [10, p. 14]; the test functions are local in their cubes; and the couplings are the same numbers on every torus, because the coefficients $\beta_{j+1}$ are defined through the limit of the vacuum polarisation "as $T^{(j+1)} \nearrow Z^d$" [8, p. 264] and the couplings of a tube are prescribed. The factors made at earlier steps, the insertions of the step $k$ inside the fluctuation integral and the merged clusters of the step $k$, differ on the two tori by at most the amounts of the clause (vi) at the step $k$, at fields which coincide on the $\iota_{T,k}/4$-neighbourhood of the dependence set; the deformation of the background by the fluctuation field vanishes outside one lift of the domain of the integral [10, p. 4], so this hypothesis holds at every field of the integral. Replacing these factors by their differences one at a time, and summing over the clusters at $b$ with one marked factor by the count which gives (7.5), we obtain for the embedding clusters the bound $2C^{(k+1)}_{\text{rel}} \mathfrak{a}_{k+1} e^{-\iota_{T,k+1}/(32M)}$, with

$$C^{(k+1)}_{\text{rel}} \le (L(1+\beta_0)^7 + \tfrac{1}{8}) e^{-(L-1)\iota_{T,k+1}/(32M)} C^{(k)}_{\text{rel}} + C_* . \tag{7.15}$$

Here the factor $L$ counts the traversals of the step $k$ in one of the step $k+1$, $(1+\beta_0)^7$ compares the amplitudes of the two steps, the exponential is the gain $e^{-\iota_{T,k}/(32M)} = e^{-L\iota_{T,k+1}/(32M)}$ of the clause (vi) at the step $k$ against the rate of the step $k+1$, and $C_*$ bounds the clusters of Bałaban's gas of polymers inside the activities which do not embed, whose sets differ between the two tori. By (7.12), $(L-1)\iota_{T,k+1}/(32M) \ge 3L^2(L-1)R_{k+1}$, so the prefactor in (7.15) is at most $1/2$, and since $C^{(1)}_{\text{rel}} = 0$ by Theorem 5.2, we get $C^{(k)}_{\text{rel}} \le 2C_*$ for every $k$. This is also the clause (vi) at the step $k+1$ for the cluster weights; for the insertions the same telescoping applies, with the deviation parts, which vanish at the reference data, bounded relative to the deviation by the Cauchy estimate of Section 6.3. This proves (iii).

*The other clusters.* If $\Gamma$ does not embed in $T$, its dependence set has diameter at least $\iota_{T,k}/4$, so that its network has diameter at least $\iota_{T,k}/4 - 2R^{(k)}_D \ge \iota_{T,k}/8$ in units of the scale

$k$, by Lemma 7.5. By (7.8) its weight is $a^\circ(\Gamma) \geq (\nu/(64ML))\iota_{T,k}/8 = \nu\iota_{T,k+1}/(512M) \geq c^Z\iota_{T,k+1}$, and the second bound of (7.5) shows that these clusters contribute at most $C_{S'}\mathfrak{a}_{k+1}e^{-c^Z\iota_{T,k+1}}$ on $T$, and the same on $T'$, where the clusters at a lift of $b$ which are not lifts of embedding clusters of $T$ are exactly those whose dependence sets have diameter at least $\iota_{T,k}/4$.

Adding the two parts gives (i), with $\bar{C}_Z \coloneqq 2C_{S'} + 4C_*$, since $c^Z \leq 1/(32M)$. For (ii), any two tori of $\mathfrak{T}$ are contained in a third one, whose periods are common multiples of theirs, and $\iota_{T_m,k+1} = L^m\iota_{T,k+1}$. By (i) the weights of a cluster along $(T_m)$ form a Cauchy sequence, whose limit does not depend on the sequence by the comparison through a common torus, and the telescoping sum gives the bound $\bar{C}_Z\mathfrak{a}_{k+1}\sum_{m\geq 0} e^{-c^Z L^m\iota_{T,k+1}} \leq 2\bar{C}_Z\mathfrak{a}_{k+1}e^{-c^Z\iota_{T,k+1}}$ for the clusters which embed in $T$, because $L^m - 1 \geq m(L-1)$ and $c^Z\iota_{T,k+1} \geq 1$; the latter follows from $\iota_{T,k+1} \geq 96L^3MR_{k+1}$, by (7.12), and from $R_{k+1} \geq (\log \hat{g}_{k+1}^{-2})^r \geq (\log \gamma_q^{-2})^r$, for $\gamma_q$ small, since $k+1 \leq K'$. The clusters which do not embed in $T$ contribute at most $C_{S'}\mathfrak{a}_{k+1}e^{-c^Z\iota_{T,k+1}}$ on $T$, and the same in the limit, by Fatou's lemma applied to the bound on each $T_m$, which holds with $\iota_{T,k+1}$ in place of $\iota_{T_m,k+1}$ because the condition is on the diameter of the dependence set. This gives (7.14). *q.e.d.*

*The exact factorisation.* With the reference gas we can now carry out the resummation of Section 6.2. On a labelled term $\ell$ we split every insertion as $Y_\mathcal{G} = c_\mathcal{G}T(\mathcal{G}) + Y_\mathcal{G}^\mathrm{d}$, where $T(\mathcal{G})$ is the product of the representations of the transports of the background field along the arc of $\mathcal{G}$, so that $Y_\mathcal{G}^\mathrm{d}$ vanishes at the reference data and is covariant with the endpoints of the arc. Since every word is multilinear in its insertions, and the transports of the scalar parts join the transports of the gaps,

$$w(\mathfrak{F}) = \sum_{D\subseteq\mathfrak{F}} \Big( \prod_{\mathcal{G}\in\mathfrak{F}\setminus D} c_\mathcal{G} \Big) w^\mathrm{d}(D) \,, \tag{7.16}$$

where $w^\mathrm{d}(D)$ is the word of the family $D$ with the insertions $Y_\mathcal{G}^\mathrm{d}$, and $w^\mathrm{d}(\emptyset) = \prod_i f_{k+1}^{(i)}$. After the merging the families of groups interact only through the hard core, so that the sum over the families $\mathfrak{F}$ of (7.16) is a sum over the pairs $(D, \mathfrak{E})$ with $\mathfrak{F} = D \cup \mathfrak{E}$, and

$$\sum_\mathfrak{F} w(\mathfrak{F}) = S^\circ(\ell)\Big[\prod_i f_{k+1}^{(i)} + \sum_{D\neq\emptyset} Q_{k+1}(D)\, w^\mathrm{d}(D)\Big] \,, \tag{7.17}$$

with

$$Q_{k+1}(D) \coloneqq \frac{S^{\circ,[D]}(\ell)}{S^\circ(\ell)} = \exp\Big(-\sum_{\Gamma\not\perp D} \Phi^\circ(\Gamma;\ell)\Big) \,, \tag{7.18}$$

the last sum over the clusters which contain a group incompatible with a member of $D$. The rearrangements are justified by the absolute convergence of $\sum_\mathfrak{F}\prod_{\mathcal{G}\in\mathfrak{F}}(|c_\mathcal{G}| + \|Y_\mathcal{G}^\mathrm{d}\|)$, which follows from (6.8); the gas is a gas of groups together with their merged clusters, and the sum over the latter is infinite even on a finite torus. The cluster series of $S^\circ(\ell)$ and of $S^{\circ,[D]}(\ell)$ converge by Lemma 7.3, which gives $S^\circ(\ell) \neq 0$ and the second expression for $Q_{k+1}(D)$, and $\sum_{\Gamma\not\perp\mathcal{G}}|\Phi^\circ(\Gamma)| \leq a^\circ(\mathcal{G}) \leq \nu\sigma_{k+1}(\mathcal{G})/16$ by [14, Theorem 5.4] and (7.7), which is the clause (i) at the step $k+1$ with $\theta \coloneqq a^\circ$. For an ensemble the groups of all loops and the cross groups form one gas; since cross groups have the activity 0, on the trivial term

$$\log S^\circ(\ell_0) = \sum_{i=1}^n F_{k+1}^\circ(C_i;T) + E_{k+1}^\times \,, \qquad F_{k+1}^\circ(C;T) \coloneqq \sum_\Gamma \Phi^\circ(\Gamma;T) \,, \tag{7.19}$$

the second sum over the clusters of the gas of the single loop $C$, and this defines the cross exponent $E_{k+1}^\times$ of (6.11). On a general term the difference $E_{k+1}^\partial(\ell) \coloneqq \log S^\circ(\ell) - \log S^\circ(\ell_0)$

consists of the clusters which see the large-field data of $\ell$; its representation as a sum of local terms $e_X^{(k+1)}$ attached to these data, with the first bound of (5.9), is part of Supplement, Proposition S.9.

### 7.2 The Coefficients

For $v \in \mathcal{V}$ let $\ell_v := \{tv \,|\, t \in \mathbb{Z}\}$ be the line through the origin in the orientation $v$, with its traversals numbered by $b \in \mathbb{Z}$, the traversal $b$ running from $bv$ to $(b+1)v$. For a perpendicular ordered pair $(v, w)$ let $B_{v,w}$ be the bent path (2.6) with $c = 0$, $v_c = v$, and $w_c = w$, with its traversals numbered by $b \in \mathbb{Z}$: for $b < 0$ the traversal from $bv$ to $(b+1)v$, and for $b \geq 0$ the traversal from $bw$ to $(b+1)w$. The *line gas* of $\ell_v$ consists of the clusters $\Gamma$ on $\mathbb{Z}^4$ whose groups have finite arcs on $\ell_v$, with the weights $\Phi_v^{\circ,\infty}(\Gamma)$ obtained as follows: we take a torus $T' \in \mathfrak{T}$ in which $\Gamma$ embeds and on it a square $C'$ at level $q$, in a coordinate plane containing $v$ and with its corners in $\Pi_{k+1}\mathbb{Z}^4$, one side of which contains the arcs of $\Gamma$ in the orientation $v$; then $\Gamma$ is a cluster of the gas of $C'$, its weight does not depend on the choice of $C'$ by Lemma 7.4(i), and $\Phi_v^{\circ,\infty}(\Gamma)$ is the limit of these weights along $(T_m)$ given by Proposition 7.6(ii). The *bent-path gas* of $B_{v,w}$ is defined in the same way, with a square in the plane spanned by $v$ and $w$ which has a corner at the origin with the incoming orientation $v$ and the outgoing orientation $w$. The squares used here need not be admissible. The proofs of Theorems 5.2 and 6.1 use the loops only through $\Lambda(C) = 2$, $P(C)/r(C) = 8$, and the torus conditions, and the torus conditions, which compare the loop with the periods, hold at the given $K$ once the periods of $T'$ are large compared with the square. The densities and the coefficients of the step $k+1$ are

$$a_v(b) := \sum_{\Gamma\colon\, b \in J_\Gamma} \frac{\Phi_v^{\circ,\infty}(\Gamma)}{|J_\Gamma|}\,, \qquad \kappa_v^{(k+1)} := \sum_{b=0}^{\Pi_{k+1}-1} a_v(b) \tag{7.20}$$

on the line, and, with $a_{v,w}(b)$ defined in the same way on $B_{v,w}$ and with $k_{v,w}(b) := a_v(b)$ for $b < 0$ and $k_{v,w}(b) := a_w(b)$ for $b \geq 0$,

$$\zeta_{v,w}^{(k+1)} := \sum_{b \in \mathbb{Z}} (a_{v,w}(b) - k_{v,w}(b)) \tag{7.21}$$

on the bent path.

**Lemma 7.7** (The Coefficients)**.** *Let $L$ and $\bar{g}$ be as in Proposition 7.6, and put $C_\zeta := 2C_{S'}(1 + 2/(e^{2c_E^T} - 1))$.*

*(i) The density $a_v$ is $\Pi_{k+1}$-periodic, $|a_v(b)| \leq C_{S'}\mathfrak{a}_{k+1}$, and $|\kappa_v^{(k+1)}| \leq \Pi_{k+1}C_{S'}\mathfrak{a}_{k+1}$.*

*(ii) $|a_{v,w}(b) - k_{v,w}(b)| \leq 2C_{S'}\mathfrak{a}_{k+1}e^{-2c_E^T|b|}$ for every $b \in \mathbb{Z}$, so that the series (7.21) converges absolutely and $|\zeta_{v,w}^{(k+1)}| \leq C_\zeta\mathfrak{a}_{k+1}$.*

*(iii) The numbers $\kappa_v^{(k+1)}$ and $\zeta_{v,w}^{(k+1)}$ do not depend on the loops, on the torus, or on the labelled term. They depend on $k$, $K$, and $\bar{g}$ through the couplings and the parameters of the tube, on the character, and on the constants of the construction.*

*Proof.* (i) By Lemma 7.4(ii), applied on each torus $T_m$, the weights of a cluster and of its translate by $\Pi_{k+1}v$ agree, and so do their limits. The bound on $a_v(b)$ is (7.5) at $R = 0$ with $|J_\Gamma| \geq 1$, valid on each $T_m$ and passed to the limit by Fatou's lemma; summing over one period gives the bound on $\kappa_v^{(k+1)}$.

(ii) Let $b \geq 0$. A cluster of the bent-path gas all of whose arcs lie on the outgoing ray is, by Lemma 7.4(i), a cluster of the line gas of $\ell_w$ with the same weight, and conversely a

cluster of the line gas at $b$ all of whose arcs lie in $\{b' \geq 0\}$ is one of the bent-path gas. The terms which survive in $a_{v,w}(b) - a_w(b)$ are therefore the clusters of the bent path at $b$ with an arc on the incoming ray, and the clusters of the line at $b$ with an arc on $\{b' < 0\}$. The network of each of them contains the position $bw$ of $b$ and a site $-tv$ or $-tw$ with $t \geq 1$, at distance at least $b$, so that (7.6) with $R = b$, in each of the two gases, gives the bound. The case $b < 0$ is the same with the two rays exchanged, the distance being at least $|b|$. Summing, with $y := e^{-2c_E^T}$,

$$|\zeta_{v,w}^{(k+1)}| \leq 2C_{S'}\mathfrak{a}_{k+1}\Big(1 + 2\sum_{b\geq 1} y^b\Big) = 2C_{S'}\mathfrak{a}_{k+1}\frac{1+y}{1-y} = C_\zeta \mathfrak{a}_{k+1} \,. \tag{7.22}$$

(iii) The coefficients are sums of weights of the line and bent-path gases on $\mathbb{Z}^4$, which by construction involve neither the loops of the ensemble, nor the torus $\mathbb{T}$, nor any term but the trivial one. The dependence on the step is through the objects of the steps $1, \dots, k+1$ on the trivial term. *q.e.d.*

The coefficients are kept separately for each orientation and each ordered pair of perpendicular orientations, since the covariance of the construction under the hypercubic group is not established; no relation between them is used. With them we define the factor of a step. At the step 1 the coefficients are those of the first step: $\Pi_1 := L$, and $\kappa_v^{(1)}$ and $\zeta_{v,w}^{(1)}$ are the perimeter and corner coefficients of [1, (5.33), (6.4)] in the orientation $v$ and at the corner $(v, w)$ (Section 5.4). A step $j$ is an **R**-*step* if $j > N_j$, so that the **R**-operation acts on the numerator at that step.

**Definition 7.8** (Step Factor)**.** For a loop $C$ with the character $\chi$ and $1 \leq j \leq K'$ let $N_v^{(j-1)}(C)$ be the number of traversals of $C$ of the orientation $v$ at the step $j-1$. The *step factor* is $\kappa_j(C,\chi) := \exp\log\kappa_j(C,\chi)$ with

$$\log\kappa_j(C,\chi) := \sum_{v\in\mathcal{V}} \frac{N_v^{(j-1)}(C)}{\Pi_j}\,\kappa_v^{(j)} + \sum_{c\in\mathcal{K}(C)} \zeta_{v_c,w_c}^{(j)} + \mathbf{1}_{\mathbf{R}}(j)\Big[\sum_{v\in\mathcal{V}} N_v^{(j-1)}(C)\,\mu_{j,v}^{\mathbf{R}} + \sum_{c\in\mathcal{K}(C)} \zeta_{j,v_c,w_c}^{\mathbf{R}}\Big] \,, \tag{7.23}$$

where $\mathbf{1}_{\mathbf{R}}(j)$ is 1 at an **R**-step and 0 otherwise, and $\mu_{j,v}^{\mathbf{R}}$, $\zeta_{j,v,w}^{\mathbf{R}}$ are the coefficients of Proposition 7.10. The coefficients depend on $\chi$ through the representation $\pi$ in the insertions.

The factor is not zero, being an exponential, and $\log\kappa_j$ has exactly the structure announced in Section 1.3: a term proportional to the length of the segments of each orientation and a term for each corner, with coefficients which are numbers of the step.

### 7.3 The Remainder of a Step

**Theorem 7.9** (The Flat Part of a Step)**.** *Let $L$ and $\bar{g}$ be as in Proposition 7.6, let $1 \leq k < K'$, and assume that the ensemble is dressed at the step $k$. For every loop $C$ of the ensemble,*

$$F_{k+1}^\circ(C;T) = \sum_{v\in\mathcal{V}} \frac{N_v^{(k)}(C)}{\Pi_{k+1}}\,\kappa_v^{(k+1)} + \sum_{c\in\mathcal{K}(C)} \zeta_{v_c,w_c}^{(k+1)} + E_{k+1}^\circ(C,T) \,, \tag{7.24}$$

*with a number $E_{k+1}^\circ(C,T)$ which satisfies, with $C_E^{\mathcal{K}} := 2C_{S'} + 2C_\zeta$,*

$$|E_{k+1}^\circ(C,T)| \leq C_E^{\mathcal{K}}\mathfrak{a}_{k+1}(P_k(C) + |\mathcal{K}(C)|)e^{-2c_E^T r_k(C)} + 3\bar{C}_Z\mathfrak{a}_{k+1}P_k(C)e^{-c^Z\iota_{T,k+1}} \,. \tag{7.25}$$

*Proof.* We allocate every cluster equally to the traversals of $J_\Gamma$,

$$F^\circ_{k+1}(C;T) = \sum_{b\in C} a_C(b;T)\,, \qquad a_C(b;T) := \sum_{\Gamma:\, b\in J_\Gamma} \frac{\Phi^\circ(\Gamma;T)}{|J_\Gamma|}\,, \tag{7.26}$$

the sum over the $P_k(C)$ traversals, which converges absolutely by (7.5). Every traversal is assigned to one corner (Section 2.4), and we compare the density at a traversal $b$ assigned to the corner $c$ with the density of the bent path $B_c = c + B_{v_c,w_c}$ at the corresponding traversal.

*Small clusters.* Let $\Gamma$ be a cluster at $b$ whose network has diameter less than $r_k(C)$. By Lemma 2.10 every site of $C$ or of $B_c$ outside $S_{\mathsf{in}}(c) \cup S_{\mathsf{out}}(c)$ is at distance at least $r_k(C)$ from the position of $b$, while the arcs of the groups of $\Gamma$ lie in its network; so these arcs lie in $S_{\mathsf{in}}(c) \cup S_{\mathsf{out}}(c)$, within distance $r_k(C)$ of the position of $b$, hence in an arc of fewer than $4r_k(C) \le P_k(C)/2$ traversals, since the chord–arc constant of two consecutive segments is 2 by Lemma 5.3(iii). The cluster does not wrap around the torus, because $r_k(C) \le \mathrm{diam}_k(C) < N^{(k)}_{\min}/3$. By Lemma 7.4(i) it is a cluster of the bent-path gas of $B_c$, and conversely a cluster of that gas at $b$ whose network has diameter less than $r_k(C)$ is a cluster of the gas of $C$. By Lemma 7.1 the corner $c$ lies in $\Pi_{k+1}\mathbb{Z}^4$, so by Lemma 7.4(ii) the weights of the gas of $B_c$ are those of $B_{v_c,w_c}$ at the translated traversal. Their weights on $T$ and on $\mathbb{Z}^4$ differ by at most $3\bar{C}_Z \mathfrak{a}_{k+1} e^{-c^Z \iota_{T,k+1}}$ in the sum at $b$, by (7.14).

*Large clusters.* The clusters at $b$ whose networks have diameter at least $r_k(C)$ contribute at most $C_{S'}\mathfrak{a}_{k+1} e^{-2c^T_E r_k(C)}$ in the gas of $C$ and the same in the gas of $B_c$, by (7.6). This includes the clusters with a group whose arc has at least $P_k(C)/2$ traversals, whose networks have diameter at least $P_k(C)/(2\Lambda(C)) \ge r_k(C)$ by Lemma 2.9 and Lemma 2.5.

Together, writing $b$ also for the corresponding traversal of $B_{v_c,w_c}$,

$$|a_C(b;T) - a_{v_c,w_c}(b)| \le 2C_{S'}\mathfrak{a}_{k+1} e^{-2c^T_E r_k(C)} + 3\bar{C}_Z \mathfrak{a}_{k+1} e^{-c^Z \iota_{T,k+1}}\,. \tag{7.27}$$

*The corner sums.* Let $\ell$ and $\ell'$ be the lengths of $S_{\mathsf{in}}(c)$ and $S_{\mathsf{out}}(c)$. The traversals assigned to $c$ are the traversals $b$ of $B_{v_c,w_c}$ with $-\lfloor \ell/2 \rfloor \le b \le \lceil \ell'/2 \rceil - 1$. By (7.21) the sum of $a_{v_c,w_c}(b)$ over them is the sum of $k_{v_c,w_c}(b)$ over them, plus $\zeta^{(k+1)}_{v_c,w_c}$, minus the sum of $a_{v_c,w_c}(b) - k_{v_c,w_c}(b)$ over the other $b$, which have $b \ge \lceil \ell'/2 \rceil \ge s_k(C)/2$ or $|b| \ge \lfloor \ell/2 \rfloor + 1 \ge (s_k(C)+1)/2$. By Lemma 7.7(ii), with $y = e^{-2c^T_E}$, this last sum is at most

$$2C_{S'}\mathfrak{a}_{k+1}\Big( \sum_{b \ge s_k(C)/2} y^b + \sum_{b \ge (s_k(C)+1)/2} y^b \Big) \le \frac{4C_{S'}\mathfrak{a}_{k+1}}{1-y} e^{-c^T_E s_k(C)} \le 2C_\zeta \mathfrak{a}_{k+1} e^{-2c^T_E r_k(C)}\,, \tag{7.28}$$

because $C_\zeta \ge 2C_{S'}/(1-y)$ and $s_k(C) \ge 2r_k(C)$.

*The segment sums.* A segment $\sigma$ from the corner $c$ to the corner $c''$, of orientation $v$ and length $\ell_\sigma$, receives the terms $k_{v_c,w_c}(b) = a_v(b)$, $0 \le b < \lceil \ell_\sigma/2 \rceil$, from $c$, and the terms $k_{v_{c''},w_{c''}}(b) = a_v(b)$, $-\lfloor \ell_\sigma/2 \rfloor \le b < 0$, from $c''$, read on the line $c + \ell_v$, since $c'' = c + \ell_\sigma v$ and both corners lie in $\Pi_{k+1}\mathbb{Z}^4$. Together these are the terms $a_v(b)$, $0 \le b < \ell_\sigma$, of the line $\ell_v$, by Lemma 7.4(ii), and since $\ell_\sigma$ is a multiple of $\Pi_{k+1}$ (Lemma 7.1) and $a_v$ is $\Pi_{k+1}$-periodic, their sum is exactly $(\ell_\sigma/\Pi_{k+1})\kappa^{(k+1)}_v$. Summing over the segments gives the first term of (7.24), since $\sum_{\sigma \text{ of orientation } v} \ell_\sigma = N^{(k)}_v(C)$.

*Conclusion.* Adding (7.27) over the $P_k(C)$ traversals and (7.28) over the $|\mathcal{K}(C)|$ corners gives (7.25), with $e^{-2c^T_E r_k(C)}$ for both corner terms. *q.e.d.*

At the steps at which the **R**-operation acts, its polymer gas linked to the loop has a flat part of the same structure, with the period $MR_j$ of the cubes of the **R**-operation in place

of $\Pi_j$, the decay $e^{-\kappa d_j}$ of its activities in place of (7.6), and a group's hull in place of its network.

**Proposition 7.10** (The Flat Part of an **R**-Step)**.** *Let $j \le K'$ be an* **R***-step. There are numbers $\mu^{\mathbf{R}}_{j,v}$, $v \in \mathcal{V}$, and $\zeta^{\mathbf{R}}_{j,v,w}$, $(v,w)$ perpendicular, independent of the loops, the torus, and the labelled term, with $|\mu^{\mathbf{R}}_{j,v}|, |\zeta^{\mathbf{R}}_{j,v,w}| \le C_E^R e^{-p_0(g_j)/4}$, such that for every loop $C$ of a dressed ensemble the flat part of the exponent which the* **R***-operation of the step $j$ contributes along $C$ is $\sum_v N_v^{(j-1)}(C)\mu^{\mathbf{R}}_{j,v} + \sum_{c\in\mathcal{K}(C)} \zeta^{\mathbf{R}}_{j,v_c,w_c} + E_j^{\circ,\mathbf{R}}(C,T)$, with*

$$|E_j^{\circ,\mathbf{R}}(C,T)| \le C_E^R M R_j e^{-p_0(g_j)/4}\Big[(P_j(C)+|\mathcal{K}(C)|)e^{-c_E^R r_j(C)/(MR_j)} + P_j(C)e^{-c_E^R \iota_{T,j}/(MR_j)}\Big]\,. \tag{7.29}$$

*Proof.* Supplement, Proposition S.10. The proof is that of Theorem 7.9 for the gas of the **R**-operation linked to the loop, whose flat activities are periodic along each segment with the period $MR_j$; every segment is a union of such periods starting at a corner, because $MR_jL^j$ divides $MR_{K'}L^{K'}$, which divides the fine length $L^{j_q-q+K'}$ of a bond of $\mathbb{T}_q$ by (3.44). *q.e.d.*

*The identity of the step.* We can now complete the step of Section 6. On every labelled term, by (7.17), (7.19), Theorem 7.9, and Proposition 7.10,

$$\prod_i \kappa^{(k)}(C_i,\chi_i)\, e^{\mathcal{E}_k} S^\circ(\ell) = \prod_i \kappa^{(k+1)}(C_i,\chi_i)\, e^{\mathcal{E}_{k+1}}\,, \tag{7.30}$$

with $\mathcal{E}_{k+1} \coloneqq \mathcal{E}_k + E^\partial_{k+1}(\ell) + \sum_i E^\circ_{k+1}(C_i,T) + E^\times_{k+1} + \mathbf{1}_{\mathbf{R}}(k+1)\sum_i E^{\circ,\mathbf{R}}_{k+1}(C_i,T)$, the last term entering at an **R**-step together with the flat part of the **R**-gas, and $E_{\mathrm{const},k+1} \coloneqq E_{\mathrm{const},k} + \sum_i E^\circ_{k+1}(C_i,T) + E^\times_{k+1} + \mathbf{1}_{\mathbf{R}}(k+1)\sum_i E^{\circ,\mathbf{R}}_{k+1}(C_i,T)$, which are numbers on each labelled term. The identity is exact, and it is the normalisation clause of the step.

### 7.4 The Normalisation

The remainders of the steps are collected in the following bound, in which $P_{j-1}(\mathcal{C}) \coloneqq \sum_i P_{j-1}(C_i)$, $|\mathcal{K}(\mathcal{C})| \coloneqq \sum_i |\mathcal{K}(C_i)|$, and $r_{j-1}(\mathcal{C}) \coloneqq \min_i r_{j-1}(C_i)$: for $1 \le j \le K'$,

$$\begin{aligned}\epsilon_j(\mathcal{C},T) \coloneqq\ & \mathfrak{a}_j\Big[C_E^{\mathcal{K}}(P_{j-1}(\mathcal{C})+|\mathcal{K}(\mathcal{C})|)e^{-2c_E^T r_{j-1}(\mathcal{C})} + 3\bar{C}_Z P_{j-1}(\mathcal{C})e^{-c^Z\iota_{T,j}} + C_{\mathrm{cr}}P_{j-1}(\mathcal{C})e^{-\nu_\times a_j/M}\Big] \\ & + \mathbf{1}_{\mathbf{R}}(j)\, C_E^R M R_j e^{-p_0(g_j)/4}\Big[(P_j(\mathcal{C})+|\mathcal{K}(\mathcal{C})|)e^{-c_E^R r_j(\mathcal{C})/(MR_j)} + P_j(\mathcal{C})e^{-c_E^R\iota_{T,j}/(MR_j)}\Big]\,.\end{aligned} \tag{7.31}$$

At $j=1$ the first line bounds the remainder of the first step, by (5.10), by the periodic comparison of [1, Appendix F], and by the cross terms of [1, Section 10]. By (7.25), (6.11), and (7.29), the constant part of the scalar exponent satisfies the bound of Definition 5.1(iv) at every step.

**Corollary 7.11** (Sum of the Remainders)**.** *Let $L$ be as in Proposition 7.6, and let $\mathcal{C}$ be a loop ensemble at level $q$ whose loops are admissible, with $\gamma_q$ as in that proposition. For every $K > j_q$,*

$$\sum_{j=1}^{K'} \epsilon_j(\mathcal{C},T) \le B_E(\mathcal{C}) \coloneqq \sum_{i=1}^{n}\Big[\Big(\frac{P(C_i)}{r(C_i)} + P_q(C_i)\Big)(C'_E\bar{\mathfrak{a}} + C''_E S^{\mathbf{R}}(\bar{g}))\Big]\,, \tag{7.32}$$

*where $C'_E$ and $C''_E$ depend on the constants of the construction, on $\Lambda_*$, and on $c_{\mathcal{C}}$, nondecreasingly in $c_{\mathcal{C}}$, and $S^{\mathbf{R}}(\bar{g}) \coloneqq \sup_K \sum_{j\le K}(MR_j)^2 e^{-p_0(g_j)/4}$ is finite and tends to 0 as $\bar{g}\to 0$. In particular $B_E(\mathcal{C})$ does not depend on $K$.*

*Proof.* We sum the four parts of (7.31) over $j$, loop by loop, with $m := K' - j + 1$, so that the lengths of the step $j-1$ are those of the stopping step multiplied by $L^m$.

*The corner part.* The ratio $P_{j-1}(C)/r_{j-1}(C) = P(C)/r(C)$ does not depend on $j$, and $x := r_{K'}(C) \geq a_{K'}/2 \geq 1$ by (3.45), since $s_{K'}, \delta_{K'} \geq a_{K'}$ and $P_{K'} \geq 4a_{K'}$. Lemma 2.11 with $c = 2c_E^T$, together with $xe^{-2c_E^T x} \leq 1/(2ec_E^T)$, $e^{-2c_E^T x} \leq 1$, and $|\mathcal{K}(C)| \leq P(C)/(2r(C))$ from Lemma 2.5, gives

$$\sum_{m\geq 1} \Big( \frac{P(C)}{r(C)} x L^m + |\mathcal{K}(C)| \Big) e^{-2c_E^T x L^m} \leq \frac{P(C)}{r(C)} \Big( \frac{1}{2ec_E^T} + \frac{1}{c_E^T} + \frac{1}{2} + \frac{1}{2c_E^T} \Big) . \tag{7.33}$$

*The torus part.* By the proof of Lemma 7.5, $\iota_{T,K'} \geq a_{K'}(L^{m_T+q} - L^{-3})/2 \geq a_{K'}$, and $\iota_{T,j} = L^{m-1}\iota_{T,K'}$, while $P_{j-1}(C) = P_q(C)a_{K'}L^m \leq P_q(C)L\,L^{m-1}\iota_{T,K'}$. Lemma 2.11 with $x = \iota_{T,K'}$ and $c = c^Z$, and $c^Z x \geq 1$, give $\sum_m P_{j-1}(C)e^{-c^Z \iota_{T,j}} \leq 3P_q(C)Lxe^{-c^Z x} \leq 3P_q(C)L/(ec^Z)$.

*The cross part.* With $a_j = a_{K'}L^{m-1}$ and $P_{j-1}(C) = P_q(C)La_j$, Lemma 2.11 with $x = a_{K'}$ and $c = \nu_\times/M$, and $\nu_\times a_{K'}/M \geq 1$, which holds for $\gamma_q$ small since $a_{K'} \geq L^3 M R_{K'}$ and $R_{K'} \geq (\log \hat{g}_{K'}^{-2})^r \geq (\log \gamma_q^{-2})^r$, give at most $3P_q(C)La_{K'}e^{-\nu_\times a_{K'}/M} \leq 3P_q(C)LM/(e\nu_\times)$.

*The* $\mathbf{R}$*-part.* Since $P_j(C) = (P(C)/r(C))r_j(C)$ and $te^{-ct/(MR_j)} \leq MR_j/(ec)$ for $t \geq 0$, the first bracket is at most $(P(C)/r(C))MR_j(1/(ec_E^R) + 1/2)$; since $\iota_{T,j} \geq a_j$ and $P_j(C) = P_q(C)a_j$, the second is at most $P_q(C)MR_j/(ec_E^R)$. The sum over $j$ is therefore at most $C_E^R(P(C)/r(C) + P_q(C))(1/2 + 2/(ec_E^R))S^{\mathbf{R}}(\bar{g})$. The number $S^{\mathbf{R}}(\bar{g})$ is finite and tends to 0 with $\bar{g}$: by Lemma 3.2 and the definition of a tube, $g_j^{-2} \geq \hat{g}_j^{-2}/\rho \geq (2/3)(\bar{g}^{-2} + b'(K-j))$, so that $e^{-p_0(g_j)/4} \leq \exp(-(A_0/4)(\log((2/3)(\bar{g}^{-2} + b'n)))^{p_0})$ with $n = K - j$, while $(MR_j)^2 \leq 4L^2M^2(\log g_j^{-2})^{2r}$ by Lemma 3.4(i), and the resulting series over $n \geq 0$ converges because $p_0 > 1$, uniformly in $K$, with terms which tend to 0 as $\bar{g} \to 0$.

The amplitudes are at most $\bar{\mathfrak{a}}$, and adding the four parts gives (7.32). *q.e.d.*

**Definition 7.12** (Normalisation)**.** The normalisation $\kappa_K^{(q)}(C,\chi)$ of a loop $C$ at level $q$ with the character $\chi$ is the product (1.5) of the step factors $\kappa_k(C,\chi)$ of Definition 7.8, $1 \leq k \leq K'$.

The normalisation removes from the expectation a perimeter term and a term for each corner, and nothing else. Since the lengths of the segments scale with the steps, $N_v^{(j-1)}(C) = L^{-(j-1)}N_v^{(0)}(C)$, the definitions give the following description, which shows which divergences are removed.

**Proposition 7.13** (Structure of the Normalisation)**.** *For a loop $C$ at level $q$ with the character $\chi$ and $K > j_q$,*

$$\log \kappa_K^{(q)}(C,\chi) = \sum_{v\in\mathcal{V}} N_v^{(0)}(C)\, m_{v,K} + \sum_{c\in\mathcal{K}(C)} z_{v_c,w_c,K} , \tag{7.34}$$

*with the numbers*

$$m_{v,K} := \sum_{j=1}^{K'} L^{-(j-1)} \Big( \frac{\kappa_v^{(j)}}{\Pi_j} + \mathbf{1}_{\mathbf{R}}(j)\mu_{j,v}^{\mathbf{R}} \Big) , \qquad z_{v,w,K} := \sum_{j=1}^{K'} (\zeta_{v,w}^{(j)} + \mathbf{1}_{\mathbf{R}}(j)\zeta_{j,v,w}^{\mathbf{R}}) , \tag{7.35}$$

*which depend on $K$, $\bar{g}$, $q$, the character, and the constants of the construction, but not on $C$ or on the torus, and satisfy $|m_{v,K}| \leq \sum_{j\leq K'} L^{-(j-1)}(C_{S'}\mathfrak{a}_j + C_E^R e^{-p_0(g_j)/4})$ and $|z_{v,w,K}| \leq \sum_{j\leq K'}(C_\zeta \mathfrak{a}_j + C_E^R e^{-p_0(g_j)/4})$.*

*Proof.* Insert $N_v^{(j-1)}(C) = L^{-(j-1)} N_v^{(0)}(C)$ into (7.23) and sum over $j$. The bounds are Lemma 7.7, its counterpart at the step 1 [1, (5.31), (6.5)], and Proposition 7.10. The dependence on $q$ is through $K' = K - j_q$ only. *q.e.d.*

The number $N_v^{(0)}(C)$ is the length of the segments of orientation $v$ in fine units, of order $L^K$, so that the first sum in (7.34) is the self-energy of the loop, proportional to the physical length of each orientation divided by the lattice spacing, and the second sum, which grows at most like $\sum_j g_j^2 (\log g_j^{-2})^{6p_0}$ in $K$, is the renormalisation of the corners.

## 8 The Identity at the Stopping Scale and the Uniform Bound

We now stop the induction and read off what it says about the renormalised expectation. Two features of the problem seem to stand in the way of a bound which is uniform in $K$. The normalisation $\prod_i \kappa_K^{(q)}(C_i, \chi_i)$ diverges as $K \to \infty$, by Proposition 7.13, so that an estimate of $\mathbb{E}_K[W_{\mathcal{C}}]$ which is not exact in the normalisation becomes useless once it is divided by it. The terms of the density with large fields, on the other hand, are not small by counting: in four dimensions the factor which a large field contributes "does not give any positive power of $\varepsilon$" [9, p. 244], while the number of places at which a large field may occur grows like $\varepsilon_K^{-4}$. Both difficulties disappear at the stopping step $K'$. The normalisation is the same number in every term of the dressed loops, so that it can be divided out exactly; and the density at the step $K'$ is a function of the block field on the lattice $\mathbb{T}_{j_q}$, whose spacing $uL^{-j_q}$ does not depend on $K$, so that the terms with large fields can be compared with the term without them as whole integrals over a fixed space, at a price which is exponential in the number of sites of $\mathbb{T}_{j_q}$ and in nothing else. The outcome is an exact identity, Theorem 8.1, and from it the bound, Theorem 8.6.

Throughout this section $\mathcal{C} = ((C_1, \chi_1), \dots, (C_n, \chi_n))$ is an ensemble as in Theorem 1.5, and

$$K \geq K_0(\mathcal{C}) \coloneqq \max(j_q + 1,\, K_{\text{tor}}(\mathcal{C}))\,, \tag{8.1}$$

so that $K' = K - j_q \geq 1$ and, by Corollary 6.2, the ensemble is dressed at the step $K'$. We use the units of the step $K'$, in which the lattice of the block field $V \coloneqq V_{K'}$ is $\mathbb{T}_{j_q}$ with the spacing 1 and

$$N_q \coloneqq \frac{T}{uL^{-j_q}} = 2L^{m_T + j_q} \tag{8.2}$$

sites per side, and the fine lattice $\mathbb{T}_K$ has the spacing $L^{-K'}$. Every loop of $\mathcal{C}$ is a path of bonds of $\mathbb{T}_q \subset \mathbb{T}_{j_q}$, with $P_{K'}(C_i) = P_q(C_i) L^3 M R_{K'-1}$ bonds of $\mathbb{T}_{j_q}$ by Lemma 3.14, and we write $P_{K'}(\mathcal{C}) \coloneqq \sum_i P_{K'}(C_i)$. We write $g_{K'}$ for the running coupling of the step $K'$; it depends on $K$, but by (3.9) with $n - m = j_q$ it satisfies $g_{K'} \leq (1 + \beta_0) g_K = (1 + \beta_0)\bar{g}$, and by Lemma 3.2(i) $\log g_{K'}^{-2} \leq \log \hat{g}_{K'}^{-2}$, where $\hat{g}_{K'} = \hat{g}_{K - j_q}$ does not depend on $K$. The parameters $\varepsilon_{K'}$, $R_{K'}$, $\alpha_{0,K'}$, $\alpha_{1,K'}$ of the step $K'$ are functions of $\hat{g}_{K'}$ (Definition 3.11), and they do not depend on $K$ either.

### 8.1 The Identity

A *labelled term* $\ell$ of $\varrho_{K'}$ is a term of the expansion (3.23) at $k = K'$ with given labels: the sequences of domains, with the refined labels of Section 5.2 at the first step, together with the labels of the **R**-operations, a renormalised component being removed from the sequences [11, (0.3), p. 176]. We write

$$\varrho_\ell(V) \coloneqq \chi_{K'}(\Omega_{K'})\, \mathbf{T}_{K'}(\ell)[\exp A_{K'}]\,, \qquad Z_\ell \coloneqq \int \varrho_\ell(V)\, \mathrm{d}V\,, \tag{8.3}$$

for the density of $\ell$ and its integral, $\mathrm{d}V$ being the normalised Haar measure on $G^{\mathcal{B}}$ with $\mathcal{B}$ the set of the positive bonds of $\mathbb{T}_{j_q}$. For a function $F$ of $V$ and of the outer variables of $\ell$, and for $Z_\ell > 0$, we put

$$\langle F \rangle_\ell := \frac{1}{Z_\ell} \int \chi_{K'}(\Omega_{K'})\, \mathbf{T}_{K'}(\ell)[F \exp A_{K'}]\, \mathrm{d}V \ . \tag{8.4}$$

The term $\ell_0$ is the one whose large-field region $Z_{K'} = \Lambda_{K'}^c$ is empty; by the nesting (3.19) it has $\Omega_j = \Lambda_j = \mathbb{T}$ for every $j \le K'$, and it collects, by [11, (0.6), p. 176], all the histories whose large-field components have been renormalised. For a labelled term $\ell$ we write $Z(\ell)$ for the set of the connected components of its large-field region at the step $K'$, so that $Z(\ell) = \emptyset$ if and only if $\ell = \ell_0$.

**Theorem 8.1** (Identity at the Stopping Scale)**.** *Under the hypotheses of Theorem 1.5, let $K \ge K_0(\mathcal{C})$. The numbers $w_\ell := Z_\ell / \sum_{\ell'} Z_{\ell'}$ are nonnegative and $\sum_\ell w_\ell = 1$. With $\widehat{\mathcal{W}}_\ell$ the dressed loop of the step $K'$ on $\ell$ divided by $\prod_i \kappa_K^{(q)}(C_i, \chi_i)$,*

$$W_K(\mathcal{C}) = \sum_{\ell \,|\, Z_\ell > 0} w_\ell \, \langle \widehat{\mathcal{W}}_\ell \rangle_\ell \ , \qquad \widehat{\mathcal{W}}_\ell = e^{\mathcal{E}_{K'}(\ell)} \Big[ \prod_{i=1}^n f_{K'}^{(i)} + \sum_{\mathfrak{B} \neq \emptyset} Q_{K'}(\mathfrak{B})\, w_{K'}(\mathfrak{B}) \Big] \ . \tag{8.5}$$

*On the term $\ell_0$ the exponent $\mathcal{E}_{K'}(\ell_0) = E_{\mathrm{const},K'}$ is a number, and $\varrho_{\ell_0}$ is a function of the field $V$ on $\mathcal{B}$ alone. Consequently, with the measure $\mu_K$ on $G^{\mathcal{B}}$ of total mass $w_{\ell_0}$ defined by*

$$\mathrm{d}\mu_K(V) := \frac{\varrho_{\ell_0}(V)\, \mathrm{d}V}{\sum_\ell Z_\ell} \ , \tag{8.6}$$

*and with $W_{\mathcal{C}}(V)$ the observable (1.4) of the field $V$ on $\mathbb{T}_{j_q}$,*

$$W_K(\mathcal{C}) = e^{E_{\mathrm{const},K'}} \int W_{\mathcal{C}}(V)\, \mathrm{d}\mu_K(V) + r_K(\mathcal{C}) \ , \tag{8.7}$$

*where*

$$r_K(\mathcal{C}) := e^{E_{\mathrm{const},K'}} \int \Big[ \prod_{i=1}^n f_{K'}^{(i)} - W_{\mathcal{C}}(V) + \sum_{\mathfrak{B} \neq \emptyset} Q_{K'}(\mathfrak{B})\, w_{K'}(\mathfrak{B}) \Big] \mathrm{d}\mu_K(V) + \sum_{\ell \neq \ell_0 \,|\, Z_\ell > 0} w_\ell \, \langle \widehat{\mathcal{W}}_\ell \rangle_\ell \ . \tag{8.8}$$

The identity is exact, and it is a convex combination: $W_K(\mathcal{C})$ is an average, with the weights $w_\ell$, of expectations of the normalised dressed loops. The measure $\mu_K$ lives on the space $G^{\mathcal{B}}$, which does not depend on $K$, and the function $W_{\mathcal{C}}(V)$ which it integrates in the leading term does not depend on $K$ either; everything that moves with $K$ is in the measure, in the number $E_{\mathrm{const},K'}$, and in the remainder.

*Proof.* We first show that every term is nonnegative. At real configurations the test functions of the construction take values in $[0,1]$ (Lemma 3.8(i)), the operation $\mathbf{T}_{K'}(\ell)$ consists of integrals against nonnegative measures (Haar measures, $\delta$-functions of the block averaging, Gaussian weights, and characteristic functions), and the effective action is real; Bałaban proves that the densities are positive, "hence the denominators are positive, and the operation $\mathbb{R}$ is well defined" [11, p. 176], and the terms which the **R**-operation produces are such a term multiplied by the exponential of a real sum [11, (0.6), p. 176]. Hence $Z_\ell \ge 0$, and (8.4) is the expectation with respect to a probability measure on the variables of $\ell$. Since every operation of the construction preserves the integral of a density (Section 5.1), $\sum_\ell Z_\ell = \int \varrho_{K'}\, \mathrm{d}V = Z_K > 0$, and the $w_\ell$ are nonnegative numbers with sum one.

Next we use the numerator. By (5.2) at $k = K'$ and Definition 5.1,

$$Z_K(\mathcal{C}) = \int N_{K'}(V)\,\mathrm{d}V = \sum_{\ell} \int \chi_{K'}(\Omega_{K'})\,\mathbf{T}_{K'}(\ell)[\mathcal{W}_{K'}(\ell)\exp A_{K'}]\,\mathrm{d}V = \sum_{\ell\,|\,Z_\ell>0} Z_\ell\,\langle \mathcal{W}_{K'}(\ell)\rangle_\ell\,, \tag{8.9}$$

where a term with $Z_\ell = 0$ contributes nothing, since it is the integral of the function $\mathcal{W}_{K'}(\ell)$, which is bounded on the support of the term by Lemma 8.5 below, against a nonnegative measure of total mass zero. The normalisation in front of the dressed loop is the same number on every term, namely $\prod_i \kappa^{(K')}(C_i,\chi_i) = \prod_i \kappa_K^{(q)}(C_i,\chi_i)$ by (5.7) and (1.5), and it is not zero. Dividing (8.9) by $Z_K$ and by this number, and using (5.1) and Definition 1.3, we obtain (8.5).

On the term $\ell_0$ there are no large-field data at any step. The carrier of the loop's integral is therefore the whole torus at every step, the distances $d^{(j)}(x)$ in (5.9) are infinite, and every local term $e_X^{(j)}$ vanishes, so that $\mathcal{E}_{K'}(\ell_0) = E_{\mathrm{const},K'}$; for $j \geq 2$ this is also the statement $E_j^{\partial}(\ell_0) = 0$ of the definition of $E_j^{\partial}$ in Section 7.1. The operation $\mathbf{T}_{K'}(\ell_0)$ holds no integrations, so that $\varrho_{\ell_0}(V) = \chi_{K'}(\mathbb{T})\exp A_{K'}(U_{K'}(V))$, where the $\mathbf{R}$-part and the boundary terms left by renormalised components are functions of $V$ [11, (0.6), p. 176], [12, p. 392]. The insertions on $\ell_0$ are functions of $V$ as well, since there are no outer variables. Separating the term $\ell_0$ in (8.5) and adding and subtracting $e^{E_{\mathrm{const},K'}} \int W_{\mathcal{C}}\,\mathrm{d}\mu_K$ gives (8.7) with the remainder (8.8). *q.e.d.*

### 8.2 The Coarse Loop

On the term $\ell_0$ the leading words $f_{K'}^{(i)} = \chi_i(U_{K'}(V)(C_i))$ are the Wilson loops of the background field, a field on the fine lattice which depends on $K$, while the leading term of (8.7) contains the Wilson loops of $V$ itself. The two differ by little, because $V$ is the $K'$-fold block average of $U_{K'}(V)$ and $U_{K'}(V)$ is close to flat on the scale of a plaquette of $\mathbb{T}_{j_q}$. To quantify this we need a comparison between an iterated average and a transport, which we now prove.

The average of [8, (0.4), p. 253] maps a field $U$ on a lattice to the field $\bar{U} = M(U)$ on the lattice whose spacing is $L$ times larger. With $d = 4$, let $B(y)$ be the set of the $L^d$ points of the finer lattice in the cube of side $L$ centred at $y$ [8, (0.3), p. 252], $\mathbf{G}(y,x)$ the family of the shortest paths from $y$ to $x$ which run through the coordinate directions in some order, one path for each order, $[x,x']$ the translate of $c$ to $x$, and $U(c)$ the transport along the straight path of $L$ bonds of $c$ [8, p. 252]. For $x \in B(c_-)$, $\Gamma \in \mathbf{G}(c_-,x)$, and $\Gamma' \in \mathbf{G}(c_+,x')$ put $\ell(x,\Gamma,\Gamma') \coloneqq \Gamma \cup [x,x'] \cup (-\Gamma') \cup (-c)$, a closed path at $c_-$, and $v(x,\Gamma,\Gamma') \coloneqq L^{-d}|\mathbf{G}(c_-,x)|^{-1}|\mathbf{G}(c_+,x')|^{-1}$. Then

$$\bar{U}(c) = \exp\Big[i \sum_{x,\Gamma,\Gamma'} v(x,\Gamma,\Gamma')\,\frac{1}{i}\log U(\ell(x,\Gamma,\Gamma'))\Big]U(c)\,, \tag{8.10}$$

and the weights $v(x,\Gamma,\Gamma')$ are nonnegative with sum one. We write $M^j$ for the $j$-fold iterate of $M$, the average of order $j$, as the averages of higher order are defined recursively in [4, (43), p. 24], and $T_c(U)$ for the transport of $U$ along the straight path of the fine bonds of a bond $c$ of a coarser lattice. Here and below $\|\cdot\|$ is the operator norm, $\mathfrak{g}$ is the Lie algebra of $G$ realised as the Hermitian matrices $X$ with $e^{iX} \in G$, $r_G \in (0, 1/10]$ is a radius such that the principal logarithm of every $W \in G$ with $\|W - 1\| \leq r_G$ lies in $i\mathfrak{g}$, which exists because the exponential map is a local diffeomorphism at 0, and for a unitary representation $\pi$ of $G$ and $Y \in \mathfrak{g}$ we write $\pi(e^{iY}) = e^{i\pi_*(Y)}$, so that $\pi_*(Y) = -i\,\mathrm{d}\pi(iY)$ and, with the number $c_\pi$ of (1.7), $\|\pi(e^{iY}) - 1\| \leq \|\pi_*(Y)\| \leq c_\pi\|Y\|$.

Three elementary facts about unitary matrices will be used. For unitary $W$ and $g$,

$\|gWg^{-1}-1\| = \|W-1\| = \|W^{-1}-1\|$, and for any matrices $a_1,\dots,a_m$,

$$\Big\|\prod_{l=1}^{m}(1+a_l)-1\Big\| \le \prod_{l=1}^{m}(1+\|a_l\|)-1\ . \tag{8.11}$$

Secondly, if $W$ is unitary and $\|W-1\| \le e^s-1$ with $0 \le s \le 1/10$, then the principal logarithm $W = e^{iX}$ has

$$\|X\| \le 1.06\, s\ : \tag{8.12}$$

an eigenvalue $e^{i\phi}$ of $W$ with $\phi \in (-\pi,\pi]$ has $2|\sin(\phi/2)| = |e^{i\phi}-1| \le e^s-1 \le 1.052\, s$, because $(e^s-1)/s$ is increasing and equals $1.0517\dots$ at $s = 1/10$, and $2\arcsin(y/2) \le y(1-y^2/4)^{-1/2} \le 1.002\, y$ for $0 \le y \le 0.11$. Thirdly, $\|e^{iX}-1\| \le \|X\|$ for Hermitian $X$, since $|e^{i\phi}-1| \le |\phi|$; consequently, for $X, F \in \mathfrak{g}$ with $s := \|X\|+\|F\| \le \min(1/10, \log(1+r_G))$, (8.11) gives $\|e^{iX}e^{iF}-1\| \le e^s-1 \le r_G$, so that $e^{iX}e^{iF} = e^{iE}$ with $E \in \mathfrak{g}$ and $\|E\| \le 1.06\, s$ by (8.12).

**Lemma 8.2** (The Coarse Loop)**.** *Let $L \ge 21$ be odd and $k \ge 1$, let $a_4 := 7.2$, and let $U$ be a $G$-valued field on a periodic lattice of spacing $\eta$, with periods which are multiples of $L^k\eta$, such that $\|U(\partial p)-1\| \le t$ for every plaquette $p$ and*

$$a_4 L^{2k} t \le \min(1/10,\ \log(1+r_G))\ . \tag{8.13}$$

*(i) For every $0 \le j \le k$ and every bond $c$ of the lattice of spacing $L^j\eta$ there is $E_c \in \mathfrak{g}$ with $(M^jU)(c) = e^{iE_c}T_c(U)$ and $\|E_c\| \le a_4 L^{2j} t$.*

*(ii) For every closed path $C$ of $P$ bonds of the lattice of spacing $L^k\eta$ and every character $\chi = (\dim\pi)^{-1}\operatorname{Tr}\pi$ of an irreducible unitary representation $\pi$,*

$$|\chi(U(C)) - \chi((M^kU)(C))| \le e^{c_\pi a_4 P L^{2k} t} - 1\ . \tag{8.14}$$

The constant $a_4 = 0.3(d-1)(d+4)$ at $d=4$ does not grow with $j$: each level of averaging loses only the factor $(d+3)/L$ on the deviation of the previous level, and the deviation of the plaquettes of the new level, which is $L^2$ times larger, is what sets the scale.

*Proof.* (i) We argue by induction on $j$, the case $j=0$ being $E_c = 0$. Let $0 \le j < k$, assume (i) at the level $j$, and write $W := M^jU$, a $G$-valued field on the lattice of spacing $L^j\eta$, which we use as the unit of length in this step. Let $c$ be a bond of the lattice of spacing $L^{j+1}\eta$, from $c_-$ to $c_+ = c_- + Le_\mu$, so that $(M^{j+1}U)(c) = e^{iX_c}W(c)$ by (8.10), with $X_c$ the convex combination in the exponent and $W(c)$ the product of $W$ over the $L$ bonds $b$ of $c$.

*The transport.* By the hypothesis of the induction, $W(c) = \prod_{b\subset c} e^{iE_b}T_b(U) = R_c\, T_c(U)$ with $R_c := \prod_{b\subset c} g_b e^{iE_b} g_b^{-1} \in G$, where $g_b$ is the transport of $U$ along the part of $c$ before $b$. By (8.11) and the third fact, $\|R_c-1\| \le e^{s_c}-1$ with $s_c := a_4 L^{2j+1} t$, and $s_c \le a_4 L^{2k-1} t \le \min(1/10, \log(1+r_G))/L$ by (8.13), since $2j+1 \le 2k-1$. Hence $R_c = e^{iF_c}$ with $F_c \in \mathfrak{g}$ and $\|F_c\| \le 1.06\, a_4 L^{2j+1} t$ by (8.12).

*The loops of the average.* Let $x = c_- + n$ with $n \in \mathbb{Z}^d$ and $|n_\nu| \le (L-1)/2$, which is the condition $x \in B(c_-)$ for odd $L$, let $\Gamma \in \mathbf{G}(c_-, x)$ and $\Gamma' \in \mathbf{G}(c_+, x')$, and let $\ell := \Gamma \cup [x,x'] \cup (-\Gamma') \cup (-c)$. The loop $\ell$ has at most $(d+2)L$ bonds: $\Gamma$ and $\Gamma'$ have $|n|_1 \le d(L-1)/2$ each, and $[x,x']$ and $c$ have $L$ each. As above, $W(\ell) = R_\ell\, U(\ell)$, where $U(\ell)$ is the transport of $U$ around the fine loop obtained by replacing each bond of $\ell$ by its straight fine path, and $\|R_\ell - 1\| \le e^{(d+2)a_4 L^{2j+1} t} - 1$. To bound $U(\ell)$ we contract $\ell$ across plaquettes of the lattice of spacing $L^j\eta$. Let $\Gamma_+$ be the translate of $\Gamma$ by $Le_\mu$, which lies in $\mathbf{G}(c_+, x')$. The paths $\Gamma'$

and $\Gamma_+$ have the same displacement $n$, and passing from the order of the directions in $\Gamma'$ to that in $\Gamma_+$ by transpositions of adjacent straight segments, each pair of directions $\alpha \neq \beta$ is transposed at most once and the transposition sweeps $|n_\alpha||n_\beta|$ plaquettes. This replaces $\Gamma'$ by $\Gamma_+$; we then move $[x, x']$ back along $\Gamma$ one bond at a time, which sweeps $L$ plaquettes for a bond of direction $\pm e_\nu$ with $\nu \neq \mu$, and none for a bond of direction $\pm e_\mu$, after which the loop is trivial. The number of plaquettes swept is therefore at most

$$\begin{aligned} A_L &\coloneqq \sum_{\alpha<\beta} |n_\alpha||n_\beta| + L(|n|_1 - |n_\mu|) \\ &\leq \frac{d(d-1)}{2}\Big(\frac{L-1}{2}\Big)^2 + \frac{(d-1)L^2}{2} \leq \frac{(d-1)(d+4)}{8}L^2 = \frac{a_4}{2.4}L^2 \,. \end{aligned} \tag{8.15}$$

Each sweep multiplies the transport around the loop, at its base point, by a conjugate of the transport of $U$ around the boundary of a plaquette of the lattice of spacing $L^j\eta$, and the latter is an ordered product of conjugates of the transports around the $L^{2j}$ fine plaquettes inside it. By (8.11) and the first fact, $\|U(\ell) - 1\| \leq (1+t)^{A_L L^{2j}} - 1 \leq e^{A_L L^{2j} t} - 1$, and therefore $\|W(\ell) - 1\| \leq e^{s_\ell} - 1$ with

$$s_\ell \coloneqq A_L L^{2j} t + (d+2) a_4 L^{2j+1} t \leq \Big(\frac{1}{2.4} + \frac{d+2}{L}\Big) a_4 L^{2j+2} t \leq 0.71\, a_4 L^{2(j+1)} t \leq a_4 L^{2k} t \,, \tag{8.16}$$

where we used (8.15) and $(d+2)/L \leq 6/21$. By (8.13), $s_\ell \leq \min(1/10, \log(1+r_G))$, so that $\|W(\ell) - 1\| \leq r_G$; since $W(\ell) \in G$, its principal logarithm lies in $i\mathfrak{g}$, and $\|(1/i)\log W(\ell)\| \leq 1.06\, s_\ell$ by (8.12). The convex combination $X_c$ therefore lies in $\mathfrak{g}$ and $\|X_c\| \leq 1.06\,(A_L L^{2j} + (d+2)a_4 L^{2j+1})t$.

*The new level.* We have $(M^{j+1}U)(c) = e^{iX_c} e^{iF_c} T_c(U)$, and

$$\begin{aligned} \|X_c\| + \|F_c\| &\leq 1.06(A_L L^{2j} + (d+3)a_4 L^{2j+1})t \\ &\leq 1.06\Big(\frac{1}{2.4} + \frac{d+3}{L}\Big) a_4 L^{2(j+1)} t \leq 0.795\, a_4 L^{2(j+1)} t \,, \end{aligned} \tag{8.17}$$

since $(d+3)/L \leq 1/3$ for $L \geq 21$. The right side is at most $0.795 \min(1/10, \log(1+r_G))$ by (8.13), so that by the third fact $e^{iX_c} e^{iF_c} = e^{iE_c}$ with $E_c \in \mathfrak{g}$ and $\|E_c\| \leq 1.06 \cdot 0.795\, a_4 L^{2(j+1)} t \leq a_4 L^{2(j+1)} t$. This is (i) at the level $j+1$; in particular $M^{j+1}U$ is $G$-valued, as the induction requires.

(ii) Let $c_1, \dots, c_P$ be the bonds of $C$ in their order, put $T_l \coloneqq T_{c_l}(U)$ and $H_l \coloneqq T_1 \cdots T_{l-1}$, and use (i) at $j = k$, with $E_{c^{-1}} = -T_c^{-1} E_c T_c$ for a bond traversed against its orientation, which has the same norm. Then

$$(M^k U)(C) = \prod_{l=1}^{P} e^{iE_{c_l}} T_l = \Big(\prod_{l=1}^{P} H_l e^{iE_{c_l}} H_l^{-1}\Big)\, U(C) \,, \tag{8.18}$$

and since $|(\dim\pi)^{-1} \operatorname{Tr}((\pi(R)-1)\pi(H))| \leq \|\pi(R) - 1\|$ for unitary $\pi(H)$, (8.11) and $\|\pi(e^{iE}) - 1\| \leq c_\pi \|E\|$ give $|\chi(U(C)) - \chi((M^k U)(C))| \leq \prod_l (1 + c_\pi a_4 L^{2k} t) - 1 \leq e^{c_\pi a_4 P L^{2k} t} - 1$. *q.e.d.*

We apply the lemma at the stopping step. Bałaban states the regularity of the background in a norm $|\cdot|$ on matrices; we let $c_h \geq 1$ be a constant with $c_h^{-1}|Y| \leq \|Y\| \leq c_h |Y|$ for all $N \times N$ matrices $Y$, and $B_3$ is the constant of [7, Theorem 1, p. 279], which depends on $d$ and $L$ only.

**Corollary 8.3** (The Background at the Stopping Step)**.** *Let $\bar{g}$ be so small that*

$$2a_4 c_h (1 + 2\kappa_s) B_3\, \varepsilon_{K'} \leq \min(1/10,\ \log(1 + r_G)) \,. \tag{8.19}$$

*Then on the term $\ell_0$, for every real $V$ in the support of $\varrho_{\ell_0}$ and every $i$,*

$$|f^{(i)}_{K'} - W_{\chi_i}(C_i; V)| \le e^{x(C_i)} - 1 \,, \qquad x(C) \coloneqq 2c_\pi a_4 c_h (1 + 2\kappa_s) B_3 \, P_{K'}(C) \, \varepsilon_{K'} \,, \tag{8.20}$$

*and $|\prod_i f^{(i)}_{K'} - W_{\mathcal{C}}(V)| \le \sum_i (e^{x(C_i)} - 1)$. The numbers $x(C_i)$ do not depend on $K$.*

*Proof.* On $\ell_0$ the determining set is the whole lattice $\mathbb{T}_{j_q}$, so that the background field $U_{K'} = U_{K'}(V)$ is a minimiser of the action on the fine fields $U$ with $M^{K'}(U) = V$ on every bond of $\mathbb{T}_{j_q}$ [9, (2.11), (2.12), p. 256]. Its plaquettes satisfy $|\partial U_{K'} - 1| < 2B_3 \varepsilon_{K'} (L^{-1}\eta)^2$ on $\Omega_{K'} = \mathbb{T}$ [9, p. 268], where $\eta = L^{-(K'-1)}$ is the fine spacing in the units of the step $K' - 1$, so that $(L^{-1}\eta)^2 = L^{-2K'}$ is the area of a fine plaquette in the units of the step $K'$. This bound is drawn from passed test functions; for the soft test functions of the construction it holds with $\varepsilon_{K'}$ replaced by $(1 + 2\kappa_s)\varepsilon_{K'}$ (Supplement, Proposition S.2). Hence the hypotheses of Lemma 8.2 hold with $k = K'$, $\eta = L^{-K'}$, and $t \coloneqq 2c_h(1+2\kappa_s)B_3\varepsilon_{K'}L^{-2K'}$, for which $a_4 L^{2K'} t$ is the left side of (8.19), and the periods of $\mathbb{T}_K$ are multiples of the spacing of $\mathbb{T}_{j_q}$. On $\ell_0$ the leading word is $f^{(i)}_{K'} = \chi_i(U_{K'}(C_i))$, and (8.14) with $P = P_{K'}(C_i)$ gives (8.20), because $c_\pi a_4 P_{K'}(C_i) L^{2K'} t = x(C_i)$. For the product we telescope: $|\prod_i a_i - \prod_i b_i| \le \sum_i |a_i - b_i|$ for complex numbers of modulus at most one, and $|\chi_i(g)| \le 1$ for $g \in G$. Finally $P_{K'}(C_i)$ and $\varepsilon_{K'}$ do not depend on $K$. *q.e.d.*

### 8.3 The Weight of the Large-Field Terms

We next show that the terms $\ell \neq \ell_0$ carry a small total weight, even after their integrals are multiplied by the factors which the dressed loop acquires near the large-field regions. Since $\mathbb{T}_{j_q}$ is fixed, this can be done by comparing whole integrals: the integral of $\ell_0$ is bounded below by the Haar volume of a ball of configurations, and the integral of every other term above by the supremum of its small-field exponent, each at a price exponential in $N_q^4$. The large-field factors of Bałaban's construction then pay for the history of each component, with half of their gain left over.

We need the following quantities, for a component $X \in Z(\ell)$. The *history* of $X$ consists of the components of the large-field regions $Z_n$, $n \le K'$, of the term which lie in $X$, together with the new large-field regions $Z'$ created at the steps $j' \le K'$ which are merged into $X$; we write $|Y|_n$ for the number of points of the scale $n$ in a set $Y$, $\#_n(Y) \coloneqq |Y|_n (M R_n)^{-4}$ for the number of cubes of side $MR_n$ of the scale $n$ in a union $Y$ of such cubes, and $d'_n(Y)$ for its linear size in terms of these cubes [12, p. 380]. Bałaban's factor [12, (1.79), p. 383] contains for each new region $Z'$ created at the step $j'$ the gain $\exp(-\frac{1}{2}\gamma_0 A_1^2 p_0^2(g_{j'})(d'_{j'}(Z') + 1))$, which holds for the construction of this paper with $\frac{1}{2}\gamma_0$ replaced by $\frac{1}{2}\gamma_0/\rho \ge \frac{1}{3}\gamma_0$ (Proposition 3.12); here $\gamma_0$, $A_1$, and $p_0(g)$ are those of [12]. The regions created at the first step carry instead the gain $\exp(-\frac{1}{2}c_4 \vartheta_V^2 A_1^2 (\log g_1^{-2})^{2p_0}(d'_1(Z') + 1))$ of the first step at the retreated region, with two constants $c_4$, $\vartheta_V$ of that step (Supplement, Proposition S.7(iii)). We put

$$P(X) \coloneqq \sum_{n \le K'} L^4 |Z_n \cap X|_n \,, \qquad G(X) \coloneqq \sum_{Z'} \Upsilon_{j'} (d'_{j'}(Z') + 1) \,, \tag{8.21}$$

with $\Upsilon_{j'} \coloneqq \frac{1}{6}\gamma_0 A_1^2 p_0^2(g_{j'})$ for $j' \ge 2$ and $\Upsilon_1 \coloneqq \frac{1}{4} c_4 \vartheta_V^2 A_1^2 (\log g_1^{-2})^{2p_0}$, the *history volume* of $X$ and half of its gains, and we let $C_P \ge 2$ be the constant of Appendix A.

**Proposition 8.4** (The Weight of the Large-Field Terms)**.** *Let $\bar{g}$ be sufficiently small, depending on the constants of the construction, and let $K > j_q$. Then $Z_{\ell_0} > 0$, and for every*

*choice of numbers $\omega(X)$, one for each component $X$ of each term $\ell \neq \ell_0$, which depend on $X$ and its history only and satisfy $1 \leq \omega(X) \leq \exp(C_P P(X) + \frac{1}{2} G(X))$,*

$$\sum_{\ell \neq \ell_0} Z_\ell \prod_{X \in Z(\ell)} \omega(X) \leq \delta_{\mathrm{LF}} Z_{\ell_0} , \qquad \delta_{\mathrm{LF}} \coloneqq e^{2C_K} (e^{s_{\mathrm{LF}}} - 1) , \tag{8.22}$$

*with*

$$C_K \coloneqq C_{\mathrm{fs}} N_q^4 (1 + \log g_{K'}^{-1}) , \qquad s_{\mathrm{LF}} \coloneqq 2N_q^4 (M R_{K'})^{-4} \exp(-c_{\mathrm{LF}}\, p_0(g_{K'})) , \tag{8.23}$$

*$c_{\mathrm{LF}} \coloneqq (\rho(1+\beta_0))^{-1}$, and $C_{\mathrm{fs}}$ the constant of* (8.29) *below. In particular $1 - w_{\ell_0} \leq \delta_{\mathrm{LF}}$, and $\delta_{\mathrm{LF}} \leq \bar{\delta}_{\mathrm{LF}}$, where $\bar{\delta}_{\mathrm{LF}}$ is the number obtained by replacing $\log g_{K'}^{-1}$ by $\frac{1}{2} \log \hat{g}_{K'}^{-2}$ and $p_0(g_{K'})$ by $p_0((1+\beta_0)\bar{g})$, which depends on $q$, $T/u$, $\bar{g}$, and the constants of the construction, and not on $K$.*

*Proof.* We follow the terms through four steps: a common normalisation, an upper bound for each $Z_\ell$, a lower bound for $Z_{\ell_0}$, and the sum over the terms. We write $g_a \coloneqq g_{K'}$, and assume $g_a \leq 1$.

*The normalisation.* Every term is written in Bałaban's completed form: for its components renormalised at the step $K'$ the missing terms of the effective action "with localization domains intersecting the domain $Z$" are added "to complete the effective action" and subtracted from the remaining expression [12, p. 377, (1.70)], and its surviving components are written with the normalised operations of [12, (1.79), p. 383], which holds for all large-field regions and not only for those which satisfy the conditions of the **R**-operation [12, p. 383–384]. A term then reads $\chi_{K'} \prod_{X \in Z(\ell)} \mathbf{T}'_{K'}(X) \exp A'_{K'}$, where the constants subtracted in the completion, "the sum of normalization terms and vacuum energy terms with localizations intersecting the large field domain $Z$" [12, p. 378], stand inside $\mathbf{T}'_{K'}(X)$. We divide every term by $e^{-E^{\ell_0}}$, with $E^{\ell_0}$ the constant of the effective action of $\ell_0$, a single number for all terms, which changes neither the $w_\ell$ nor (8.22). The constant $E^\ell$ of a term is obtained by subtracting localised one-step vacuum energies from the initial constant [9, p. 262], so that $E^{\ell_0} - E^\ell$ is the sum of the localised normalisation and vacuum-energy terms whose localisations meet the history of a component of $\ell$; a normalisation constant is $O(\log g_n^{-2})$ per point of the scale $n$ and the vacuum energies are $O(1)$ per point [12, p. 380], [9, p. 264], and with a constant $C_\lambda$ of $d$, $G$, and these bounds which absorbs the enlargement of the localisations by a layer of cubes and their exponential tails,

$$|E^{\ell_0} - E^\ell| \leq \sum_{X \in Z(\ell)} \sum_{n \leq K'} C_\lambda (1 + \log g_n^{-2}) L^4 |Z_n \cap X|_n . \tag{8.24}$$

*The upper bound.* The action is nonnegative, so that $-A(1/g_{K'}^2, U_{K'}) \leq 0$. On the support of the term, the regular part, the **R**-part, and the boundary part are bounded by [9, (2.45), (2.46), (2.48), pp. 263–264] by $C_{\mathcal{E}} \sum_n |\Gamma_n|$, with $C_{\mathcal{E}} \coloneqq E_1 (1 - L^{-\beta})^{-1} + 1 + 2B_\partial$ and $B_\partial$ the constant printed $B_1$ in [9, (2.47)]. The couplings of the completed action to the exterior of a renormalised component are at most $\alpha e^{-(1+2\beta)\kappa d_{k,Z(Y)}}$, with $\alpha$ arbitrarily small for $\gamma$ small [12, (1.68), p. 377], and summed over the localisations which meet a cube of side $M R_{K'}$ they are at most a constant $C_\partial \leq 1$, so that they contribute at most $C_\partial N_q^4$. The last layer $\Gamma_{K'}$ of the determining set lies in $\mathbb{T}_{j_q}$ and has at most $N_q^4$ points, and every earlier layer $\Gamma_n \subset \Omega_{n+1}^c \subset Z_{n+1}$ lies in a component of $Z(\ell)$, with $|\Gamma_n \cap X|_n \leq L^4 |Z_{n+1} \cap X|_{n+1}$, since the large-field regions grow and merge until they are renormalised, and a renormalised component

leaves the sequence [11, (0.3), p. 176]. Together with (8.24), the exponent of the term, less the common constant, is at most $(C_{\mathcal{E}}+C_{\partial})N_q^4+\sum_{X\in Z(\ell)}P^c(X)$ with the *history cost*

$$P^c(X)\coloneqq\sum_{n\le K'}c_nL^4|Z_n\cap X|_n\;,\qquad c_n\coloneqq C_{\mathcal{E}}+C_\lambda(1+\log g_n^{-2})\;. \tag{8.25}$$

Since $\mathbf{T}'_{K'}(X)$ is a positive operation, $\mathbf{T}'_{K'}(X)F\le(\mathbf{T}'_{K'}(X)1)\sup F$ for $F\ge0$ [12, (1.73), p. 380], and integrating over $V$ with the normalised Haar measure,

$$Z_\ell\le M_\ell\coloneqq e^{C_K}\prod_{X\in Z(\ell)}\Big[\sup\mathbf{T}'_{K'}(X)1\cdot e^{P^c(X)}\Big]\;, \tag{8.26}$$

where we used $(C_{\mathcal{E}}+C_{\partial})N_q^4\le C_K$, which holds for the constant (8.29) below.

*The lower bound.* Let $\mathfrak{B}\coloneqq\{V\,|\,\|V(b)-1\|<r_{\mathfrak{B}}\text{ for every }b\in\mathcal{B}\}$ with $r_{\mathfrak{B}}\coloneqq g_a/(10c_hB_3)$. The averaged extension $M^{\cdot}(Q^{s*}_{K'}V)$ of [9, (2.16), p. 257], from which the minimisers are built, takes on each bond either the value 1 or a value $V(c)^{\pm1}$: the field $Q^{s*}_{K'}V$ equals 1 on the bonds inside the blocks and $V(c)$ on one bond joining the blocks of $c$, and in (8.10) the paths $\Gamma$, $\Gamma'$ stay inside blocks, while $[x,x']$ and the straight path of a bond each cross the face between two blocks at most once. Hence on $\mathfrak{B}$ every plaquette of the block data is a product of at most four factors within $r_{\mathfrak{B}}$ of 1, so that by (8.11) its deviation is at most $c_h((1+r_{\mathfrak{B}})^4-1)\le5c_hr_{\mathfrak{B}}=g_a/(2B_3)$ in the norm $|\cdot|$. Then [7, Theorem 1, p. 279], with its $\varepsilon_1=g_a/(2B_3)\le a_1$ for $g_a$ small, puts the local minimisers and the minimiser $U_{K'}(V)$ in the space of the configurations with $|U(\partial p)-1|<B_3\varepsilon_1\eta^2=(g_a/2)\eta^2$ for every fine plaquette, $\eta=L^{-K'}$. So every test function of the term $\ell_0$ passes, because $g_a/2\le(\rho^{1/2}/2)\hat g_{K'}<\hat g_{K'}p_0(\hat g_{K'})=\varepsilon_{K'}$ by Lemma 3.2(i), and $\chi_{K'}(\mathbb{T})=1$ on $\mathfrak{B}$. For unitary $W$, $1-\operatorname{Re}\operatorname{tr}W=\frac{1}{2N}\operatorname{Tr}((W-1)^*(W-1))\le\frac12\|W-1\|^2$, and there are $6N_q^4L^{4K'}$ fine plaquettes, so that on $\mathfrak{B}$

$$\frac{1}{g_a^2}A(U_{K'}(V))\le\frac{1}{g_a^2}\cdot6N_q^4L^{4K'}\cdot\frac{c_h^2}{2}\Big(\frac{g_a}{2}\Big)^2L^{-4K'}=\frac34c_h^2N_q^4\;; \tag{8.27}$$

and $\mathbf{E}_{K'}+\mathbf{R}_{K'}\ge-C_{\mathcal{E}}N_q^4$ by [9, (2.45), (2.46)] with $\sum_n|\Gamma_n|=N_q^4$. The Haar measure of the ball of radius $r\le1$ about 1 in $G$ is at least $(c_Gr)^{\dim G}$, with a constant $c_G$ of $G$, by the exponential coordinates, and $\mathcal{B}$ has $4N_q^4$ bonds. Therefore

$$Z_{\ell_0}\ge\exp\Big(-\frac34c_h^2N_q^4-C_{\mathcal{E}}N_q^4\Big)(c_Gr_{\mathfrak{B}})^{4N_q^4\dim G}\ge e^{-C_K}\;, \tag{8.28}$$

with

$$C_{\mathrm{fs}}\coloneqq1+c_h^2+C_{\partial}+C_{\mathcal{E}}+4\dim G(1+\log_+(10c_hB_3/c_G))\;, \tag{8.29}$$

a constant of $d$, $L$, $G$, and the constants of Bałaban's theorems, independent of $K$; for the last inequality of (8.28) we used $\log(c_Gr_{\mathfrak{B}})\ge-\log g_a^{-1}-\log_+(10c_hB_3/c_G)$ and $g_a\le1$.

*The component factors.* Let $X\in Z(\ell)$, and let $\kappa^{\mathrm{tr}}(X)$ be the exact exponent of the factor of $X$ in [12, (1.79)], so that $\sup\mathbf{T}'_{K'}(X)1\le\exp(-\kappa^{\mathrm{tr}}(X)-(2/\rho)p_0(g_{j(X)}))$, with $j(X)$ the step of the first large-field region in $X$ [12, p. 384]. By Supplement, Proposition S.11, which is Bałaban's induction for [12, (1.80), p. 384] with half of the gains set aside and the cost of each step raised,

$$\kappa^{\mathrm{tr}}(X)=\hat\kappa(X)+G(X)+H(X)\;,\qquad H(X)\coloneqq\sum_{n\le K'}\lambda_n\sum_{Z''}M^4R_n^5d'_n(Z'')\;, \tag{8.30}$$

with $\lambda_n\coloneqq1+3\cdot2^3L^4C_\lambda(1+\log g_n^{-2})$, the inner sum over the components $Z''$ of $Z_n$ in $X$, where $\hat\kappa(X)\ge0$, and $\hat\kappa(X)\ge\hat c_0\#_{K'}(X)$ if $X$ survives the stopping step, that is, if it fails one of the conditions of the **R**-operation, with $\hat c_0\coloneqq M^4R_{K'+1}^5(3\cdot2^3)^{-1}(2L^2)^{-4}$.

The history volume and the history cost are paid by $H(X)$ and half of $G(X)$:

$$P^c(X) + C_P P(X) \le H(X) + \frac{1}{2} G(X) \,, \tag{8.31}$$

provided that, for every $n \le K'$,

$$3 \cdot 2^3 L^4 (C_{\mathcal{E}} + C_P) \le R_n \qquad \text{and} \qquad 2^4 L^4 M^4 R_n^4 (c_n + C_P) \le \frac{1}{2} \Upsilon_n \,. \tag{8.32}$$

Indeed, a component $Z''$ of $Z_n$ in $X$ with $d'_n(Z'') \neq 0$ has $\#_n(Z'') \le 3 \cdot 2^3 d'_n(Z'')$ [12, p. 384], so that its points cost $(c_n + C_P) L^4 |Z''|_n \le 3 \cdot 2^3 L^4 (c_n + C_P) M^4 R_n^4 d'_n(Z'') \le \lambda_n M^4 R_n^5 d'_n(Z'')$, by the first condition of (8.32) and $R_n \ge 1$. A component with $d'_n(Z'') = 0$ has at most $2^4$ cubes, because a tree of length zero is a point and at most $2^4$ cubes of a partition contain a given point. It contains no image $S(\cdot)$ of an earlier region, since $S$ adds ten layers of cubes [12, p. 384], so it consists of new regions created at the step $n$, whose gain enters $G(X)$ once; its points cost at most $2^4 L^4 M^4 R_n^4 (c_n + C_P) \le \frac{1}{2} \Upsilon_n$, which is half of the contribution of one new region to $G(X)$, by the second condition. The conditions (8.32) hold for $\bar g$ small, since $R_n \ge (\log g_n^{-2})^r$ and $R_n \le 2L (\log g_n^{-2})^r$ by Lemma 3.4, while $\Upsilon_n$ is a multiple of $(\log g_n^{-2})^{2p_0}$ and $2p_0 > 4r + 1$ (Appendix A).

By (8.30), (8.31), and $p_0(g_j) \ge (1 + \beta_0)^{-1} p_0(g_a)$ for $j \le K'$, which follows from $g_j \le (1 + \beta_0) g_a$ by (3.9) for $\bar g$ small,

$$\sup \mathbf{T}'_{K'}(X) 1 \cdot e^{P^c(X)} \le \exp\Bigl( -\hat\kappa(X) - C_P P(X) - \frac{1}{2} G(X) - 2 c_{\mathrm{LF}}\, p_0(g_a) \Bigr) \,. \tag{8.33}$$

Let $C_{\mathrm{ent}}$ be a constant such that at most $C_{\mathrm{ent}}^m$ connected unions of $m$ cubes contain a given cube. If $X$ survives the stopping step, then $\hat\kappa(X) \ge \hat c_0 \#_{K'}(X) \ge (\log C_{\mathrm{ent}} + 1) \#_{K'}(X)$ for $\bar g$ small, since $\hat c_0$ grows with $R_{K'+1}$. If $X$ is renormalised at the stopping step, then $\#_{K'}(X) \le 100^4$ by the first condition of the **R**-operation, and $c_{\mathrm{LF}} p_0(g_a) \ge (\log C_{\mathrm{ent}} + 1) 100^4$ for $\bar g$ small. In both cases, for every $\omega(X) \le \exp(C_P P(X) + \frac{1}{2} G(X))$,

$$\omega(X) \sup \mathbf{T}'_{K'}(X) 1 \cdot e^{P^c(X)} \le \exp\Bigl( -c_{\mathrm{LF}}\, p_0(g_a) - (\log C_{\mathrm{ent}} + 1) \#_{K'}(X) \Bigr) \,. \tag{8.34}$$

*The sum.* The components of $Z(\ell)$ are disjoint connected unions of the $N_q^4 (M R_{K'})^{-4}$ cubes of side $M R_{K'}$ of $\mathbb{T}_{j_q}$, each of one of the two classes. The sum over the labels with given components and classes, the sequences of domains and the labels of the **R**-operations, is contained in Bałaban's bound, since "the summations over the admissible sequences can be replaced by the factors $\exp O(1) (M R_j)^{-d} |Z_j|$" in the first exponential of [12, (1.79)] [12, p. 383], and (8.34) is a bound on the resulting sum for each component. Dropping the disjointness of the components, and with the factor 2 for the two classes,

$$\begin{aligned} \sum_{\ell \neq \ell_0} M_\ell \prod_{X \in Z(\ell)} \omega(X) &\le e^{C_K} \sum_{m \ge 1} \frac{1}{m!} \Bigl( \frac{2 N_q^4}{(M R_{K'})^4} e^{-c_{\mathrm{LF}} p_0(g_a)} \sum_{m' \ge 1} C_{\mathrm{ent}}^{m'} e^{-(\log C_{\mathrm{ent}} + 1) m'} \Bigr)^m \\ &\le e^{C_K} (e^{s_{\mathrm{LF}}} - 1) \,, \end{aligned} \tag{8.35}$$

because $\sum_{m' \ge 1} e^{-m'} < 1$. With $Z_\ell \le M_\ell$ from (8.26) and $e^{C_K} \le e^{2 C_K} Z_{\ell_0}$ from (8.28), this is (8.22). Taking $\omega = 1$ gives $\sum_{\ell \neq \ell_0} Z_\ell \le \delta_{\mathrm{LF}} Z_{\ell_0} \le \delta_{\mathrm{LF}} \sum_\ell Z_\ell$, that is, $1 - w_{\ell_0} \le \delta_{\mathrm{LF}}$. Finally $C_K$ increases with $\log g_a^{-1} \le \frac{1}{2} \log \hat g_{K'}^{-2}$, and $s_{\mathrm{LF}}$ decreases with $p_0(g_a) \ge p_0((1 + \beta_0) \bar g)$, while $N_q$ and $R_{K'}$ do not depend on $K$; so $\delta_{\mathrm{LF}} \le \bar\delta_{\mathrm{LF}}$. *q.e.d.*

At a fixed level the number $\bar\delta_{\mathrm{LF}}$ tends to zero with $\bar g$, although it is not needed for the bound. Indeed, $L^{j_q} = L^{q+3} M R_{K'-1}$ grows like $(\log \bar g^{-2})^r$, so that $C_K$ is of the order of $(\log \bar g^{-2})^{4r+1}$, while $c_{\mathrm{LF}} p_0((1 + \beta_0) \bar g)$ is of the order of $(\log \bar g^{-2})^{p_0}$ with $p_0 > 4r + 1$. This smallness is used in Section 12.

### 8.4 The Uniform Bound

It remains to bound the normalised dressed loops on each term. On $\ell_0$ the bound is a number which does not depend on $K$, because the number of traversals at the stopping step is fixed; on the other terms the dressed loop acquires, in addition, the costs of the insertions attached to the large-field regions and of the traversals near them, and these are paid as in Proposition 8.4. We put

$$\bar\rho \coloneqq C_\rho(1+\beta_0)^2\bar g^2\Big(1+A_1(\log((1+\beta_0)^{-2}\bar g^{-2}))^{p_0}\Big)^4 , \tag{8.36}$$

$S_w \coloneqq \sum_{t\ge1}(t+1)^3e^{-\eta_w t}$, and $S_{\mathrm{wr}}(b) \coloneqq \sum_{i\ge0} L^i e^{-bL^i}$ for $b>0$, so that $\rho_j \le \bar\rho$ for every $j\le K$ when $\bar g$ is small, because $g_j \le (1+\beta_0)\bar g$ by (3.9) and $x\mapsto x(1+A_1(\log x^{-1})^{p_0})^4$ is increasing for small $x$. With the constants of (5.8), let

$$I_*(\mathcal{C}) \coloneqq P_{K'}(\mathcal{C})\Big[\frac{2c_\sigma C_\chi(1+2\kappa_s)B_3S_w\,\bar\rho\,\varepsilon_{K'}}{\min(\alpha_{0,K'},\alpha_{1,K'})} + C^2\bar\rho\,S_{\mathrm{wr}}\Big(\frac{c_{\mathrm{att}}N_q}{M}\Big)\Big] , \tag{8.37}$$

a number which does not depend on $K$.

For a component $X\in Z(\ell)$ we collect the costs of its history in

$$\omega_A(X) \coloneqq \exp\Big(\sum_{n\le K'} C_{A,n}\#_n(Z_n\cap X)\Big) , \qquad C_{A,n} \coloneqq 2C_A + 8C_e\Lambda_*MR_n\hat\tau_n^2 , \tag{8.38}$$

with $C_A$ the constant of (6.9), $\hat\tau_n$ as in (6.10), and $C_e$ the constant of the cost of an exposed traversal in Section 6.4.

**Lemma 8.5** (The Terms at the Stopping Step)**.** *Let $\bar g$ satisfy* (8.19)*. For every labelled term $\ell$ of $\varrho_{K'}$ and every real configuration of its variables in the support of its indicators,*

$$|\widehat{\mathcal{W}}_\ell| \le e^{B_E(\mathcal{C})+I_*(\mathcal{C})}\prod_{X\in Z(\ell)}\omega_A(X) , \tag{8.39}$$

*with $B_E(\mathcal{C})$ of Corollary 7.11. On $\ell_0$, moreover, $|\sum_{\mathfrak{B}\neq\emptyset}Q_{K'}(\mathfrak{B})w_{K'}(\mathfrak{B})| \le e^{I_*(\mathcal{C})}-1$. If, for every $n\le K'$,*

$$C_{A,n} \le L^4(MR_n)^4 , \tag{8.40}$$

*which holds for $\gamma_q$ small, depending on $c_{\mathcal{C}}$, then $1\le\omega_A(X)\le\exp(C_PP(X))$.*

*Proof.* We first bound the sum over the families. For a family $\mathfrak{B}$ the word $w_{K'}(\mathfrak{B})$ is the normalised trace of a product of unitary transports, which contain the background and, on large-field regions, outer variables of the term, and of the insertions of $\mathfrak{B}$, so that $|w_{K'}(\mathfrak{B})| \le \prod_{W\in\mathfrak{B}}\|Y_W\|$; with Definition 5.1(i), and since the families are sets of groups of the term, of which there are finitely many,

$$\Big|\prod_i f_{K'}^{(i)} + \sum_{\mathfrak{B}\neq\emptyset}Q_{K'}(\mathfrak{B})\,w_{K'}(\mathfrak{B})\Big| \le \prod_W(1+e^{\theta(W)}\|Y_W\|) \le \exp\Big(\sum_W e^{\theta(W)}\|Y_W\|\Big) , \tag{8.41}$$

where we used $|f_{K'}^{(i)}|\le1$, and where the sum without the leading term is at most the right side minus one. Every group contains a traversal, so that the sum over the groups which are not attached is at most $\sum_{\mathfrak{n}}\nu_{\mathfrak{n}}^{(K')}$, the sum over the $P_{K'}(\mathcal{C})$ traversals of the step $K'$.

*The term $\ell_0$.* On $\ell_0$ there are no large-field data at the step $K'$, so that $d_{K'}(\mathfrak{n})=\infty$ and $\mathfrak{m}_{K'}(\mathfrak{n})=0$ in (5.8), and there are no attached groups, since the attached parts of

the renormalised components have become deviation insertions of the class (Section 6.4 and Supplement, Proposition S.10). By (6.1) at $k+1 = K'$, with $\varepsilon_{K'}$ enlarged by the factor $1+2\kappa_s$ of the soft test functions as in the proof of Corollary 8.3, $\delta^*_{\mathfrak{n}}(U_{K'}) \le 2c_\sigma(1+2\kappa_s)B_3\varepsilon_{K'}S_w$, and $\tilde\rho_{K'} \le \bar\rho/\min(\alpha_{0,K'},\alpha_{1,K'})$. The smallest period of the torus in units of the scale $j$ is $N^{(j)}_{\min} = N_qL^{K'-j}$, so that with $i = K'-j$ the last term of (5.8) is at most $C^2\bar\rho\sum_{i\ge0}L^ie^{-c_{\mathrm{att}}N_qL^i/M} = C^2\bar\rho\,S_{\mathrm{wr}}(c_{\mathrm{att}}N_q/M)$. Summing over the traversals, $\sum_{\mathfrak{n}}\nu^{(K')}_{\mathfrak{n}} \le I_*(\mathcal{C})$, and (8.41) gives both assertions on $\ell_0$, since $|e^{\mathcal{E}_{K'}(\ell_0)}| = e^{\mathrm{Re}\,E_{\mathrm{const},K'}} \le e^{B_E(\mathcal{C})}$ by Definition 5.1(iv) and Corollary 7.11.

*The other terms.* On a term $\ell \ne \ell_0$ three further contributions enter, each attached to a component of a large-field region $Z_n$, $n \le K'$, which lies in a component $X \in Z(\ell)$. The groups attached at the step $n$ to a component $Z$ of $Z_n$, including the second and third terms of (5.8) at $n = K'$, have total weight at most $A_n(Z) \le C_A(d'_n(Z)+1) \le 2C_A\#_n(Z)$ by (6.9), where $d'_n(Z)+1 \le 2\#_n(Z)$ for a connected union of cubes, since a tree through the centres of adjacent cubes meets all of them and has edges of length at most twice the side. At the step $n$ the traversals within a cube of side $MR_n$ of $Z_n$ are exposed, at most $4\Lambda_*MR_n$ per cube by Lemma 5.3(i), since such a cube meets one loop by Lemma 3.14; each costs at most $C_e\hat\tau^2_n$, and so does each traversal in the local terms $e^{(n)}_X$ of $E^\partial_n(\ell)$ attached to $Z_n$ (Supplement, Proposition S.9), which gives at most $8C_e\Lambda_*MR_n\hat\tau^2_n$ per cube. The constant part of $\mathcal{E}_{K'}(\ell)$ is bounded by $B_E(\mathcal{C})$ as on $\ell_0$. These contributions add to the exponent in (8.41) and to $\mathrm{Re}\,\mathcal{E}_{K'}(\ell)$ at most $\sum_{X\in Z(\ell)}\sum_{n\le K'}C_{A,n}\#_n(Z_n\cap X)$, which gives (8.39).

Finally, $\omega_A(X) \ge 1$ trivially, and under (8.40) $\sum_n C_{A,n}\#_n(Z_n\cap X) \le \sum_n L^4|Z_n\cap X|_n = P(X) \le C_PP(X)$. The condition (8.40) holds for $\gamma_q$ small, depending on $c_{\mathcal{C}}$, because both terms of $C_{A,n}$ are at most $\frac12L^4(MR_n)^4$ for $n \le K'$. Indeed, $R_n \ge (\log\hat g_n^{-2})^r \ge (\log\gamma_q^{-2})^r$ by Definition 3.3 and (3.46), so that $4C_A \le L^4(MR_n)^4$ once $L^4M^4(\log\gamma_q^{-2})^{4r} \ge 4C_A$. Moreover $R_n \le 2L(\log g_n^{-2})^r$ by Lemma 3.4(i) and $\hat\tau^2_n \le Cg^2_n(\log g_n^{-2})^{6p_0}$, so that $16C_e\Lambda_*MR_n\hat\tau^2_n \le C'\gamma_q^2(\log\gamma_q^{-2})^{6p_0+r}$, since $x \mapsto x(\log x^{-1})^{6p_0+r}$ is increasing for small $x$, and the right side is at most $1 \le L^4(MR_n)^4$ for $\gamma_q$ small. *q.e.d.*

**Theorem 8.6** (Uniform Bound)**.** *Under the hypotheses of Theorem 1.5, let $\bar g$ be so small that* (8.19) *and the hypothesis of Proposition 8.4 hold, and $\gamma_q$ so small that* (8.40) *holds. For every $K \ge K_0(\mathcal{C})$,*

$$|W_K(\mathcal{C})| \le B(\mathcal{C}) := e^{B_E(\mathcal{C})+I_*(\mathcal{C})}(1+\bar\delta_{\mathrm{LF}})\,, \tag{8.42}$$

*and the remainder* (8.8) *satisfies*

$$|r_K(\mathcal{C})| \le e^{B_E(\mathcal{C})}\Big[\sum_{i=1}^n(e^{x(C_i)}-1)+e^{I_*(\mathcal{C})}-1\Big] + e^{B_E(\mathcal{C})+I_*(\mathcal{C})}\,\bar\delta_{\mathrm{LF}}\,. \tag{8.43}$$

*The number $B(\mathcal{C})$ does not depend on $K$; it depends on $\mathcal{C}$ through $\sum_i(P(C_i)/r(C_i)+P_q(C_i))$, through $P_q(\mathcal{C})$, and through the characters, and on $q$, $T/u$, $\bar g$, and the constants of the construction.*

*Proof.* By (8.5), the triangle inequality, and Lemma 8.5,

$$|W_K(\mathcal{C})| \le e^{B_E(\mathcal{C})+I_*(\mathcal{C})}\Big(w_{\ell_0}+\sum_{\ell\ne\ell_0}w_\ell\prod_{X\in Z(\ell)}\omega_A(X)\Big)\,, \tag{8.44}$$

where we used that $\omega_A(X)$ is a number of the label, so that $\langle\prod_X\omega_A(X)\rangle_\ell = \prod_X\omega_A(X)$. By Lemma 8.5 the numbers $\omega(X) := \omega_A(X)$ satisfy the hypothesis of Proposition 8.4, which gives $\sum_{\ell\ne\ell_0}w_\ell\prod_X\omega_A(X) \le \delta_{\mathrm{LF}}Z_{\ell_0}/\sum_\ell Z_\ell \le \bar\delta_{\mathrm{LF}}$; together with $w_{\ell_0} \le 1$ this is (8.42). For the remainder, the integrand of the first term of (8.8) is at most $\sum_i(e^{x(C_i)}-1)+e^{I_*(\mathcal{C})}-1$ by

Corollary 8.3 and Lemma 8.5, the measure $\mu_K$ has mass $w_{\ell_0} \le 1$, and $|e^{E_{\text{const},K'}}| \le e^{B_E(\mathcal{C})}$; the second term is bounded as above. The dependence of $B(\mathcal{C})$ is read from Corollary 7.11, (8.37), and Proposition 8.4: $P_{K'}(\mathcal{C}) = P_q(\mathcal{C})L^3 M R_{K'-1}$, and $N_q$, $\varepsilon_{K'}$, $R_{K'}$, $\alpha_{i,K'}$ are functions of $q$, $T/u$, $\bar{g}$, and the constants of the construction. *q.e.d.*

The bound (8.42) is the first half of (1.10). Two features of it deserve a comment. First, no condition on the coupling in this section involves the size of the loops or their number: (8.19) and the conditions of Proposition 8.4 concern the construction alone, and (8.40) is a condition on $\gamma_q$ which involves the shape constant $\Lambda_*$ and the representations through $c_{\mathcal{C}}$. The size and the number of the loops enter only the value of $B(\mathcal{C})$, through $B_E(\mathcal{C})$ and $I_*(\mathcal{C})$, which are sums over the loops. Secondly, the bound does not say that the large-field terms are small: $\bar{\delta}_{\text{LF}}$ is a fixed number at a fixed coupling. What makes the terms with large fields harmless is that their weights, multiplied by the costs of the dressed loop, are summable against the term without large fields on the fixed lattice $\mathbb{T}_{j_q}$.

## 9 The Difference of the Walk Factors

The limit $K \to \infty$ is obtained by comparing the construction for the target $(K, \bar{g})$ with the construction for $(K+1, \bar{g})$. The two share every lattice but the finest, and on a shared lattice they perform the same operations, with the same parameters, on data which differ by little; what differs is the fine lattice, on which every background field, every propagator, and every term of the effective action is computed. This section and the next prove the two estimates of the comparison which are not contained in Bałaban's analysis of a single run. The present one concerns the propagators. The effective actions are built from the inverses of the quadratic forms of the fluctuation integrals, the propagators of [5], through their expansions into generalised random walks, and the comparison needs a bound on the difference between the terms of one walk in the two constructions which is small in the ratio of the fine spacing to the spacing of the scale on which the walk lives. The difficulty is that the two terms do not act on the same space: one maps sources on the lattice of spacing $\varepsilon$ to fields on that lattice, the other does the same on the lattice of spacing $\varepsilon/L$. We compare them through the one-step average and its right inverses. The comparison is then exact up to a bilinear defect, and this defect is small only between fields which are regular at the scale of their shell; it is the decay of the propagators across the shells which turns this into a bound of the order of a power of $\varepsilon/a_j$.

### 9.1 Two Runs on Common Lattices

Throughout this section and the next two, $\mathcal{C}$ is an ensemble as in Theorem 1.5, $K \ge K_0(\mathcal{C})$, and $K' = K - j_q$. *Run A* is the construction for the target $(K, \bar{g})$, started at the bare coupling $g_0(K, \bar{g})$ of Theorem 4.1 and stopped after $K'$ steps; *run B* is the construction for $(K+1, \bar{g})$, started at $g_0(K+1, \bar{g})$ and stopped after $K'+1$ steps. Both are constructions for tubes with the same numbers $b$, $b'$, and $C_W$, which Theorem 4.1 provides independently of $K$. We write $\varepsilon \coloneqq \varepsilon_K = uL^{-K}$ for the fine spacing of run A and

$$a_k \coloneqq \varepsilon L^k \,, \qquad \text{so that} \qquad \frac{\varepsilon}{a_k} = L^{-k} \,. \tag{9.1}$$

The step $k$ of run A maps a density of the field on the lattice of spacing $a_{k-1}$ to one on the lattice of spacing $a_k$, and so does the step $k+1$ of run B, because $(k+1)-(K+1) = k-K$. We call $k$ the *shared scale* of these two steps, $0 \le k \le K'$; the step 1 of run B, which maps

the lattice of spacing $\varepsilon/L$ to that of spacing $\varepsilon$, is its *extra step*. The couplings at the shared scale $k$ are $g_k^A$ and $g_{k+1}^B$, and we put

$$\delta_k := (g_{k+1}^B)^{-2} - (g_k^A)^{-2} \,. \tag{9.2}$$

**Lemma 9.1** (Shared Parameters)**.** *For every integer $k \le K$ the numbers $u_k$ and $\ell_k$ of* (3.3) *of run A are those of run B at $k+1$. Consequently:*

*(i) every parameter of Definition 3.3, the depth $N_k$ of Definition 3.9, and every threshold, radius, cube size, and soft width of the construction at the shared scale $k$ are the same in both runs;*

*(ii) the stopping index $j_q$, the stopping lattice $\mathbb{T}_{j_q}$, and the number $N_q$ of* (8.2) *are the same in both runs, and run B's stopping step is $K'+1$;*

*(iii) for $0 \le k \le K'$, $|\delta_k| \le u_k - \ell_k \le \frac{1}{2}\min((g_k^A)^{-2}, (g_{k+1}^B)^{-2})$.*

*Proof.* By (3.3), $u_k$ for run A is $\bar{g}^{-2} + b'(K-k) + C_W$, and $u_{k+1}$ for run B is $\bar{g}^{-2} + b'((K+1) - (k+1)) + C_W$, the same number; the same holds for $\ell$. (i) The parameters of Definition 3.3 are functions of $\hat{g}_k = u_k^{-1/2}$, and $N_0(k)$ and $N_k'$ in Definition 3.9 are defined through $\hat{g}_k$ and through the cube sizes $R_{k'}$ at the integers $k' \le k$, which correspond to each other in the same way; the remaining parameters are, by Definition 3.11(i), functions of $\hat{g}_k$ as well. (ii) By Definition 3.13, $j_q$ is defined through the numbers $R^{[n]} = R_{K-n}$, which depend on $n$ and not on $K$. (iii) The squared inverse couplings $(g_k^A)^{-2}$ and $(g_{k+1}^B)^{-2}$ are coordinates of histories of the two tubes (Theorem 4.1), so both lie in $[\ell_k, u_k]$, which gives the first inequality, and $u_k - \ell_k = (b'-b)(K-k) + 2C_W$. Since $b' \le \frac{3}{2}b$ and $C_W \le \beta_0 t/4 \le t/5$ by (3.2), this is at most $\frac{1}{2}(t + b(K-k) - C_W) = \frac{1}{2}\ell_k$, and both squared inverse couplings are at least $\ell_k$. *q.e.d.*

The lemma is the reason for the tubes. With Bałaban's own parameters the cube size $R_k$ of the two runs at a shared scale would differ wherever a power of $L$ lies between $(\log(g_k^A)^{-2})^r$ and $(\log(g_{k+1}^B)^{-2})^r$, and then the two expansions would have different terms, which could not be compared one by one. With the parameters of the tubes the two runs perform the same decompositions on every shared lattice, and they differ only through the fine lattice and through the couplings, the latter by the numbers $\delta_k$, whose size is determined in Section 11.

We need one more object, which carries a field of run A to the fine lattice of run B. For a field $U$ on the lattice of spacing $\varepsilon$ whose plaquettes are small, let $\iota(U)$ be the minimiser of the Wilson action on the lattice of spacing $\varepsilon/L$ among the fields $W$ with $M(W) = U$, the minimal configuration of [9, (2.11), (2.12), p. 256] for one averaging step. By [7, Theorem 1, p. 279] it exists, it is unique up to gauge transformations, it is regular, and it is an analytic and gauge covariant function of $U$; moreover $M\iota(U) = U$ and $\iota(1) = 1$. For a domain $X$ we write $\iota_X$ for the minimiser localised to $X$ in the sense of [9, (2.14)–(2.16), p. 257], which depends on $U$ on a neighbourhood $X^\sim$ of $X$ and satisfies $M\iota_X(U) = U$ there and $\iota_X(1) = 1$. Finally, for a field $U^B$ on the fine lattice of run B we call

$$\tilde{U} := MU^B \tag{9.3}$$

the *aligned background*: a field on the fine lattice of run A, which satisfies run A's constraints whenever $U^B$ satisfies run B's, since the iterated averages of run B are those of run A composed with $M$.

### 9.2 Propagators and Their Walk Expansions

We recall the objects to be compared. The fluctuation integral of a step is Gaussian to leading order, with a quadratic form built from the covariant Laplacian at the background, the block averages, and the gauge fixing, and its inverse is expressed through three operators of [5]: $G'$, the inverse of the scalar operator $\Delta'_a$; $(Q'G'^2Q'^*)^{-1}$, an operator on the block data; and $G$, the inverse of the one-form operator $\Delta_a$. Each of them is approximated by local inverses: for a cube $\square$ of the scale $j$, Bałaban takes a sequence of cubes $\Omega_{j+1}(\square) \subset \Omega_j(\square) = \tilde{\square}^4 \subset \Omega_{j-1}(\square) \subset \cdots \subset \Omega_0(\square) \subset \tilde{\square}^5$ with the same centre and

$$\operatorname{dist}(\Omega_n(\square)^c, \Omega_{n+1}(\square)) = 2R_0M_0L^n\eta \,, \tag{9.4}$$

on which the operators are taken with Dirichlet conditions outside $\Omega_0(\square)$ and with the averaging of the scale $n$ on the shell $\Omega_n(\square) \setminus \Omega_{n+1}(\square)$, and he calls $G'_\square$, $C_\square = (Q'G'^2_\square Q'^*)^{-1}$, and $G_\square$ the resulting operators [5, pp. 408–409]. Here $\eta$ is the fine spacing and $R_0M_0$ the smallest size for which the results of [3] hold, with $M = KR_0M_0$ [5, p. 408]. With a partition of unity $\sum_\square h_\square^2 = 1$ he puts $G'_0 = \sum_\square h_\square G'_\square h_\square$, and similarly $C_0$ and $G_0$ [5, (3.87), p. 409], writes $\Delta'_a G'_0 = I - R'$ with the small remainder $R' = \sum_\square K(h_\square)G'_\square h_\square$ [5, (3.88)–(3.89), p. 409], and expands $G' = G'_0 \sum_n R'^n$ into a sum over walks of cubes [5, Theorem 3.7, (3.90), p. 409]. The same construction, with more kinds of factors, gives the expansion of $G$,

$$G = \sum_\omega R_0(X_0)R_{\alpha_1}(X_1)\cdots R_{\alpha_n}(X_n) \,, \tag{9.5}$$

over walks $\omega = ((0, X_0), (\alpha_1, X_1), \ldots, (\alpha_n, X_n))$ with $X_{i-1} \cap X_i \neq \emptyset$, whose term depends on the configuration on $\tilde{X}_0^5 \cup \cdots \cup \tilde{X}_n^5$ and satisfies

$$|(R_0(X_0)\cdots R_{\alpha_n}(X_n)J)(x)| \leq O(1)(L^j\eta)^2 O(M^{-1/2})^{|\omega|} M^{-\frac{1}{2}|\omega|} e^{-\frac{1}{2}\delta_0 d(\omega,y,y')}|J| \tag{9.6}$$

for $x \in \Delta(y)$, $y \in \Lambda_j$, and $\operatorname{supp} J \subset \Delta(y')$, together with the corresponding bounds in the norms of [5, (3.42)–(3.47)] [5, Theorem 3.10, (3.107), (3.108), p. 416]. The operators $H$, $H_0$, and $\tilde{G}$ through which the fixed point of the fluctuation integral is written [10, (1.2)–(1.4), p. 2] have expansions of the same form, "a propagator is represented by the sum (3.107) [13]", and the term of a walk is bounded "as (3.108) [13]" in "all operator norms in formulations of theorems in [13]" [10, (1.6), (1.7), p. 3]. Bałaban states these bounds for whole terms of walks and not for their factors, of which he says only that each "satisfies a bound of the type (3.89), possibly with an additional power of $L^j\eta$" [5, p. 413]; the comparison below is accordingly a comparison of whole terms.

What the two runs share is read off from these definitions. The cubes of the scale $j$ are cells of the lattice $ML^j\eta\mathbb{Z}^4$, and $ML^j\varepsilon = ML^{j+1}(\varepsilon/L)$, so that for $\square$ at the shared scale $j$ the cubes $\square$, $\tilde{\square}^n$, the partitions of unity as functions on the torus, the localisation domains, and the walks are the same in both runs. The two local sequences are related by

$$\Omega^B_{n+1}(\square) = \Omega^A_n(\square) \,, \qquad 0 \leq n \leq j+1 \,, \tag{9.7}$$

because, by (9.4), $\operatorname{dist}(\Omega_n(\square)^c, \tilde{\square}^4) = 2R_0M_0L^n\eta(L^{j-n}-1)/(L-1)$ [5, p. 408], and the value for run B at $n+1$, with $\eta = \varepsilon/L$ and $j+1$ in place of $j$, is the value for run A at $n$. So the shells of the two runs coincide at every common spacing $L^n\varepsilon$, and run B has one more shell $\Omega^B_0 \setminus \Omega^B_1$, of width $2R_0M_0\varepsilon/L$ at the spacing $\varepsilon/L$, where run A's operators have their Dirichlet exterior. What the runs do not share is the space on which the operators act: on $\Omega_0(\square)$ every factor of run A acts on fields and sources on the lattice of spacing $\varepsilon$, every factor of run B on the lattice of spacing $\varepsilon/L$.

### 9.3 The Galerkin Identities

Fix $\Box$ at the shared scale $j$ and one of the local operators: $\mathcal{O}^X$ is $\Omega_0\Delta_a\Omega_0$ or $\Omega_0\Delta'_a\Omega_0$ of run $X \in \{A, B\}$, acting on the space $\mathfrak{F}^X$ of fields (or functions) on the bonds (or sites) of the fine lattice of run $X$ in $\Omega_0^X(\Box)$, at run B's background $U^B$ for $X = B$ and at the aligned background $\tilde{U}$ for $X = A$, and $G^X := (\mathcal{O}^X)^{-1}$ on $\mathfrak{F}^X$. The pairings $\langle\cdot,\cdot\rangle_X$ carry the weights $\eta_X^4$ of [5, p. 393] and are bilinear, so that $\mathcal{O}^X$ is symmetric on the complex domain of [5, (3.37)], as the analytic extension of a symmetric form. Let $Q_1$ be the linearised one-step average at $U^B$, which maps fields of run B to fields of run A, let $\chi_A$ be the characteristic function of $\Omega_0^A(\Box)$, and let $\mathcal{L}$ be a right inverse of $Q_1$ with

$$\mathcal{L}\mathfrak{F}^A \subset \mathfrak{F}^B \ . \tag{9.8}$$

The *bilinear defect* of $\Box$ is

$$\beta_\Box(\mathcal{F}, \mathcal{A}) := \langle \chi_A Q_1 \mathcal{F}, \mathcal{O}^A \chi_A Q_1 \mathcal{A}\rangle_A - \langle \mathcal{F}, \mathcal{O}^B \mathcal{A}\rangle_B \ , \qquad \mathcal{F}, \mathcal{A} \in \mathfrak{F}^B \ , \tag{9.9}$$

and $\beta_\Box(\cdot, \mathcal{A})^\sharp$ denotes the source of run B which represents the linear form $\mathcal{F} \mapsto \beta_\Box(\mathcal{F}, \mathcal{A})$ on $\mathfrak{F}^B$. The defect measures how far the quadratic form of run A, pulled back by the average, is from that of run B.

**Lemma 9.2** (Galerkin Identities)**.** *Under* (9.8)*, for every source $g$ of run $B$ and every source $f$ of run $A$,*

$$\chi_A Q_1 G^B g - G^A \chi_A \mathcal{L}^* g = G^A \chi_A \mathcal{L}^* \beta_\Box(\cdot, G^B g)^\sharp \ , \tag{9.10}$$

$$\mathcal{L} G^A f - G^B Q_1^* \chi_A f = -G^B \beta_\Box(\cdot, \mathcal{L} G^A f)^\sharp \ . \tag{9.11}$$

*Proof.* For (9.10) put $u_B := G^B g$ and $u_A := G^A \chi_A \mathcal{L}^* g$. For $X \in \mathfrak{F}^A$ we have $\mathcal{L}X \in \mathfrak{F}^B$ by (9.8) and $\chi_A Q_1 \mathcal{L} X = X$, so that

$$\begin{aligned} \langle X, \mathcal{O}^A u_A\rangle_A = \langle \mathcal{L}X, g\rangle_B = \langle \mathcal{L}X, \mathcal{O}^B u_B\rangle_B \ , \\ \langle X, \mathcal{O}^A \chi_A Q_1 u_B\rangle_A = \langle \chi_A Q_1 \mathcal{L}X, \mathcal{O}^A \chi_A Q_1 u_B\rangle_A \ . \end{aligned} \tag{9.12}$$

The difference of the two right sides is $\beta_\Box(\mathcal{L}X, u_B) = \langle X, \chi_A \mathcal{L}^* \beta_\Box(\cdot, u_B)^\sharp\rangle_A$. Both $\chi_A Q_1 u_B - u_A$ and $G^A \chi_A \mathcal{L}^* \beta_\Box(\cdot, u_B)^\sharp$ lie in $\mathfrak{F}^A$, on which $X$ is arbitrary and $\mathcal{O}^A$ is invertible, and (9.10) follows. For (9.11) put $u_A := G^A f$ and $u_B := G^B Q_1^* \chi_A f$. For $Y \in \mathfrak{F}^B$, since $\chi_A Q_1 \mathcal{L} u_A = u_A$,

$$\langle Y, \mathcal{O}^B u_B\rangle_B = \langle \chi_A Q_1 Y, f\rangle_A = \langle \chi_A Q_1 Y, \mathcal{O}^A \chi_A Q_1 \mathcal{L} u_A\rangle_A = \langle Y, \mathcal{O}^B \mathcal{L} u_A\rangle_B + \beta_\Box(Y, \mathcal{L} u_A) \ , \tag{9.13}$$

so that $\langle Y, \mathcal{O}^B(\mathcal{L}u_A - u_B)\rangle_B = -\beta_\Box(Y, \mathcal{L}u_A)$ for every $Y \in \mathfrak{F}^B$, and $\mathcal{L}u_A - u_B \in \mathfrak{F}^B$ by (9.8). *q.e.d.*

The identities are exact, and the whole difficulty is in the size of the defect. This is not small as a bilinear form: for a field $\mathcal{A}$ which alternates at the spacing $\varepsilon/L$ inside one block, $Q_1\mathcal{A} = 0$ and $\beta_\Box(\mathcal{A}, \mathcal{A}) = -\langle \mathcal{A}, \mathcal{O}^B \mathcal{A}\rangle_B$, which is of the order $(\varepsilon/L)^{-2}$ times the square of the norm of $\mathcal{A}$. In the identities, however, the defect is paired with $G^B g$ or $\mathcal{L}G^A f$, which are regular at the scale of their shell by [5, (3.43)], and between regular fields it is small: on a shell of spacing $L^n\eta$ the lattices of the two runs are averages of each other, and the defect of two regular fields is an oscillation of relative order $L^{-n}$. To make this precise we use, for a block $y$ of the shell $n$, the norm

$$\|\mathcal{A}\|_{1,\beta;y} := \sup((L^n\eta)^{-1}|\mathcal{A}| + |\nabla\mathcal{A}|) + (L^n\eta)^\beta \|\nabla\mathcal{A}\|_\beta \ , \tag{9.14}$$

the supremum and the Hölder quotient of [5, (3.40)] taken over the blocks within weighted distance 2 of $y$. We index the shells by run A's index $n$, so that run A's shell $n$ is run B's shell $n+1$ by (9.7), and run B's extra shell has the index $-1$.

**Proposition 9.3** (The Defect of the Local Operators). *Let $\beta \in [\frac{3}{4}, 1)$, and let $R_0 M_0$ satisfy the conditions of Appendix C. There is a constant $C_{\mathrm{df}}(\beta)$, independent of the shell, of $j$, $k$, $K$, the torus, the block, and the field, such that for each of the operators of $G'_\square$, $C_\square$, and $G_\square$, at aligned backgrounds on the complex domain of [5, (3.37)], and for each $\mathcal{A} \in \mathfrak{F}^B$, there are fields $s_0$, $s_1$, $s_2$ of run B with*

$$\beta_\square(\mathcal{F}, \mathcal{A}) = \langle D^*\mathcal{F}, s_2\rangle_B + \langle \nabla\mathcal{F}, s_1\rangle_B + \langle \mathcal{F}, s_0\rangle_B \qquad \text{for every } \mathcal{F} \in \mathfrak{F}^B \,, \tag{9.15}$$

*and, on every block $y$ of the shell $n$,*

$$\sup_{\Delta(y)} (|s_1| + |s_2| + L^n\eta|s_0|) \le C_{\mathrm{df}}(\beta)\varpi_n \|\mathcal{A}\|_{1,\beta;y} \,, \qquad \varpi_n := \begin{cases} L^{-\beta n}, & n \ge 1, \\ 1, & n \in \{0, -1\}, \end{cases} \tag{9.16}$$

*together with a Hölder bound of $s_1$ and $s_2$ at the scale of their shell; the same holds with the two arguments of $\beta_\square$ exchanged.*

*Proof.* We show the mechanism on the averaging terms, which can be computed exactly, and refer to Supplement, Proposition S.12, for the other terms. The quadratic form of [5, (3.16)] contains $Q^*aQ$ with the weight $(L^n\eta)^{d-2}$ on the bonds of the shell $n$. Run B's shell $n+1$ is run A's shell $n$, with the weight $(L^{n+1}\varepsilon/L)^{d-2} = (L^n\varepsilon)^{d-2}$, and run B's averaging of the scale $n+1$ is run A's averaging of the scale $n$ composed with the one-step average, $Q^B_{n+1}(U^B) = Q^A_n(\tilde{U})Q_1$ (Supplement, Proposition S.12(a)), while $\chi_A = 1$ on run A's shells. Hence the averaging part of $\beta_\square$ is exactly $-a\sum_b (\varepsilon/L)^{d-2}\langle \mathcal{F}(b), \mathcal{A}(b)\rangle$, the sum over the bonds of run B's extra shell, which has the form (9.15) with $s_0$ supported there and $\varpi_{-1} = 1$. For the scalar operator the coefficients of the averaging on the shells are $a_n$, with $1/a_n = a^{-1}\sum_{i=0}^{n-1} L^{-2i}$ for $n \ge 1$ [5, p. 394]; run A's shell $n$ carries $a_n$ and run B's the same shell $a_{n+1}$, and $|a_{n+1} - a_n|/a_n \le L^{-2n}/(1 - L^{-2}) \le 2L^{-\beta n}$, which is again of the form (9.16). The covariant Laplacian, the curvature term, and the gauge-fixing term are not compositions of averages; on run A's shells $n \ge 1$ their defect between regular fields is an oscillation of relative order $L^{-n}$, of which the Hölder exponent retains $L^{-\beta n}$, and on the two shells of the finest spacing nothing relates the two runs and $\varpi = 1$. The derivation of these bounds, for $G'_\square$, for $C_\square$, which needs an identity of its own because the data spaces of the two runs differ by run B's extra layer of data, and for $G_\square$, whose gauge-fixing part is nonlocal through $P = G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'$, is Supplement, Proposition S.12(b)–(d). *q.e.d.*

The following lemma shows why the shells of the finest spacing, on which the defect is of order one, cost only a factor $(\varepsilon/a_j)^\beta$: a propagator of the scale $j$ reaches them only after crossing $j$ shells, each of width $2R_0M_0$ in its own units.

**Lemma 9.4** (Reduction over the Shells). *Let $\mathcal{L}$ satisfy (9.8), have the range of one block of run A, and satisfy $\sum_{z\in\Delta(y)} \eta_B^4(|\nabla\mathcal{L}v(z)| + (L^n\eta)^{-1}|\mathcal{L}v(z)|) \le C_{\mathcal{L}} \sum_{y'''} \sum_{w\in\Delta(y''')} \eta_A^4(|\nabla v(w)| + (L^n\eta)^{-1}|v(w)|)$, the sum over the $3^4$ blocks $y'''$ next to $y$. Assume $\frac{1}{6}\delta_0 R_0 M_0 > 8\log c_0(\frac{1}{6}) + 1$ and $4\delta_0 R_0 M_0 \ge 3(1+\beta)\log L$, with $c_0$ the constant of [3, (2.58)]. Then for blocks $y$, $y'$ of $\square$ in the layers $j$ and $j+1$ of the local sequence, $x \in \Delta(y)$, and a source $g$ of run B with* $\operatorname{supp} g \subset \Delta(y')$,

$$|(\chi_A Q_1 G^B_\square - G^A_\square \chi_A \mathcal{L}^*)g(x)| \le C_\square(\beta)\Big(\frac{\varepsilon}{a_j}\Big)^\beta (L^j\eta)^2 e^{-\frac{1}{3}\delta_0 d(y,y')}|g| \,, \tag{9.17}$$

*with* $C_\Box(\beta) \coloneqq 6c_0^4(\frac{1}{6})c_\beta C_3$, $C_3 \coloneqq 9\cdot 3^4 e^{4\delta_0}(1+L)C_{\mathcal{L}}C_{\mathrm{df}}(\beta)B_0B_0(\beta)$, *and* $c_\beta \coloneqq L^{1+\beta}(3+L^\beta + L^{1+\beta}/(L-1))$, *where* $B_0$ *and* $B_0(\beta)$ *are the constants of [5, (3.42), (3.43)].*

*Proof.* By (9.10) and Proposition 9.3 with $\mathcal{A} = u_B \coloneqq G^B_\Box g$, the left side is $|(G^A\chi_A\mathcal{L}^*(Ds_2 + \nabla^* s_1 + s_0))(x)|$, and we split the sources over the blocks $y''$ of the shells. We bound the contribution of a block $y''$ of the shell $n$ in two legs.

*The leg from* $y'$ *to* $y''$. The bounds [5, (3.42), (3.43)] for $G^B_\Box$, which hold for the local sequences by [5, Corollary 3.6, p. 408], give $\|u_B\|_{1,\beta;y''} \le 3e^{2\delta_0}B_0(\beta)(L^n\eta)e^{-\delta_0 d(y'',y')}|g|$, where $e^{2\delta_0}$ accounts for the blocks within distance 2 and $B_0 \le B_0(\beta)$. By (9.16),

$$\sup_{\Delta(y'')}(|s_1| + |s_2| + L^n\eta|s_0|) \le 3e^{2\delta_0}C_{\mathrm{df}}(\beta)\varpi_n B_0(\beta)(L^n\eta)e^{-\delta_0 d(y'',y')}|g| \,. \tag{9.18}$$

*The leg from* $y''$ *to* $y$. For a source $\sigma$ on $\Delta(y'')$, the symmetry of $G^A$ gives $(G^A\chi_A\mathcal{L}^*\nabla^*\sigma)(x) = \langle \nabla\mathcal{L}[G^A(\cdot,x)],\sigma\rangle_B$, which is at most $\sup|\sigma|$ times the block sum of $\nabla\mathcal{L}G^A(\cdot,x)$. By the hypothesis on $\mathcal{L}$ and [5, (3.42)] for $G^A$, this is at most

$$\sup|\sigma|\cdot 3^4 e^{2\delta_0}C_{\mathcal{L}}B_0[(L^j\eta) + (L^j\eta)^2(L^n\eta)^{-1}]e^{-\delta_0 d(y,y'')} \,, \tag{9.19}$$

and the same holds with $D$ in place of $\nabla^*$, while for $s_0$ the factor is $3^4e^{2\delta_0}C_{\mathcal{L}}B_0(L^j\eta)^2e^{-\delta_0 d(y,y'')}$, that is $(L^j\eta)^2(L^n\eta)^{-1}$ times $L^n\eta|s_0|$. Since $n \le j+1$, $L^j\eta \le L(L^j\eta)^2(L^n\eta)^{-1}$.

*The product and the sum.* The factor $(L^n\eta)^{-1}$ of the second leg cancels the factor $L^n\eta$ of the first, so that the shell $n$ contributes at most $C_3\varpi_n(L^j\eta)^2\sum_{y''}e^{-\delta_0[d(y,y'')+d(y'',y')]}|g|$. One third of the decay gives $e^{-\frac{1}{3}\delta_0 d(y,y')}$ by the triangle inequality [3, (2.54)]; one third gives the count $\sum_{y''}e^{-\frac{1}{3}\delta_0 d(y,y'')} \le 6c_0^4(\frac{1}{6})$ over one shell, by [3, (2.58)] with $\alpha = \frac{1}{3}$ under the first condition; and the last third is at most $e^{-\frac{4}{3}\delta_0 R_0 M_0(j-n-1)_+} \le L^{-(1+\beta)(j-n-1)_+}$, because the weighted distance from a block of $\Box$ to the shell $n$ is at least $2R_0M_0(j-n-1)$ [3, (2.60)], both legs cross the shells in between, and the second condition holds. Summing over the shells with (9.16): the shells $j$ and $j+1$ give at most $2L^{-\beta j}$; the shells $1 \le n \le j-1$ give $\sum_n L^{-\beta n}L^{-(1+\beta)(j-n-1)} \le L^{1+\beta}L^{-\beta j}/(L-1)$; the shell 0 gives $L^{-(1+\beta)(j-1)} \le L^\beta L^{-\beta j}$; and run B's extra shell, at distance at least $2R_0M_0 j$, gives at most $L^{-\beta j}$. Hence $\sum_n \varpi_n L^{-(1+\beta)(j-n-1)_+} \le c_\beta L^{-\beta j}$, with room for the intersected sequences of the cubes next to a coarser layer [5, p. 408], and $L^{-j} = \varepsilon/a_j$ by (9.1). *q.e.d.*

### 9.4 The Walk Terms

We can now state the comparison of the walk terms. Let $W^X_\omega$ be the term of the walk $\omega$ in run $X$'s expansion of one of the operators $H$, $H_0$, or $\tilde{G}$ of [10, (1.2)–(1.4), p. 2], at run $X$'s background, on a set of the **R**-operation whose layers lie at shared scales; for these operators the walks of run A and of run B are the same, because the cubes are. Let $\iota_Y$ be the localised one-step lift of a localisation domain $Y$ of the walk, and let $\mathcal{L}_\iota$ be the part of its derivative in the data at the aligned background which lies in the Landau gauge, a right inverse of $Q_1$ on the data of the shell of the finest spacing (Supplement, Proposition S.12(e)). The operator $\mathcal{J}'$ is the identity on data, for $H$ and $H_0$, and $\mathcal{L}^*_\iota$ on sources, for $\tilde{G}$; $m(\omega)$ is the finest scale of a layer of the set which meets $\bigcup_i \tilde{X}^5_i$.

**Theorem 9.5** (Walk-Factor Difference)**.** *Let the constants satisfy the conditions of Appendix C and* $\bar{g}$ *be sufficiently small. For every walk* $\omega$ *of the expansions of* $H$, $H_0$, *and* $\tilde{G}$ *at the backgrounds of the two runs on a set whose layers lie at the shared scales* $j \ge K'/2 - 1$,

$$|(W^B_\omega - \mathcal{L}_\iota W^A_\omega \mathcal{J}')\lambda(x)| \le \delta_w(\omega)\cdot O(1)(L^j\eta)^p O(M^{-1/2})^{|\omega|}M^{-\frac{1}{2}|\omega|}e^{-\frac{1}{6}\delta_0 d(\omega,y,y')}|\lambda| \,, \tag{9.20}$$

*in run B's norms, where $(L^j\eta)^p$ is the power of* (9.6) *for the norm in question, and*

$$\delta_w(\omega) \le C_w \Big(\frac{\varepsilon}{a_{m(\omega)}}\Big)^{\gamma_w}, \qquad \gamma_w := \min\Big(\beta, \frac{1-\varepsilon_0}{3}\Big), \tag{9.21}$$

*with a constant $C_w$ of the constants of the construction and of $\varepsilon_0$, $\varepsilon_1$, and $\beta$, which does not depend on $K$, $k$, $\varepsilon$, or the torus. If run A's term is evaluated at the aligned background $\tilde U$ in place of its own, the same holds with the exponent $\beta$ in place of $\gamma_w$. Applying $Q_1$ to* (9.20) *gives the same bound for $Q_1 W_\omega^B - W_\omega^A \mathcal{L}_\iota^*$, with $\delta_w(\omega)$ multiplied by the norm of $Q_1$.*

*Proof.* The proof has three parts: the factors at the aligned background, the change from the aligned background to run A's own, and the nonlinear functions through which $H$, $H_0$, and $\tilde G$ are built from the propagators.

*The factors.* A term of a walk is a product of factors, each of which is a local operator $G'_\square$, $C_\square$, $G_\square$, multiplied by partition functions, commutators with them, and averaging operators [5, pp. 409–416]. For two products $a_1 \cdots a_n$ and $b_1 \cdots b_n$,

$$a_1 \cdots a_n - b_1 \cdots b_n = \sum_{i=1}^{n} a_1 \cdots a_{i-1}(a_i - b_i) b_{i+1} \cdots b_n \,, \tag{9.22}$$

and we apply this to run B's factors and to run A's factors conjugated by $Q_1$ and $\mathcal{L}_\iota$, inserting $\mathcal{L}_\iota Q_1$ between consecutive factors of run A where it acts as the identity. Each difference of local propagators is bounded by Lemma 9.4 in its coarse form and by (9.11) in its fine form, which costs the factor $C_\square(\beta)(\varepsilon/a_j)^\beta$ and lowers the rate $\frac12\delta_0$ to $\frac16\delta_0$; the explicit factors, multiplications by the partition functions and commutators with them, are functions on the torus which both runs take to be the same, and their defect between the two lattices is of relative order $\varepsilon/a_j$ by the same oscillation (Supplement, Proposition S.12(f)). The sum (9.22) has $|\omega|+1$ terms, which the factor $O(M^{-1/2})^{|\omega|}$ absorbs after a change of the constant. Since a cube of the walk lies at a scale $j \ge m(\omega)$, $(\varepsilon/a_j)^\beta \le (\varepsilon/a_{m(\omega)})^\beta$, and the shells at the scales $n < j$ inside $\tilde\square^5$, which are not layers of the set, enter only through the constant $c_\beta$ of Lemma 9.4. This gives the bound at the aligned background, with the exponent $\beta$.

*The background.* Run A's own background is $U^A = e^{i\varepsilon A}\tilde U$ with $a_{j(x)}|A(x)| + a_{j(x)}^2|\nabla A(x)| \le \bar d(x) := C_U(\varepsilon/a_{j(x)})^{(1-\varepsilon_0)/3}\varepsilon_1$, where $j(x)$ is the layer of $x$ and $C_U$ a constant of $M$ (Proposition 10.6). Put $\bar d(\omega) := \sup \bar d(x)$ over $\bigcup_i \tilde X_i^5$ and $U(t) := e^{it\varepsilon A}\tilde U$. For $|t| \le r := \alpha'_*/(2\bar d(\omega))$, with $\alpha'_* := \min(\alpha_1, \alpha_J/\hat x_J)$ the radius of the class of pairs of a configuration and a current of Supplement, Proposition S.12(g), the configuration $U(t)$ lies in the domain of [5, (3.37)] layer by layer, where the local propagators are analytic with the bounds of [5, Theorems 3.1–3.3] [5, Theorem 3.4, p. 400], and so is the term of a walk, since the proof of (9.6) uses only these bounds [5, p. 410]. Under $4\bar d(\omega) \le \alpha'_*$ we have $r \ge 2$, and Cauchy's estimate gives

$$|W_\omega^A(U^A) - W_\omega^A(\tilde U)| \le \sup_{|t|\le 1} |\partial_t W_\omega^A(U(t))| \le \frac{\sup_{|t|=r}|W_\omega^A(U(t))|}{r-1} \le \frac{4\bar d(\omega)}{\alpha'_*} \tag{9.23}$$

times the bound (9.6), in each of its norms. With $\bar d(\omega) \le C_U\varepsilon_1(\varepsilon/a_{m(\omega)})^{(1-\varepsilon_0)/3}$ this is of the form (9.21).

*The nonlinear functions.* The operators $H$, $H_0$, and $\tilde G$ are built from the propagators through the fixed point $\mathbf{A}_0$ of [10, (1.4), p. 2], the operator $D$ of [10, (1.3)], and the solution of a Neumann series; their walk expansions are composite walks with vertices at which three chains meet [5, p. 427]. The comparison of these functions between the two runs, at complex values of the decoupling parameters $s(\Delta)$ on the circles $|s(\Delta)| = e^{\kappa_1}$ on which they enter

Cauchy integrals [10, (1.23), p. 7], is Supplement, Proposition S.12(h); it costs the further constants collected in $C_w$ and no further power of $\varepsilon/a_j$. *q.e.d.*

The proof shows where the exponent comes from. The local operators are compared at the rate $(\varepsilon/a_j)^\beta$ for every $\beta < 1$, and the loss is in the change of the background, whose rate $(1-\varepsilon_0)/3$ is that of the distance between the minimisers of the two runs in the norm of the analyticity domains. This distance is the subject of the next section.

## 10 The Large-Field Terms of Two Cutoffs

We now turn to the terms of the two densities which carry large fields. Here the obvious approach, a comparison of the two runs term by term, meets two obstacles. The first is that most of these terms cannot be compared at all with an error that is small relative to the term: a large-field region which was created at a fine scale has been integrated with the test functions, cutoffs, and Gaussian weights of the fine lattice of its run, and the two fine lattices are different. The second is that the test functions are evaluated on fields of the two runs which differ, so that a term of one run may be supported where the corresponding term of the other vanishes. The resolution of the first obstacle is that we do not need to compare the terms with an old large-field region at all: such a term carries the gain of every scale through which the region has lived, and the weight of Proposition 8.4 shows that the terms of this kind have a total weight which tends to zero with $K$, in each run separately. The terms that remain have their large fields at scales coarser than $K'/2$, where the fine lattice is invisible up to a power of $\varepsilon/a_{K'/2} = L^{-K'/2}$. The second obstacle is removed by the soft test functions, whose change is proportional to that of their arguments.

### 10.1 The Matched Decomposition

Both densities at the stopping step have the form (3.23), a sum over the admissible sequences of domains, with one pair $(\Omega_j, \Lambda_j)$ for each step, and with the components which the **R**-operation has renormalised removed from the sequences [11, (0.3), p. 176]. The sequences of run B have one entry more, that of the extra step.

**Lemma 10.1** (Matched Decomposition)**.** *Every labelled term of run B at the step $K'+1$ is of exactly one of two kinds:*

*(M) its pair of domains at the extra step is $(\mathbb{T}, \mathbb{T})$; then the sequence obtained by deleting this pair is a labelled term of run A at the step $K'$, and this is a bijection $\tau^+ \mapsto \tau$ between the terms of kind (M) and the labelled terms of run A;*

*(U) it has a component of its large-field region at the step $K'+1$ whose history contains a large-field region created at the extra step.*

*Proof.* A component which is renormalised at some step leaves the sequences, so a term whose pair at the extra step is not $(\mathbb{T}, \mathbb{T})$ has a large-field region created at the extra step which belongs to a component alive at the step $K'+1$, which is the kind (U). For the kind (M), the admissibility conditions on the sequences [9, pp. 255–256] and the conditions of the **R**-operation [11, p. 177, (i), (ii)] are conditions on the domains, stated with the cube sizes, the distances $MR_j$, and the depth $N_j$ of the scale; by Lemma 9.1 these are the same at a shared scale, so that the conditions of run B at the steps $2, \dots, K'+1$ are those of run A at the steps $1, \dots, K'$. *q.e.d.*

For a matched pair $(\tau^+, \tau)$ the two terms are the same operations applied to ingredients which depend on the run. The pair is the *all-small-field pair* if $\tau = \ell_0$. Otherwise let $j_0$ be the least scale at which one of its terms has a nonempty large-field region; since every nonempty large-field region of a term belongs, after the **R**-operations, to the history of a component which is alive at the stopping step, $j_0$ is the scale at which the oldest of these components was born, and $\Lambda_i = \Omega_i = \mathbb{T}$ for $i < j_0$ in both terms. With

$$m \coloneqq \lfloor K'/2 \rfloor \,, \tag{10.1}$$

the pair is *old* if $K' - j_0 \geq m$ and *young* otherwise. A young pair has all its large fields at the shared scales $i \geq j_0 - 1 \geq K' - m \geq K'/2$; this is the case in which a comparison is possible.

The terms which are not compared are disposed of by the following lemma, which uses only the weight of Section 8.3. We write

$$\mathfrak{b}(\mathcal{C}) \coloneqq e^{B_E(\mathcal{C}) + I_*(\mathcal{C})} \,, \tag{10.2}$$

so that $\mathfrak{b}(\mathcal{C}) \leq B(\mathcal{C})$, and we use the notation of Theorem 8.1 for either run, with $Z$ the integral of the density and $N \coloneqq \sum_\ell Z_\ell \langle \widehat{\mathcal{W}}_\ell \rangle_\ell$, so that $W = N/Z$.

**Lemma 10.2** (Terms with an Old Component)**.** *Let $\bar{g}$ be as in Theorem 8.6, let $p \geq 0$, and let $\mathcal{S}$ be a set of labelled terms of one run, none of them $\ell_0$, each of which has a component $X$ of its large-field region at the stopping step with $P(X) \geq p$. Then*

$$\sum_{\ell \in \mathcal{S}} Z_\ell \leq e^{-C_P p} \delta_{\mathrm{LF}} Z_{\ell_0} \,, \qquad \sum_{\ell \in \mathcal{S}} Z_\ell |\langle \widehat{\mathcal{W}}_\ell \rangle_\ell| \leq \mathfrak{b}(\mathcal{C}) e^{-(C_P - 1)p} \delta_{\mathrm{LF}} Z_{\ell_0} \,, \tag{10.3}$$

*and removing the terms of $\mathcal{S}$ from $N$ and $Z$ changes $W = N/Z$ by at most $2B(\mathcal{C}) e^{-(C_P-1)p} \bar{\delta}_{\mathrm{LF}}$.*

*Proof.* For each $\ell \in \mathcal{S}$ choose a component $X_\ell$ with $P(X_\ell) \geq p$. For the first bound let $\omega(X_\ell) \coloneqq e^{C_P P(X_\ell)}$ and $\omega(X) \coloneqq 1$ for the other components; these numbers satisfy the hypothesis of Proposition 8.4, so that

$$\sum_{\ell \in \mathcal{S}} Z_\ell \leq e^{-C_P p} \sum_{\ell \in \mathcal{S}} Z_\ell \, \omega(X_\ell) \leq e^{-C_P p} \sum_{\ell \neq \ell_0} Z_\ell \prod_{X \in Z(\ell)} \omega(X) \leq e^{-C_P p} \delta_{\mathrm{LF}} Z_{\ell_0} \,. \tag{10.4}$$

For the second we use Lemma 8.5, $|\langle \widehat{\mathcal{W}}_\ell \rangle_\ell| \leq \mathfrak{b}(\mathcal{C}) \prod_X \omega_A(X)$, and the numbers $\omega(X_\ell) \coloneqq \omega_A(X_\ell) e^{(C_P - 1)P(X_\ell)}$ and $\omega(X) \coloneqq \omega_A(X)$ otherwise. Since $1 \leq \omega_A(X) \leq e^{P(X)}$ under (8.40), as the proof of Lemma 8.5 shows, they satisfy $1 \leq \omega(X) \leq e^{C_P P(X)}$, and the same computation gives the second bound. Finally, if $N_{\mathcal{S}}$ and $Z_{\mathcal{S}}$ are the contributions of $\mathcal{S}$, then $Z - Z_{\mathcal{S}} \geq Z_{\ell_0}$ and

$$\Big| \frac{N - N_{\mathcal{S}}}{Z - Z_{\mathcal{S}}} - \frac{N}{Z} \Big| = \Big| \frac{N_{\mathcal{S}} - W Z_{\mathcal{S}}}{Z - Z_{\mathcal{S}}} \Big| \leq \frac{|N_{\mathcal{S}}| + |W| Z_{\mathcal{S}}}{Z_{\ell_0}} \leq (\mathfrak{b}(\mathcal{C}) + B(\mathcal{C})) e^{-(C_P - 1)p} \delta_{\mathrm{LF}} \,, \tag{10.5}$$

by (10.3) and Theorem 8.6, and $\delta_{\mathrm{LF}} \leq \bar{\delta}_{\mathrm{LF}}$. *q.e.d.*

**Corollary 10.3** (The Unmatched Terms and the Old Pairs)**.** *Under the hypotheses of Lemma 10.2, removing the terms of kind (U) from run B changes $W_{K+1}(\mathcal{C})$ by at most $2B(\mathcal{C}) e^{-(C_P-1)(K'+1)} \bar{\delta}_{\mathrm{LF}}$, and removing the old pairs from both runs changes $W_{K+1}(\mathcal{C}) - W_K(\mathcal{C})$ by at most $4B(\mathcal{C}) e^{-(C_P-1)(m+1)} \bar{\delta}_{\mathrm{LF}}$.*

*Proof.* The large-field regions grow and merge until they are renormalised [11, (0.3), p. 176], so a component alive at the stopping step whose history begins at the scale $j$ meets the large-field region at every scale from $j$ to the stopping step, and its history volume (8.21), in which every scale with a nonempty intersection contributes at least $L^4$, is at least the number of these scales. For a term of kind (U) this number is $K' + 1$, and for each term of an old pair it is at least $m + 1$. Lemma 10.2 gives the assertions. *q.e.d.*

### 10.2 The Shells of the Test Functions

On a young pair the two terms are compared ingredient by ingredient, and the first ingredients are the test functions. Let $\Pi_X$ be the product of the test functions and characteristic functions of run $X$'s term and $\varrho_X$ the rest of its integrand. Since all factors of $\Pi_X$ take values in $[0,1]$,

$$\Pi_A\varrho_A - \Pi_B\varrho_B = \min(\Pi_A,\Pi_B)(\varrho_A-\varrho_B) + (\Pi_A-\Pi_B)_+\varrho_A - (\Pi_B-\Pi_A)_+\varrho_B \;, \tag{10.6}$$

which puts each of the last two pieces on the support of one run's own term, where that run's bounds apply.

**Lemma 10.4** (Shells of the Soft Test Functions)**.** *Let a soft test function $\bar\chi(f_\square/t)$ of a cube $\square$ have the same threshold $t$ in both runs, and let $|f^A_\square - f^B_\square| \le \delta_\square t$ wherever $\min(f^A_\square, f^B_\square) < (1+2\kappa_s)t$. Then $|\bar\chi(f^A_\square/t) - \bar\chi(f^B_\square/t)| \le \delta_\square/\kappa_s$, and the same holds for $1-\bar\chi$. Consequently, if $\Pi_A$ and $\Pi_B$ consist of such soft test functions and of characteristic functions which are the same in both runs,*

$$(\Pi_A-\Pi_B)_+\varrho_A \le \Big(\sum_\square \frac{\delta_\square}{\kappa_s}\Big)\,\Pi^\sharp_A\varrho_A \;, \tag{10.7}$$

*where $\Pi^\sharp_A$ is the product of the sharp test functions of run $A$ at the thresholds $(1+\kappa_s)t$ for the passed and $t$ for the failed test functions; for a test function of the cascade of [11, (1.22)–(1.24), p. 181] the factor $\delta_\square/\kappa_s$ is replaced by $(N+1)\delta_\square/\kappa_s$ and the thresholds by those of Supplement, Proposition S.2.*

*Proof.* Where $\min(f^A_\square, f^B_\square) \ge (1+2\kappa_s)t$ both values of $\bar\chi$ vanish. Elsewhere $|f^A_\square/t - f^B_\square/t| \le \delta_\square$, and $\bar\chi$ is $\kappa_s^{-1}$-Lipschitz by Lemma 3.8(iii). Replacing the factors of $\Pi_A$ by those of $\Pi_B$ one at a time, with all factors in $[0,1]$, gives $(\Pi_A-\Pi_B)_+ \le \mathbf{1}\{\Pi_A>0\}\sum_\square \delta_\square/\kappa_s$, and on $\{\Pi_A>0\}$ every passed test function of run A has $f^A_\square < (1+\kappa_s)t$ and every failed one $f^A_\square > t$, by Lemma 3.8(ii). The cascade test functions are soft with the width $\kappa_s/(N+1)$ (Supplement, Proposition S.2), which gives the factor $N+1$. *q.e.d.*

The right side of (10.7) is a term of run A with sharp test functions at enlarged thresholds, and Bałaban's bounds hold for it by Proposition 3.12: a shell costs, relative to Bałaban's bound of the term, the factor $\delta_\square/\kappa_s$. The characteristic functions which restrict the fluctuation fields are sharp, and their thresholds differ between the runs through the factor $g_j$ of Definition 3.11(ii). For them we use the density of the Gaussian weight.

**Lemma 10.5** (Shells of the Coupling)**.** *Let $|\delta_j| \le \frac12 (g^A_j)^{-2}$, and let the fluctuation field $A_j(b)$ of a bond be restricted in the two runs by $|A_j(b)| < t$ and $|A_j(b)| < t'$ with $t = \vartheta_j/g_j$ and $t' = \vartheta_j/g'_j$, $g_j$ and $g'_j$ the two couplings at the shared scale $j$. Then $|t'-t| \le 2tg_j^2|\delta_j|$, and the mass of the shell between the two radii under the normalised Gaussian weight of the step, with the other variables fixed, is at most*

$$C_{\mathrm{sh}}\, p_0(\hat g_j)^{\dim G} g_j^2 |\delta_j| \;, \tag{10.8}$$

*with a constant $C_{\mathrm{sh}}$ of $L$, $G$, and $A_1/A_0$. The same holds for the field $B_j$ of [11, (1.55), p. 188] with $\delta'_j$ in place of $\vartheta_j$.*

*Proof.* With $x := g_j^{-2}$ and $x' := g_j'^{-2}$ we have $t'/t = g_j/g'_j = (x'/x)^{1/2}$ and $|x'-x| = |\delta_j| \le \frac12 x$, so that $x'/x \in [\frac12, \frac32]$, and $|u^{1/2}-1| \le |u-1|$ for $u>0$ gives $|t'/t - 1| \le |\delta_j|/x = g_j^2|\delta_j|$; exchanging the runs gives the bound with either coupling and the factor 2. The Gaussian weight of the fluctuation field has the precision $Q = C^*\Delta^{(j)}C - C^*\Delta^{(j)}CC^{(j)}C^*\Delta^{(j)}C \le$

$C^*\Delta^{(j)}C$ of [9, (2.21), p. 258], because $C^{(j)} \geq 0$. The conditional variance of $A_j(b)$, the other variables fixed, is $(Q_{bb})^{-1} \geq \|C^*\Delta^{(j)}C\|^{-1}$, so that the conditional density of $A_j(b) \in \mathfrak{g}$ is at most $(\|C^*\Delta^{(j)}C\|/2\pi)^{\dim G/2}$, a constant of the lattice. The shell $\{\min(t,t') \leq |A_j(b)| < \max(t,t')\}$ has Lebesgue measure at most $C|t'-t|\max(t,t')^{\dim G-1}$, and $t \leq (A_1/A_0)p_0(\hat{g}_j)\cdot \hat{g}_j/g_j \leq (A_1/A_0)p_0(\hat{g}_j)$ by Lemma 3.2(i), which gives (10.8). The same argument applies to $B_j$ under the Gaussian form of [11, p. 188], and to the complement of a characteristic function, whose shell is the same set. *q.e.d.*

The positivity of the quadratic form is the same fact by which Bałaban bounds the integrals over the fluctuation fields in the large-field operations [12, p. 383], and so a shell term is at most (10.8) times his bound of the term. Two features of these shell bounds matter for what follows. The plaquette shells cost $\delta_\square/\kappa_s$, which will be a power of $\varepsilon/a_j$; the coupling shells cost a multiple of $g_j^2|\delta_j|$, which is controlled in Section 11 together with the other differences of the couplings.

### 10.3 The Configurations of the Two Runs

The functions $f_\square$ of the plaquette and reference test functions, and all integrands of the large-field operations, are computed from minimisers of the action on the fine lattice of the run. At equal data the minimisers of the two runs differ by a lattice artefact, which is the content of the following proposition. For a field $W$ on the fine lattice of run B we write $MW$ for its average on the fine lattice of run A, and for a point $x$ of a determining set $j(x)$ is the layer of $x$.

**Proposition 10.6** (Minimiser Distance)**.** *Let $0 < \varepsilon_0 < 1/125$, let the constants satisfy the conditions of Appendix C, and let $\bar{g}$ be sufficiently small. There are constants $C_G(\varepsilon_0)$ and $C_U$, independent of $K$, $k$, $\varepsilon$, and the torus, with the following properties.*

*(i) Let $U^A$ and $U^B$ be the minimisers of the two runs for a determining set at the shared scales with the same data $V$, which satisfy the hypothesis (7) of [7, Theorem 1, p. 279] with the parameter $\varepsilon_1$. Then, in a gauge, $U^A = e^{i\varepsilon A'}MU^B$ with*

$$a_{j(x)}|A'(x)| + a_{j(x)}^2|\nabla A'(x)| \leq 2\bar{\rho}(x)\,, \qquad \bar{\rho}(x) \coloneqq C_G(\varepsilon_0)\Big(\frac{\varepsilon}{a_{j(x)}}\Big)^{1-\varepsilon_0}\varepsilon_1\,. \tag{10.9}$$

*(ii) For each of the configurations $U_k''$, $U_k^0$, $U_{k,\Lambda}$, $U_k$ of the* **R***-operation of [12, §1] on a young history born at the scale $n$, at the data of the two runs, $c^A = e^{i\varepsilon A_c}Mc^B$ with*

$$a_{j(x)}|A_c(x)| + a_{j(x)}^2|\nabla A_c(x)| \leq \bar{d}(x) \coloneqq C_U\Big(\frac{\varepsilon}{a_{\min(j(x),n)}}\Big)^{(1-\varepsilon_0)/3}\varepsilon_1\,. \tag{10.10}$$

*Here $\varepsilon_1 \leq \bar{\varepsilon}_1$, the number of* (10.11) *below, which depends on $\bar{g}$ and the constants of the construction and not on $K$.*

The proof is Supplement, Proposition S.13. In (i), which is the case of a single run of [7, Theorem 1] on two lattices, $MU^B$ satisfies run A's constraint, and the difference of the two actions at $MU^B$ is the blocking artefact $b(W) \coloneqq A_\varepsilon(MW) - A_{\varepsilon/L}(W)$, whose first variation is a tangential current of order $\varepsilon/a_j$; the gap of the action at the minimum and the bounds of [5] for the Green operator of the constrained problem turn it into (10.9), with the loss $\varepsilon_0$ in the Hölder norm. In (ii) the data themselves differ, on the region $\Lambda$ where the **R**-operation integrates, through the minimiser $U_0$ of [12, (1.3), p. 357]; the minimiser of the

nested problem of [11, Proposition 1, p. 194] is compared by the growth of its action away from the minimum. If $\Delta$ is the distance between the minimising data of the two runs, the quadratic growth $\frac{1}{2}\kappa_\Lambda\Delta^2$ is at most the change of the difference of the two runs' actions between the two minimisers, a sum over blocks of the smaller of a bound of order $(\varepsilon/a_n)^{1-\varepsilon_0}$ and a bound linear in $\Delta$; with $\min(u,v) \le (uv)^{1/2}$ this gives $\Delta^{3/2} \le C(\varepsilon/a_n)^{(1-\varepsilon_0)/2}$, whence the exponent $(1-\varepsilon_0)/3$.

The data parameter of the regularity of the history is

$$\begin{aligned} \varepsilon_1 &\coloneqq \max\Big(C\max_{j\le K'}(\varepsilon_j+\vartheta_j),\, C_M R_{K'}\varepsilon_{K'}\Big) \\ &\le \bar\varepsilon_1 \coloneqq \max(C(1+A_1/A_0)\bar g p_0(\bar g),\, C_M L A_0 \bar g(\log \bar g^{-2})^{p_0+r}) \,, \end{aligned} \tag{10.11}$$

where $C$ is the constant with which [11, p. 178] bounds the data of a large-field region by its thresholds and $C_M$ a constant of $M$ [11, (1.83), p. 197]; the bound holds because $\varepsilon_j+\vartheta_j = (1+A_1/A_0)\hat g_j p_0(\hat g_j)$ with $\hat g_j \le \bar g$, because $R_{K'}\varepsilon_{K'} < LA_0\hat g_{K'}(\log \hat g_{K'}^{-2})^{p_0+r}$ by Lemma 3.4(i), and because $g \mapsto g p_0(g)$ and $g \mapsto g(\log g^{-2})^{p_0+r}$ increase for small $g$. The point of (i) is the exponent: the tested quantities of a cube at the scale $j$ differ between the runs by a power of $\varepsilon/a_j$, which is small precisely because the young histories live at the scales $j \ge K'/2-1$. In terms of the thresholds:

**Corollary 10.7** (The Tested Quantities)**.** *Under the hypotheses of Proposition 10.6, let $f_\square^X$ be the quantity which a plaquette or reference test function of a cube $\square$ at the scale $j$ of a young history evaluates in run $X$, at the threshold $t$. Wherever $\min(f_\square^A, f_\square^B) < (1+2\kappa_s)t$, at equal data,*

$$|f_\square^A - f_\square^B| \le \delta_j^s t \,, \qquad \delta_j^s \coloneqq \delta_j^f + 7g_j^2|\delta_j| \,, \qquad \delta_j^f \coloneqq C_f\Big(\frac{\varepsilon}{a_j}\Big)^{1-2\varepsilon_0} \,, \tag{10.12}$$

*with a constant $C_f$ which is independent of $K$, $k$, $\varepsilon$, and the torus, and at most polynomial in $j_q$.*

*Proof.* For the test functions of the local minimisers of the cubes [11, (1.3), (1.4), (1.7), (1.8), p. 178], [9, (2.17), p. 257], the plaquettes of $U^A$ and of $MU^B$ differ by the curl of $\varepsilon A'$ with transports, at most $a_j^2|\nabla A'| + C(a_j|A'|)^2 + C(a_j|A'|)B_3\varepsilon_1 \le 3\bar\rho$ in the units of the scale for $\bar g$ small, by (10.9) and [7, (8)]; the plaquettes of $MU^B$ and of $U^B$ differ by the blocking artefact, of the same order. Relative to the threshold, which is at least $(1-\beta)\varepsilon_j \ge \delta_j'$, this is at most a constant times $(\varepsilon/a_j)^{1-\varepsilon_0}\varepsilon_1/\delta_j'$. On a young history the scales satisfy $K'-j+1 \le 8(1+j_q)(j+1)$, and $\varepsilon_1/\delta_j'$ is at most a polynomial in $j$ and $\log \hat g_j^{-2}$ with coefficients polynomial in $j_q$, by (10.11), Lemma 3.2, and $p_1 < p_0$; this polynomial is absorbed by $(a_j/\varepsilon)^{\varepsilon_0} = L^{\varepsilon_0 j}$, which gives $\delta_j^f$. The test functions of the multiscale minimisers, those of the cascade [11, (1.22)–(1.24), p. 181], of the reference configurations [11, (1.27), p. 182], and of [11, (1.75), p. 193], are bounded in the same way, with the layer of the tested plaquette in place of $j$ (Supplement, Proposition S.13(c)). The coupling part $7g_j^2|\delta_j|$ enters only where a tested quantity contains the scaled field $g_j B_j$, as in [11, (1.27), (1.59)], by the computation of Lemma 10.5. *q.e.d.*

### 10.4 The Large-Field Comparison

We can now state the comparison of the large-field terms of a young pair. Let $\tilde\theta \in (0,1)$ be the rate of (11.16) in Section 11; in this section we use only $\tilde\theta \ge L^{-(1-\varepsilon_0)/4}$ and $\tilde\theta \ge L^{-\theta_V/8}$, with $\theta_V$ of (11.8). The ingredients of a term are the factors of the integrands of its one-step

operations and the terms of its effective action. For a young pair we distinguish three kinds of them at the scales $i \geq j_0 - 1$:

($\alpha$) the ingredients of the small-field integration: the frozen terms of the effective action, the Gaussian weights and their normalisations, the function [11, (1.6), p. 178] of the block variables, and the localised constants;

($\beta$) the test functions of the plaquettes and of the reference configurations, and the characteristic functions of the fluctuation fields;

($\gamma$) the large-field integrands: the action and the quadratic form of [12, (1.70), p. 377], the minimiser $U_0$ and the operators $H_{1,k}$ and $\Delta_1(\zeta_0)$ of the denominators of the renormalisation quotients [12, (1.1)–(1.2), p. 356], and the factors $V(Y)$ of [12, (1.91)] and the boundary terms of the steps at which a large-field region existed, as functions of the configurations and the propagators of the run, the terms of the effective action from which they are built being held fixed.

The dependence of the factors $V(Y)$ and of the boundary terms on the terms of the effective action from which they are built is of the kind ($\alpha$). The ingredients ($\alpha$) are compared in Section 11, by the recursion which also compares the all-small-field pair. The ingredients ($\beta$) and ($\gamma$) are compared by the following theorem, which says that their run dependence costs, per point of the history of a component, an amount which decays geometrically in the scale at which the component was born. For a component $X$ of a young pair let $n(X)$ be its birth scale and $|Z_i \cap X|_i$ the number of points of the scale $i$ of its history at the scale $i$, as in (8.21). Write $M_\tau^A$ for the majorant (8.26) of run A's term and $M_{\tau^+}^B$ for that of run B's.

**Theorem 10.8** (Large-Field Comparison)**.** *Let the constants satisfy the conditions of Appendix C, and let $\bar{g}$ be sufficiently small. There are a constant $C_M$ of $L$ and $G$, a constant $C_h$ which is independent of $K$, $\varepsilon$, and the torus and at most polynomial in $j_q$, and an index $k_{\mathrm{LF}}$ with the following property. For $K' \geq k_{\mathrm{LF}}$ and every young matched pair $(\tau^+, \tau)$, let $\varrho^{A,B}[\tau]$ be run A's term with its ingredients of the kinds ($\beta$) and ($\gamma$) replaced by run B's, except for the factors* (10.6) *of the shells. Then*

$$\int |\varrho^{A,B}[\tau] - \varrho^A[\tau]| \, \mathrm{d}V \leq \Big( \exp\Big( \sum_X \sum_i C_M s_{n(X)}^{\mathrm{LF}} L^4 |Z_i \cap X|_i \Big) - 1 + \sigma_\delta \Big) M_\tau^A \,, \qquad s_n^{\mathrm{LF}} := C_h \tilde{\theta}^n \,, \tag{10.13}$$

*the sum over the components $X$ of the pair and the scales $i$ of their histories, where $\sigma_\delta := \sum_X P(X) \sup_{i \geq K'-m} |\delta_i|$ collects the coupling parts of the shells; the shell pieces of* (10.6) *satisfy the same bound, with $M_{\tau^+}^B$ for the piece on run B's support. The same bounds hold for the numerators and the denominators of the renormalisation quotients of [12, (1.1), (1.2), p. 356] at the complex arguments of the inductive class, relative to Bałaban's majorants of their moduli, and linearly in the norms of the classes of the effective actions of the two runs.*

*Proof.* We treat the two kinds of ingredients in turn and then sum.

*The ingredients ($\beta$).* By the redundancy of Bałaban's test functions [9, (3.6)–(3.9), p. 265–266; p. 268–269], a test function of the scale $j$ which is not identically one on the support of the other characteristic functions of the term lies in the large-field region of the term or in its collar, so on a young pair it lies on a cube of a history at a scale $j \geq j_0 - 1$, or on a cube of the stopping scale. By Lemma 10.4 and Corollary 10.7 each of them contributes the factor $\delta_j^s / \kappa_s$, and $(N+1)\delta_j^s / \kappa_s$ for a cascade test function; by Lemma 10.5 each characteristic function of a fluctuation field contributes at most $C_{\mathrm{sh}} p_0(\hat{g}_j)^{\dim G} g_j^2 |\delta_j|$. A cube of the scale $j$ carries the test

functions of the step $j$ and of the **R**-operations of the steps $k' \geq j$ with $k' - N_{k'} \leq j$, at most 14 of each step and two of the cascade, and there are at most $s_T(j) \coloneqq 2 + N_j$ such steps, because $N$ is nonincreasing (Lemma 3.10). Since $N_j \leq \log_L C_{\mathrm{dep}} + \nu_L \log_L \log g_j^{-2}$ by (3.39), the first parts of all the test functions of a cube are at most a constant times $(j+1)s_T(j)L^{-(1-2\varepsilon_0)j}$, and with $L^{-(1-2\varepsilon_0)j} \leq \tilde{\theta}^j L^{-(3-7\varepsilon_0)j/4}$, which holds because $\tilde{\theta} \geq L^{-(1-\varepsilon_0)/4}$, and $j \geq n(X) - 1$, this is at most $s^{\mathrm{LF}}_{n(X)}$ once $C_h$ contains $\tilde{\theta}^{-1} \sup_j (j+1)s_T(j)L^{-(3-7\varepsilon_0)j/4}$, which is finite since $s_T(j)$ grows like $\log\log j$. The coupling parts of all the test functions and characteristic functions of a cube sum to at most $(7/\kappa_s)16 s_T(j)^2 g_j^2 |\delta_j| + 8 s_T(j) C_{\mathrm{sh}} p_0(\hat{g}_j)^{\dim G} g_j^2 |\delta_j| \leq |\delta_j|$ for $\bar{g}$ small, and the cubes of the histories are counted by $P(X)$.

*The ingredients ($\gamma$).* By Proposition 10.6, the configurations of the **R**-operation of the two runs differ, in the units of their layers, by $(\varepsilon/a_n)^{(1-\varepsilon_0)/3}$ times a constant, where $a_n \geq a_{n(X)-1}$ is the spacing of the finest layer of the history. Each large-field integrand is a function of these configurations whose two-run difference is bounded per point of the history by a constant times $g_j^{-2}(\varepsilon/a_n)^{(1-\varepsilon_0)/3}\varepsilon_1^2$, for the actions and quadratic forms, by a constant times $(\varepsilon/a_n)^{(1-\varepsilon_0)/3} N_k$, for the forms of the denominators, and, for the factors $V(Y)$ relative to their bound $|V(Y)| \leq \alpha e^{-(1+2\beta)\kappa d(Y)}$ of [12, (1.68), p. 377], by a sum over the generations $j$ of the terms which they re-sum of powers $\theta_V$ of the configuration distance relative to the radii of analyticity, with the factor $L^{-(\beta'/2)(m-j)}$ of the age, where Theorem 9.5 compares the propagators through which they are built; the boundary terms are compared through the recursion of their generation, under the third condition of (C.2). These bounds are Supplement, Proposition S.14. The factors $g_j^{-2}$, $\varepsilon_1^2/\varepsilon_j^2$, $N_k$, and the polynomial dependence on the scale are absorbed into the geometric factor, as in the first part, since $(1-\varepsilon_0)/3 > (1-\varepsilon_0)/4$, and for the factors $V(Y)$ since $\tilde{\theta} \geq L^{-\theta_V/8}$.

*The sum.* Every ingredient of the two kinds enters per point of the history of a component at a scale $i \geq n(X) - 1$, and its contribution relative to Bałaban's bound is at most $C_M s^{\mathrm{LF}}_{n(X)}$ per point plus the coupling parts, with $C_M$ the number of ingredients per point, a constant of $L$ and $G$. Bałaban's bound of a term is proved by estimating the integrals over the group variables, with their normalised weights, by one, and those over the fluctuation fields by the positivity of their quadratic forms [12, p. 383]. A product of factors which each change by a relative amount $x_\iota \geq 0$ changes by at most $\prod_\iota (1 + x_\iota) - 1 \leq \exp(\sum_\iota x_\iota) - 1$ relative to the product of the bounds, which gives (10.13). For the renormalisation quotients the same argument applies to the moduli of the numerators and denominators at complex arguments, where the shells are those of the real integration variables, and the dependence on the effective actions of the two runs enters linearly through the norms of their classes (Supplement, Proposition S.14(e)). *q.e.d.*

The theorem is the per-point form in which the large fields enter the recursion of Section 11: the factor $s^{\mathrm{LF}}_{n(X)}$ is attached to the birth scale of the component and not to the scale of the point, which is what makes its sum over all points of a history, counted by $P(X)$, controllable by the weight of Proposition 8.4. The stopping-scale test functions of the all-small-field pair are the one place where a shell does not lie on a history, and they are treated directly.

**Proposition 10.9** (The Stopping-Scale Shells)**.** *Under the hypotheses of Theorem 10.8, replacing the stopping-scale test function $\chi_{K'}(\mathbb{T})$ of run A's all-small-field term by run B's, in the numerator and in the normalisation, and the same for run B, changes $W_{K+1}(\mathcal{C}) - W_K(\mathcal{C})$ by at most $b_K^{(S)} B(\mathcal{C})$, with*

$$b_K^{(S)} \coloneqq 4\kappa_s^{-1} e^{2C_K} N_q^4 \delta^f \,, \qquad \delta^f \coloneqq C_f L^{-(1-2\varepsilon_0)K'} \,, \tag{10.14}$$

*which is summable in $K$.*

*Proof.* The term $\ell_0$ carries the test function (3.22) of the stopping step, a product over at most $N_q^4$ cubes, built on each run's fine lattice at the common threshold $\varepsilon_{K'}$, and no other test function which is not identically one (Supplement, Proposition S.2). By Corollary 10.7 at $j = K'$, where $\varepsilon/a_{K'} = L^{-K'}$ and $\delta_{K'}$ does not enter, and by Lemma 10.4, the shell of each run is at most $N_q^4 \delta^f/\kappa_s$ times that run's term with sharp test functions at the enlarged thresholds, whose integral is at most $e^{C_K}$ by the proof of Proposition 8.4. Relative to $Z_{\ell_0} \geq e^{-C_K}$ this is at most $N_q^4 \delta^f \kappa_s^{-1} e^{2C_K}$, and the change of the numerator is at most $\mathfrak{b}(\mathcal{C}) \leq B(\mathcal{C})$ times this by Lemma 8.5. Arguing as in (10.5) for the numerator and the normalisation of each run gives the factor $4B(\mathcal{C})$. The sum over $K$ converges because $C_K$ and $N_q$ do not depend on $K$. *q.e.d.*

## 11 The Limit along the Full Sequence

We can now compare $W_{K+1}(\mathcal{C})$ with $W_K(\mathcal{C})$. By Theorem 8.1 each of them is the average of the normalised dressed loops against the density of its run on the common space $G^{\mathcal{B}}$, so that the comparison is a comparison of two densities and of two families of functions on one space. The difference between the densities is, at first sight, of order one: the exponents of the two runs contain the effective actions of all the steps, and the terms of the first steps, produced on the fine lattices, differ by quantities which do not tend to zero. What saves the comparison is the decay of Bałaban's effective actions in the age of their terms, by which a term produced $m$ steps before the stopping step contributes at most $E_1 L^{-\beta m}$ per point of the stopping lattice [9, (2.43), p. 263]. The old terms can therefore be bounded without comparing them, and only the recent ones, produced at scales at which the fine lattice is far away, have to be compared. The comparison of the recent terms is a recursion: the difference of the terms produced by one step is bounded by the differences of the terms which enter it, through a contraction which is Bałaban's, and by sources which are the lattice artefacts of Sections 9 and 10. Its solution decays geometrically in the scale, up to a term from the fine scales which is a negative power of $K$.

**Theorem 11.1** (Comparison of Two Cutoffs)**.** *Under the hypotheses of Theorem 1.5 there are an index $K_1 \geq K_0(\mathcal{C})$ and nonnegative numbers $a_K$, given for $K \geq K_1$ by* (11.26) *and for $K_0(\mathcal{C}) \leq K < K_1$ by $a_K := 2B(\mathcal{C})$, such that $\sum_K a_K < \infty$ and $|W_{K+1}(\mathcal{C}) - W_K(\mathcal{C})| \leq a_K$ for every $K \geq K_0(\mathcal{C})$. The index $K_1$ depends on $\mathcal{C}$ through the quantities on which $B(\mathcal{C})$ depends, and on $q$, $T/u$, $\bar{g}$, and the constants of the construction.*

For the finitely many $K < K_1$ there is nothing to prove, since $|W_K(\mathcal{C})| \leq B(\mathcal{C})$ for every $K \geq K_0(\mathcal{C})$ by Theorem 8.6. The rest of this section proves the bound for $K \geq K_1$, first for the all-small-field pair and then for the other terms. Throughout, the two runs are those of Section 9.1.

### 11.1 The Exponents of the All-Small-Field Pair

On the term $\ell_0$ of either run the large-field regions are empty, the coupling (3.26) is constant on the last domain, and the exponent of the density at $V$ is, by (3.25)–(3.27),

$$-\frac{1}{g_{\text{stop}}^2} A_{\text{fine}}(U(V)) + \sum_j \text{irr}_j(U(V)) + \sum_j [\mathbf{R}^{(j)}(U(V)) - \mathbf{R}^{(j)}(1)] - E \,, \tag{11.1}$$

with $\mathrm{irr}_j(U) \coloneqq \mathbf{E}^{(j)}(U) - \mathbf{E}^{(j)}(1) - \beta_j A(\phi_j, U)$ the quantity bounded by (3.31), where for run A $g_{\text{stop}} = g^A_{K'}$, $A_{\text{fine}} = A_\varepsilon$, and $U = U^A(V)$ is the minimiser on the lattice of spacing $\varepsilon$ with $M^{K'}U = V$, and for run B $g_{\text{stop}} = g^B_{K'+1}$, $A_{\text{fine}} = A_{\varepsilon/L}$, and $U = U^B(V)$ with $M^{K'+1}U = V$. To compare the terms of the same shared scale we carry run A's arguments to run B's fine lattice. For a localisation domain $X$ and a term $\mathbf{E}^{B,(j+1)}(X, \cdot, z)$ of run B let

$$\mathcal{G}_j(X, (\mathbf{U}, \mathbf{J}), z) \coloneqq \mathbf{E}^{B,(j+1)}(X, (\iota_X \mathbf{U}, Q_1^{\mathrm{pl}}(\iota_X \mathbf{U})^* \mathbf{J}), z) \,, \qquad \Delta_j \coloneqq \mathcal{G}_j - \mathbf{E}^{A,(j)} \,, \tag{11.2}$$

where $Q_1^{\mathrm{pl}}(W)$ is the plain linearised one-step average at $W$ and $\mathbf{J}$ is the current variable of the spaces of [8, (1.10)–(1.16), p. 262]; both are evaluated at the current $\mathbf{J} = J(U)$ of the configuration, so that $\Delta_j$ is a functional of fields on the lattice of spacing $\varepsilon$. We write $\mathrm{irr}(\Delta_j)(U) \coloneqq \Delta_j(U) - \Delta_j(1) - \Pi^\Delta A(h_z, U) - N_j(U)$, with $\Pi^\Delta$ the isotropic coefficient of $\Delta_j$ in the sense of [9, (3.57)–(3.61), pp. 281–282], $h_z$ the tent functions of [9, (3.65), p. 283], and $N_j$ the part of the second moment of $\mathcal{G}_j$ off its isotropic coefficient: since $\mathcal{G}_j$ evaluates every domain at its own lift, the Euclidean covariance by which Bałaban shows that this part vanishes [9, (3.60)–(3.61), p. 282] is not available for it, and $N_j$ is kept as a term of its own, a sum of 36 plane-pair actions whose coefficients are boundary terms of the localised lift. Similarly $\mathrm{irr}_j^{\mathcal{G}}(U) \coloneqq \mathcal{G}_j(U) - \mathcal{G}_j(1) - \Pi^{\mathcal{G}} A(h_z, U) - N_j(U)$, and $\mathcal{G}_j(1) = \mathbf{E}^{B,(j+1)}(1)$ because $\iota_X 1 = 1$ and $J(1) = 0$.

**Lemma 11.2** (Exponent Difference)**.** *At every $V$ in the support of both terms $\ell_0$, the exponent of run B minus that of run A is*

$$D(V) = -\delta_{K'} A_{\varepsilon/L}(U^B) - (g^A_{K'})^{-2}[A_{\varepsilon/L}(U^B) - A_\varepsilon(U^A)] + \mathrm{irr}_1^B(U^B) + \sum_{j=1}^{K'} \mathrm{irr}(\Delta_j)(U^A) + \sum_{j=1}^{K'} [\mathrm{irr}_{j+1}^B(U^B) - \mathrm{irr}_j^{\mathcal{G}}(U^A)] + D^{\mathbf{R}}(V) - (E^B - E^A) \,, \tag{11.3}$$

*where $D^{\mathbf{R}}$ consists of the same three groups for the $\mathbf{R}$-terms and $E^B - E^A$ is a number.*

*Proof.* The coupling of run B at its stopping step is $(g^B_{K'+1})^{-2} = (g^A_{K'})^{-2} + \delta_{K'}$ by (9.2), which gives the first two terms. The regular part of run B consists of its terms of the extra step, $\mathrm{irr}_1^B$, and of those of the steps $j+1$, $1 \le j \le K'$. For the latter, $\mathrm{irr}_{j+1}^B(U^B) - \mathrm{irr}_j^A(U^A) = [\mathrm{irr}_{j+1}^B(U^B) - \mathrm{irr}_j^{\mathcal{G}}(U^A)] + [\mathrm{irr}_j^{\mathcal{G}}(U^A) - \mathrm{irr}_j^A(U^A)]$, and the second bracket is $\mathrm{irr}(\Delta_j)(U^A)$, because the subtractions at $U = 1$, of the isotropic part, and of $N_j$ are linear and $\Pi^\Delta = \Pi^{\mathcal{G}} - \beta_j^A$, the coefficient of run A being absorbed into the coupling by (3.26). The same holds for the $\mathbf{R}$-terms. *q.e.d.*

The groups of (11.3) are of four kinds: the coupling term, which is controlled by the recursion through $\delta_{K'}$; the difference of the two constrained minima of the action, a lattice artefact; the terms of the extra step and the old terms, which are bounded directly; and the differences of the recent terms of the same shared scale, which are the subject of the recursion. Since $MU^B$ satisfies run A's constraint, the second group is

$$A_{\varepsilon/L}(U^B) - A_\varepsilon(U^A) = [A_\varepsilon(MU^B) - A_\varepsilon(U^A)] - b(U^B) \,, \qquad b(W) \coloneqq A_\varepsilon(MW) - A_{\varepsilon/L}(W) \,, \tag{11.4}$$

the gap between two configurations of run A's problem and the blocking artefact.

**Proposition 11.3** (Classical Depth)**.** *Let $0 < \varepsilon_0 < 1/2$. There are $k_0(\varepsilon_0)$ and $C_b$ such that for $K' \ge k_0(\varepsilon_0)$ and every $V$ in the support of both terms $\ell_0$,*

$$|A_{\varepsilon/L}(U^B(V)) - A_\varepsilon(U^A(V))| \le 2 C_b L^{-K'} \varepsilon_1^2 N_q^4 \,. \tag{11.5}$$

*Proof.* The blocking artefact is a sum over the coarse plaquettes of local Taylor remainders, each bounded by the regularity of its cube [7, Theorem 1, (9), (10), p. 279], which gives $|b(U^B)| \le C_b L^{-K'} \varepsilon_1^2 N_q^4$, and the gap is at most $C_{\mathrm{gap}}(\varepsilon_0) L^{-(2-2\varepsilon_0)K'} \varepsilon_1^2 N_q^4$, which is at most the same bound for $K' \ge k_0(\varepsilon_0)$; both are Supplement, Proposition S.13(a), (b). *q.e.d.*

### 11.2 The Direct Bounds

The terms of the extra step and the terms of large age are bounded without comparing them.

**Lemma 11.4** (The Extra Step and the Old Tail)**.** *Let $\bar g$ be as in Theorem 1.5, and let $1 \le m \le K'$. On the support of both terms $\ell_0$,*

$$|\mathrm{irr}_1^B(U^B)| + |\mathbf{R}^{B,(1)}(U^B) - \mathbf{R}^{B,(1)}(1)| \le (E_1 L^{-\beta K'} + R_1(bK - C_W)^{-\kappa_0/2}) N_q^4 \,, \tag{11.6}$$

*and the differences of the regular and $\mathbf{R}$-parts of the shared scales $j \le K' - m$ satisfy*

$$\sum_{j \le K'-m} |\mathrm{irr}_{j+1}^B(U^B) - \mathrm{irr}_j^A(U^A)| \le \frac{2E_1 N_q^4 L^{-\beta m}}{1 - L^{-\beta}} \,, \qquad \sum_{j \le K'-m} |\Delta \mathbf{R}_j| \le C_{\mathbf{R}} N_q^4 (1+m)^{1-\kappa_0/2} \,, \tag{11.7}$$

*with $\Delta \mathbf{R}_j$ the difference of the $\mathbf{R}$-parts of the shared scale $j$ and $C_{\mathbf{R}}$ a constant of $R_1$, $b$, and $\kappa_0$.*

*Proof.* On $\ell_0$ the determining set is the stopping lattice itself, $\Gamma_{K'} = \mathbb{T}_{j_q}$ with $N_q^4$ points, so that in (3.31) at the generation $j$ only the term $n = K'$ (respectively $K'+1$ for run B) occurs, and it gives $E_1 L^{-\beta(K'-j)} N_q^4$ for the regular part and $R_1 g_j^{\kappa_0} N_q^4$ for the $\mathbf{R}$-part. At the extra step $j = 1$ of run B this is $E_1 L^{-\beta K'} N_q^4$, and $(g_1^B)^{-2} \ge \bar g^{-2} + bK - C_W$ by (4.1) for the target $(K+1, \bar g)$, which gives (11.6). For (11.7) the triangle inequality and the same bound for each run give $\sum_{j \le K'-m} 2E_1 L^{-\beta(K'-j)} N_q^4$, and the geometric series gives the first bound. For the second, $g_j^{-2} \ge g_{K'}^{-2} + b(K'-j) - C_W$ by (4.1) gives $g_j^{\kappa_0} \le C(1 + K' - j)^{-\kappa_0/2}$ for $\bar g$ small, and $\sum_{i \ge m} (1+i)^{-\kappa_0/2} \le C(1+m)^{1-\kappa_0/2}$ since $\kappa_0 > 2$. *q.e.d.*

We use (11.7) at the age $m_K := \lfloor K^{1/2} \rfloor$, where it is summable in $K$ because $\kappa_0 \ge 11$ (Appendix A), and it uses no information on the differences of the terms. The terms of the shared scales $j > K' - m_K$ are compared, and this is where Bałaban's proof of (3.31) has to be run for a difference.

### 11.3 The Frozen Differences

The terms $\mathbf{E}^{A,(j)}$ and $\mathcal{G}_j$ are two families of terms of the generation $j$ on the same lattices, with the localisation, analyticity, gauge invariance, and bound of Section 3.2, for $\mathcal{G}_j$ on a common polydisc of reduced radii $\alpha'_{i,m} := \alpha_{i,m}/(4B_3)$. For their difference we use the *class norm* $d_j$, the least number with $|\Delta_j(X, (\mathbf{U}, J(\mathbf{U})), z)| \le d_j E_0 e^{-\kappa d_j(X)}$ on the evaluation sets of this polydisc, and $D_k := \sum_{j \le k} L^{-\beta'(k-j)} d_j$ for its carried sum. Here

$$\beta' := \tfrac{1}{2} \min(\beta_H, 1 - \beta_E) \ge \tfrac{19}{40} \,, \qquad \theta_V := \frac{\beta'}{2(4+\beta')} \,, \tag{11.8}$$

with $\beta_H \ge 19/20$ the Hölder index in the regularity bounds of [8, (3.31), p. 276], which Bałaban states for every index less than one and which hold up to the index one for the minimisers [7, Theorem 1, (9), p. 279], and $\beta_E \le 1/20$ the positive number of [9, (3.67), p. 283], which may be taken arbitrarily small at the cost of a larger constant. We write $d_j^{\mathbf{R}}$ for the class norm of the difference of the $\mathbf{R}$-terms of the shared scale $j$ and $S_k := \sum_{j \le k} d_j^{\mathbf{R}}$.

**Proposition 11.5** (Theorem 2 for a Difference)**.** *Let the constants satisfy the conditions of Appendix C and $\bar{g}$ be small. There are constants $E_1'$, $E_V$, and $C_\alpha$, independent of $j$, $k$, $K$, and the torus, such that for $z \in \Lambda_j^0 \cap (\Omega_n \setminus \Omega_{n+1})$ and every regular background $U$ of the class of the step,*

$$|\mathrm{irr}(\Delta_j)(U)| \le d_j\, C_\alpha E_0 (L^{j-n})^{4+\beta'} \,, \tag{11.9}$$

*and on the all-small-field pair, at $n = K'$,*

$$|\mathrm{irr}^B_{j+1}(U^B) - \mathrm{irr}^{\mathcal{G}}_j(U^A)| \le E_V \Big[ \Big( \frac{\hat\sigma_{K',j}}{r_{0,K'}} \Big)^{\theta_V} + (C_\Pi(j) \varepsilon_1^2 e^{-\delta_0 M_1 L^j})^{\theta_V} \Big] (L^{j-K'})^{4+\beta'/2} \,, \tag{11.10}$$

*where $\hat\sigma_{K',j}$ is the distance between $U^B$ and the lift $\iota(U^A)$ in the norm of the evaluation sets, bounded by Proposition 10.6 and the tail of the localised lift, $r_{0,K'}$ the radius of these sets, and $C_\Pi(j) e^{-\delta_0 M_1 L^j}$ the bound of the boundary terms of the lift in the coefficients of $N_j$. The same holds for the $\mathbf{R}$-terms with $d_j^{\mathbf{R}}$ and without the factor of the age.*

*Proof.* For (11.9) we run Bałaban's proof of [9, Theorem 2, (2.43)], which occupies [9, pp. 279–283] and rests on [8, Sects. 3–4], with the difference $\Delta_j$ in place of a single family; the proof is linear in the family, and it uses of the family only its localisation, its analyticity on the polydiscs of [8, (3.13)–(3.14)] with the radii of the step, and its bound, all of which $\Delta_j$ has with $d_j E_0$ in place of $E_0$, except the covariance which removes the plane-resolved moment, which is kept as $N_j$. The Hölder regularity of the background, of index $\beta_H$, gives the power $4 + \beta'$ in place of 4. This is Supplement, Proposition S.15(a)–(b). For (11.10) the two values are the irrelevant parts of one term of run B at two configurations, $U^B$ and the lift $\iota(U^A)$, each bounded by the age bound for one term; interpolating the configuration along a line of length $\hat\sigma_{K',j}$ in the norm of the evaluation sets and using Cauchy's estimate on a disc of radius $r_{0,K'}$, then interpolating between the bound of the age and the bound of the distance with the weight $\theta_V$, gives (11.10); this is Supplement, Proposition S.15(c). *q.e.d.*

### 11.4 The Generated Difference of One Step

At a shared scale $k$ the two runs apply the same step to inputs on the same lattice, with the same block data and the same fluctuation variable, and the step's output is the logarithm of a fluctuation integral. The two integrals differ by the interaction, which collects the fluctuation parts of the frozen terms and of the fine actions, by the Gaussian form, which is built from the propagators of the run, and by the cutoff of the fluctuation field. Let $\mathbf{P}^A$ and $\mathbf{P}^B$ be the two interactions, $\Delta_A^{(k)}$ and $\Delta_A^{(k)} + \delta\Delta^{(k)}$ the two quadratic forms of the fluctuation field, and for $t \in [0,1]$ put $\mathbf{P}_t \coloneqq \mathbf{P}^A + t(\mathbf{P}^B - \mathbf{P}^A)$, $\Delta_t \coloneqq \Delta_A^{(k)} + t\delta\Delta^{(k)}$, and

$$Z_t \coloneqq \int \chi^{A,(k)} \exp\Big[ -\frac{1}{2} \langle A, C^* \Delta_t C A \rangle + \mathbf{P}_t \Big] \,\mathrm{d}A \,, \tag{11.11}$$

with run A's cutoff in both, so that run B's integral is $Z_1$ plus the shell term of Lemma 10.5.

**Lemma 11.6** (Interpolation Identity)**.** *With $\langle\cdot\rangle_t$ the expectation of the normalised integrand of $Z_t$,*

$$\log Z_1 - \log Z_0 = \int_0^1 \Big[ \langle \mathbf{P}^B - \mathbf{P}^A \rangle_t - \frac{1}{2} \langle\langle A, C^* \delta\Delta^{(k)} C A \rangle\rangle_t \Big] \,\mathrm{d}t \,, \tag{11.12}$$

*and for every $t$ the measure $\langle\cdot\rangle_t$ is of the kind which Bałaban's cluster expansion of the step treats.*

*Proof.* The function $t \mapsto \log Z_t$ is differentiable, because the derivative of the integrand in $t$ is the integrand times $\mathbf{P}^B - \mathbf{P}^A - \frac{1}{2}\langle A, C^*\delta\Delta^{(k)}CA\rangle$, which on the support of the cutoff is the sum of a bounded function and a quadratic form in $A$, and is therefore dominated, uniformly in $t$, by an integrable function, since $\Delta_t$ is bounded below by a positive multiple of the identity uniformly in $t$; its derivative is the expectation of the derivative of the exponent, which gives (11.12), the device by which Bałaban separates the terms of a step [9, (3.26)–(3.27), p. 271]. The interaction $\mathbf{P}_t$ is a convex combination of two interactions which satisfy the inductive bounds [8, (1.18), p. 263], [9, (2.31), (2.42)], and these bounds define convex sets; the quadratic form $\Delta_t$ is a convex combination of two positive forms with the bounds of [5, Theorems 3.1–3.3]. *q.e.d.*

The right side of (11.12) is a cluster expansion with one marked activity, the piece of $\mathbf{P}^B - \mathbf{P}^A$ or of the quadratic insertion on a domain, and the localised terms of the difference of the step's outputs are the localised terms of this marked expansion. The marked activity consists of the fluctuation parts of the frozen differences of the generations $j \le k$, of size $D_k + S_k$ in the class norm; of the coupling, through the factors $g_k$ in the backgrounds and the coefficients $g_k^{n-2}$ of the terms of order $n \ge 3$, whose two values differ relatively by at most $g_k^2|\delta_k|$; of the depth sources $e_{k+1}$, the classical depth of Proposition 11.3 at the fluctuated block field and the Gaussian depth, the difference $\delta\Delta^{(k)}$ of the quadratic forms, which is at most

$$r_k \le C_{\delta\Delta}[L^{-\gamma_H(k-1)} + L^{-(1-\varepsilon_0)k}] , \qquad \gamma_H \coloneqq \frac{\beta\gamma_w}{1+\beta} , \tag{11.13}$$

times the bound of the form itself, by Theorem 9.5 and Supplement, Proposition S.17; and of the shell of the cutoff, at most $C|\delta_k|g_k^2 e^{-cp_0(\hat g_k)^2}$ per point by Lemma 10.5 and the Gaussian tail beyond the radius $\vartheta_k/g_k$. The marked-cluster bound of Supplement, Proposition S.16, then gives, per point of the scale $k+1$,

$$d_{k+1} \le \varepsilon_k^*(D_k + S_k) + C_g\varepsilon_k^* g_k^2|\delta_k| + e_{k+1} , \qquad \varepsilon_k^* \coloneqq C_X C_{\mathrm{pt}} C_{\mathrm{KP}} L^4 C_\alpha^{\mathbf{A}}[C_T\mu_k + 4a_k] , \tag{11.14}$$

where $\mu_k$ is the square of the ratio of the cutoff's supremum to the radius of analyticity of the frozen terms, a negative power of $\log g_k^{-2}$, $a_k$ the bound of the interaction's activities, a positive power of $g_k$ times a power of $\log g_k^{-2}$, and the other constants are those of Supplement, Proposition S.16, independent of $k$ and $K$. The coefficient of the flow is the marginal part of the step's output, so that $|\beta^B_{k+2} - \beta^A_{k+1}| \le C_\beta d_{k+1}$ by Cauchy's estimate, and from $\delta_K = 0$, which holds because both runs are pinned at $g^A_K = g^B_{K+1} = \bar g$, the coupling differences are $|\delta_j| \le C_\beta \sum_{i>j} d_i$; the steps between the stopping step and the unit scale, which both runs perform for the coupling prescription, obey the same recursion. The **R**-terms have no decay in their age beyond the volume factor [9, p. 283], and their recursion, Supplement, Proposition S.16(c), together with the renormalisation quotients of the **R**-operation, in which Theorem 10.8 enters through $h_n \coloneqq C_h\tilde\theta^n$, gives

$$S_k \le C_S\Big[\sum_{j\le k} c_j(\mathcal{D}'_{[j]} + D_{j-1}) + \sum_{j\le k} c_j^B\mathcal{D}_j^{*\prime} + \sum_{j\le k} c_j^V\mathcal{D}_j^{V\prime} + C_\nu w_K\Big] , \tag{11.15}$$

with $c_j$, $c_j^B$, $c_j^V$ constants times $(100MR_j)^4 g_j^{\kappa_0}$, whose sum over $j$ is bounded uniformly in $K$ for $\kappa_0 > 2$, $\mathcal{D}'_{[j]}$, $\mathcal{D}_j^{*\prime}$, $\mathcal{D}_j^{V\prime}$ the maxima over the history of the integrands' differences per point, built from the $d_i$, the $|\delta_i|$, the $e_i$, and the $h_i$ (Supplement, Proposition S.16(d)), and $w_K$ of (11.17) below.

We now solve the recursion. Put

$$\theta \coloneqq L^{-\beta'} + \varepsilon^* , \qquad \tilde\theta \coloneqq \max\Big(\frac{1+\theta}{2}, L^{-\gamma_H}, L^{-(1-2\varepsilon_0)/3}, L^{-(1-\varepsilon_0)/4}, L^{-\theta_V/8}\Big) , \tag{11.16}$$

where $\varepsilon^* := \sup_{g\le(1+\beta_0)\gamma}\varepsilon^*(g)$ is the supremum of the coefficient of (11.14) as a function of the coupling, which bounds $\varepsilon_k^*$ at every shared scale because $g_k \le (1+\beta_0)\bar g$ by (3.9); and put

$$\ell_K := (\log(bK+\bar g^{-2}))^{4r+2p_0}\,, \qquad \mu_K := K^{2-\kappa_0/2}\ell_K\,, \qquad w_K := (K+1)(\tilde\theta^{K'/8}+2^{-K'/16})+\mu_K\,. \tag{11.17}$$

**Proposition 11.7** (Solution of the Recursion)**.** *Let $\gamma$ satisfy the conditions (T1) of* (C.2)*. Then $\tilde\theta < 1$, and for every shared scale $i$*

$$d_i \le 2C_s(i\tilde\theta^i+w_K)\,, \qquad S_i \le C_S'' w_K\,, \qquad |\delta_i| \le 2C_sC_\beta[2(1-\tilde\theta)^{-2}(i+1)\tilde\theta^{i+1}+(K+1-i)w_K]\,, \tag{11.18}$$

*with constants $C_s$ and $C_S''$ which do not depend on $K$. In particular, at the stopping step,*

$$D_{K'}+S_{K'}+C\varepsilon_{K'}^2|\delta_{K'}|+e_{K'} \le C_a[(K+1)\tilde\theta^{K'}+(j_q+2)w_K]\,, \tag{11.19}$$

*with a constant $C_a$ which does not depend on $K$ and depends on $j_q$ at most as $L^{C\varepsilon_0 j_q}$ times a polynomial.*

*Proof.* Substituting (11.15) and the bound on $|\delta_i|$ into (11.14) gives $d \le \mathcal{T}d + s$, with an operator $\mathcal{T}$ on sequences which is positive, monotone, and positively homogeneous, and a source $s$ built from the $e_i$, the $h_i$, and $C_\nu w_K$, with $s_1 := 2$ for the first step, at which both runs' outputs satisfy the bound of their class. Put $m_i := i\tilde\theta^i + w_K$ and $\|u\|_m := \sup_i u_i/m_i$. Since $\tilde\theta > L^{-\beta'}$, $\sum_{j\le i} L^{-\beta'(i-j)}m_j \le (1-L^{-\beta'}/\tilde\theta)^{-1}m_i$, and $m_i \le \tilde\theta^{-1}m_{i+1}$; the sums over the fine scales are paid by the factors $c_j$, which for $j \le K/2$ are at most $C(bK/2)^{-\kappa_0/2}\ell_K$, and the sums over the coarse scales by $\tilde\theta^j$, which together give at most a constant times $\mu_K \le w_K$; and the coupling differences enter through $g_i^2\sum_{l>i} m_l \le (2(1-\tilde\theta)^{-2}+b^{-1})m_{i+1}$, which uses $g_i^2(K-i) \le b^{-1}$ by (4.1). Hence $\|\mathcal{T}\|_m \le \varepsilon^* C_{\mathcal{T}} \le \frac14$ by the first condition of (C.2), and since the system is finite, $\|d\|_m < \infty$ and $\|d\|_m \le \|\mathcal{T}\|_m\|d\|_m + \|s\|_m \le \frac14\|d\|_m + C_s$. This gives the first bound of (11.18), and the others follow from (11.15) and the bound on $|\delta_i|$. The first condition implies $\varepsilon^* \le (1-L^{-\beta'})/4$, because $C_{\mathcal{T}} \ge (1-L^{-\beta'})^{-1}$, so that $(1+\theta)/2 \le (3+L^{-\beta'})/4 < 1$, and the other entries of (11.16) are less than one. At $i = K'$, $K'\tilde\theta^{K'} \le (K+1)\tilde\theta^{K'}$ and $(K+1-K')w_K = (j_q+1)w_K$, which gives (11.19); the details, the constants, and the depth sources $e_i \le C_e\tilde\theta^i$ are Supplement, Proposition S.16(e). *q.e.d.*

The bound (11.19) is summable in $K$, because $(K+1)\tilde\theta^{K'}$ and $(j_q+2)(K+1)(\tilde\theta^{K'/8} + 2^{-K'/16})$ decay exponentially at fixed $j_q$, and $\mu_K = K^{2-\kappa_0/2}\ell_K$ with $\kappa_0 \ge 11$. The second condition of (C.2) is the one which allows the recent **R**-terms, which are carried through $N_k' \le N'(\hat g_k)$ steps without contraction [9, pp. 270–271], to be summed by a window lemma; the memory $N'$ grows as a double logarithm of the inverse coupling, while $\hat\varepsilon^*$ decays as a negative power of its logarithm. The same recursion holds for the deviation insertions of the dressed loops, which are produced by the same fluctuation integral with the loop's words as further marked activities, with their class norms $d_j^{\text{ins}}$ in place of $d_j$ and an additional term $\varepsilon_k^Q D_k^{\text{ins}}$ from the old vertices of the groups, which enter as numbers, with $\varepsilon^Q(g)$ of order $g^2$ times a power of $\log g^{-2}$ (Supplement, Proposition S.16(f)). The coefficient $\varepsilon^Q$ depends on the representations, but the recursion of the insertions runs over the steps $k \le K'$ only, at which $g_k \le \gamma_q$ by (3.46); under the condition

$$\varepsilon^Q(g)C_{\mathcal{T}} \le \frac14 \tag{11.20}$$

for every $g \le \gamma_q$, which holds for $\gamma_q$ small, depending on $c_{\mathcal{C}}$, the operator of this recursion has the norm at most $(\varepsilon^* + \sup_{g\le\gamma_q}\varepsilon^Q(g))C_{\mathcal{T}} \le \frac12$, by the first condition of (C.2), and its solution is $d_i^{\text{ins}} \le 2C_{\text{ins}}^* m_i$.

### 11.5 The All-Small-Field Pair

On the common support of the two terms $\ell_0$ the groups of (11.3), less the number $E^B - E^A$, are bounded at every $V$ by

$$X_K \coloneqq X_K^{\mathrm{dir}} + N_q^4 C_a[(K+1)\tilde{\theta}^{K'} + (j_q+2)w_K] + X_K^{\mathrm{var}} + X_K^{\mathbf{R}} + X_K^{\mathrm{cl}} \,, \qquad x_K \coloneqq e^{X_K} - 1 \,, \tag{11.21}$$

where $X_K^{\mathrm{dir}}$ is the sum of the right sides of (11.6) and (11.7) at $m = m_K$, the second term is (11.19) summed over the $N_q^4$ points of the stopping lattice, $X_K^{\mathrm{var}}$ is the sum over $j$ of (11.10), $X_K^{\mathbf{R}}$ the bound of the corresponding group of the **R**-terms of the ages less than $m_K$, and $X_K^{\mathrm{cl}} \coloneqq 2C_b(g_{K'}^A)^{-2}L^{-K'}\varepsilon_1^2 N_q^4$ by (11.5), together with its counterpart on the young pairs (Supplement, Proposition S.13(d)). Each of these is summable in $K$: the variation group by $\hat{\sigma}_{K',j} \le C(\varepsilon/a_{K'})^{(1-\varepsilon_0)/3}\varepsilon_1$ and the factor of the age, and the coupling factor $(g_{K'}^A)^{-2} \le \bar{g}^{-2} + b' j_q + C_W$ of the classical group is bounded. The loop's terms are compared on the same support in four parts.

*The leading words.* By Proposition 10.6(i), $U^A = e^{i\varepsilon A'} M U^B$ in a gauge, with $\varepsilon|A'(c)| \le 4C_G(\varepsilon_0)(\varepsilon/a_{K'})^{2-\varepsilon_0}\varepsilon_1$ on every bond $c$ of the fine lattice of run A, and by (8.10) $MU^B(c) = e^{iY_c}W(c)$, with $W(c)$ the transport of $U^B$ along the $L$ fine bonds of $c$ and $|Y_c| \le C_Y(\varepsilon/a_{K'})^2\varepsilon_1$ by Lemma 8.2. So the holonomies of $C_i$ in the two backgrounds are $\prod_c W(c)$ and, up to a gauge transformation which leaves the trace unchanged, $\prod_c e^{i\varepsilon A'(c)}e^{iY_c}W(c)$. Telescoping inside the normalised trace, with $\|\pi(e^{iX}) - 1\| \le c_\pi\|X\| \le c_\pi c_h|X|$ for the unitary representation $\pi$ (Section 8.2), and summing over the $P_{K'}(C_i)L^{K'}$ fine bonds of $C_i$,

$$\Big|\prod_i f^{(i),B} - \prod_i f^{(i),A}\Big| \le \lambda_K \coloneqq \sum_i c_{\pi_i} c_h (4C_G(\varepsilon_0) + C_Y)\varepsilon_1 P_{K'}(C_i) L^{-(1-\varepsilon_0)K'} \,, \tag{11.22}$$

because every factor has modulus at most one and $L^{K'}(\varepsilon/a_{K'})^{2-\varepsilon_0} = L^{-(1-\varepsilon_0)K'}$. The words with insertions are covariant at the slots of their insertions, and the same telescoping with the insertions held fixed changes the normalised dressed loop by at most $\lambda_K \prod_b(1 + \|Y_b\|) \le \lambda_K \mathfrak{b}(\mathcal{C})$.

*The deviation insertions.* By the recursion for the insertions their difference changes the normalised dressed loop by at most $\mathfrak{b}(\mathcal{C})X_K$.

*The wrap insertions and the interaction weights.* The insertions which wrap around the torus have no deviation bound and are not contracted by the recursion, but their mass decays as $e^{-cN_{\min}^{(j)}/M}$ in the smallest period of the torus at their scale, $N_{\min}^{(j)} \ge c(T/u)L^{K'-j+j_q}$. Those of age at least $m_K^w$, the least $m$ with $a_w L^m \ge 3\log(K+2)$, are not compared and have total mass at most $\tau_K^w \coloneqq 2A_w \max(L, 3L\log(K+2)/a_w)(K+2)^{-3}$ in each run; the younger ones are compared at their birth, by the interpolation of the constituents of their step described below for the loop constant, and then carried, with a difference which grows by at most $2L$ per step. This gives a change of the normalised dressed loop of at most $x_K^w \mathfrak{b}(\mathcal{C})$, with $x_K^w$ summable in $K$; the interaction weights $Q_{K'}(\mathfrak{B})$ of the families, cluster activities of the reference gas of the stopping step, are compared in the same way, with a change at most $x_K^Q \mathfrak{b}(\mathcal{C})$. Both are Supplement, Proposition S.16(g).

*The loop constant.* The constant $E_{\mathrm{const}} = \sum_{j \le K'} E_j^\circ$ of the dressed loop is a sum of per-step terms, which we index by the age $m \coloneqq K' + 1 - j$ in run A and $m \coloneqq K' + 2 - j$ in run B, so that the two terms of the same shared scale have the same $m$; each is at most $A_E L^{m/2} e^{-a_E L^{m/2}}$, by the proof of Corollary 7.11, with constants $A_E$ and $a_E$ which do not depend on $K$. The terms of age $m \ge m_K^E$, the least $m$ with $a_E L^{m/2} \ge 2\log(K+2)$, are not compared; their sum is at most $\tau_K^E \coloneqq 2A_E \max(L^{1/2}, 2L^{1/2}\log(K+2)/a_E)(K+2)^{-2}$ in each

run, and for $K \geq K_1$ the extra step of run B is among them. For the younger terms, which are produced at **R**-steps of both runs, we interpolate: every constituent $x$ of the step, an activity of the effective action, a marked activity of the loop, a factor $V(Y)$, or the operation of a young component, is replaced by $x^\varsigma := x^A + \varsigma(x^B - x^A)$, $\varsigma \in \mathbb{C}$, after the components born before the scale $K' - m$ have been removed by multiplying their operations by a parameter $\mathfrak{z}$ with $|\mathfrak{z}| \leq e^{C_P K'/4}$. The relative differences of the constituents are at most $\rho_K^E$, a quantity built from (11.18), from Theorem 10.8, and from the shells, which tends to zero; for $|\varsigma| \leq 1/(2\rho_K^E)$ the constituents are at most twice their bounds, the cluster expansion of the step converges, and the per-step term is analytic in $\varsigma$ with the bound $A^{E,2} L^{m/2} e^{-a^{E,2} L^{m/2}}$ at the enlarged amplitudes and the lowered decay. Cauchy's estimate on the disc of radius $1/(2\rho_K^E) \geq 2$ gives

$$|E_j^\circ(1) - E_j^\circ(0)| \leq 4\rho_K^E A^{E,2} L^{m/2} e^{-a^{E,2} L^{m/2}} , \tag{11.23}$$

and the old components cost at most $2e^{-C_P K'/4} A_E L^{m/2} e^{-a_E L^{m/2}}$ in each run, by Cauchy's estimate in $\mathfrak{z}$. Summing over $m$ with $\sum_{m \geq 1} \Lambda^m e^{-a\Lambda^m} \leq e^{-a}(1 + 6/a)$ for $\Lambda \geq 2$, we obtain $|E^B_{\text{const}} - E^A_{\text{const}}| \leq \Delta E_K$, with $\Delta E_K \leq \min(2\tau_K^E + \Delta_K^E, 2B_E(\mathcal{C}))$ summable in $K$. The details are Supplement, Proposition S.16(h).

**Lemma 11.8** (The All-Small-Field Pair)**.** *For $K \geq K_1$, with run A's constant on both runs, the terms $\ell_0$ change $W_{K+1}(\mathcal{C}) - W_K(\mathcal{C})$ by at most $(3x_K + \lambda_K + x_K^w + x_K^Q)\, B(\mathcal{C})$ on their common support, and by at most $b_K^{(S)} B(\mathcal{C})$ through the shells of the stopping-scale test functions.*

*Proof.* On the common support, run B's density is $e^{D(V) + E^B - E^A}$ times run A's, and $|D(V) + E^B - E^A| \leq X_K$, so that $\int |\varrho^B_{\ell_0} - \varrho^A_{\ell_0}|\, \mathrm{d}V \leq x_K Z^A_{\ell_0}$, where the number $E^B - E^A$ cancels between the numerator and the normalisation. The four parts above give $\int |\widehat{\mathcal{W}}^B \varrho^B_{\ell_0} - \widehat{\mathcal{W}}^A \varrho^A_{\ell_0}|\, \mathrm{d}V \leq \mathfrak{b}(\mathcal{C})(x_K + X_K + \lambda_K + x_K^w + x_K^Q) Z^A_{\ell_0}$. As in (10.5), $N'/Z' - N/Z = (\Delta N - W' \Delta Z)/Z$ with $|W'| \leq B(\mathcal{C})$ and $Z \geq Z^A_{\ell_0}$, and $X_K \leq x_K$. The shells are Proposition 10.9. *q.e.d.*

### 11.6 The Pairs with Large Fields

For a young matched pair Theorem 10.8 compares the ingredients $(\beta)$ and $(\gamma)$; the ingredients $(\alpha)$ are compared by (11.18) per point of a history at a scale $i \geq K' - m$: the frozen exponents and the depth terms by $C_a[(i+1)\tilde{\theta}^i + w_K]$; the coupling term of the action by $C\varepsilon_i^2 |\delta_i| \leq |\delta_i|$; the function [11, (1.6), p. 178], $z^{-1} \exp[-g_i^{-2}(1 - \operatorname{Re} \operatorname{tr} V_i)]$, relatively by at most $e^{4|\delta_i|} - 1 \leq 10|\delta_i|$, because $0 \leq 1 - \operatorname{Re} \operatorname{tr} V \leq 2$ and the normalisation $z$ has the common cutoff $\varepsilon_i$; the Gaussian form by $C r_i |A_i(b)|^2$ per variable, by (11.13), which on the cutoff is at most $C r_i p_0(\hat{g}_i)^2$ and where a large-field function leaves $A_i$ unbounded is absorbed by the positivity of the form; and the localised normalisation constants, of order $\log g_i^{-2}$ per variable [12, p. 380], whose dependence on the coupling is the term $-\frac{1}{2} \dim G \log g_i^{-2}$ per variable of the normalisation of a Gaussian integral over a variable scaled by $g_i$, by $\frac{1}{2} \dim G |\log(1 + g_i^2 \delta_i)| \leq |\delta_i|$. Collecting, with $C_M$ the number of ingredients per point,

$$\begin{aligned} s'_K := {} & 2C_a[(K'+1)\tilde{\theta}^{K'-m} + w_K] + 10 \sup_{i \geq K'-m} |\delta_i| \\ & + 4C(1 + C_{\text{cov}}) C_{\delta\Delta} \ell_K [L^{-\gamma_H (K'-m-1)} + L^{-(1-\varepsilon_0)(K'-m)}] , \end{aligned} \tag{11.24}$$

$s_K := s'_K + L^4 C_h \tilde{\theta}^{K'/2}$, and $\omega^s(X) := (1 + C_M P(X)) e^{C_M s_K P(X)}$, which is at most $e^{C_P P(X)}$ when $C_P \geq 2C_M$ and $s_K \leq 1$.

**Proposition 11.9** (The Young Pairs)**.** *Let $k_*$ be the least integer with $k_* \geq k_{\text{LF}}$ such that $s'_K \leq \frac{1}{2}$ and $L^4 C_h \tilde{\theta}^{K'/2} \leq \frac{1}{2}$ for $K' \geq k_*$. For $K' \geq k_*$ the young matched pairs change*

*$W_{K+1}(\mathcal{C}) - W_K(\mathcal{C})$ by at most $\bar{\delta}_{\mathrm{LF}}(5s_K + 3x_K + s_K^W)B(\mathcal{C})$, where $s_K^W$, summable in $K$, bounds the change of the dressed loops near the large-field regions (Supplement, Proposition S.14(f)). The threshold $k_*$ exists, and $s_K$ is summable in $K$.*

*Proof.* By (10.6), (10.13), and the bounds on the ingredients ($\alpha$), each of whose contributions enters per point of a history and is counted by $P(X)$, with the small-field exponent on the last domain bounded by $X_K$ as for the all-small-field pair,

$$\begin{aligned} \int |\varrho^{B'}[\tau^+] - \varrho^A[\tau]| \, \mathrm{d}V &\le (s_K + x_K) \prod_X \omega^s(X) \, M_\tau^A \,, \\ \int (\Pi_B - \Pi_A)_+ \varrho_B \, \mathrm{d}V &\le s_K \prod_X \omega^s(X) \, M_{\tau^+}^B \,, \end{aligned} \tag{11.25}$$

where $\varrho^{B'} := \min(\Pi_A, \Pi_B)\varrho_B$ and we used $(e^a - 1) + (e^b - 1) \le e^{a+b} - 1$, $e^{\sum \phi_X} - 1 \le (\sum \phi_X) e^{\sum \phi_X}$, and $\sum_X C_M P(X) \le \prod_X (1 + C_M P(X))$. Since $Z_\ell \le M_\ell$ and the weights $\omega^s(X)$ are admissible in Proposition 8.4, by the proof of that proposition the sums over the young pairs are at most $\delta_{\mathrm{LF}}(s_K + x_K) Z_{\ell_0}^A$ and $\delta_{\mathrm{LF}} s_K Z_{\ell_0}^B$. Against the normalised dressed loops the first bound holds with $\mathfrak{b}(\mathcal{C})(2s_K + 2x_K + s_K^W)$ in place of $s_K + x_K$, the deviation insertions at the points of the histories being compared by the recursion and the dressings near the large-field regions by Supplement, Proposition S.14(f). Removing run B's own-support pieces changes $W_{K+1}$ by at most $2B(\mathcal{C}) s_K \delta_{\mathrm{LF}}$, as in (10.5), and the rest changes $W_{K+1} - W_K$ by at most $B(\mathcal{C})\delta_{\mathrm{LF}}(3s_K + 3x_K + s_K^W)$ through the numerators and the normalisations. Finally $s'_K \to 0$: $(K'+1)\tilde{\theta}^{K'-m} \to 0$ since $K' - m \ge K'/2$, the covariance terms decay geometrically against the polylogarithm $\ell_K$, and for $i \ge K'/2$ the bound (11.18) gives $|\delta_i| \le C[K'\tilde{\theta}^{K'/2} + K w_K]$, where $K w_K$ is summable because $K\mu_K = K^{3-\kappa_0/2}\ell_K$ and $\kappa_0 > 8$. *q.e.d.*

### 11.7 Proof of the Comparison

Let $K_1 \ge K_0(\mathcal{C})$ be the least index such that every $K \ge K_1$ satisfies $K' \ge \max(k_0(\varepsilon_0), k_*)$, $\max(m_K^E, m_K^w) \le \lfloor K'/4 \rfloor$, $\rho_K^E \le \frac{1}{4}$, and the further conditions of Supplement, Proposition S.16, each of which holds for $K$ large because $m_K^E$ and $m_K^w$ grow as $\log\log K$ while $\rho_K^E$, $x_K^*$, and $s_K$ tend to zero. For $K \ge K_1$ put

$$a_K := \Big[3x_K + \lambda_K + x_K^w + x_K^Q + b_K^{(S)} + \bar{\delta}_{\mathrm{LF}}(a_K^{(U)} + b_K^{(M)})\Big] B(\mathcal{C}) + (e^{\Delta E_K} - 1) B(\mathcal{C}) \,, \tag{11.26}$$

with $a_K^{(U)} := 2e^{-(C_P - 1)(K'+1)}$ and $b_K^{(M)} := 4e^{-(C_P-1)(\lfloor K'/2 \rfloor + 1)} + 5s_K + 3x_K + s_K^W$.

*Proof of Theorem 11.1.* For $K < K_1$ the bound $|W_{K+1} - W_K| \le 2B(\mathcal{C})$ is Theorem 8.6. Let $K \ge K_1$. We first replace run B's constant $E_{\mathrm{const}}^B$ by run A's: with $W^{B,A} := e^{E_{\mathrm{const}}^A - E_{\mathrm{const}}^B} W_{K+1}(\mathcal{C})$,

$$W_{K+1} - W_K = (W^{B,A} - W_K) + (e^{E_{\mathrm{const}}^B - E_{\mathrm{const}}^A} - 1) W^{B,A} \,, \tag{11.27}$$

and $|W^{B,A}| \le B(\mathcal{C})$ because the bound of Lemma 8.5 holds for run B's dressed loop with run A's constant, so that the second term is at most $(e^{\Delta E_K} - 1)B(\mathcal{C})$. For the first, the terms of kind (U) change $W^{B,A}$ by at most $\bar{\delta}_{\mathrm{LF}} a_K^{(U)} B(\mathcal{C})$ and the old pairs change the difference by at most $4e^{-(C_P-1)(m+1)}\bar{\delta}_{\mathrm{LF}} B(\mathcal{C})$, by Corollary 10.3; the young pairs change it by at most $\bar{\delta}_{\mathrm{LF}}(5s_K + 3x_K + s_K^W)B(\mathcal{C})$, by Proposition 11.9; and the all-small-field pair by at most $(3x_K + \lambda_K + x_K^w + x_K^Q + b_K^{(S)})B(\mathcal{C})$, by Lemma 11.8. This gives $|W_{K+1} - W_K| \le a_K$. Each term of (11.26) is summable in $K$: $x_K \le X_K e^{X_K}$ with $X_K$ summable and bounded; $\lambda_K$, $a_K^{(U)}$,

and the first term of $b_K^{(M)}$ decay geometrically; $b_K^{(S)}$ by Proposition 10.9; $s_K$ and $s_K^W$ by Proposition 11.9; $x_K^w$, $x_K^Q$, and $\Delta E_K$ by the bounds above; and $e^{\Delta E_K} - 1 \le \Delta E_K e^{2B_E(\mathcal{C})}$. Since $B(\mathcal{C})$ and $\bar{\delta}_{\mathrm{LF}}$ do not depend on $K$, $\sum_K a_K < \infty$. *q.e.d.*

*Proof of Theorem 1.5. The thresholds.* We sort the requirements on the coupling of Sections 3–12 into the two kinds of Section 1.3. Those of the first kind are finitely many upper bounds on $\bar{g}$ in terms of the constants of the construction and of $\Lambda_*$, among them $\bar{g} \le \gamma'$ of Theorem 4.1, and we let $\gamma_0 > 0$ be the smallest of these bounds. Each requirement of the second kind, for an ensemble with $c_{\mathcal{C}} = c$, bounds a nondecreasing function of $\gamma_q$ from above by a positive quantity which does not increase with $c$, because the constants which it contains are nondecreasing functions of $c_{\mathcal{C}}$ (Definition 5.1(v)); it therefore holds whenever $\gamma_q$ lies below a positive number which does not increase with $c$. We let $\gamma_\sharp(c) > 0$ be the smallest of these numbers, a nonincreasing function of $c$ which depends only on the constants of the construction and on $\Lambda_*$, and we put

$$q_\sharp(c, \bar{g}) \coloneqq \max\Big(0,\ \Big\lceil \frac{\gamma_\sharp(c)^{-2} - \bar{g}^{-2} + C_W}{b} \Big\rceil\Big)\ , \tag{11.28}$$

with the constants $b$ and $C_W$ of Theorem 4.1. This function is nondecreasing in $c$, because $\gamma_\sharp$ does not increase, and nondecreasing in $\bar{g}$; it vanishes as soon as $\bar{g}^{-2} \ge \gamma_\sharp(c)^{-2} + C_W$, and it depends only on the constants of the construction and on $\Lambda_*$. If $q \ge q_\sharp(c, \bar{g})$, then $bq \ge \gamma_\sharp(c)^{-2} - \bar{g}^{-2} + C_W$, and the first inequality of (3.47) gives

$$\gamma_q^{-2} \ge \bar{g}^{-2} + bq - C_W \ge \gamma_\sharp(c)^{-2}\ , \tag{11.29}$$

that is, $\gamma_q \le \gamma_\sharp(c)$.

*The bound and the limit.* Let $\bar{g} \in (0, \gamma_0]$, and let $\mathcal{C}$ be a loop ensemble with admissible loops at a level $q \ge q_\sharp(c_{\mathcal{C}}, \bar{g})$. Then $\bar{g}$ satisfies the requirements of the first kind, and $\gamma_q$ satisfies those of the second kind for $\mathcal{C}$ by (11.29), so that $\bar{g}$ and $q$ are as in Theorem 1.5 for $\mathcal{C}$ in the sense of Section 1.3. The bound of (1.10) is therefore Theorem 8.6 with $K_0 \coloneqq K_0(\mathcal{C})$ and $B \coloneqq B(\mathcal{C})$, and the summability of the differences is Theorem 11.1. A sequence with summable differences is a Cauchy sequence, which gives the limit (1.11). The dependence of $K_0$ and $B$ is read from (8.1), Theorem 5.2, and Theorem 8.6. *q.e.d.*

### 11.8 Several Loops

The proof has been written for an ensemble of $n$ loops throughout, and it remains to say where the number of loops enters. Every comparison above is made per traversal of a bond of the stopping step, per insertion, per cube, or per point of a history, and a cube of side less than the separation $a_k$ of Lemma 3.14 meets one loop. The number of loops therefore enters only through sums over the loops: the leading words through $\lambda_K$, a sum over the loops, because every factor of $\prod_i f^{(i)}$ has modulus at most one; the transports and the insertions through $\mathfrak{b}(\mathcal{C})$, the exponential of a sum over the loops; the wraps through the total perimeter $P_{K'}(\mathcal{C})$ in their constants $A_w$; and the loop constant through the sum over the loops of the single-loop constants and the constants of the cross groups, whose per-step terms are bounded as those of one loop with the total perimeter in place of the perimeter (Corollary 7.11). The cross groups, which contain insertions on two loops, are produced by the same fluctuation integrals, are compared at their birth by the same interpolation, and are carried by the same recursion, since the recursion does not distinguish the species of a group; the only new constants are the counts of the positions of a cross group, which are bounded by the counts of one loop because two loops are at distance at least $a_k$ (Supplement, Proposition S.16(i)). No condition on the

coupling or on $L$ involves $n$; the index $K_1$ involves $\mathcal{C}$ through $B(\mathcal{C})$, the total perimeter, and the thresholds of the single loops.

## 12 Symmetries, Reflection Positivity, and Nonvanishing

The symmetries and the reflection positivity are properties of the lattice measures at every cutoff, and one expects them to pass to the limits without effort, since a limit of equal numbers is an equality and a limit of positive semidefinite matrices is positive semidefinite. The difficulty lies in the normalisation. The renormalised expectations are the expectations divided by the products $\prod_i \kappa_K^{(q)}(C_i, \chi_i)$, so that a symmetry $h$ of the lattice measures survives the division only if $\kappa_K^{(q)}(hC, \chi) = \kappa_K^{(q)}(C, \chi)$ for every loop; and the positivity of a matrix $(\mathbb{E}_K[F_a\, \Theta F_b])_{a,b}$ survives it only if the number by which the entry $(a, b)$ is divided factorises as a number of $a$ times the complex conjugate of the same number of $b$, which for the ensembles $\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b$ means that the normalisation of a reflected loop with the conjugate character must be the complex conjugate of the normalisation of the loop. The coefficients of the normalisation are defined by cluster expansions in which these symmetries are not visible, so we obtain them from the exact values of the flat parts, whose symmetries follow from those of the construction, and from the extraction of the coefficients along dilations of the loop. The construction is symmetric under the reflections in coordinate hyperplanes, but it is not known to be symmetric under the exchange of two axes, and this is where our covariance statements stop. The nonvanishing is then a consequence of the identity at the stopping scale: at small coupling the measure $\mu_K$ has almost full mass, the background is almost flat on its support, and all the corrections tend to zero with $\bar{g}$.

### 12.1 Symmetries and Reflection Positivity on the Lattice

A symmetry $h$ of level $q$ (Section 1.3) acts on the bonds of $\mathbb{T}_K$, $K \geq q$, by $h(x, y) \coloneqq (hx, hy)$, on paths bond by bond, and on gauge fields by $(hU)(b) \coloneqq U(hb)$, so that $(hU)(\gamma) = U(h\gamma)$ for every path $\gamma$. For a function $F$ of the gauge field we put $(F \circ h)(U) \coloneqq F(hU)$; in particular $W_\chi(C) \circ h = W_\chi(hC)$ for a loop $C$ at level $q$, and $W_{\mathcal{C}} \circ h = W_{h\mathcal{C}}$ for an ensemble.

**Lemma 12.1** (Invariance of the Lattice Measures)**.** *Let $K \geq q$ and let $h$ be a symmetry of level $q$. For every bounded measurable function $F$ of the gauge field on $\mathbb{T}_K$, $\mathbb{E}_K[F \circ h] = \mathbb{E}_K[F]$ and $\mathbb{E}_K[\bar{F}] = \overline{\mathbb{E}_K[F]}$.*

*Proof.* We write $s_U(p) \coloneqq 1 - \operatorname{Re}\operatorname{tr} U(\partial p)$. The holonomy $(hU)(\partial p) = U(h\partial p)$ is that of $U$ around the plaquette $hp$, from one of its corners and in one of its two orientations, and Re tr is invariant under conjugation and, on unitary matrices, under inversion; hence $s_{hU}(p) = s_U(hp)$. Since $h$ permutes the plaquettes, the action of (1.2) is invariant under $U \mapsto hU$. The map $h$ permutes the unoriented bonds, and the value of $hU$ on a positive bond $b$ is the value of $U$ on $hb$ if $hb$ is positive and the inverse of its value on $(hb)^{-1}$ otherwise; since the Haar measure is invariant under inversion, the bijection $U \mapsto hU$ preserves the product Haar measure. The change of variables $U \mapsto hU$ in the integral defining $\mathbb{E}_K[F \circ h]$ therefore gives the first assertion, and the second holds because $\mu_{K,g_0}$ is a positive measure. *q.e.d.*

For the reflection positivity we use the reflections of level $q$, for which the positivity of the Wilson action was observed by Osterwalder and Seiler [16]; we give the short proof, since it is the basis of everything in this section. Let $\mu \in \{0, \dots, 3\}$ and $a \in uL^{-q}\mathbb{Z}$. Since $a$ and $T/2 = L^{m_T} u$ are multiples of $\varepsilon_K$ for $K \geq q$, the reflection $\theta_{\mu,a}$ is an involution of the

sites of $\mathbb{T}_K$, which fixes the two hyperplanes $P_{\mu,a} \coloneqq \{x_\mu \equiv a\} \cup \{x_\mu \equiv a + T/2\}$ pointwise and exchanges the open half-tori $H_{\mu,a} \coloneqq \{a < x_\mu < a + T/2\}$ and $\theta_{\mu,a} H_{\mu,a}$. For a bounded measurable function $F$ of the gauge field we put $(\Theta_{\mu,a} F)(U) \coloneqq \overline{F(\theta_{\mu,a} U)}$; the map $\Theta_{\mu,a}$ is antilinear and multiplicative.

Let $\bar\Lambda_+$ be the set of the sites with $x_\mu - a \in [0, T/2]$ modulo $T$, and $\bar\Lambda_- \coloneqq \theta_{\mu,a}\bar\Lambda_+$, so that $\bar\Lambda_+ \cap \bar\Lambda_- = P_{\mu,a}$. Let $E_0$ be the set of the bonds with both ends in $P_{\mu,a}$, $E_+$ the set of the other bonds with both ends in $\bar\Lambda_+$, and $E_- \coloneqq \theta_{\mu,a} E_+$. Every bond belongs to exactly one of these sets: a bond orthogonal to $e_\mu$ has both ends on one hyperplane $\{x_\mu = \text{const}\}$, and a bond along $e_\mu$ joins two neighbouring hyperplanes, which lie both in $\bar\Lambda_+$ or both in $\bar\Lambda_-$, because the two hyperplanes of $P_{\mu,a}$ are hyperplanes of $\mathbb{T}_K$ at distance $T/2 \geq 2\varepsilon_K$. The bonds of $E_0$ are orthogonal to $e_\mu$, so $\theta_{\mu,a}$ fixes each of them with its orientation. We write $\mathcal{A}_+$ for the set of the bounded measurable functions of the restriction of $U$ to $E_+ \cup E_0$; for $F \in \mathcal{A}_+$ the function $\Theta_{\mu,a} F$ depends on the restriction to $E_- \cup E_0$.

**Lemma 12.2** (Reflection Positivity on the Lattice)**.** *Let $K \geq q$, $\mu \in \{0, \dots, 3\}$, and $a \in uL^{-q}\mathbb{Z}$. For every bounded measurable $F$, $\mathbb{E}_K[\Theta_{\mu,a} F] = \overline{\mathbb{E}_K[F]}$, and for every $F \in \mathcal{A}_+$, $\mathbb{E}_K[F\, \Theta_{\mu,a} F] \geq 0$.*

*Proof.* We write $\theta$ for $\theta_{\mu,a}$, which is a symmetry of level $q$, and $s_U(p) \coloneqq 1 - \operatorname{Re}\operatorname{tr} U(\partial p)$ as in the proof of Lemma 12.1. That lemma, applied to $\bar F$, gives $\mathbb{E}_K[\Theta F] = \mathbb{E}_K[\bar F \circ \theta] = \mathbb{E}_K[\bar F] = \overline{\mathbb{E}_K[F]}$, which is the first assertion.

For the second, a plaquette lies in $\bar\Lambda_+$ or in $\bar\Lambda_-$, by the argument given for the bonds, and in both exactly when it lies in $P_{\mu,a}$. Put

$$B(U) \coloneqq \sum_{p \subset \bar\Lambda_+,\ p \not\subset P_{\mu,a}} s_U(p) + \frac{1}{2} \sum_{p \subset P_{\mu,a}} s_U(p)\,, \tag{12.1}$$

a real function in $\mathcal{A}_+$. The reflection maps the plaquettes of $\bar\Lambda_+$ which do not lie in $P_{\mu,a}$ bijectively onto those of $\bar\Lambda_-$, and each plaquette of $P_{\mu,a}$ onto itself, so that, since $s_{\theta U}(p) = s_U(\theta p)$, $B(\theta U)$ is the sum (12.1) with $\bar\Lambda_-$ in place of $\bar\Lambda_+$, and the action is $B + B \circ \theta$. With $f \coloneqq F e^{-g_0^{-2} B} \in \mathcal{A}_+$ and since $B$ is real, the integrand of $\mathbb{E}_K[F\, \Theta F]$ against the Haar measure is therefore

$$F(U)\overline{F(\theta U)}\, e^{-g_0^{-2}(B(U) + B(\theta U))} = f(U)\, \overline{f(\theta U)}\,. \tag{12.2}$$

Write $U = (U_+, U_0, U_-)$ for the restrictions to $E_+$, $E_0$, and $E_-$. The restriction of $\theta U$ to $E_+ \cup E_0$ is $(\varphi(U_-), U_0)$, where $\varphi(U_-)(b) \coloneqq U(\theta b)$ for $b \in E_+$ defines a map $\varphi\colon G^{E_-} \to G^{E_+}$ which maps the product Haar measure to the product Haar measure, as in the proof of Lemma 12.1. By Fubini's theorem, the integral of (12.2) is

$$\int \Big( \int f(U_+, U_0)\, \mathrm{d}U_+ \Big) \overline{\Big( \int f(\varphi(U_-), U_0)\, \mathrm{d}U_- \Big)}\, \mathrm{d}U_0 = \int \Big| \int f(U_+, U_0)\, \mathrm{d}U_+ \Big|^2 \mathrm{d}U_0 \geq 0\,, \tag{12.3}$$

and $\mathbb{E}_K[F\, \Theta F]$ is this number divided by $Z_{K,g_0} > 0$. *q.e.d.*

The lemma makes $\langle F, G \rangle \coloneqq \mathbb{E}_K[F\, \Theta_{\mu,a} G]$ a positive semidefinite sesquilinear form on $\mathcal{A}_+$, linear in $F$ and antilinear in $G$; such a form is Hermitian, by polarisation. If $F \in \mathcal{A}_+$ has the mean $c \coloneqq \mathbb{E}_K[F]$, then the constant $c$ lies in $\mathcal{A}_+$, $\mathbb{E}_K[\Theta_{\mu,a} F] = \bar c$ by the lemma, and

$$0 \leq \langle F - c, F - c \rangle = \mathbb{E}_K[F\, \Theta_{\mu,a} F] - |\mathbb{E}_K[F]|^2\,. \tag{12.4}$$

The Wilson loops behave as follows. Let $C$ be a loop whose sites lie in $H_{\mu,a}$, with the character $\chi$ of the representation $\pi$. Its bonds have both ends in $H_{\mu,a}$, hence lie in $E_+$,

and $W_\chi(C) \in \mathcal{A}_+$. Since $\pi$ is unitary, $\overline{\chi(g)} = \bar\chi(g)$ for the character $\bar\chi$ of the conjugate representation, which is irreducible and unitary, and therefore

$$(\Theta_{\mu,a} W_\chi(C))(U) = \overline{\chi(U(\theta_{\mu,a} C))} = W_{\bar\chi}(\theta_{\mu,a} C; U) \,, \tag{12.5}$$

where $\theta_{\mu,a} C$ is the image of $C$ with the image orientation, a loop at level $q$ with the same chord–arc constant, ratio $P/r$, and number $P_q$ as $C$ (Section 1.3). For an ensemble $\mathcal{D}$ all of whose loops lie in $H_{\mu,a}$, the ensemble $\theta_{\mu,a}\bar{\mathcal{D}}$ consists of the images with the conjugate characters, and by multiplicativity $\Theta_{\mu,a} W_{\mathcal{D}} = W_{\theta_{\mu,a}\bar{\mathcal{D}}}$.

## 12.2 The Covariance of the Normalisation

The division by the normalisations keeps the symmetries of the lattice measures only where the normalisations share them, and the following proposition says where they do.

**Proposition 12.3** (Covariance of the Normalisation)**.** *Let $L$ and $\Lambda_*$ be as in Theorem 1.5, let $C$ be an admissible loop at level $q$ with the character $\chi$ of a representation $\pi$, and let $\bar g$ and $q$ be as in Theorem 1.5 for $((C,\chi))$. For every $K > j_q$ and every $g \in \mathfrak{G}_q$,*

$$\kappa_K^{(q)}(gC,\chi) = \kappa_K^{(q)}(C,\chi) \,, \qquad \kappa_K^{(q)}(C,\bar\chi) = \overline{\kappa_K^{(q)}(C,\chi)} \,, \qquad \kappa_K^{(q)}(C^{-1},\chi) = \kappa_K^{(q)}(C,\bar\chi) \,. \tag{12.6}$$

*In particular $\kappa_K^{(q)}(\theta_{\mu,a} C,\bar\chi) = \overline{\kappa_K^{(q)}(C,\chi)}$ for every $\mu$ and every $a$ with $2a \in uL^{-q}\mathbb{Z}$.*

For the translations in $\mathfrak{G}_q$ the first identity can be read off from Proposition 7.13, since a translate of $C$ has the same numbers $N_v^{(0)}$ and the same pairs $(v_c, w_c)$ at its corners as $C$. The structure of the normalisation does not help with the reflections $\theta_\mu$, nor with the other two identities: the loops $\theta_\mu C$ and $C^{-1}$ have different numbers of segments of a given orientation and different pairs at their corners, and the coefficients of $\chi$ and of $\bar\chi$ are different numbers, while nothing relates the coefficients of different orientations, or of conjugate characters, to each other. The coefficients of Definition 7.8 are sums of cluster weights, and (12.6) would follow from symmetries of the individual weights; but the expansions which define the weights involve many choices whose behaviour under the reflections we do not wish to follow. We use instead the exact values of the flat parts, which do not depend on any choice. For a torus $T' \in \mathfrak{T}$, a loop $C$ at level $q$ on $T'$, and $0 \le j \le K'$, let $\widehat{\mathcal{W}}_j(C,\chi;T')$ be the value at the reference data of the dressed loop of the step $j$ of the single-loop ensemble $((C,\chi))$ on the trivial term, with $\widehat{\mathcal{W}}_0 := 1$, whenever this ensemble is dressed at the step $j$.

**Lemma 12.4** (Flat Values under the Symmetries)**.** *Let $T' \in \mathfrak{T}$, let $C$ be a loop at level $q$ on $T'$ with the character $\chi$, let $1 \le j \le K'$, and let $\mu \in \{0,\dots,3\}$. If the ensemble $((C,\chi))$ is dressed at the steps $1,\dots,j$, then*

$$\widehat{\mathcal{W}}_j(C,\chi;T') = \kappa^{(j)}(C,\chi)\, e^{E_{\mathrm{const},j}(C,\chi;T')} \,, \tag{12.7}$$

*where $E_{\mathrm{const},j}(C,\chi;T')$ is the constant of the scalar exponent of the step $j$ on the trivial term, and*

$$\widehat{\mathcal{W}}_j(\theta_\mu C,\chi;T') = \widehat{\mathcal{W}}_j(C,\chi;T') \,, \qquad \widehat{\mathcal{W}}_j(C,\bar\chi;T') = \overline{\widehat{\mathcal{W}}_j(C,\chi;T')} \,, \qquad \widehat{\mathcal{W}}_j(C^{-1},\chi;T') = \widehat{\mathcal{W}}_j(C,\bar\chi;T') \,, \tag{12.8}$$

*each identity holding whenever the single-loop ensembles which occur in it are dressed at the steps $1,\dots,j$.*

*Proof.* At the reference data the leading word is $\chi(1) = 1$, and the insertions of the nonempty families vanish: after the exact factorisation (7.17), and after its counterpart at the first step in the proof of Theorem 5.2, the insertions of the families are the parts $Y^{\mathrm{d}}_{\mathcal{G}}$, which vanish at the reference data, because for a single loop every insertion takes a scalar value there, by Schur's lemma. On the trivial term the scalar exponent is the number $E_{\mathrm{const},j}$, as in the proof of Theorem 8.1. This gives (12.7).

For (12.8) we express the flat values through the construction itself. The reference data of the trivial term are those at the block field $V_j = 1$, which lies in the support of all its test functions, and on the trivial term the operation $\mathbf{T}_j$ holds no integrations. Let $N_j[F]$ be the density obtained from $F\varrho_0$ by the operations of the first $j$ steps on $T'$, and $N_j[F](1;\ell_0)$ the value of its term $\ell_0$ at $V_j = 1$. Since the operations which produce the terms of the numerator are those which produce the terms of the density, applied to a function which carries one more factor, Definition 5.1 gives

$$\widehat{\mathcal{W}}_j(C,\chi;T') = \frac{N_j[W_\chi(C)](1;\ell_0)}{N_j[1](1;\ell_0)} \,, \tag{12.9}$$

where $N_j[1](1;\ell_0) = \exp A_j(1) > 0$. The right side depends on the loop and its character only through the function $W_\chi(C)$. The map $F \mapsto N_j[F](1;\ell_0)$ has two properties, which are proved in Supplement, Proposition S.18. It is *real*: $N_j[\bar{F}](1;\ell_0) = \overline{N_j[F](1;\ell_0)}$, because at real fields every operation of the construction is a sum, a multiplication by a real function, or an integral against a real kernel (Haar and Gaussian measures, $\delta$-functions of the block averaging, test functions, and exponentials of real actions [9, (3.27), p. 271], [12, p. 380]). And it is *invariant under each* $\theta_\mu$: $N_j[F\circ\theta_\mu](1;\ell_0) = N_j[F](1;\ell_0)$, because $\theta_\mu$ maps every lattice and every partition of the construction to itself, the partitions being centred at the origin (Section 7.1), maps the trivial term and the trivial field to themselves, and commutes with every operation: with the averages, which Bałaban defines so as to be "symmetric with respect to lattice Euclidean transformations" [8, p. 252], with the gauge fixing, introduced "to preserve the Euclidean invariance" [9, p. 247], with the fluctuation integral [8, (2.17), (2.18), p. 269], whose fluctuation fields are written first in an axial gauge [6, (1.19), p. 79] with contours which start at the centre of a block and change the coordinates in a fixed order [2, (1.7), p. 18], an order which the reflection of one coordinate about the centre does not change, with the soft test functions, which are functions of plaquette deviations in norms invariant under conjugation and inversion, and with the **R**-operation, whose gauge-fixing graphs are averaged over "$2^d$ reflections in subsets of coordinates" [11, (1.100), p. 201], [12, p. 379].

The three identities now follow from (12.9). Since $W_\chi(C)\circ\theta_\mu = W_\chi(\theta_\mu C)$, the invariance gives $N_j[W_\chi(\theta_\mu C)](1;\ell_0) = N_j[W_\chi(C)](1;\ell_0)$. Since $\pi$ is unitary, $\overline{\chi(V)} = \chi(V^{-1}) = \bar{\chi}(V)$ for $V \in G$, so that $W_{\bar\chi}(C) = \overline{W_\chi(C)}$, and the reality gives $N_j[W_{\bar\chi}(C)](1;\ell_0) = \overline{N_j[W_\chi(C)](1;\ell_0)}$. Finally $U(C^{-1}) = U(C)^{-1}$, so that $W_\chi(C^{-1}) = \chi(U(C)^{-1}) = \bar\chi(U(C)) = W_{\bar\chi}(C)$ is the same function as $W_{\bar\chi}(C)$. Dividing by the positive number $N_j[1](1;\ell_0)$ gives (12.8). *q.e.d.*

*Proof of Proposition 12.3.* We first reduce the first identity to the reflections $\theta_\mu$. Every $g \in \mathfrak{G}_q$ is a composition $g = g_1\cdots g_r$ of translations by vectors of $uL^{-q}\mathbb{Z}^4$ and reflections $\theta_\mu$. Each loop $g_{i+1}\cdots g_r C$ is an admissible loop at level $q$ with the character $\chi$ (Section 1.3), so that $\bar g$ and $q$ are as in Theorem 1.5 for it, the conditions of that theorem involving the loop only through $c_\pi$ and $\Lambda_*$; and the normalisation of a loop is unchanged by a translation in $\mathfrak{G}_q$, as noted after the proposition. It therefore suffices to prove the first identity for $g = \theta_\mu$, and the last assertion follows from the first two identities, since $\theta_{\mu,a} \in \mathfrak{G}_q$ when $2a \in uL^{-q}\mathbb{Z}$.

The three remaining identities have a common form. Let $(C',\chi',\chi^\circ,\epsilon)$ be one of the three

quadruples

$$(\theta_\mu C, \chi, \chi, \mathsf{id}) \,, \qquad (C, \bar\chi, \chi, \mathsf{conj}) \,, \qquad (C^{-1}, \chi, \bar\chi, \mathsf{id}) \,, \tag{12.10}$$

where $\mathsf{id}$ is the identity of $\mathbb{C}$ and $\mathsf{conj}$ the complex conjugation, so that the identity to be proved reads $\kappa_K^{(q)}(C', \chi') = \epsilon(\kappa_K^{(q)}(C, \chi^\circ))$. The pairs $(C, \chi^\circ)$ and $(C', \chi')$ satisfy the hypotheses of the proposition, because $c_{\bar\pi} = c_\pi$ and $\Lambda(C') = \Lambda(C)$. It suffices to prove $\kappa_j(C', \chi') = \epsilon(\kappa_j(C, \chi^\circ))$ for every step $1 \le j \le K'$, and since $\kappa_j$ is the exponential of (7.23) and $\epsilon$ commutes with the exponential, it suffices to prove that

$$D_j(C) \coloneqq \log \kappa_j(C', \chi') - \epsilon(\log \kappa_j(C, \chi^\circ)) \in 2\pi i \mathbb{Z} \,. \tag{12.11}$$

Fix $j$. Let $c_0$ be a corner of $C$, and for an integer $m \ge 1$ let $C_m$ be the image of $C$ under the dilation $x \mapsto c_0 + m(x - c_0)$, read on a torus $T'_m \in \mathfrak{T}$ whose period we choose below. The path $C_m$ visits no site twice, its corners lie in $c_0 + uL^{-q}\mathbb{Z}^4$, its segments have the orientations of those of $C$ and $m$ times their lengths, and its corners have the same pairs $(v_c, w_c)$ as those of $C$. Hence $C_m$ is a loop at level $q$ on $T'_m$ as soon as $3\,\mathrm{diam}(C_m) < T'_m$, with $\Lambda(C_m) = \Lambda(C)$, $P(C_m)/r(C_m) = P(C)/r(C)$, $P_{j-1}(C_m) = mP_{j-1}(C)$, and $r_{j-1}(C_m) = mr_{j-1}(C)$. We choose the period of $T'_m$ so large that, in addition, $K \ge K_{\mathrm{tor}}$ for $C_m$ on $T'_m$, which is possible because the torus conditions of the first step read $(P_0(C_m)/L)e^{-c\iota_{T'_m}} \le 1$ in the proof of Theorem 5.2, and that $mP_{j-1}(C)e^{-c_2\iota_{T'_m,j}} \le 1/m$, with the constant $c_2 > 0$ of (12.15) below. Let $C'_m$ be obtained from $C_m$ as $C'$ is obtained from $C$, that is, $C'_m$ is $\theta_\mu C_m$, $C_m$, or $C_m^{-1}$; it is the dilation of $C'$ by the factor $m$ about $\theta_\mu c_0$, $c_0$, or $c_0$, since $\theta_\mu$ is linear. Then the ensembles $((C_m, \chi^\circ))$ and $((C'_m, \chi'))$ are dressed at every step up to $K'$, by Theorems 5.2 and 6.1, applied on $T'_m$; their hypotheses hold because $\Lambda(C'_m) = \Lambda(C_m) = \Lambda(C)$, because $C'_m$ has the same numbers $P_i$ and $r_i$ as $C_m$ at every step $i$, so that the torus conditions hold for both, and because $c_{\bar\pi} = c_\pi$, so that $\bar g$ and $q$ are as in Theorem 1.5 for both ensembles.

By (7.23), $\log \kappa_j(C_m, \chi^\circ)$ is an affine function of $m$, and so is $\log \kappa_j(C'_m, \chi')$, because $C'_m$ is a dilation of $C'$ by the factor $m$. Hence

$$D_j(C_m) = mA + B \tag{12.12}$$

with numbers $A$ and $B$ which do not depend on $m$, and $D_j(C) = A + B$. On the other hand, by (12.7) at the steps $j$ and $j-1$, for a single loop on the trivial term,

$$\frac{\widehat{\mathcal{W}}_j(C_m, \chi^\circ; T'_m)}{\widehat{\mathcal{W}}_{j-1}(C_m, \chi^\circ; T'_m)} = \exp(\log \kappa_j(C_m, \chi^\circ) + E_j(C_m, \chi^\circ; T'_m)) \,, \tag{12.13}$$

where $E_j \coloneqq E_{\mathrm{const},j} - E_{\mathrm{const},j-1}$ is the remainder of the step, which for a single loop is $E_j^\circ + \mathbf{1}_{\mathbf{R}}(j) E_j^{\circ,\mathbf{R}}$ by (7.30) at $j \ge 2$, and the remainder bounded in (5.10) and by the periodic comparison of [1, Appendix F] at $j = 1$. The same identity holds for $(C'_m, \chi')$, and by (12.8) at the steps $j$ and $j-1$ the left side for $(C'_m, \chi')$ is the image under $\epsilon$ of the left side for $(C_m, \chi^\circ)$. Since $\epsilon$ commutes with the exponential, taking exponentials apart gives

$$D_j(C_m) \in 2\pi i \mathbb{Z} + \epsilon(E_j(C_m, \chi^\circ; T'_m)) - E_j(C'_m, \chi'; T'_m) \,. \tag{12.14}$$

The remainders tend to zero as $m \to \infty$. Indeed, by (7.25) and (7.29) at $j \ge 2$, and by (5.10) and the periodic comparison of [1, Appendix F] at $j = 1$, and since $P_j \le P_{j-1}$ and $r_{j-1} \ge r_j$, each of them is at most a constant times

$$(mP_{j-1}(C) + |\mathcal{K}(C)|)e^{-c_1 m r_j(C)} + mP_{j-1}(C)\, e^{-c_2 \iota_{T'_m,j}} \,, \tag{12.15}$$

with positive constants $c_1$ and $c_2$ which depend on $j$ but not on $m$ (at $j \ge 2$ the smaller of $2c_E^T$ and $c_E^R/(MR_j)$, and of $c^Z$ and $c_E^R/(MR_j)$), and with the constants of $\chi$ and $\bar\chi$; the first

term tends to zero because $r_j(C) > 0$, and the second is at most $1/m$ by the choice of $T'_m$. Let $\delta_m$ be the modulus of the difference of the two remainders in (12.14), so that $\delta_m \to 0$. By (12.12), $A = D_j(C_{m+1}) - D_j(C_m)$ lies within the distance $\delta_m + \delta_{m+1}$ of $2\pi i\mathbb{Z}$ for every $m$, and since $2\pi i\mathbb{Z}$ is closed, $A \in 2\pi i\mathbb{Z}$. Then $B = D_j(C_m) - mA$ lies within the distance $\delta_m$ of $2\pi i\mathbb{Z}$ for every $m$, and $B \in 2\pi i\mathbb{Z}$. Hence $D_j(C) = A + B \in 2\pi i\mathbb{Z}$, which is (12.11). *q.e.d.*

The argument determines the logarithms of the normalisations only modulo $2\pi i$, which is all that (12.6) asks for; it does not show that the coefficients themselves are real, nor that they are invariant under the reflections, since the dilations of one loop see only the combinations of the coefficients which occur in that loop.

### 12.3 Proof of the Symmetries

*Proof of Corollary 1.6.* Each of the ensembles $\mathcal{C}'$ of the statement, namely $\mathcal{C}$ with its loops in another order, $g\mathcal{C}$, $\bar{\mathcal{C}}$, $\mathcal{C}$ with one loop reversed and its character conjugated, and $h\mathcal{C}$, consists of admissible loops at level $q$ which are pairwise vertex-disjoint, the maps $g$ and $h$ being bijections, and has $c_{\mathcal{C}'} = c_{\mathcal{C}}$, because $c_{\bar{\pi}} = c_\pi$. So $q \geq q_\sharp(c_{\mathcal{C}'}, \bar{g})$, the limit $W(\mathcal{C}')$ exists by Theorem 1.5, and $\bar{g}$ and $q$ are as in that theorem for every single loop of $\mathcal{C}$ with its character, since $c_{\pi_i} \leq c_{\mathcal{C}}$ and $q_\sharp$ is nondecreasing. It suffices to prove the identities for $W_K$ at a fixed $K > j_q$.

(i) Both the observable (1.4) and the product of the normalisations in (1.6) are products over the loops, which do not depend on the order of the factors.

(ii) By Lemma 12.1, $\mathbb{E}_K[W_{g\mathcal{C}}] = \mathbb{E}_K[W_{\mathcal{C}} \circ g] = \mathbb{E}_K[W_{\mathcal{C}}]$, and by Proposition 12.3, $\kappa_K^{(q)}(gC_i, \chi_i) = \kappa_K^{(q)}(C_i, \chi_i)$ for every $i$. Dividing gives $W_K(g\mathcal{C}) = W_K(\mathcal{C})$.

(iii) Since $W_{\bar{\chi}_i}(C_i) = \overline{W_{\chi_i}(C_i)}$, as in the proof of Lemma 12.4, Lemma 12.1 gives $\mathbb{E}_K[W_{\bar{\mathcal{C}}}] = \overline{\mathbb{E}_K[W_{\mathcal{C}}]}$, and Proposition 12.3 gives $\kappa_K^{(q)}(C_i, \bar{\chi}_i) = \overline{\kappa_K^{(q)}(C_i, \chi_i)}$; dividing gives $W_K(\bar{\mathcal{C}}) = \overline{W_K(\mathcal{C})}$. If $C_i$ is replaced by $C_i^{-1}$ and $\chi_i$ by $\bar{\chi}_i$, the observable does not change, since $W_{\bar{\chi}_i}(C_i^{-1}) = W_{\chi_i}(C_i)$, and neither does the normalisation, since $\kappa_K^{(q)}(C_i^{-1}, \bar{\chi}_i) = \kappa_K^{(q)}(C_i, \chi_i)$ by the third identity of (12.6) applied to $(C_i, \bar{\chi}_i)$.

(iv) By Lemma 12.1, $\mathbb{E}_K[W_{h\mathcal{C}}] = \mathbb{E}_K[W_{\mathcal{C}}]$, and by Proposition 12.3, $\kappa_K^{(q)}(hC_i, \chi_i) = \kappa_K^{(q)}(g_iC_i, \chi_i) = \kappa_K^{(q)}(C_i, \chi_i)$ for every $i$. *q.e.d.*

The proof of (iv) shows what is missing for a general symmetry $h$ which exchanges two axes. Writing $h(x) = Ox + t$, the image $hC$ has $N_v^{(0)}(hC) = N_{O^{-1}v}^{(0)}(C)$ and the pairs $(Ov_c, Ow_c)$ at its corners, so that Lemma 12.1 and Proposition 7.13 give

$$\frac{W_K(h\mathcal{C})}{W_K(\mathcal{C})} = \prod_{i=1}^{n} \exp\Bigl(\sum_{v\in\mathcal{V}} N_v^{(0)}(C_i)(m_{v,K} - m_{Ov,K}) + \sum_{c\in\mathcal{K}(C_i)} (z_{v_c,w_c,K} - z_{Ov_c,Ow_c,K})\Bigr) \quad (12.16)$$

whenever $W_K(\mathcal{C}) \neq 0$, with the coefficients of the characters $\chi_i$. The covariance of the limits under $h$ therefore follows from the equality of the coefficients of the orientations $v$ and $Ov$ and of the corner pairs $(v, w)$ and $(Ov, Ow)$, and the method of Proposition 12.3 would give it from the invariance $N_j[F \circ h](1; \ell_0) = N_j[F](1; \ell_0)$ of the flat values. We do not know whether the construction has this invariance. The averages, the first gauge fixing, the designated bonds [8, p. 267, ll. 19–20], and the **R**-operation, whose graphs are averaged over the permutations of the coordinates as well [11, p. 201, ll. 34–38], are symmetric under the exchange of two axes; but the fluctuation fields of the steps $k \geq 2$ are written in the axial gauge of [6, (1.19), p. 79],

whose contours change the coordinates in one fixed order [2, (1.7), p. 18], and the Landau gauge into which they are then transformed is defined relative to this preliminary choice, since the restricted gauge transformations do not form a group and the orbit "depends on an element $U$ chosen to generate the set" [6, pp. 80–81]. The test functions of the fluctuation fields [9, (3.16), p. 268], and with them the term $\ell_0$, are therefore not known to be invariant under the exchange of two axes. Bałaban states the covariance of his effective actions under every Euclidean transformation which preserves the lattices [8, p. 269, ll. 9–26], [9, (3.39), p. 274], but the operations from which we would derive the invariance of the numerator include this axial gauge, and we have not found an argument which removes it. An axial gauge in the averaged variables, as for the first gauge fixing, would give the symmetry, at the price of proving Bałaban's existence, uniqueness, and bounds for the Landau gauge again from it.

The positivity has no counterpart for the reflections which exchange two axes, for a reason which has nothing to do with the normalisations. Such a reflection is $x \mapsto Ox + t$ with $O$ the reflection in the hyperplane orthogonal to $e_\mu - e_\nu$ or to $e_\mu + e_\nu$. Its fixed points, if there are any, satisfy one equation $x_\mu \mp x_\nu \equiv c$ modulo $T$, so that they form a single hyperplane of $\mathbb{T}$, whose complement is connected: it is the image of the connected set $(0,T) \times (\mathbb{R}/T\mathbb{Z})^3$ under the continuous map which sends $(s,y)$ to the point whose coordinates other than $x_\mu$ are those of $y$ and for which $x_\mu \mp x_\nu = c + s$. There are therefore no two half-tori which the reflection exchanges and whose union with the fixed points is the torus, and the proof of Lemma 12.2 has nothing to apply to.

### 12.4 Proof of the Reflection Positivity

*Proof of Corollary 1.7.* Let $\theta = \theta_{\mu,s}$ with $s \in uL^{-q}\mathbb{Z}$. The two half-tori bounded by its fixed points are $H_{\mu,s}$ and $H_{\mu,s+T/2}$, and $\theta_{\mu,s+T/2} = \theta_{\mu,s}$, because $2s + T \equiv 2s$ modulo $T$; since $T/2 = L^{m_T} u \in uL^{-q}\mathbb{Z}$, we may therefore assume $H_+ = H_{\mu,s}$, and we write $\Theta \coloneqq \Theta_{\mu,s}$ and $H_- \coloneqq \theta H_+$. For $0 \le a, b \le m$ the family $\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b$ is a loop ensemble at level $q$ with admissible loops: the loops of $\mathcal{D}_a$ lie in $H_+$ and those of $\theta\bar{\mathcal{D}}_b$ in $H_-$, which is disjoint from $H_+$, so that they are pairwise vertex-disjoint, and the images are admissible loops at level $q$ by Section 1.3. Its number $c_{\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b} = \max(c_{\mathcal{D}_a}, c_{\mathcal{D}_b})$ is at most $c$, because $c_{\theta\bar{\mathcal{D}}_b} = c_{\mathcal{D}_b}$, so that $q \ge q_\sharp(c, \bar{g}) \ge q_\sharp(c_{\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b}, \bar{g})$. By Theorem 1.5 the limits $W(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b)$ exist, with $W_K(\emptyset) = 1$ for every $K$.

Fix $K > j_q$, write $F_a \coloneqq W_{\mathcal{D}_a}$, with $F_0 \coloneqq 1$, and $\kappa_a$ for the product of the normalisations $\kappa_K^{(q)}$ of the loops of $\mathcal{D}_a$, with $\kappa_0 \coloneqq 1$. By the last assertion of Proposition 12.3, which applies to each loop of $\mathcal{D}_b$ because its number $c_\pi \le c$ gives $q \ge q_\sharp(c_\pi, \bar{g})$, the product of the normalisations of the loops of $\theta\bar{\mathcal{D}}_b$ is $\bar{\kappa}_b$. Since $F_a \in \mathcal{A}_+$ and $\Theta F_b = W_{\theta\bar{\mathcal{D}}_b}$ by Section 12.1, the functions $G_a \coloneqq F_a/\kappa_a \in \mathcal{A}_+$ satisfy, by Definition 1.3,

$$W_K(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) = \frac{\mathbb{E}_K[F_a\, \Theta F_b]}{\kappa_a \bar{\kappa}_b} = \mathbb{E}_K[G_a\, \Theta G_b] = \langle G_a, G_b \rangle \;, \tag{12.17}$$

because $\Theta$ is antilinear. For $\xi \in \mathbb{C}^m$ we obtain $\sum_{a,b} \xi_a \bar{\xi}_b W_K(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) = \langle \sum_a \xi_a G_a, \sum_b \xi_b G_b \rangle \ge 0$ by Lemma 12.2, and the matrix is Hermitian because the form is. Letting $K \to \infty$ entry by entry, the limit matrix is Hermitian and positive semidefinite, which is (i).

For (ii), $W_K(\theta\bar{\mathcal{D}}_a) = \mathbb{E}_K[\Theta F_a]/\bar{\kappa}_a = \overline{\mathbb{E}_K[F_a]/\kappa_a} = \overline{W_K(\mathcal{D}_a)}$ by Lemma 12.2, and the limit gives $W(\theta\bar{\mathcal{D}}_a) = \overline{W(\mathcal{D}_a)}$. The mean of $G_a$ is $w_a \coloneqq W_K(\mathcal{D}_a)$, and the mean of $\Theta G_b$ is $\bar{w}_b$, so that

$$\langle G_a - w_a, G_b - w_b \rangle = W_K(\mathcal{D}_a \cup \theta\bar{\mathcal{D}}_b) - w_a \bar{w}_b \;, \tag{12.18}$$

and the matrix of the right sides is Hermitian and positive semidefinite by the same argument; its limit is the matrix of (ii). *q.e.d.*

### 12.5 Nonvanishing

The connected functions of Corollary 1.7(ii), and the logarithms of Theorem 1.8, are useful only if the limits do not vanish, and we now show that they do not, on the ensembles whose loops are not too long for the coupling.

**Proposition 12.5** (Nonvanishing)**.** *Let $L$ and $\Lambda_*$ be as in Theorem 1.5, and let $q \geq 0$. There are $\bar g_1 \in (0, \gamma_0]$ and a function $\ell\colon [0,\infty) \times (0, \bar g_1] \to [0,\infty]$, nonincreasing in its first argument and with $\ell(c, \bar g) \to \infty$ as $\bar g \to 0$ for every $c$, which depend only on the constants of the construction, $\Lambda_*$, $q$, and $T/u$, such that the following holds for every $\bar g \in (0, \bar g_1]$. If $\mathcal{C}$ is a loop ensemble at level $q$ whose loops are admissible and whose lengths satisfy*

$$\sum_{i=1}^{n} \Big( P_q(C_i)^2 + \frac{P(C_i)}{r(C_i)} \Big) \leq \ell(c_{\mathcal{C}}, \bar g) \,, \tag{12.19}$$

*then $q \geq q_\sharp(c_{\mathcal{C}}, \bar g)$, the limit $W(\mathcal{C})$ exists, and, for every $K \geq K_0(\mathcal{C})$,*

$$|W_K(\mathcal{C})| \geq \frac{1}{2} \,, \qquad \text{so that} \qquad |W(\mathcal{C})| \geq \frac{1}{2} \,. \tag{12.20}$$

The condition (12.19) bounds the lengths of the loops by a number which depends on their representations through $c_{\mathcal{C}}$ and grows without limit as the coupling decreases. Unlike the conditions of Theorem 1.5, it does restrict the representations at a given coupling: since a loop has at least four bonds, it fails whenever $\ell(c_{\mathcal{C}}, \bar g) < 16$. It involves the number of loops only through the sum, and for the ensembles which are generated from a fixed one by reflections in coordinate hyperplanes it is needed only for the ensemble from which they are generated (Proposition 12.7).

For the lower bound we need that the Wilson loop of a nearly flat field is close to one, with a bound in terms of the length of the loop and the size of a box containing it.

**Lemma 12.6** (Holonomy of a Nearly Flat Field)**.** *Let $U$ be a $G$-valued field on a periodic lattice of spacing $\eta$, let $B$ be a box of side $D\eta$ in it, with $D \geq 1$ an integer smaller than every period in lattice units, and let $C$ be a closed path of $P$ bonds in $B$. If $\|U(\partial p) - 1\| \leq t$ for every plaquette $p$ of $B$ and $s := 3DPt \leq \min(1/10, \log(1 + r_G))$, then for every irreducible unitary representation $\pi$ of $G$ with the character $\chi$,*

$$|\chi(U(C)) - 1| \leq 1.06\, c_\pi s \,. \tag{12.21}$$

*Proof.* Let $o$ be the corner of $B$ with the smallest coordinates, and for a site $y$ of $B$ let $\gamma_y$ be the path from $o$ to $y$ which changes first the coordinate 0, then 1, 2, and 3. For a bond $b = (y, y')$ of $B$ put $\ell_b := \gamma_y\, b\, \gamma_{y'}^{-1}$, a closed path at $o$, and $U^\gamma(b) := U(\ell_b)$. Then $U^\gamma$ is the gauge transform of $U$ by $y \mapsto U(\gamma_y)^{-1}$, so that $U^\gamma(C)$ is conjugate to $U(C)$ and $\chi(U^\gamma(C)) = \chi(U(C))$.

We first bound $U^\gamma(b)$. It suffices to take $b$ in a positive direction $e_\nu$, since $U^\gamma(b^{-1}) = U^\gamma(b)^{-1}$. Let $p$ be the path formed by the legs of $\gamma_y$ in the directions $\nu' > \nu$, a path of $n \leq 3D$ bonds from a site $z$ to $y$. Then $\gamma_y = \gamma_z\, p$ and $\gamma_{y'} = \gamma_z\, e\, (p + e_\nu)$, with $e$ the bond from $z$ to $z + e_\nu$, so that $\ell_b = \gamma_z\, p\, b\, (p + e_\nu)^{-1} e^{-1} \gamma_z^{-1}$, and $U(\ell_b)$ is conjugate to the holonomy $H$ of the boundary of the strip swept by $p$ in the direction $e_\nu$. With $p = (p_1, \dots, p_n)$ and $q_l$

the plaquette spanned by $p_l$ and $e_\nu$, telescoping along the strip writes $H = \lambda_n \cdots \lambda_1$, where $\lambda_l$ is the holonomy of $q_l$, from one of its corners and in one of its orientations, conjugated by the transport along $p_1 \cdots p_{l-1}$. Each $\lambda_l$ satisfies $\|\lambda_l - 1\| = \|U(\partial q_l) - 1\| \le t$, since the operator norm of $W - 1$ is invariant under unitary conjugation and inversion, and (8.11) gives

$$\|U^\gamma(b) - 1\| = \|H - 1\| \le (1+t)^{3D} - 1 \,. \tag{12.22}$$

The path $C$ has $P$ bonds, and (8.11) applied to $U^\gamma(C) = \prod_{b\in C} U^\gamma(b)$ gives $\|U^\gamma(C) - 1\| \le (1+t)^{3DP} - 1 \le e^s - 1$. Since $s \le \min(1/10, \log(1+r_G))$, the principal logarithm $U^\gamma(C) = e^{iX}$ has $X \in \mathfrak{g}$ and $\|X\| \le 1.06\, s$ by (8.12), so that $\|\pi(U^\gamma(C)) - 1\| \le c_\pi \|X\|$ by (1.7). Finally $|\chi(W) - 1| = |(\dim \pi)^{-1} \operatorname{Tr}(\pi(W) - 1)| \le \|\pi(W) - 1\|$. *q.e.d.*

At the stopping step we apply the lemma to the background field. For a loop $C$ at level $q$ with the character of $\pi$ put

$$y(C) \coloneqq 3.18\, c_\pi c_h (1 + 2\kappa_s) B_3\, P_{K'}(C)^2\, \varepsilon_{K'} \,, \tag{12.23}$$

which does not depend on $K$, since $P_{K'}(C)$ and $\varepsilon_{K'}$ do not (Section 8).

*Proof of Proposition 12.5.* We first bound the leading words on $\ell_0$. On the support of $\varrho_{\ell_0}$ every fine plaquette of the background satisfies $\|U_{K'}(\partial p) - 1\| \le t \coloneqq 2c_h(1+2\kappa_s)B_3 \varepsilon_{K'} L^{-2K'}$, as in the proof of Corollary 8.3. In fine units the loop $C_i$ has $P \coloneqq P_{K'}(C_i) L^{K'}$ bonds, and it lies in a box of side $D \le P/2$, since a closed path of $P$ bonds has $\ell^\infty$-diameter at most $P/2$; the box embeds in the torus because $3\,\mathrm{diam}(C_i) < T$. Hence $3DPt \le 3c_h(1 + 2\kappa_s) B_3 P_{K'}(C_i)^2 \varepsilon_{K'}$, and Lemma 12.6 gives $|f^{(i)}_{K'} - 1| \le 1.06\, c_{\pi_i} \cdot 3DPt \le y(C_i)$ whenever $3c_h(1 + 2\kappa_s) B_3 P_{K'}(C_i)^2 \varepsilon_{K'} \le \min(1/10, \log(1 + r_G))$. Since $|f^{(i)}_{K'}| \le 1$, telescoping gives $|\prod_i f^{(i)}_{K'} - 1| \le \sum_i y(C_i)$.

Next we bound the term $\ell_0$ of the identity (8.5) from below. On $\ell_0$ the normalised dressed loop is $e^{E_{\mathrm{const},K'}} [\prod_i f^{(i)}_{K'} + \sum_{\mathfrak{B}\neq\emptyset} Q_{K'}(\mathfrak{B}) w_{K'}(\mathfrak{B})]$, where $|E_{\mathrm{const},K'}| \le B_E(\mathcal{C})$ by Definition 5.1(iv) and Corollary 7.11, and the sum over the families has modulus at most $e^{I_*(\mathcal{C})} - 1$ by Lemma 8.5. Hence

$$|\langle \widehat{\mathcal{W}}_{\ell_0} \rangle_{\ell_0}| \ge e^{-B_E(\mathcal{C})} \Big( 2 - e^{I_*(\mathcal{C})} - \sum_i y(C_i) \Big) \,. \tag{12.24}$$

The other terms are bounded as in the proof of Theorem 8.6: $\sum_{\ell\neq\ell_0} w_\ell |\langle \widehat{\mathcal{W}}_\ell \rangle_\ell| \le e^{B_E(\mathcal{C}) + I_*(\mathcal{C})} \bar\delta_{\mathrm{LF}}$, and $w_{\ell_0} \ge 1 - \bar\delta_{\mathrm{LF}}$ by Proposition 8.4. By (8.5) and the triangle inequality, for every $K \ge K_0(\mathcal{C})$,

$$|W_K(\mathcal{C})| \ge (1 - \bar\delta_{\mathrm{LF}}) e^{-B_E(\mathcal{C})} \Big( 2 - e^{I_*(\mathcal{C})} - \sum_i y(C_i) \Big) - e^{B_E(\mathcal{C}) + I_*(\mathcal{C})} \bar\delta_{\mathrm{LF}} \,. \tag{12.25}$$

It remains to choose $\bar g_1$ and $\ell$ so that the four quantities $B_E(\mathcal{C})$, $I_*(\mathcal{C})$, $\sum_i y(C_i)$, and $\bar\delta_{\mathrm{LF}}$ are small. Each of them is a sum over the loops of a number of the loop, multiplied by a constant which is a nondecreasing function of $c_{\mathcal{C}}$ and by a function of $\bar g$ which tends to zero, at fixed $q$ and $T/u$, except for $\bar\delta_{\mathrm{LF}}$, which does not depend on the ensemble. For $B_E(\mathcal{C})$ this is (7.32), with the constants $C'_E$ and $C''_E$, since $\bar{\mathfrak{a}} \le C \bar g^2 (\log \bar g^{-2})^{6p_0}$ by (7.4) and $S^{\mathbf{R}}(\bar g) \to 0$; for $\bar\delta_{\mathrm{LF}}$ it is the remark after Proposition 8.4. For the other two we note that $\hat g^{-2}_{K'} = \bar g^{-2} + b' j_q + C_W$ by (3.3), so that $\hat g_{K'} \le \bar g$, and $\log \hat g^{-2}_{K'} \le 2 \log \bar g^{-2}$ for $\bar g$ small, because $j_q$ grows only like $\log\log \bar g^{-2}$ by (3.44). The numbers $P_{K'}(C) = P_q(C) L^3 M R_{K'-1}$ are at most

a constant times $P_q(C)(\log \hat g_{K'}^{-2})^r$ by Lemma 3.4; by Definition 3.11, $\varepsilon_{K'} = \hat g_{K'} p_0(\hat g_{K'})$ is $\hat g_{K'}$ times a power of $\log \hat g_{K'}^{-2}$, and $\varepsilon_{K'}/\min(\alpha_{0,K'}, \alpha_{1,K'})$ is a power of $\log \hat g_{K'}^{-2}$; the number $\bar\rho$ of (8.36) is $\bar g^2$ times a power of $\log \bar g^{-2}$; and $S_{\mathrm{wr}}(b) \le e^{-b}(1+2/b)$ by Lemma 2.11 with $x=1$. Hence, apart from the factors $P_q(C_i)$, $c_{\pi_i}$, and the constants $C_\chi$ and $C$ of (5.8), the numbers (12.23) and (8.37) are at most a positive power of $\bar g$ times a power of $\log \bar g^{-2}$. Collecting, there are functions $h_E$, $h_I$, and $h_y$ on $(0,\gamma_0]$ which tend to zero as $\bar g \to 0$, and nondecreasing functions $F_E$ and $F_I$ on $[0,\infty)$, all of which depend only on the constants of the construction, $\Lambda_*$, $q$, and $T/u$, such that

$$\begin{aligned} B_E(\mathcal{C}) &\le F_E(c_{\mathcal{C}}) h_E(\bar g) \sum_i \Big( \frac{P(C_i)}{r(C_i)} + P_q(C_i) \Big) , \\ I_*(\mathcal{C}) &\le F_I(c_{\mathcal{C}}) h_I(\bar g) \sum_i P_q(C_i) , \qquad y(C_i) \le c_{\pi_i} h_y(\bar g) P_q(C_i)^2 , \end{aligned} \tag{12.26}$$

and such that $3c_h(1+2\kappa_s)B_3 P_{K'}(C_i)^2 \varepsilon_{K'} \le h_y(\bar g) P_q(C_i)^2$. We let $\bar g_1 \in (0,\gamma_0]$ be so small that $\bar\delta_{\mathrm{LF}} \le 1/20$ for $\bar g \le \bar g_1$, put $m_G := \min(1/10, \log(1+r_G))$, and define $\ell(c,\bar g) := 0$ if $q < q_\sharp(c,\bar g)$ and otherwise

$$\ell(c,\bar g) := \min\Big( \frac{1}{10 F_E(c) h_E(\bar g)}, \ \frac{1}{20 F_I(c) h_I(\bar g)}, \ \frac{1}{20\, c\, h_y(\bar g)}, \ \frac{m_G}{h_y(\bar g)} \Big) , \tag{12.27}$$

with the convention $1/0 := \infty$. Each entry of the minimum does not increase with $c$, and neither does the indicator of $q \ge q_\sharp(c,\bar g)$, since $q_\sharp$ is nondecreasing in $c$; so $\ell$ does not increase with $c$. For fixed $c$ we have $q_\sharp(c,\bar g) = 0 \le q$ for $\bar g$ small, by Theorem 1.5, and every entry of the minimum tends to infinity as $\bar g \to 0$; so $\ell(c,\bar g) \to \infty$.

Let now $\bar g \le \bar g_1$ and let $\mathcal{C}$ satisfy (12.19). A closed path which visits no site twice has at least four bonds, so that the left side of (12.19) is at least 16; hence $\ell(c_{\mathcal{C}},\bar g) > 0$, which by definition means $q \ge q_\sharp(c_{\mathcal{C}},\bar g)$, and Theorem 1.5 gives the limit $W(\mathcal{C})$. Since $P_q(C_i) \le P_q(C_i)^2$, the bounds (12.26) and (12.27) give $B_E(\mathcal{C}) \le 1/10$, $I_*(\mathcal{C}) \le 1/20$, $\sum_i y(C_i) \le c_{\mathcal{C}} h_y(\bar g) \sum_i P_q(C_i)^2 \le 1/20$, and the condition of Lemma 12.6 for every loop, and $\bar\delta_{\mathrm{LF}} \le 1/20$ by the choice of $\bar g_1$. Then (12.25) gives

$$|W_K(\mathcal{C})| \ge 0.95\, e^{-0.1}(2 - e^{0.05} - 0.05) - 0.05\, e^{0.15} \ge 0.71 \ge \frac{1}{2} , \tag{12.28}$$

and in the limit $|W(\mathcal{C})| \ge 1/2$. For a loop $C$ at level $q$ in $H_+$ for which $\mathcal{D} := ((C,\chi))$ satisfies (12.19), we have $q \ge q_\sharp(c_\pi,\bar g)$, as just shown, and Corollary 1.7 with $m=1$, $\mathcal{D}_1 = \mathcal{D}$, and $c = c_\pi$ gives $W(\mathcal{D} \cup \theta\bar{\mathcal{D}}) - |W(\mathcal{D})|^2 \ge 0$, so that $W(\mathcal{D} \cup \theta\bar{\mathcal{D}}) \ge |W(\mathcal{D})|^2 \ge 1/4$. *q.e.d.*

The condition (12.19) involves the number of loops, because the bounds on $B_E(\mathcal{C})$, $I_*(\mathcal{C})$, and $\sum_i y(C_i)$ are bounds on sums over the loops. For ensembles which are generated from a fixed one by reflections this dependence disappears. We say that an ensemble $\mathcal{D}$ at level $q$ is *generated by reflections from* an ensemble $\mathcal{B}$ if $\mathcal{D} = \mathcal{B}$, or if $\mathcal{D} = \mathcal{D}' \cup \theta_{\mu,a}\bar{\mathcal{D}}'$ for an ensemble $\mathcal{D}'$ which is generated by reflections from $\mathcal{B}$ and all of whose loops lie in $H_{\mu,a}$, for some $\mu$ and some $a \in uL^{-q}\mathbb{Z}$. The *blocks* of $\mathcal{B}$ are $\{\mathcal{B}\}$, and those of $\mathcal{D}' \cup \theta_{\mu,a}\bar{\mathcal{D}}'$ are the blocks of $\mathcal{D}'$ and their images under $\theta_{\mu,a}$ with the conjugate characters. Every block is the image of $\mathcal{B}$ under a composition of reflections, with all its characters conjugated or none, so that it has the same chord–arc constants, ratios $P/r$, and numbers $P_q$ as $\mathcal{B}$, and either the characters of $\mathcal{B}$ or their conjugates. In particular $c_{\mathcal{B}_i} = c_{\mathcal{B}} = c_{\mathcal{D}}$ for every block $\mathcal{B}_i$ of $\mathcal{D}$, because $c_{\bar\pi} = c_\pi$, and every block satisfies (12.19) if $\mathcal{B}$ does.

**Proposition 12.7** (Products over Reflections)**.** *Let $L$ and $\Lambda_*$ be as in Theorem 1.5, let $\mathcal{B}$ be a loop ensemble at level $q$ whose loops are admissible, with $\bar g$ and $q$ as in Theorem 1.5 for $\mathcal{B}$,*

*and let $\mathcal{D}$ be generated by reflections from $\mathcal{B}$, with the blocks $\mathcal{B}_1, \dots, \mathcal{B}_N$. Then $|W_K(\mathcal{D})| \geq \prod_{i=1}^N |W_K(\mathcal{B}_i)|$ for every $K > j_q$. If $\bar{g} \leq \bar{g}_1$ and $\mathcal{B}$ satisfies* (12.19)*, then $|W(\mathcal{D})| \geq 2^{-N}$; the condition does not involve $\mathcal{D}$, and in particular not its number of loops.*

*Proof.* We argue by induction along the generation. For $\mathcal{D} = \mathcal{B}$ there is nothing to prove. Let $\mathcal{D} = \mathcal{D}' \cup \theta_{\mu,a}\bar{\mathcal{D}}'$, with the loops of $\mathcal{D}'$ in $H_{\mu,a}$, and write $\Theta := \Theta_{\mu,a}$ and $F := W_{\mathcal{D}'} \in \mathcal{A}_+$. Then $W_{\mathcal{D}} = F\,\Theta F$ by Section 12.1, and (12.4) with Lemma 12.2 gives

$$|\mathbb{E}_K[W_{\mathcal{D}}]| = \mathbb{E}_K[F\,\Theta F] \geq |\mathbb{E}_K[F]|^2 = |\mathbb{E}_K[W_{\mathcal{D}'}]|\,|\mathbb{E}_K[W_{\theta_{\mu,a}\bar{\mathcal{D}}'}]|\,. \tag{12.29}$$

The normalisation of $\mathcal{D}$ is the product of those of $\mathcal{D}'$ and $\theta_{\mu,a}\bar{\mathcal{D}}'$, and dividing by its modulus gives $|W_K(\mathcal{D})| \geq |W_K(\mathcal{D}')||W_K(\theta_{\mu,a}\bar{\mathcal{D}}')|$. By the induction hypothesis $|W_K(\mathcal{D}')|$ is at least the product over the blocks of $\mathcal{D}'$; and since $\theta_{\mu,a}\bar{\mathcal{D}}'$ is generated by reflections from $\theta_{\mu,a}\bar{\mathcal{B}}$, with the reflections conjugated by $\theta_{\mu,a}$, and has the images of the blocks of $\mathcal{D}'$ as its blocks, the same holds for it. This proves the first assertion. If $\bar{g} \leq \bar{g}_1$ and $\mathcal{B}$ satisfies (12.19), then so does every block, so that $|W_K(\mathcal{B}_i)| \geq 1/2$ for $K \geq K_0(\mathcal{B}_i)$ by Proposition 12.5, and the limit gives $|W(\mathcal{D})| \geq 2^{-N}$; the limit exists by Theorem 1.5, because the loops of $\mathcal{D}$ are admissible and $c_{\mathcal{D}} = c_{\mathcal{B}}$, so that $\bar{g}$ and $q$ are as in Theorem 1.5 for $\mathcal{D}$. *q.e.d.*

The lower bound decays with the number of blocks, but it is positive at a coupling which is fixed once $\mathcal{B}$ is. The number of loops which can be generated in this way is limited only by the torus: starting from one loop of diameter $h$ and doubling in each direction at a hyperplane at distance $h$ from the current configuration, the extent in the direction $\mu$ after $k_\mu$ doublings is at most $2^{k_\mu} \cdot 3h - 2h$, so that of the order of $4\log_2(T/h)$ doublings fit, and the number of loops is of the order of $(T/h)^4$.

## 13 Non-Gaussianity

We now prove Theorem 1.8. The proof computes the number $\mathfrak{Y}$ of (1.13) in two ways. For a Gaussian charge model it vanishes identically, by an algebraic cancellation which uses nothing but two facts: $\log|W|$ is a quadratic function of the covariance, and the logarithm of a normalisation is a linear function of the lengths and the corners (Section 13.2). For the limits of Yang–Mills theory we compute $\mathfrak{Y}$ to the order $\bar{g}^2$ at the stopping scale, where the identity of Theorem 8.1 expresses $W_K$ through a measure on the fixed lattice $\mathbb{T}_{j_q}$ which, at small coupling, is concentrated near the flat fields.

Three things have to be shown, and each meets a difficulty of its own. The normalisations have to drop out of $\mathfrak{Y}$, which they do because $\mathfrak{Y}$ was built to annihilate every function of the form of Proposition 7.13 (Section 13.3). The pair terms have to be of the fourth order. Under the measure of the stopping scale the Wilson loop of each of two separated rectangles depends, up to corrections which decay with their distance, on the field near that rectangle only, and its deviation from one has an $L^2$-norm of the order $\bar{g}^2$ rather than $\bar{g}$, because its part of first order is a trace of an element of the Lie algebra and vanishes; the covariance of two such deviations is then of the order $\bar{g}^4$ (Section **??**). Finally the third difference of the rectangles has to be bounded below by a positive multiple of $\bar{g}^2$, and this is the heart of the matter. After the insertions are shown to cancel in the third difference, by the translation invariance which made the coefficients of the normalisation independent of the loop (Section **??**), the third difference is a combination of the second moments of the fluxes of the background field through the rectangles $I_a$. An integration by parts against the density of the stopping scale identifies these second moments, to the leading order, with the covariances of the fluxes of a

free field at the block field closest to a flat one (Section **??**), and the third difference of those covariances is computed explicitly (Section 13.4).

Throughout the section $q$, the plane spanned by $e_\mu$ and $e_\nu$, $t$, $s$, $o$, and $\pi$ are fixed as in Section 1.3, with $\Lambda_* \geq 1 + \max(s/t, 4t/s)$ and $L$ as in Theorem 1.5, and the eleven diagrams are those defined there. Every requirement in this section that $\bar g$ be small is an upper bound on $\bar g$ in terms of $q$, $t/u$, $s/u$, $T/u$, $\pi$, and the constants of the construction, and $\bar g_0$ in Theorem 1.8 is the least of these bounds.

### 13.1 The Witness

For an ensemble $\mathcal{D}$ we write $|\mathcal{D}|$ for the number of its loops, $N_v(\mathcal{D}) \coloneqq \sum_{C\in\mathcal{D}} N_v(C)$ for the total length of its segments of orientation $v \in \mathcal{V}$, and, for a perpendicular pair $(v, w)$, $K_{v,w}(\mathcal{D})$ for the number of the corners $c$ of its loops with $(v_c, w_c) = (v, w)$. The number (1.13) is

$$\mathfrak{Y}(W) = \sum_{\mathcal{D}} n_{\mathcal{D}} \log|W(\mathcal{D})| = \mathfrak{y}_S(W) - \mathfrak{y}_2(W) + \mathfrak{y}_3(W) \,, \tag{13.1}$$

the sum over the six diagrams $R_1$, $R_2$, $R_3$, $R_4$, $P_2$, $P_3$ with the weights $n_{\mathcal{D}} = 1, -3, 3, -1, -1, 1$, where $\mathfrak{y}_S(W) \coloneqq \sum_{n=1}^4 n_{R_n} \log|W(R_n)|$ is the *shape term* and $\mathfrak{y}_k(W) \coloneqq \log|W(P_k)/W(R_1)^2|$ are the pair terms. To each diagram we assign the set $\mathrm{ind}(\mathcal{D})$ of the indices $a$ of the rectangles $I_a$ which tile it, or which are its loops: $\mathrm{ind}(I_a) \coloneqq \{a\}$, $\mathrm{ind}(R_n) \coloneqq \{0, \dots, n-1\}$, $\mathrm{ind}(R_n^{(1)}) \coloneqq \{1, \dots, n\}$, and $\mathrm{ind}(P_k) \coloneqq \{0, k\}$. The Gaussian models are tested by a second set of weights on all eleven diagrams,

$$n_{\mathcal{D}}^{\mathrm{sym}} \coloneqq \begin{cases} 0, 1, 1, -1 & \text{for } \mathcal{D} = I_0, I_1, I_2, I_3 \,, \\ -1, -2, 2, 1 & \text{for } \mathcal{D} = R_2, R_2^{(1)}, R_3, R_3^{(1)} \,, \\ -1, -1, 1 & \text{for } \mathcal{D} = R_4, P_2, P_3 \,, \end{cases} \tag{13.2}$$

and we put $\mathfrak{Y}^{\mathrm{sym}}(W) \coloneqq \sum_{\mathcal{D}} n_{\mathcal{D}}^{\mathrm{sym}} \log|W(\mathcal{D})|$, the sum over the eleven diagrams.

**Lemma 13.1** (Counts and Index Sums)**.** *(i) A rectangle of width at and height $s$ among the nine has $N_{\pm e_\mu} = at$, $N_{\pm e_\nu} = s$, and $N_v = 0$ for the other $v$, and it has exactly one corner of each of the types $(e_\mu, e_\nu)$, $(e_\nu, -e_\mu)$, $(-e_\mu, -e_\nu)$, $(-e_\nu, e_\mu)$ and no other corner.*

*(ii) Let $f$ be a function on the eleven diagrams such that $f(\mathcal{D})$ depends only on $\mathcal{D}$ through $|\mathcal{D}|$, the numbers $N_v(\mathcal{D})$, and the numbers $K_{v,w}(\mathcal{D})$, and linearly. Then $\sum_{\mathcal{D}} n_{\mathcal{D}} f(\mathcal{D}) = 0$, $\sum_n n_{R_n} f(R_n) = 0$, $f(P_k) = 2f(R_1)$, and $\sum_{\mathcal{D}} n_{\mathcal{D}}^{\mathrm{sym}} f(\mathcal{D}) = 0$.*

*(iii) If $f(I_a) = f(I_0)$ for every $a$ and $f(R_n^{(1)}) = f(R_n)$ for $n \in \{2, 3\}$, then $\sum_{\mathcal{D}} n_{\mathcal{D}}^{\mathrm{sym}} f(\mathcal{D}) = \sum_{\mathcal{D}} n_{\mathcal{D}} f(\mathcal{D})$, the left sum over the eleven diagrams and the right one over the six.*

*(iv) For $0 \leq a \leq b \leq 3$, the sum $\mathsf{s}(a, b)$ of $n_{\mathcal{D}}^{\mathrm{sym}}$ over the diagrams with $a, b \in \mathrm{ind}(\mathcal{D})$ vanishes.*

*Proof.* (i) The rectangle is traversed from its corner of least coordinates in the directions $e_\mu$, $e_\nu$, $-e_\mu$, $-e_\nu$, in this order, which gives the four corners and the lengths.

(iii) The weights $n^{\mathrm{sym}}$ sum to $0 + 1 + 1 - 1 = 1$ over $I_0, \dots, I_3$, to $-1 - 2 = -3$ over $R_2$, $R_2^{(1)}$, and to $2 + 1 = 3$ over $R_3$, $R_3^{(1)}$, which are the weights $n$ of $R_1$, $R_2$, and $R_3$, and they agree with $n$ on $R_4$, $P_2$, $P_3$.

(ii) It suffices to take for $f$ each of the functions $|\mathcal{D}|$, $N_v(\mathcal{D})$, $K_{v,w}(\mathcal{D})$. On the six diagrams $R_1, R_2, R_3, R_4, P_2, P_3$, by (i), $|\mathcal{D}| = 1, 1, 1, 1, 2, 2$, $N_{\pm e_\mu}(\mathcal{D}) = t, 2t, 3t, 4t, 2t, 2t$, $N_{\pm e_\nu}(\mathcal{D}) =$

$s, s, s, s, 2s, 2s$, the other $N_v$ vanish, and $K_{v,w}(\mathcal{D}) = |\mathcal{D}|$ for the four types of (i) and 0 for the others. Hence

$$\sum_{\mathcal{D}} n_{\mathcal{D}} |\mathcal{D}| = 1-3+3-1-2+2 = 0 \;, \qquad \sum_{\mathcal{D}} n_{\mathcal{D}} N_{\pm e_\mu}(\mathcal{D}) = (1-6+9-4-2+2)t = 0 \;, \quad (13.3)$$

and $\sum_{\mathcal{D}} n_{\mathcal{D}} N_{\pm e_\nu}(\mathcal{D}) = (1-3+3-1-2+2)s = 0$. On the rectangles alone the same sums are $1-3+3-1 = 0$, $(1-6+9-4)t = 0$, and 0. The loops of $P_k$ have the lengths and the corners of $R_1$, which gives $f(P_k) = 2f(R_1)$. Finally $f$ takes the same value on a rectangle and on its translates, so that (iii) and the first assertion give the last one.

(iv) The diagrams which contain a given pair of indices, with their weights $n^{\mathrm{sym}}$, are: for $(0,0)$, $I_0$, $R_2$, $R_3$, $R_4$, $P_2$, $P_3$, with $0-1+2-1-1+1$; for $(1,1)$, $I_1$, $R_2$, $R_2^{(1)}$, $R_3$, $R_3^{(1)}$, $R_4$, with $1-1-2+2+1-1$; for $(2,2)$, $I_2$, $R_2^{(1)}$, $R_3$, $R_3^{(1)}$, $R_4$, $P_2$, with $1-2+2+1-1-1$; for $(3,3)$, $I_3$, $R_3^{(1)}$, $R_4$, $P_3$, with $-1+1-1+1$; for $(0,1)$, $R_2$, $R_3$, $R_4$, with $-1+2-1$; for $(0,2)$, $R_3$, $R_4$, $P_2$, with $2-1-1$; for $(0,3)$, $R_4$, $P_3$, with $-1+1$; for $(1,2)$, $R_2^{(1)}$, $R_3$, $R_3^{(1)}$, $R_4$, with $-2+2+1-1$; and for $(1,3)$ and $(2,3)$, $R_3^{(1)}$, $R_4$, with $1-1$. Each sum is zero. *q.e.d.*

### 13.2 Gaussian Charge Models

The Gaussian charge models of Section 1.3 fail the test for an algebraic reason, which the next proposition records.

**Proposition 13.2** (Gaussian Charge Models)**.** *For every Gaussian charge model, its function $W$ does not vanish on the eleven diagrams and $\mathfrak{Y}^{\mathrm{sym}}(W) = 0$. If $W_m$ are the functions of a sequence of Gaussian charge models, each with its own charge, Gaussian vector, and normalisation, and $W_m(\mathcal{D}) \to W(\mathcal{D}) \neq 0$ for each of the eleven diagrams, then $\mathfrak{Y}^{\mathrm{sym}}(W) = 0$.*

*Proof.* For a real Gaussian variable $Z$ we have $|\mathbb{E}[e^{iZ}]| = e^{-\operatorname{Var} Z/2}$. Hence $W(\mathcal{D}) \neq 0$ and

$$\log|W(\mathcal{D})| = -\frac{\mathsf{g}^2}{2} \sum_{a,b \in \mathrm{ind}(\mathcal{D})} \operatorname{Cov}(\mathsf{X}_a, \mathsf{X}_b) - \sum_{C \in \mathcal{D}} \log|\kappa(C)| \;. \quad (13.4)$$

We multiply by $n_{\mathcal{D}}^{\mathrm{sym}}$ and sum over the eleven diagrams. In the first term the coefficient of $\operatorname{Cov}(\mathsf{X}_a, \mathsf{X}_b)$, $a \le b$, is $-\frac{1}{2}\mathsf{g}^2$ or $-\mathsf{g}^2$ times $\mathsf{s}(a,b)$, which vanishes by Lemma 13.1(iv). The second term is a function of $\mathcal{D}$ which is linear in the numbers $N_v(\mathcal{D})$ and $K_{v,w}(\mathcal{D})$, by the definition of a Gaussian charge model, so that its weighted sum vanishes by Lemma 13.1(ii). Hence $\mathfrak{Y}^{\mathrm{sym}}(W) = 0$. For the limit, $\mathfrak{Y}^{\mathrm{sym}}(W) = \lim_m \mathfrak{Y}^{\mathrm{sym}}(W_m) = 0$, since $\log|\cdot|$ is continuous on $\mathbb{C} \setminus \{0\}$. *q.e.d.*

The cancellation can be read off in the models whose covariance is invariant under the translation by $te_\mu$, $\operatorname{Cov}(\mathsf{X}_a, \mathsf{X}_b) = c^{\mathrm{G}}_{|a-b|}$. There $\operatorname{Var} \mathsf{X}(R_n) = nc_0^{\mathrm{G}} + 2\sum_{1 \le k < n}(n-k)c_k^{\mathrm{G}}$ and $\operatorname{Var} \mathsf{X}(P_k) = 2c_0^{\mathrm{G}} + 2c_k^{\mathrm{G}}$, so that, the normalisations cancelling by Lemma 13.1(ii),

$$\mathfrak{y}_S(W) = \mathsf{g}^2(c_3^{\mathrm{G}} - c_2^{\mathrm{G}}) \;, \qquad \mathfrak{y}_k(W) = -\mathsf{g}^2 c_k^{\mathrm{G}} \;, \qquad \mathfrak{Y}(W) = \mathfrak{Y}^{\mathrm{sym}}(W) = 0 \;: \quad (13.5)$$

the covariances of the separated rectangles which make up the shape term also make up the pair terms.

*Remark* 13.3 (Gaussian Fields with Values in the Lie Algebra). The models of Proposition 13.2 are abelian, and the number $\mathfrak{Y}$ does not distinguish Yang–Mills theory from Gaussian fields with values in $\mathfrak{g}$ and path-ordered holonomies. Let $(T_\alpha)$ be a basis of $\mathfrak{g}$ which is

orthonormal for an Ad-invariant inner product, with $\sum_\alpha \pi_*(T_\alpha)^2 = C\cdot 1$, let $B = \sum_\alpha B^\alpha T_\alpha$ on the bonds of a lattice containing the rectangles, with $B^\alpha$ real, centred, and jointly Gaussian, $\mathrm{Cov}(B^\alpha(e), B^\beta(f)) = \delta^{\alpha\beta}G(e,f)$, and $G$ invariant under the translation by $te_\mu$, and let $W(\mathcal{D}) := \mathbb{E}[\prod_{C\in\mathcal{D}}\chi(\prod_{e\in C} e^{i\mathsf{g}B(e)})]$. Expanding the ordered product to the second order in $\mathsf{g}$, the term of first order of $\chi(\prod_e e^{i\mathsf{g}B(e)})$ is $i\mathsf{g}(\dim\pi)^{-1}\operatorname{Tr}\pi_*(\sum_e B(e))$, which vanishes because $\operatorname{Tr}\circ\pi_*$ vanishes on $\mathfrak{g}$ (Lemma **??** below), the term of second order has the mean $-\frac{1}{2}\mathsf{g}^2 C\operatorname{Var}\mathsf{X}^1(C)$ with $\mathsf{X}^1(C) := \sum_{e\in C}B^1(e)$, the terms of odd order have mean zero, and the remainder of fourth order has a finite moment. Hence the pair terms are $O(\mathsf{g}^4)$ and $\mathfrak{Y}(W) = \mathsf{g}^2C(c_3 - c_2) + O(\mathsf{g}^4)$ with $c_k := \mathrm{Cov}(\mathsf{X}^1(I_0), \mathsf{X}^1(I_k))$, which for the free covariance is positive by Lemma 13.5 below. These fields share with Yang–Mills theory the feature on which the theorem rests.

### 13.3 The Normalisations Drop Out

We next show that the limits exist on the eleven diagrams and that $\mathfrak{Y}(W)$ is the limit of the same number computed from the expectations $N_K(\mathcal{D}) := \mathbb{E}_K[W_\mathcal{D}]$, in which no normalisation appears.

**Proposition 13.4** (The Normalisations Drop Out)**.** *(i) Let $K > j_q$. If $N_K(\mathcal{D}) \neq 0$ on the six diagrams $R_1,\dots,R_4,P_2,P_3$, then each of $\mathfrak{Y}$, $\mathfrak{y}_S$, $\mathfrak{y}_2$, and $\mathfrak{y}_3$ takes the same value on $W_K$ and on $N_K$.*

*(ii) There is $\bar g_1' > 0$ such that for $0 < \bar g \le \bar g_1'$ the limits $W(\mathcal{D})$ exist on the eleven diagrams, $|W_K(\mathcal{D})| \ge 1/2$ for $K \ge K_0(\mathcal{D})$, $|W(\mathcal{D})| \ge 1/2$, $W_K(I_a) = W_K(I_0)$ and $W_K(R_n^{(1)}) = W_K(R_n)$ for $K > j_q$, and $W(I_a) = W(I_0)$ and $W(R_n^{(1)}) = W(R_n)$. For these $\bar g$, $\mathfrak{Y}^{\mathrm{sym}}(W) = \mathfrak{Y}(W) = \lim_K \mathfrak{Y}(N_K)$, $\mathfrak{y}_S(W) = \lim_K \mathfrak{y}_S(N_K)$, and $\mathfrak{y}_k(W) = \lim_K \mathfrak{y}_k(N_K)$.*

*(iii) Let $\bar g \le \bar g_1'$ and $q' \ge q$. If the limits $W^{(q')}$ of the renormalised expectations with the normalisation of the level $q'$ exist and do not vanish on the six diagrams, then $\mathfrak{Y}(W^{(q')}) = \mathfrak{Y}(W)$.*

*Proof.* (i) By Proposition 7.13, $\log|\kappa_K^{(q)}(C,\chi)| = \sum_v N_v^{(0)}(C)\operatorname{Re} m_{v,K} + \sum_{c\in\mathcal{K}(C)}\operatorname{Re} z_{v_c,w_c,K}$, where $N_v^{(0)}(C) = N_v(C)/\varepsilon_K$ is the number of bonds of $\mathbb{T}_K$ of orientation $v$ on $C$ and the coefficients do not depend on $C$. Hence $\sum_{C\in\mathcal{D}}\log|\kappa_K^{(q)}(C,\chi)|$ is a linear function of the numbers $N_v(\mathcal{D})$ and $K_{v,w}(\mathcal{D})$, and by Lemma 13.1(ii) its weighted sums with the weights of $\mathfrak{Y}$ and of $\mathfrak{y}_S$ vanish, and its value on $P_k$ is twice its value on $R_1$. Since $\log|W_K(\mathcal{D})| = \log|N_K(\mathcal{D})| - \sum_{C\in\mathcal{D}}\log|\kappa_K^{(q)}(C,\chi)|$, this is (i).

(ii) Each of the eleven diagrams is an ensemble at level $q$ of at most two rectangles with the representation $\pi$. A rectangle with sides $a \ge b$ has $s(C) = b$, $\delta(C) = b$, and $P(C) = 2(a+b)$, so that $\Lambda(C) \le \max(2, 1 + a/b)$ by Lemma 2.5, $r(C) = \min(b/2, b, (a+b)/4) = b/2$, and $P(C)/r(C) = 4(1 + a/b)$; for the nine rectangles $a/b \le \max(s/t, 4t/s)$, so that they are admissible and $P(C)/r(C) \le 4\Lambda_*$, and $P_q(C) \le 2(4t+s)L^q/u$. The left side of (12.19) is therefore at most $\ell_{\mathrm{w}} := 8(4t+s)^2L^{2q}/u^2 + 8\Lambda_*$ for each diagram. Let $\bar g_1$ and $\ell$ be those of Proposition 12.5 at the level $q$, and let $\bar g_1' \le \bar g_1$ be so small that $\ell(c_\pi, \bar g) \ge \ell_{\mathrm{w}}$ for $\bar g \le \bar g_1'$, which is possible because $\ell(c_\pi, \bar g) \to \infty$ as $\bar g \to 0$. Proposition 12.5 then gives $q \ge q_\sharp(c_\pi, \bar g)$, the limits, and the lower bounds. The diagrams $I_a$ and $R_n^{(1)}$ are the images of $I_0$ and $R_n$ under the translations by $ate_\mu$ and $te_\mu$, which belong to $\mathfrak{G}_q$ because $t \in uL^{-q}\mathbb{N}$, so that Corollary 1.6(ii) gives the equalities for $W_K$ and for $W$. By the lower bounds $N_K(\mathcal{D}) \neq 0$ for $K$ large, and (i) together with the continuity of $\log|\cdot|$ on $\mathbb{C}\setminus\{0\}$ gives the three limits. Finally $\mathfrak{Y}^{\mathrm{sym}}(W) = \mathfrak{Y}(W)$ by Lemma 13.1(iii) applied to $f = \log|W|$.

(iii) By (1.12) and Proposition 7.13 at the level $q'$, $\log|\kappa_K^{(q')}(C,\chi)|$ is again a linear function of the lengths and the corner numbers of $C$ with coefficients independent of $C$, so that (i) holds with $W_K^{(q')}$ in place of $W_K$. Since $W_K^{(q')}(\mathcal{D}) \to W^{(q')}(\mathcal{D}) \neq 0$, also $N_K(\mathcal{D}) \neq 0$ for $K$ large, and $\mathfrak{Y}(W^{(q')}) = \lim_K \mathfrak{Y}(N_K) = \mathfrak{Y}(W)$ by (ii). *q.e.d.*

In other words, $\mathfrak{Y}(W)$ does not depend on the renormalisation prescription within the class of perimeter and corner normalisations, and in particular not on the level at which the diagrams are read; any change of the masses $m_{v,K}$, of the corner constants, or of a constant factor per loop leaves it unchanged.

### 13.4 The Free Field

The lower bound on the shape term rests on an explicit computation for the free field. At the stopping scale the field is close to a flat field, whose holonomies around the cycles of the torus need not be trivial, and in the components of the Lie algebra on which these holonomies act diagonally the free field acquires twisted periodicity conditions. We therefore compute the covariance of the fluxes for every twist, and we need a lower bound which is uniform in it.

Let $\eta > 0$ and $\mathsf{T} \in \eta\mathbb{N}$, let $\theta \in [0,1)^4$ be a *twist*, and let $\mathbb{T}^*_{\eta,\theta}$ be the set of the $(\mathsf{T}/\eta)^4$ momenta $p = (2\pi/\mathsf{T})(n+\theta)$, $n \in \mathbb{Z}^4$, taken modulo $(2\pi/\eta)\mathbb{Z}^4$, with the momentum $p = 0$ removed if $\theta = 0$. Put $\hat{p}_\alpha := (2/\eta)\sin(\eta p_\alpha/2)$ and $\hat{p}^2 := \sum_\alpha \hat{p}_\alpha^2$, which is positive on $\mathbb{T}^*_{\eta,\theta}$, and define for $z \in \eta\mathbb{Z}^4$ the kernel

$$k_\theta(z) := \frac{1}{\mathsf{T}^4} \sum_{p \in \mathbb{T}^*_{\eta,\theta}} e^{ip\cdot z}\, \frac{\hat{p}_\mu^2 + \hat{p}_\nu^2}{\hat{p}^2} \,. \tag{13.6}$$

For a rectangle $\mathsf{I}$ of plaquettes of the lattice $\eta\mathbb{Z}^4$ in the plane of $e_\mu$ and $e_\nu$, which we identify with the set of the base points of its plaquettes, and for $\mathsf{w} \in \eta\mathbb{Z}$, put

$$c^\theta(\mathsf{w}) := \eta^4 \sum_{x \in \mathsf{I}} \sum_{y \in \mathsf{I} + \mathsf{w}e_\mu} \operatorname{Re} k_\theta(x-y) \,. \tag{13.7}$$

At $\theta = 0$ this is the covariance of the fluxes $\mathsf{X}(\mathsf{I}) = \oint_{\partial \mathsf{I}} \mathsf{B}$ and $\mathsf{X}(\mathsf{I}+\mathsf{w}e_\mu)$ of the free field, the centred Gaussian 1-form on the periodic lattice of spacing $\eta$ and period $\mathsf{T}$ with $\mathbb{E}[\mathsf{B}_\alpha(x)\mathsf{B}_\beta(y)] = \delta_{\alpha\beta}\mathsf{T}^{-4}\sum_p e^{ip\cdot(x-y)}/\hat{p}^2$: by the lattice Stokes formula $\mathsf{X}(\mathsf{I}) = \eta^2 \sum_{x \in \mathsf{I}} \mathsf{F}(x)$ with the field strength $\mathsf{F} = \partial_\mu \mathsf{B}_\nu - \partial_\nu \mathsf{B}_\mu$, and $\mathbb{E}[\mathsf{F}(x)\mathsf{F}(y)] = k_0(x-y)$ because $|e^{i\eta p_\alpha} - 1|^2 = \eta^2 \hat{p}_\alpha^2$.

**Lemma 13.5** (The Free Field's Third Difference)**.** *Let $\mathsf{I}$ have width $t$ along $e_\mu$ and height $s$ along $e_\nu$, with $t, s \in \eta\mathbb{N}$, $12t \leq \mathsf{T}$, $s \leq \mathsf{T}/2$, and $\eta \leq \mathsf{T}/4$. Then for every twist $\theta$,*

$$c^\theta(3t) - c^\theta(2t) \geq \frac{s^2 t^3}{112\,\mathsf{T}^5} \,. \tag{13.8}$$

*Proof.* We split the multiplier of (13.6) as $(\hat{p}_\mu^2 + \hat{p}_\nu^2)/\hat{p}^2 = 1 - \sum_{\alpha \notin \{\mu,\nu\}} \hat{p}_\alpha^2/\hat{p}^2$ and treat the two parts separately.

*The constant part.* For $z \in \eta\mathbb{Z}^4$ the sum of $e^{ip\cdot z}$ over all the $(\mathsf{T}/\eta)^4$ momenta $(2\pi/\mathsf{T})(n+\theta)$ factorises over the coordinates, and in each coordinate it equals $(\mathsf{T}/\eta)e^{2\pi i\theta_\alpha z_\alpha/\mathsf{T}}$ if $z_\alpha \in \mathsf{T}\mathbb{Z}$ and 0 otherwise. We take lifts in which $\mathsf{I}+\mathsf{w}e_\mu$ is the translate of $\mathsf{I}$ by $\mathsf{w}e_\mu$, with $t \leq \mathsf{w} \leq \mathsf{T}-t$. For $x \in \mathsf{I}$ and $y \in \mathsf{I} + \mathsf{w}e_\mu$ we then have $0 < y_\mu - x_\mu < \mathsf{w} + t \leq \mathsf{T}$, so that $x - y \notin \mathsf{T}\mathbb{Z}^4$, and the constant part contributes 0 to $k_\theta(x-y)$ if $\theta \neq 0$, and $-\mathsf{T}^{-4}$, the removed momentum,

if $\theta = 0$. Either way its contribution to $c^\theta(\mathsf{w})$ does not depend on $\mathsf{w}$ in this range, which contains $2t$ and $3t$ because $12t \le \mathsf{T}$.

*The resolvent in one direction.* In the second part we fix $p_\perp := (p_\alpha)_{\alpha\neq\mu}$ with $\sum_{\alpha\notin\{\mu,\nu\}} \hat p_\alpha^2 > 0$, the other terms vanishing, put $\mathsf{m}^2 := \sum_{\alpha\neq\mu} \hat p_\alpha^2 > 0$, and let $\vartheta > 0$ with $\cosh\vartheta = 1 + \eta^2\mathsf{m}^2/2$. The sum over $p_\mu$ is then

$$\mathsf{h}(z) := \frac{1}{\mathsf{T}} \sum_{p_\mu} \frac{e^{ip_\mu z}}{\hat p_\mu^2 + \mathsf{m}^2} = \frac{\eta}{2\sinh\vartheta} \sum_{n\in\mathbb{Z}} e^{2\pi i\theta_\mu n} e^{-\vartheta|z-n\mathsf{T}|/\eta} , \qquad z \in \eta\mathbb{Z} , \tag{13.9}$$

the sum over $p_\mu \in (2\pi/\mathsf{T})(\mathbb{Z}+\theta_\mu)$ modulo $2\pi/\eta$. Indeed, both sides satisfy $h(z+\mathsf{T}) = e^{2\pi i\theta_\mu} h(z)$ and solve $\eta^{-2}(2h(z) - h(z+\eta) - h(z-\eta)) + \mathsf{m}^2 h(z) = \eta^{-1}\sum_n e^{2\pi i\theta_\mu n}\delta_{z,n\mathsf{T}}$, whose solution in the space of such $h$ is unique because the operator has the eigenvalues $\hat p_\mu^2 + \mathsf{m}^2 > 0$ there. For the left side this is the sum of $e^{ip_\mu z}$ over the momenta, computed above in one coordinate. For the right side, $g(z) := (\eta/(2\sinh\vartheta))e^{-\vartheta|z|/\eta}$ solves the equation with the source $\eta^{-1}\delta_{z,0}$: at $z \neq 0$ because $2\cosh\vartheta = 2 + \eta^2\mathsf{m}^2$, and at $z = 0$ because $2 - 2e^{-\vartheta} + \eta^2\mathsf{m}^2 = 2\cosh\vartheta - 2e^{-\vartheta} = 2\sinh\vartheta$; the sum $\sum_n e^{2\pi i\theta_\mu n} g(z - n\mathsf{T})$ converges absolutely and has the periodicity and the sources. Since $g$ is real and even, $\mathsf{h}(-z) = \overline{\mathsf{h}(z)}$, and $\operatorname{Re}\mathsf{h}$ is even.

*Monotonicity.* For $0 \le z \le \mathsf{T}$ put $y := \vartheta z/\eta$, $\mathsf{N} := \vartheta\mathsf{T}/\eta$, $\mathsf{q} := e^{-\mathsf{N}}$, and $\psi := 2\pi\theta_\mu$. Summing the two geometric series over $n \le 0$ and $n \ge 1$,

$$\frac{2\sinh\vartheta}{\eta} \operatorname{Re}\mathsf{h}(z) = \frac{S(y)}{|1 - \mathsf{q}e^{i\psi}|^2} , \qquad S(y) := e^{-y}(1 - \mathsf{q}\cos\psi) + \mathsf{q}e^{y}(\cos\psi - \mathsf{q}) . \tag{13.10}$$

We claim that $-S'(y) \ge (1-\mathsf{q})e^{-y}(1 - e^{2y-\mathsf{N}})$ for $0 \le y \le \mathsf{N}/2$. Indeed $-S'(y) = e^{-y}(1 - \mathsf{q}\cos\psi) - \mathsf{q}e^{y}(\cos\psi - \mathsf{q})$. If $\cos\psi \le \mathsf{q}$, the second term is nonnegative and the claim follows from $1 - \mathsf{q}\cos\psi \ge 1 - \mathsf{q}$. Otherwise $\cos\psi - \mathsf{q} \le 1 - \mathsf{q}\cos\psi$, the difference being $-(1 - \cos\psi)(1+\mathsf{q})$, and $\mathsf{q}e^{y} = e^{2y-\mathsf{N}}e^{-y}$, so that $-S'(y) \ge e^{-y}(1 - \mathsf{q}\cos\psi)(1 - e^{2y-\mathsf{N}})$, which gives the claim. Since $|1 - \mathsf{q}e^{i\psi}|^2 \le 4$, for $0 \le z < z' < \mathsf{T}/2$,

$$\operatorname{Re}\mathsf{h}(z) - \operatorname{Re}\mathsf{h}(z') \ge \frac{\eta(1-\mathsf{q})}{8\sinh\vartheta} \int_{\vartheta z/\eta}^{\vartheta z'/\eta} e^{-y}(1 - e^{2y-\mathsf{N}})\,\mathrm{d}y > 0 , \tag{13.11}$$

so that $\operatorname{Re}\mathsf{h}$ is strictly decreasing on $[0, \mathsf{T}/2)$, for every twist.

*The difference.* For $x \in \mathsf{I}$ and $y \in \mathsf{I} + \mathsf{w}e_\mu$ the coordinates other than $x_\mu$ and $x_\nu$ agree, and the two rectangles have the same rows. The second part of $c^\theta(\mathsf{w})$ is therefore $-\mathsf{T}^{-3}\sum_{p_\perp}(\sum_{\alpha\notin\{\mu,\nu\}}\hat p_\alpha^2)|\eta\sum_{x_\nu} e^{ip_\nu x_\nu}|^2\eta^2\sum_{x_\mu,y_\mu}\operatorname{Re}\mathsf{h}(x_\mu - y_\mu - \mathsf{w})$, with $x_\nu$ over the rows and $x_\mu$, $y_\mu$ over the columns of $\mathsf{I}$, and by the first step

$$c^\theta(3t) - c^\theta(2t) = \frac{1}{\mathsf{T}^3}\sum_{p_\perp}\Big(\sum_{\alpha\notin\{\mu,\nu\}}\hat p_\alpha^2\Big)\Big|\eta\sum_{x_\nu} e^{ip_\nu x_\nu}\Big|^2\eta^2\sum_{x_\mu,y_\mu}[\operatorname{Re}\mathsf{h}(z+2t) - \operatorname{Re}\mathsf{h}(z+3t)] , \tag{13.12}$$

with $z := y_\mu - x_\mu \in (-t, t)$, where we used that $\operatorname{Re}\mathsf{h}$ is even. Since $t < z+2t < z+3t < 4t \le \mathsf{T}/3$, every bracket is positive by (13.11), and every term of (13.12) is nonnegative. We keep one of them. Let $\alpha$ and $\alpha'$ be the directions other than $\mu$ and $\nu$, and choose $p_\nu$, $p_\alpha$, $p_{\alpha'}$ in their classes with $|p_\nu|, |p_{\alpha'}| \le \pi/\mathsf{T}$ and $\pi/\mathsf{T} \le |p_\alpha| \le 2\pi/\mathsf{T}$, which is possible because every class $(2\pi/\mathsf{T})(\mathbb{Z}+\theta_\beta)$ meets both $[-\pi/\mathsf{T}, \pi/\mathsf{T}]$ and $\{\pi/\mathsf{T} \le |p| \le 2\pi/\mathsf{T}\}$. Since $\eta \le \mathsf{T}/4$, these momenta have $|\eta p/2| \le \pi/4$, and $\sin y \ge (2\sqrt2/\pi)y$ on $[0, \pi/4]$ gives $\hat p_\alpha^2 \ge 8/\mathsf{T}^2$, $\mathsf{m}^2 \le 6\pi^2/\mathsf{T}^2$ because $\hat p_\beta \le |p_\beta|$, and, with $s \le \mathsf{T}/2$,

$$\Big|\eta\sum_{x_\nu} e^{ip_\nu x_\nu}\Big| = \eta\,\Big|\frac{\sin(sp_\nu/2)}{\sin(\eta p_\nu/2)}\Big| \ge \frac{2\sqrt2}{\pi}\,s , \tag{13.13}$$

the left side being $s$ if $p_\nu = 0$.

*The bracket.* Now $\eta\mathsf{m} \le \sqrt{6}\pi/4 < 2$, so that $\vartheta = 2\,\mathrm{arsinh}(\eta\mathsf{m}/2) \in [\eta\mathsf{m}/2, \eta\mathsf{m}]$, because $y/2 \le \mathrm{arsinh}\, y \le y$ on $[0,1]$, and $\sinh\vartheta = \eta\mathsf{m}(1+\eta^2\mathsf{m}^2/4)^{1/2} \le \sqrt{2}\,\eta\mathsf{m}$. In (13.11) with $z+2t$ and $z+3t$ the interval of integration starts at most at $3\vartheta t/\eta \le 3\mathsf{m}t$ and has length $\vartheta t/\eta \ge \mathsf{m}t/2$; on it $2y - \mathsf{N} \le (\vartheta/\eta)(8t-\mathsf{T}) \le -\vartheta\mathsf{T}/(3\eta) \le -\mathsf{m}\mathsf{T}/6$; and $1-\mathsf{q} \ge 1 - e^{-\mathsf{m}\mathsf{T}/2}$. With $1-e^{-x} \ge x(1-x/2)$ for $x \ge 0$, the bracket is at least

$$\frac{1}{8\sqrt{2}\,\mathsf{m}}(1-e^{-\mathsf{m}\mathsf{T}/2})(1-e^{-\mathsf{m}\mathsf{T}/6})e^{-3\mathsf{m}t}\,\frac{\mathsf{m}t}{2}\Big(1-\frac{\mathsf{m}t}{4}\Big) \ge \frac{t}{512\sqrt{2}} \ge \frac{t}{725}\,, \tag{13.14}$$

because $2\sqrt{2} \le \mathsf{m}\mathsf{T} \le \sqrt{6}\pi$ and $\mathsf{m}t \le \sqrt{6}\pi/12$ give $e^{-\mathsf{m}\mathsf{T}/2} \le e^{-\sqrt{2}} \le 1/4$, $e^{-\mathsf{m}\mathsf{T}/6} \le e^{-\sqrt{2}/3} \le 5/8$, $e^{-3\mathsf{m}t} \ge e^{-2} \ge 2/15$, and $1-\mathsf{m}t/4 \ge 5/6$, and $(3/4)(3/8)(2/15)(5/6) = 1/32$. The kept term of (13.12) is therefore at least $\mathsf{T}^{-3}(8/\mathsf{T}^2)(8s^2/\pi^2)\,t^2\,t/725 = 64s^2t^3/(725\pi^2\mathsf{T}^5)$, since the sum over $x_\mu$ and $y_\mu$ has $(t/\eta)^2$ terms, and $64/(725\pi^2) \ge 1/112$. *q.e.d.*

For the free field with the charge $\mathsf{g}$ the fluxes $\mathsf{X}_a := \oint_{I_a} \mathsf{B}$ form a Gaussian charge model whose covariance is invariant under the translation by $te_\mu$, with $c^{\mathsf{G}}_k = c^0(kt)$. By (13.5) its number $\mathfrak{Y}$ vanishes, while its shape term is $\mathsf{g}^2(c^0(3t) - c^0(2t)) > 0$ by the lemma: in the abelian theory the positive shape term is cancelled exactly by the pair terms.

## 14 Notes on the Papers

We record the places at which Bałaban's papers sketch or defer an argument which we use, the places at which we read a printed formula otherwise than it is printed, and the conventions in which our notation differs from his.

*Deferred and sketched arguments.* The asymptotic behaviour of the couplings, [8, Theorem 2, (0.31), p. 259], is stated with the remark that "a proof of this theorem, based on perturbative calculations, will be given in a separate paper"; the bounds [9, (2.6), (2.7), p. 255] on the running couplings are stated to "follow from the renormalization group equations (0.20) [I]". We do not use these statements: Section 4 proves what we need for the construction of this paper. The bounds on the walk factors of the propagators are given "rather briefly" [5, p. 409], and the bounds on the factors of the walk expansions are stated up to powers of the constants [5, p. 413]; we use them as printed, and the comparison of two fine lattices in Section 9 is proved in Supplement, Proposition S.12. Bałaban sketches a second proof of [9, Theorem 2] by gauge invariance [9, pp. 283–284]; we use the first.

*Formulae read otherwise than printed.* Theorem 1 of [9, p. 262] assumes that the couplings satisfy "the inequalities (I.0.33)", and [9, p. 246] replaces $g_0$ in the logarithmic factor of the threshold by the bound $\gamma_0 \ge g_0$, $\gamma_0^{-2} = \gamma^{-2} + \beta\log\varepsilon^{-1}$, "coming from the inequality (0.33) [I]". The numbered displays of [8, Sect. 0] end with (0.32) on p. 260, and the only inequality among them which concerns the sequence of the couplings is the two-sided bound (0.31) of Theorem 2 [8, p. 259], of which the bound on p. 246 is the case $k = 0$, while [12, p. 355] cites it as "the basic inequality (0.31)", "the result of Theorem 2" of [8], with the reference printed as the second paper of the series; we read (0.33) as (0.31) in both places, and Theorem 3.6 is quoted with this reading. The decomposition of unity [9, (3.16), p. 268] for the fluctuation field of the step $k$ prints the threshold $\delta_k$, while its counterpart at the step 0 [9, (1.18), p. 250] prints $g_0^{-1}\delta_0$ for the same scaled variable, which enters the configuration as $\exp ig_0 A$ [9, (1.19), p. 250]; we read the threshold of [9, (3.16)] as $g_k^{-1}\delta_k$, which is our $\vartheta_k/g_k$. In the same way [11, (1.55), p. 188] prints $\delta'_j$ for the threshold of the scaled field $B_j$, while [11, p. 183] defines

$\delta'_j = g_j A_1 p_1(g_j)$ and [12, p. 383] uses $g_j^{-1}\delta'_j = A_1 p_1(g_j)$; we read $g_j^{-1}\delta'_j$. In the displays of [12, pp. 380–383] the symbol $p_0(g)$ stands for $(\log g^{-2})^{p_0}$, without the factor $A_0$ of (3.1), as in $p_1(g) = (\log g^{-2})^{p_1}$ [11, p. 183]: only in this reading does the bound of $\exp(-\frac{1}{4}g_m^{-2}B_3^{-2}\frac{1}{2}\varepsilon_m^2)$ by $\exp(-\frac{1}{8}B_3^{-2}(1+\beta_0)^{-2}A_0^2 p_0^2(g_j))$ [12, p. 381] follow from $\varepsilon_m = g_m p_0(g_m)$ and (2.7). The transcription of [9, (1.8), p. 247] which we used reads $\delta_0 = g_0(A_1/A_0)p_0(g_0)$, with the factor $A_0^{-1}$ inserted to agree with $\varepsilon_0 = g_0 p_0(g_0)$ and with the hierarchy of the thresholds of [9, (3.7)–(3.9)]; we did not compare this line with the printed page. In [9, Theorem 2] the exponent of the age bound (3.31) is written $\beta < 1$, and the proof gives it in the form $1-\beta$ with $\beta > 0$ arbitrary [9, p. 283]; we write the small number as $\beta_E$.

*Iterated averages.* We read the averaging operation of order $j$ as the $j$-fold iterate of the average of order one, as the averages of higher order are defined recursively in [4, (43), p. 24].

*Free choices.* The papers centre the cubes of the families $\pi_j$ on a lattice of spacing $M$ [8, p. 257] and leave the position of the other partitions free; we centre all of them at the origin (Section 7.1), which is compatible with the blocks of the averages [8, (0.3), p. 252] because all sides are odd, and which makes every partition invariant under the reflections $x_\mu \mapsto -x_\mu$. Every object of a step which the papers define by reference to a coordinate system, the axial contours of the fluctuation fields, the gauge-fixing graphs, and the designated bonds of the averages, is attached to its block or cube through the centre and the fixed orientation of the axes, so that a translation which preserves the partitions maps the object of a cube to the object of its image. This is the rule under which Lemma 7.4(ii) holds; the reflections $\theta_\mu$ of Section 12 map the objects of a cube to those of its image because the averages use the contours of all orders [8, p. 252], and the axial contours, which start at the centre of the block and change the coordinates in a fixed order [2, (1.7), p. 18], [8, p. 275], are mapped to the axial contours of the image block by a reflection of one coordinate, which does not change that order. A permutation of the axes changes that order, which is why the covariance of the normalisations is proved for the reflections and not for the exchange of two axes (Section 12.3).

*Notation.* We write $\vartheta_k$ for the threshold of the fluctuation field which Bałaban writes $\delta_k$ [9, (3.19), p. 268], because $\delta_k$ is the difference of the couplings of the two runs in Sections 9–11. The constant of the boundary terms which [9, (2.47), p. 264] prints $B_1$ is our $B_\partial$. In [9, (2.5), p. 255] the letter $r$ denotes both the exponent of the powers of $L$ among which $R_j$ is chosen and the fixed exponent of $\log g_j^{-2}$; we keep $r$ for the second. Our couplings, thresholds, cube sizes, and depths are those of the tube of a target (Definition 3.11), evaluated at $\hat{g}_j$, where Bałaban's are evaluated at $g_j$; Proposition 3.12 states that his bounds hold for them.

## Appendix A The Constants in the Order of Choices

The estimates of the paper contain many constants, and their consistency rests on the order in which they are chosen: a smallness condition can be met only if the quantity which it constrains is chosen after the constants on which the condition depends. We fix the constants in seven stages, numbered from 0 to 6, so that a constant of a stage depends only on the constants of the earlier stages. Every condition of the paper is then of one of two kinds, a lower bound on a quantity of its own stage in terms of earlier ones, or an upper bound on a quantity of a later stage in terms of earlier ones, and conditions of both kinds are met by taking each quantity large or small enough when its stage comes. The *constants of the construction* are those of the stages 0 to 5, together with the constants computed from them. The last stage chooses the bound $\gamma$ of Section 3.1, and with it the coupling $\bar{g}$ and the target $(K, \bar{g})$; every requirement that $\bar{g}$ be sufficiently small which involves no representation is a condition of this stage.

*Stage* 0*: the dimension, the group, and the exponents.* These numbers are fixed once and for all.

| | |
|---|---|
| $d = 4$, $G \subset U(N)$ | the dimension and the gauge group, with the numbers $\kappa$ of (4.2), $c$ of (4.3), and $\dim G$, the radius $r_G$ of the logarithmic chart, the constant $c_G$ of the Haar measure of balls (Section 8.3), and the constant $c_h$ which compares the two norms of Appendix B |
| $r = 2$ | the exponent of the cube sizes $R_j$, for which [9, p. 245] requires $r \geq 2$ |
| $p_0$ | the exponent of the thresholds, $p_0 \geq 5r$ [9, p. 246] |
| $p_1$ | the exponent of the thresholds of the **R**-operation, $p_1 < p_0$ [11, p. 183], $p_0 - p_1 - 1 \geq \lceil r \rceil / 2$ [11, (1.94), p. 198], and $2p_1 - (d+5)r > p_0$ [12, p. 383], where the exponent printed $r_0$ is read as $r$, the only exponent of the cube sizes in the construction |
| $q_0 = q_1$ | the exponents of the radii of analyticity, integers greater than 1 [9, (2.28), p. 259], with $q_0 > p_0 + 3r$ |
| $\nu_L$ | $p_0 + q_1 + 7r + 1$, the exponent of the depth (Definition 3.9) |
| $\beta_0 = 1/7$ | the constant of Bałaban's inequalities (2.6), which "can be chosen arbitrarily small, if $g$ is sufficiently small" [9, p. 255]; it satisfies $\beta_0 \leq \sqrt{2} - 1$ (Section 3.1) and $(1 + \beta_0)^8 \leq 3 < L/4$ (Section 6.3) |
| $\beta = 1/8$ | the constant of Bałaban's spaces of the boundary terms [9, (2.34)–(2.39), p. 261], a small positive number "but not too small" [9, p. 261]; with it $\kappa_s = \beta/30 = 1/240$ in (3.37) |
| $\varepsilon_0$ | $0 < \varepsilon_0 < 1/125$, the loss in the exponents of the comparison of the two runs (Proposition 10.6) |
| $\beta_H$, $\beta_E$, $\beta'$, $\theta_V$ | $\beta_H \geq 19/20$, $\beta_E \leq 1/20$, and $\beta'$, $\theta_V$ of (11.8) |
| Hölder exponent | a number in $[\frac{3}{4}, 1)$, written $\beta$ in Section 9 only, with $\gamma_w$ of (9.21) and $\gamma_H$ of (11.13) |
| $\Lambda_*$ | a number $\Lambda_* \geq 2$, the bound on the chord–arc constants of the admissible loops (Definition 1.4) |

*Stage* 1*: the blocking factor.* The integer $L$ is odd and satisfies (C.1), which bounds it from below in terms of $d$, $\varepsilon_0$, and $\Lambda_*$. The constant $C_{\text{dep}} = 16L^4(L+1)^2$ of Lemma 3.10 is computed here.

*Stage* 2*: the constants computed from d, L, and G.* These are Bałaban's constants which depend on $d$ and $L$ only: the decay rate $\delta_0$ and the constants $B_0$, $B_0(\beta)$ of the propagators [5, Theorem 3.1, (3.42), (3.43), p. 397], the constant $c_0$ of [3, (2.58)], and the constant $B_3$ of the minimisers [7, Theorem 1, p. 279]. With them come the constant $C_W$ of Theorem 4.4, the numbers $b_* = c \ln L$ and $b^* = 3c \ln L$ of (4.4), the constant $C_M$ of Theorem 10.8, which counts the ingredients of a term per point, and

$$C_P \coloneqq \max(2, 2C_M) \,, \tag{A.1}$$

the constant of Proposition 8.4; it satisfies the two requirements $C_P \geq 2$ and $C_P \geq 2C_M$ which Sections 8.3 and 11.6 place on it.

*Stage* 3*: the decay rates and the sizes.* First Bałaban's decay rate $\kappa$ of the inductive bounds, large at fixed $\beta$ [9, p. 260], and the exponent $\kappa_0 \geq 11$ of (3.29), which "can be chosen arbitrarily large, similarly as $\kappa$, if the other parameters are fixed" [9, p. 260]; then the rate $\kappa_1$ of the decoupling parameters of [10], which is bounded below in terms of $\kappa$ and $L$. Then

the sizes, each bounded below in terms of the earlier constants and the sizes before it, in the order: the size $R_0M_0$ of [5, p. 408], enlarged beyond Bałaban's smallest choice, which leaves his results valid because "in [4] we have proved all theorems under the assumption that $R$, $M$ are sufficiently large" [5, p. 408]; the integer $K$ and the block size $KR_0M_0$ of [5]; the size $M_1$ of the cubes for which the theorems of [5, 7, 6] hold, with $M_2 = L^2M_1$ [9, p. 245]; the separation $R$ of the walk problems [3, (2.2)]; and the cube size $M = L^m$, which is "sufficiently large" [9, p. 245].

*Stage* 4*: the regularity parameters of the propagators.* The parameters $\alpha_0$ and $\alpha_1$ of the domains of [5, (3.35)–(3.37), p. 396], with $\alpha_J \coloneqq KR_0M_0\alpha_0$ and the radius $\alpha'_* = \min(\alpha_1, \alpha_J/\hat{x}_J)$ of Supplement, Proposition S.12(g), bounded above in terms of the stages 0–3.

*Stage* 5*: the parameters of the variational problem.* The parameter $\varepsilon_3$ of [7, Proposition 4, p. 292], and then the data parameter $\varepsilon_1$ of [7, Theorem 1, p. 279] at which its theorems are applied, bounded above in terms of the stages 0–4. In Sections 10 and 11 the letter $\varepsilon_1$ denotes the data parameter (10.11) of a history, a function of $\bar{g}$ which lies below this fixed value once $\bar{g}$ is small.

*Stage* 6*: the coupling.* First Bałaban's large constants: $A_1$ and then $A_0$ of the thresholds [9, p. 246, p. 265], with $A_1/A_0$ small [9, p. 247], and $C_0^\alpha$, $C_1^\alpha$ of the radii [9, (2.28), p. 259], each bounded below in terms of the stages 0–5. Then the bound $\gamma$, small in terms of everything before it: the conditions (3.7) and (3.38), $\gamma \le \gamma_1$ of Proposition 4.3, the conditions (T1) of Appendix C, and the requirements "$\bar{g}$ sufficiently small" of the statements which involve no representation, where $\bar{g} \le \gamma$ by (3.2). The requirements which involve the representations are upper bounds on the number $\gamma_q$ of (3.46), and they are met by the choice of the level $q$ of the ensemble, which comes after the coupling. The integer $K$ and the tube are chosen last. The thresholds $\varepsilon_j$, $\vartheta_j$, $\delta'_j$, the radii $\alpha_{0,j}$, $\alpha_{1,j}$, the cube sizes $R_j$, and the depths $N_k$ of Definition 3.3 and Definition 3.9 are functions of the tube.

*The derived constants.* Every other constant of the paper is computed from those above and is introduced where it first occurs. They fall into five groups, which differ in what they may depend on.

(a) The constants of Bałaban's theorems, $E_0$, $E_1$, $E_\pm$, $R_1$, the constant $B_0$ of the boundary terms, and $B_\partial$ (Section 3.2), depend on the constants of the construction and not on the tube, the step, $K$, or the torus (Proposition 3.12). The constants $C_\Lambda$, $C_3$, $C_{\mathrm{rem}}$, $C_R$, and $\gamma_1$ of Proposition 4.3 are of the same kind.

(b) The constants of the inductive class (Definition 5.1): the constants $C_\tau$, $c_\tau$, $C_\rho$, $C_t$ of (5.5), the box constant $C_{\mathrm{box}} = 2916$, the reserve $\nu$ of the first step [1], taken with $\nu \le \eta(\mathcal{C})$ of the proof of Theorem 5.2, the extent rate $\nu_e \coloneqq \nu/(2C_{\mathrm{box}})$ and the cross rate $\nu_\times \le \nu_e$, the rates $\eta_w$, $c_{\mathrm{att}}$, $c'_*$, $\kappa_e$, $c_{\mathrm{th}}$, and the constants $C_\chi$, $C_{\mathrm{new}}$, $C_{\mathrm{rel}}$, $C_{\mathrm{cr}}$, $C_{S'}$. For ensembles with admissible loops they depend on the constants of the construction, among them $\Lambda_*$, and on the representations through $c_{\mathcal{C}}$, as nondecreasing functions of it, and on nothing else (Definition 5.1(v)).

(c) The constants of the stopping scale, $C_{\mathcal{E}}$, $C_\partial$, $C_\lambda$, $C_{\mathrm{fs}}$, $C_A$, $C_e$, and $C_{\mathrm{ent}}$ of Section 8, which are of the kind (a), and the constant $c_\pi$ of (1.7), which depends on the representation. The numbers $B_E(\mathcal{C})$ and $I_*(\mathcal{C})$ of Section 8.4 are not constants of a stage: they are bounds attached to an ensemble, which depend on $\bar{g}$ and not on $K$.

(d) The constants of the comparison of two cutoffs: $C_{\mathrm{df}}(\beta)$ and $C_\square(\beta)$ of Section 9.3, $C_w$ of Theorem 9.5, $C_G(\varepsilon_0)$ and $C_U$ of Proposition 10.6, $C_f$, $C_{\mathrm{sh}}$, $C_h$ of Section 10, $E'_1$, $E_V$,

$C_\alpha$ of Proposition 11.5, and $C_{\mathcal{T}}$, $C_s$, $C''_S$, $C_a$, $C_\beta$ of Section 11.4. They do not depend on $K$, the step, the lattice spacing, or the torus; some of them depend on $\bar{g}$ through the stopping index $j_q$ and through the bound $\bar{\varepsilon}_1$ of (10.11), as stated where they are introduced.

(e) The rates of the sums over $K$: $\tilde{\theta}$ of (11.16), which depends on $\gamma$ through $\varepsilon^*$, and $\ell_K$, $\mu_K$, $w_K$ of (11.17).

## Appendix B Conventions

*Lattices and units.* The lattice of the scale $j$ of a run with $K$ steps has the spacing $a_j = L^j\varepsilon$, with $\varepsilon = uL^{-K}$ the finest spacing, and a length "in units of the scale $j$" is a length divided by $a_j$. Distances are $\ell^\infty$-distances unless stated otherwise. The bonds and the plaquettes of a set are those which intersect it [8, p. 251].

*Cubes and partitions.* For a cube $\square$ of the family $\pi_j$, of side $M$, the cube $\tilde{\square}^n$ is the cube of side $(1+2n)M$ with the same centre [8, p. 257]. The operation $\sim$ applied to a set adds one layer of the cubes of the family in terms of which the set is defined [9, p. 245], a cube being added when it has a point in common with the set, and $X^{\sim -1} \coloneqq ((X^c)^\sim)^c$ [9, p. 247]. Partitions are compatible, every cube of a coarser partition being a union of cubes of a finer one [9, p. 245], and every partition consists of cubes centred at the points of $s\mathbb{Z}^4$, $s$ its side (Section 7.1).

*Norms.* On matrices $\|\cdot\|$ is the operator norm (Section 8.2), which is invariant under multiplication by unitary matrices, and $|\cdot|$ is the norm in which Bałaban states the regularity of the fields; the constant $c_h \geq 1$ of Section 8.2 compares the two. The trace $\mathrm{tr} = N^{-1}\,\mathrm{Tr}$ is normalised.

*Letters with several meanings.* A few letters keep the meanings which they have in the papers that we cite, and the context separates them.

(i) $\beta$ is Bałaban's constant $1/8$ of the spaces of the boundary terms in Sections 3–8 and in the coefficients $1-\beta(1-2^{-(k-m)})$ of the radii; the exponent $\beta < 1$ of the age bound (3.31), which is $1-\beta_E$, in Section 3 and in Lemma 11.4; and the Hölder exponent of stage 0 in Section 9.

(ii) $M$ is the cube size $L^m$ of [9, p. 245] and [8, p. 257], except in Section 9.2 and in the bounds (9.6), (9.20), where it is the block size $KR_0M_0$ of [5, p. 408]; $K$ is the number of steps, except in $KR_0M_0$.

(iii) $\kappa$ is the number of Lemma 4.2 in Section 4 and Bałaban's decay rate elsewhere.

(iv) $B_0$ is the constant of the boundary terms [9, (2.42)] in Sections 3 and 10, and the constant of the propagators [5, (3.42)] in Section 9.

(v) $\delta'$ without an index is the deviation of the coefficients in Theorem 4.1 and Proposition 4.3; $\delta'_j$ is the threshold of the **R**-operation.

(vi) $r$ is the exponent of stage 0; $r(C)$ and $r_0(C)$ are the lengths of a loop of Section 2, $r_k$ the radius of the fluctuation cutoff, and $r_G$ the radius of the logarithmic chart.

## Appendix C The Conditions and Their Sources

We list the conditions on the constants by the stage of Appendix A to which they belong, with the place where each is used and, where it is Bałaban's, where he imposes it. Each of them bounds a quantity of its stage from below, or a quantity of a later stage from above, in terms of the earlier stages; so all of them hold once the stages are chosen in order, and none of them involves the number of loops, their sizes, or $K$.

*Stage* 1. The blocking factor $L$ is an odd integer, as in [8, p. 251], with

$$L \geq L_0(\Lambda_*) \,, \qquad L_0(\Lambda) \coloneqq \max(2^{52},\, 4\Lambda^2 L_0^\times) \,, \qquad L_0^\times \coloneqq 4 \cdot 10^4 \cdot 640(2^{d+2}+8) \,, \tag{C.1}$$

so that $L_0^\times = 1.8432 \cdot 10^9$ at $d = 4$. The two terms have different origins. The term $2^{52} = 4^{26}$ gives the inequality $L^{5(1-\varepsilon_0)/128} \geq 4$, because $128/(5(1-\varepsilon_0)) < 128 \cdot 125/620 < 26$ for $\varepsilon_0 < 1/125$; this inequality is used in Supplement, Proposition S.14, for the variation of the terms of the effective action between the scales of a young history, where a factor $2^n$ from the radii of the analyticity domains has to be beaten by the decay $L^{-5(1-\varepsilon_0)n/128}$. The term $4\Lambda^2 L_0^\times$ is the condition of the counts of the inductive step (Section 6.3): the shares of the costs of a group of the step in the reserve $\nu$ and in the extent rate $\nu_e$ of Definition 5.1 grow linearly in the chord–arc constant $\Lambda$ through two factors, the counts of Lemma 5.3 and the extent of the insertions made by the **R**-operation, and they remain within their margins when $L \geq 4\Lambda^2 L_0^\times$, and this bound contains the lower bounds $5000\Lambda$, $3(1+\Lambda)$, and $20\Lambda$ of the same counts. Since $L_0$ increases and every admissible loop has $\Lambda(C) \leq \Lambda_*$, the condition (C.1) covers all of them at once. It covers as well the other lower bounds on $L$ of the paper and of the supplement, all of which are below $2^{52}$: $L > 11$ [8, p. 251], $L \geq 3$ (Lemma 3.10), $L \geq 5$ in (7.10), $(1+\beta_0)^8 \leq L/4$ (Section 6.3), $L \geq 21$ (Lemma 8.2), $L \geq 195$ (Lemma 7.5), and $L \geq 12d+6$, $L^2 > 9d$, $L \geq 128(1+\beta_0)$ (Supplement, Propositions S.12–S.14).

*Stage* 3. The rate $\kappa_1$ satisfies $\frac{1}{2}(\kappa_1 - 1) \geq 2L\kappa$ [10, p. 20]. The size $R_0 M_0$ satisfies the two conditions of Lemma 9.4, $\frac{1}{6}\delta_0 R_0 M_0 > 8 \log c_0(\frac{1}{6}) + 1$ and $4\delta_0 R_0 M_0 \geq 3(1+\beta)\log L$, with the Hölder exponent $\beta$, and the conditions of the same form $c\delta R_0 M_0 \geq C \log L + C'$, with a rate $\delta$ of stage 2 and numbers $c$, $C$, $C'$ depending on $d$ and on the Hölder exponent, of Supplement, Proposition S.12; Bałaban's results hold for every such enlargement of $R_0 M_0$ [5, p. 408]. The integer $K$ of [5, p. 408] is a power of $L$ with $K \geq 4$, and the separation of the walk problems is $R \geq 12dL$. The cube sizes satisfy Bałaban's lower bounds $M \geq M(\kappa)$ [8, Theorem 3, p. 264], $\delta_0 M \geq \kappa$ [10, p. 21], $\delta_1 M \geq \kappa_1$ [10, p. 5], and $\delta(M/M_1) \geq 2$ [11, p. 186], and the lower bounds on $M_1$ and $M$ of Supplement, Propositions S.12–S.14, each of which has the form $c\delta M_1 \geq C \log L + C'$ or $M \geq C(d, L) M_1$.

*Stage* 4. The parameters $\alpha_0$ and $\alpha_1$ are so small that "$O(1)M\alpha_0$ is still a sufficiently small number" [5, p. 396], with $M$ the block size $KR_0M_0$, and that the upper bounds of Supplement, Proposition S.12, hold; these involve the constants of the stages 0–3 only.

*Stage* 5. The parameters of the variational problem satisfy $\varepsilon_3 \leq a_3$ [7, Proposition 4, p. 292] and $\varepsilon_1 \leq a_1$, and every cube of the parameter $M_V$ at which [7, Theorem 1] is applied has $M_V \leq M(\varepsilon_1) = R_1 M_1 a_1/\varepsilon_1$ [7, p. 279], an upper bound on $\varepsilon_1$; the further upper bounds on $\varepsilon_3$ and $\varepsilon_1$ in terms of $\alpha_0$, $\alpha_1$, and the earlier constants are those of Supplement, Propositions S.12 and S.13.

*Stage* 6. The constants of the thresholds satisfy $A_1 \geq 48(15/8)^{1/2}$ and $A_0^2 \geq \frac{15}{8} A_1^2 \max(8(1+\beta_0)^2 B_3^2,\, 4O(1)B_3 M^2)$, with the constant $O(1)$ of [12, p. 381], which keep the comparisons of [12, pp. 381–382] valid when the coefficient of the action is reduced by the factor 15/8 (Supplement, Proposition S.1). The bound $\gamma$ satisfies (3.7) and (3.38), $\gamma \leq \gamma_1$ of

Proposition 4.3, the bound $\delta' \le (2/5)c \ln L$ of Theorem 4.1, which holds for $\gamma$ small because $\delta' \to 0$ with $\gamma$, and $C_W \gamma^2 \le 11/40$ (Supplement, Lemma 2.1). Moreover $\gamma$ satisfies the following conditions, which we call (T1): for every $g \le (1+\beta_0)\gamma$,

$$\varepsilon^*(g) C_{\mathcal{T}} \le \frac{1}{4}\,, \qquad 4\hat{\varepsilon}^*(g) N'\Big(\frac{g}{1+\beta_0}\Big) \le 1\,, \qquad 4 C_{\mathrm{ov}} \varepsilon^{*B}(g) N_0^+(g) W^\sharp \le 1\,. \tag{C.2}$$

Here $\varepsilon^*(g)$ is the coefficient $\varepsilon_k^*$ of (11.14) as a function of the coupling, $C_{\mathcal{T}} \ge (1 - L^{-\beta'})^{-1}$ the constant of Supplement, Proposition S.16(e), $\hat{\varepsilon}^*(g) := \sup_{g' \le 2g} \varepsilon^*(g')$, and $N'(g) := \log_L(MR(g)) + 3$, with $R(g)$ the smallest power of $L$ which is at least $(\log g^{-2})^r$, the bound on the number of steps through which a recent **R**-term is carried without contraction. In the third condition $\varepsilon^{*B}(g)$ is the coefficient of the recursion of the boundary terms, a constant times $g p_0(g)$, $N_0^+(g) := 1 + 2(\log(32L(L+1)) + 2\log\log g^{-2})/\log L$ bounds the depth, $C_{\mathrm{ov}} = 2$ counts the generations of created boundary terms at a point, and $W^\sharp$ is a constant of $L$ and $d$ (Supplement, Proposition S.14). Each condition holds for $\gamma$ small: $\varepsilon^*(g)$ decays as a negative power of $\log g^{-2}$, $\varepsilon^{*B}(g)$ as $g$ times a power of it, while $N'$ and $N_0^+$ grow as double logarithms; the running couplings of the histories which the conditions concern are at most $(1+\beta_0)\bar{g} \le (1+\beta_0)\gamma$ by (3.9). None of the conditions on $\gamma$ involves a loop or a representation, so that the construction, and with it the step factors of the normalisation, is the same for all the ensembles.

Every statement of the paper which requires the coupling to be sufficiently small imposes one of two kinds of conditions (Section 1.3). If it involves no representation, it is an upper bound on $\bar{g} \le \gamma'$ in terms of the constants of the construction and of $\Lambda_*$, and, where it concerns a given ensemble, of the data of the ensemble which the statement names. If it involves the representations of an ensemble $\mathcal{C}$, it concerns the steps $k \le K'$ at which the loops are carried, and it is an upper bound on the number $\gamma_q$ of (3.46) in terms of the constants of the construction, of $\Lambda_*$, and of $c_{\mathcal{C}}$: these are the conditions of Theorem 5.2, of Sections 6.3 and 6.4, of Lemma 7.3, Proposition 7.6, and Corollary 7.11, the condition (8.40), and the condition (11.20) on the coefficient $\varepsilon^Q(g)$ of the old vertices in the recursion of the insertions (Supplement, Proposition S.16(f)), together with those of Supplement, Propositions S.9 and S.10. They are not conditions on the constants of a stage: they are met by the choice of the level $q$ in Theorem 1.5.

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